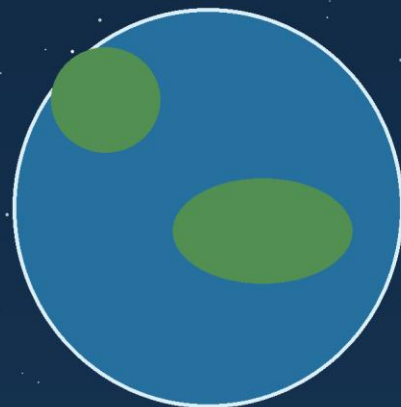


ISO OLYMPIAD PREPARATION JUNIOR LEVEL

SPACE SCIENCE • CLASSES 5-6

2026-2027 EDITION



INDIAN SPACE SCHOOL

Preparation for the Indian Space Olympiad (ISO)

ISO OLYMPIAD PREPARATION

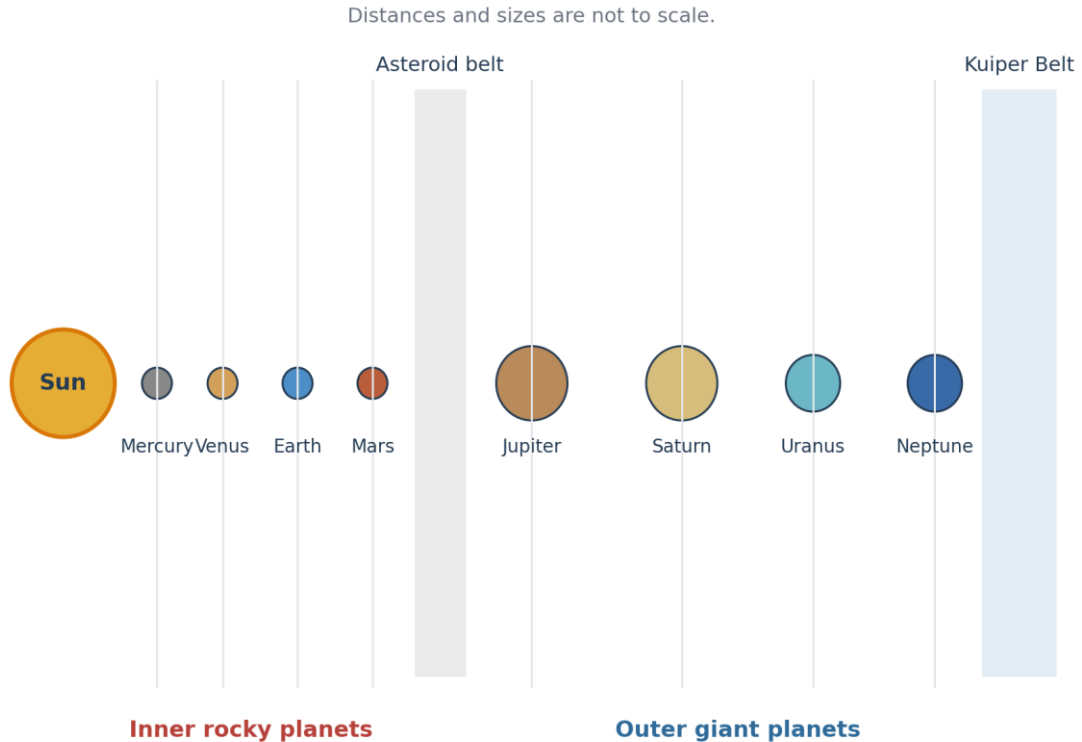
JUNIOR LEVEL

Space Science • Classes 5-6

ISO Olympiad Preparation - Junior Level

Space Science for Classes 5-6 | 2026-2027 Edition

The Solar System: Orbits and Regions



An original overview illustration for the junior-level course.

Preparation for the Indian Space Olympiad (ISO)

Indian Space School

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Preface

Space science connects observations made from a school courtyard with ideas that reach across the Solar System and beyond the Milky Way. This textbook begins with the visible sky, then builds the physical explanations needed to understand planets, stars, satellites, rockets and missions. The sequence is deliberate: learners first practise describing evidence, then use models and basic physics to explain what they observe.

The book follows the supplied Indian Space Olympiad junior-level syllabus without turning it into a list of isolated facts. Every chapter develops a small set of connected ideas, uses an original diagram or structured visual, and ends with questions that move from recall to reasoning. Activities use common materials and include limits or safety notes. Current mission information is dated because space programmes change.

Olympiad preparation should deepen understanding rather than reward hurried memorisation. A student who can explain why a phase occurs, interpret a graph, identify a misleading claim and justify a mission design is better prepared than one who has memorised many numbers without relationships. This edition therefore gives equal attention to concepts, communication and evidence.

Editorial convention

SI units are used. “Moon” with a capital M refers to Earth’s natural satellite. Mission status words such as planned, launched, operational and completed are used precisely and with dates where needed.

About This Book

This is a complete foundation-level textbook for Classes 5-6 and learners approximately 10-13 years old preparing for a junior space-science Olympiad. It is suitable for classroom teaching, guided home study and structured revision. The explanations assume ordinary school mathematics and science, while introducing selected terms used in astronomy and space technology.

Intended learners

- students beginning formal astronomy and space-science study
- Olympiad candidates who need conceptual rather than fact-only preparation
- teachers seeking a coherent sequence with safe activities
- parents supporting observation, reading and problem solving

What the book does

- covers every item in the supplied 11-part syllabus
- uses original diagrams designed for print and grayscale readability
- separates stable concepts from dated mission awareness
- provides chapter exercises, unit reviews, Olympiad challenges and two full practice papers
- includes a complete answer section and brief guidance for descriptive responses

Important limitation No textbook can replace live mission updates. For current affairs, always check the date and an official agency source before an examination.

How to Use This Textbook

Stage	Method
Before reading	Study the opening question and diagram. Write what you already think, including uncertainty.
During reading	Pause at “Check your understanding” prompts. Explain the idea without looking back.
After a section	Redraw the diagram from memory and label only relationships you can explain.
Chapter practice	Complete questions in order. Mark the reason for each MCQ option you reject.
Unit review	Mix concepts from several chapters and identify which principle each problem uses.
Current awareness	Write the source date beside every mission update. Never convert “planned” into “completed.”

A weekly study rhythm

1. Read one concept page and make a one-minute oral summary.
2. Complete the activity or analyse its expected result.
3. Attempt chapter questions without notes.
4. Correct errors using the explanation, not only the answer letter.
5. Finish with one mixed reasoning problem or a current mission source check.

Guidance for Teachers and Parents

The most useful support is to ask for evidence and reasoning. Avoid supplying the answer immediately. Ask: “What did you observe?”, “Which diagram supports that?”, “What would change your conclusion?” and “Is this claim dated?”

Observation

Choose safe, dark locations; permit clouded-out entries; never reward unsafe Sun viewing.

Models

Require learners to state what is not to scale and which real features the model omits.

Numerical work

Insist on units, estimation and a final reasonableness check before calculator use.

Olympiad practice

Discuss distractors. The reason an option is wrong often reveals the misconception being tested.

Current affairs

Prefer official agency pages and primary research summaries. Record the retrieval date.

Inclusive learning

Allow oral explanation, labelled sketches and structured tables alongside extended prose where appropriate.

Safety Activities are written for common school settings, but adults must review local conditions, allergies, electrical risks, moving objects and outdoor access before use.

Syllabus Overview

Unit	Syllabus focus	Chapters
1. Fundamentals of Astronomy	Learning to read the sky begins with careful observation, shared directions and tools that extend human sight.	1, 2, 3
2. The Solar System	The Sun and the worlds that travel around it form a connected system shaped by gravity, motion and a common origin.	4, 5, 6, 7
3. Stars, Galaxies and the Universe	Stars are distant suns. In groups and galaxies, they reveal a universe far larger and older than the Solar System.	8, 9, 10
4. Earth and Moon	Earth's motions and its partnership with the Moon produce familiar cycles: day, year, seasons, phases, eclipses and tides.	11, 12, 13
5. Satellites and Space Technology	Satellites turn orbital motion into practical services for communication, navigation, weather study and observation of Earth.	14, 15
6. Rockets and Launch Vehicles	A launch vehicle must overcome gravity, control its path and deliver a payload to the correct orbit.	16, 17

The first six units establish observing language, Solar System structure, Earth–Moon cycles, satellite orbits and launch physics. They provide the prerequisites for understanding missions rather than treating mission names as isolated facts.

Syllabus Overview (continued)

Unit	Syllabus focus	Chapters
7. ISRO and India's Space Missions	India's space programme combines launch systems, science missions and applications that serve society.	18, 19, 20
8. Space Exploration	Robotic spacecraft and human crews explore places that cannot be studied fully from Earth.	21, 22, 23
9. Basic Physics for Space Science	Forces, energy, waves, electricity and magnetism explain how spacecraft move, communicate and operate.	24, 25, 26
10. Scientific Reasoning and Logical Aptitude	Olympiad success depends on observing evidence, recognising patterns and explaining how a conclusion follows.	27, 28, 29
11. General Space Awareness	Space science changes rapidly, so mission claims must always be connected to reliable sources and dates.	30

Learning progression

- Observe: directions, patterns, instruments and records
- Organise: objects, systems, categories and scales
- Explain: motion, gravity, energy, waves and technology
- Apply: mission design, data interpretation and problem solving
- Verify: current status, sources, uncertainty and answer checking

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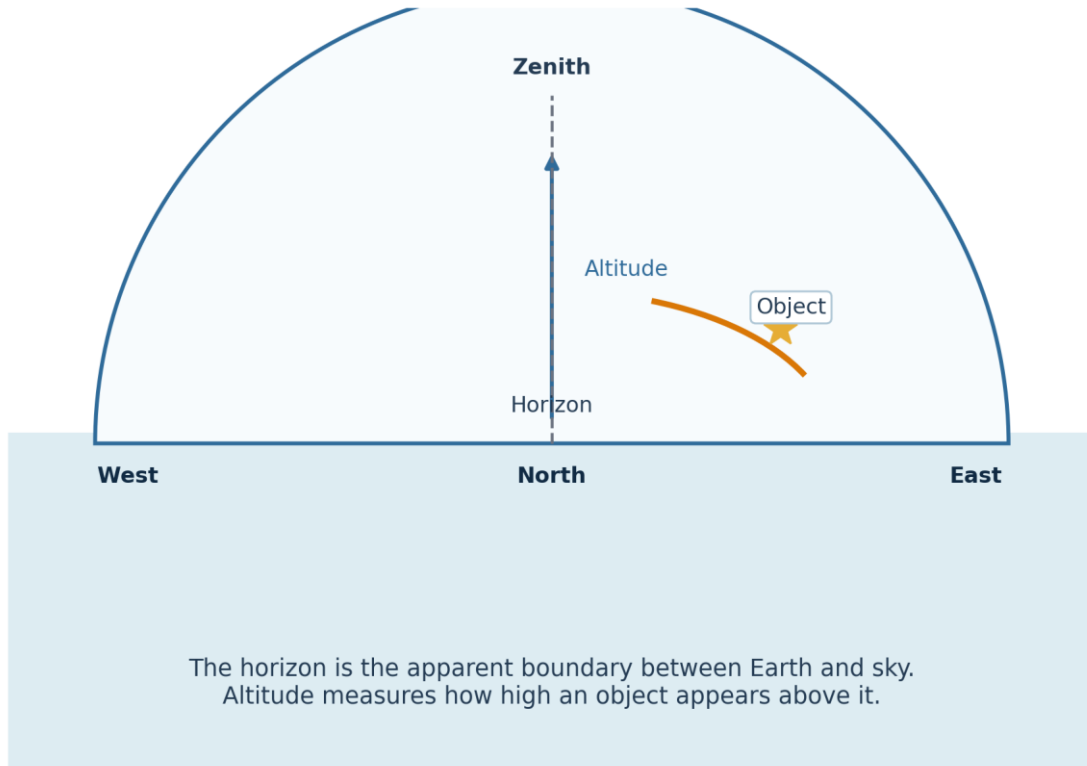
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UNIT 1

Fundamentals of Astronomy

Learning to read the sky begins with careful observation, shared directions and tools that extend human sight.

Finding Directions in the Sky



Unit 1 opening visual - original illustration.

Major concepts

- Chapter 1: Astronomy and the Night Sky
- Chapter 2: Directions, Constellations and the Celestial Sphere
- Chapter 3: Telescopes and Astronomical Events

Expected unit outcomes

- Distinguish astronomy from astrology.
- Use cardinal directions and altitude.
- Compare refracting and reflecting telescopes.

CHAPTER 1

Astronomy and the Night Sky

Astronomy is the scientific study of objects and events beyond Earth. It began with people noticing repeated patterns: sunrise, Moon phases, seasonal star groups and the wandering paths of planets. Modern astronomy still begins with observation, but it also uses measurement, physics, mathematics and instruments that detect forms of light invisible to our eyes. A clear night sky is therefore both a view and a source of evidence.

Opening question When two observers describe the same bright object, how can they make sure they are pointing to the same place?

Learning outcomes

- Distinguish astronomy from astrology.
- Classify common celestial objects.
- Describe daily apparent motion in the sky.
- Keep a simple observation record.

Finding Directions in the Sky

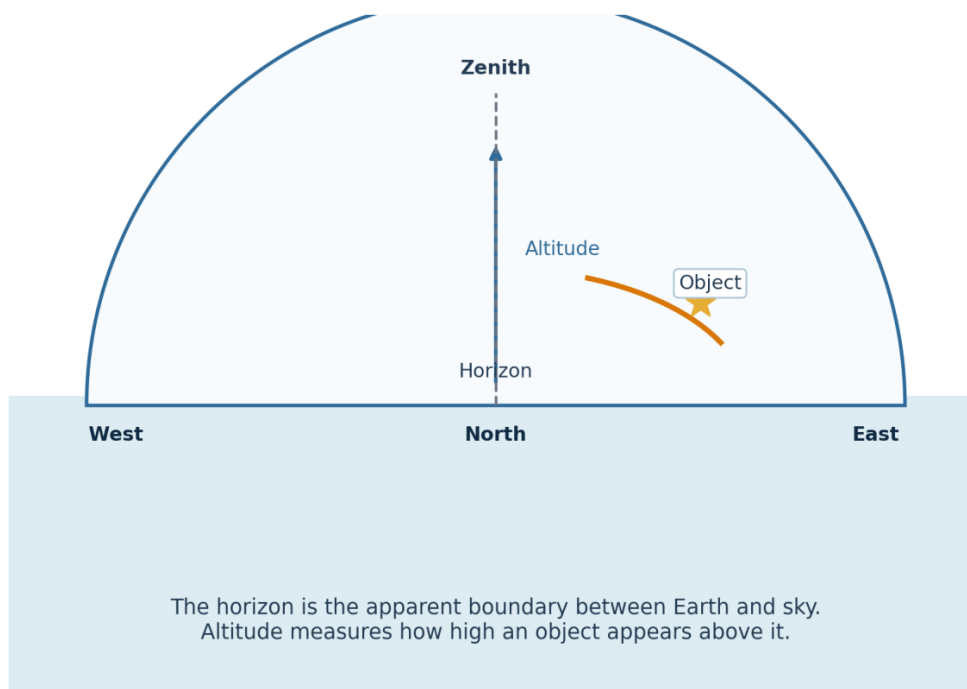


Figure 1.1 Astronomy as a science. Original illustration; not to scale where stated.

Astronomy as a science

A science does more than collect names. It asks testable questions, records evidence and changes explanations when better evidence appears.

Astronomy

Astronomy means the study of celestial objects, space and the physical universe. It works because astronomers compare observations from different times, instruments and places to test explanations. A concrete case is measuring how a planet changes position night after night. By contrast, astrology uses evidence; astrology makes personal claims that are not tested in the same scientific way.

Observation

A careful, purposeful use of senses or instruments to gather information. The underlying relationship is that good observations include time, place, conditions and a description that another person could check. This can be seen when writing “8:10 p.m., clear western sky, bright point above the trees” instead of “I saw a star”. An observation reports what was detected; an inference explains what it may mean.

Check your understanding Why is a dated sketch more useful than a memory of what the sky looked like?

What we see above us

The word celestial means “related to the sky or space.” Celestial objects differ greatly in size, distance and behaviour.

Celestial object

A natural object located beyond most of Earth’s atmosphere. The underlying relationship is that objects may produce light, reflect light or block light, and their apparent motion depends on both their own movement and Earth’s movement. This can be seen when the Sun is a star, the Moon is a natural satellite and Venus is a planet. A bright point is not automatically a star; planets can also look star-like.

Apparent motion

The phrase Apparent motion refers to motion that an observer sees, which may be caused partly by the observer’s own motion. Here, Earth’s rotation makes most sky objects appear to travel from east to west during a night. Consider the following case: a constellation rises in the east and later sets in the west although its stars have not circled Earth that night. The careful interpretation is that apparent motion describes the view; true motion describes movement in space.

Check your understanding Name one object that makes its own light and one that mainly reflects light.

Recording the sky

Repeated observations reveal patterns that a single glance cannot show. A sky journal is a basic scientific instrument.

Sky journal

An organised record of observations made over several nights or weeks. The process can be followed as using the same location and similar time reduces unnecessary differences and makes patterns easier to compare. For instance, drawing the Moon's shape and position every evening for a month. A common overstatement is avoided by recognising that a journal is not an art competition; labels, dates and consistency matter more than decoration.

Light pollution

Light pollution concerns unnecessary or poorly directed artificial light that brightens the night sky. The key cause-and-effect link is that scattered light reduces contrast, so faint stars and the Milky Way become harder to see. This appears in a rural sky may reveal many more stars than a city centre even on equally clear nights. At the same time, clouds can make city glow worse, while darkness alone cannot overcome thick cloud or haze.

Check your understanding Which details should every sky-journal entry contain?

Activity: Make a seven-night sky journal

Aim	to recognise repeated change in one selected sky object
Materials	notebook, pencil, clock and a safe observing place
Steps	1. Choose the Moon or one bright object that is visible from the same place. 2. Observe at nearly the same time on seven suitable evenings. 3. Record date, time, direction, height above the horizon and weather. 4. Add a small labelled sketch and one sentence describing change.
Observe	Look for change in position, brightness or visible shape, and note nights when weather prevents observation.
Expected learning	Repeated records allow evidence-based comparison.
Safety	Observe with an adult from a safe location; never stare at the Sun.

Chapter Review

Key ideas

- The study of celestial objects, space and the physical universe.
- A natural object located beyond most of earth's atmosphere.
- An organised record of observations made over several nights or weeks.
- "Astronomy is mainly a list of interesting facts." Correct this by describing its methods.

Important terms

astronomy: scientific study of space and celestial objects	celestial object: natural object beyond most of Earth's atmosphere
observation: careful collection of information	inference: an explanation based on observations
apparent motion: motion as it seems from an observer's viewpoint	sky journal: dated record of sky observations
light pollution: artificial brightening of the night sky	horizon: apparent boundary between Earth and sky

A. Choose the correct answer

1. Which term names scientific study of space and celestial objects?

- A. celestial object
- B. astronomy
- C. inference
- D. sky journal

3. Choose the vocabulary item defined as careful collection of information.

- A. inference
- B. sky journal
- C. horizon
- D. observation

5. In this chapter, what is defined as motion as it seems from an observer's viewpoint?

- A. sky journal
- B. apparent motion
- C. horizon
- D. celestial object

2. A learner describes natural object beyond most of Earth's atmosphere. What is the correct term?

- A. observation
- B. apparent motion
- C. celestial object
- D. light pollution

4. Which expression would a scientist use for an explanation based on observations?

- A. inference
- B. apparent motion
- C. light pollution
- D. astronomy

Exercises (continued)

6. Which statement best explains Astronomy?

- A. good observations include time, place, conditions and a description that another person could check
- B. Earth's rotation makes most sky objects appear to travel from east to west during a night
- C. astronomers compare observations from different times, instruments and places to test explanations
- D. astronomy uses evidence; astrology makes personal claims that are not tested in the same scientific way

8. Which statement correctly avoids a common misunderstanding about Celestial object?

- A. a bright point is not automatically a star; planets can also look star-like
- B. apparent motion describes the view; true motion describes movement in space
- C. an organised record of observations made over several nights or weeks
- D. objects may produce light, reflect light or block light, and their apparent motion depends on both their own movement and Earth's movement

10. Which example most directly illustrates Sky journal?

- A. a rural sky may reveal many more stars than a city centre even on equally clear nights
- B. measuring how a planet changes position night after night
- C. drawing the Moon's shape and position every evening for a month
- D. a journal is not an art competition; labels, dates and consistency matter more than decoration

7. Which example most directly illustrates Observation?

- A. the Sun is a star, the Moon is a natural satellite and Venus is a planet
- B. a constellation rises in the east and later sets in the west although its stars have not circled Earth that night
- C. an observation reports what was detected; an inference explains what it may mean
- D. writing "8:10 p.m., clear western sky, bright point above the trees" instead of "I saw a star"

9. Which statement best explains Apparent motion?

- A. using the same location and similar time reduces unnecessary differences and makes patterns easier to compare
- B. Earth's rotation makes most sky objects appear to travel from east to west during a night
- C. astronomers compare observations from different times, instruments and places to test explanations
- D. apparent motion describes the view; true motion describes movement in space

B. Answer in one or two sentences

1. Define astronomy and celestial object in your own words.
2. Explain how Observation works. Use one cause-and-effect link.
3. Compare Celestial object and Apparent motion. State one similarity or connection and one difference.
4. Use the example of Sky journal to explain why context or measurement matters.

C. Answer in detail

1. Explain the main ideas of "Astronomy and the Night Sky" as a connected account. Include at least four important terms from the chapter.
2. A student says, "That bright point must be the brightest star." Explain what additional observations would be needed.

D. Apply and reason

1. If Earth stopped rotating but still orbited the Sun, the nightly east-to-west sweep would not occur in the same way.
2. "Astronomy is mainly a list of interesting facts." Correct this by describing its methods.

E. Activity or project

Create a one-page guide for reducing unnecessary outdoor lighting around a home or school.

CHAPTER 2

Directions, Constellations and the Celestial Sphere

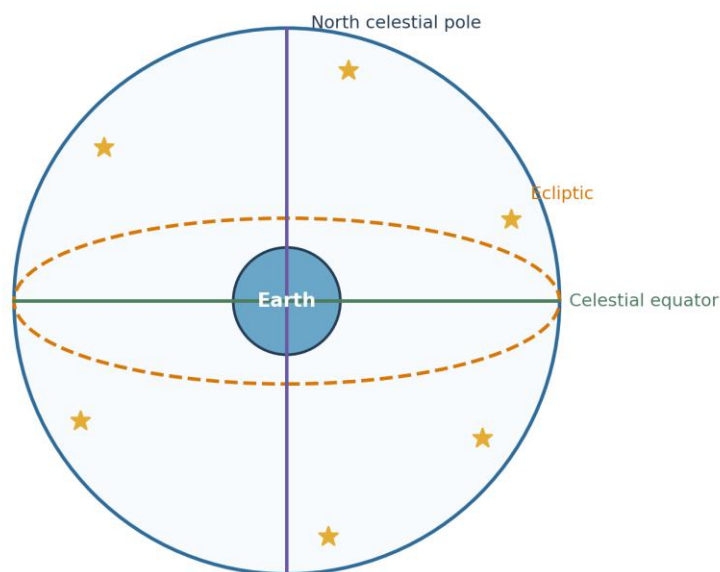
The sky seems like a vast dome surrounding an observer. Astronomers use directions, angles and an imaginary celestial sphere to describe positions on that dome. Constellations divide the sky into recognised regions and help observers communicate. These patterns are useful, but the stars that appear side by side may be separated by enormous distances.

Opening question How can a sky pattern remain recognisable even though its stars are not physically connected?

Learning outcomes

- Use cardinal directions and altitude.
- Explain the celestial-sphere model.
- Distinguish a constellation from an asterism.
- Use a simple star map.

The Celestial Sphere: A Useful Model



The sphere is imaginary. It helps observers describe apparent positions on the sky.

Figure 2.1 Directions on the sky. Original illustration; not to scale where stated.

Directions on the sky

Sky directions begin with the same north, south, east and west used on the ground, then add angular height.

Horizon

The apparent boundary where the ground or sea seems to meet the sky. The underlying relationship is that every direction around an observer meets the horizon; objects rise above it and later approach it again. This can be seen when a star just above a distant building has low altitude. The horizon is an apparent circle, not a physical edge of Earth.

Altitude

The phrase Altitude refers to the angle of a celestial object above the horizon. Here, altitude is measured from 0 degrees at the horizon to 90 degrees at the zenith. Consider the following case: an object halfway from horizon to overhead has an altitude near 45 degrees. The careful interpretation is that altitude is an angle, not the object's distance from Earth.

Check your understanding What is the altitude of an object at the zenith?

The celestial-sphere model

A model can be useful even when it is not a literal structure.

Celestial sphere

The phrase Celestial sphere refers to an imaginary sphere centred on the observer or Earth on which sky positions are pictured. Here, astronomers project directions to stars and planets onto the sphere to organise coordinates and apparent paths. Consider the following case: a star map represents part of this imaginary sphere on a flat page. The careful interpretation is that stars are not attached to a shell and are not all equally far away.

Celestial poles and equator

Sky directions obtained by extending Earth's axis and equator outward. In practice, the apparent daily rotation of the sky is organised around the celestial poles. An example is near the north celestial pole, stars trace small circles during a long exposure. However, the celestial equator is not Earth's equator in the atmosphere; it is a projected reference circle.

Check your understanding Why is the celestial sphere called a model rather than an object?

Patterns and maps

Humans remember shapes easily, so star patterns became practical signposts.

Constellation

Constellation concerns one of the officially defined regions of the sky, traditionally associated with a star pattern. The key cause-and-effect link is that modern boundaries ensure every sky position belongs to one constellation region. This appears in Orion is recognised by a set of bright stars and also names a larger defined region. At the same time, a constellation is a viewing pattern, not a close cluster whose stars travel together.

Asterism

A familiar star pattern that may be part of one constellation or may cross constellation boundaries. Evidence for the idea comes from situations such as the Big Dipper is an asterism within Ursa Major. The explanation is asterisms are informal but useful for finding other objects. Its boundary is equally important: an asterism is not one of the official sky regions.

Check your understanding How does rotating a star map help match it to the real sky?

Activity: Estimate angular distances with your hand

Aim	to use a rough angular scale for sky observations
Materials	your hand and a distant outdoor view
Steps	1. Stretch one arm fully. 2. Compare the width of one little finger, three middle fingers and a closed fist against distant objects. 3. Use these hand spans to estimate the altitude of a safe object such as the Moon. 4. Repeat and compare estimates with classmates.
Observe	Different hands give slightly different values, but keeping the arm fully extended improves consistency.
Expected learning	Angles allow positions to be compared without knowing distance.
Safety	Never use this activity to look at or near the Sun.

Chapter Review

Key ideas

- The apparent boundary where the ground or sea seems to meet the sky.
- An imaginary sphere centred on the observer or earth on which sky positions are pictured.
- One of the officially defined regions of the sky, traditionally associated with a star pattern.
- “All stars in a constellation are neighbours.” Explain the line-of-sight error.

Important terms

cardinal directions: north, south, east and west	altitude: angular height above the horizon
zenith: point directly overhead	celestial sphere: imaginary sphere used to map the sky
celestial pole: projected direction of Earth’s rotation axis	celestial equator: projection of Earth’s equator on the sky
constellation: officially defined region of the sky	asterism: recognisable informal star pattern

A. Choose the correct answer

1. Which term names north, south, east and west?

- A. altitude
- B. celestial sphere
- C. cardinal directions
- D. celestial equator

3. Choose the vocabulary item defined as point directly overhead.

- A. zenith
- B. celestial sphere
- C. celestial equator
- D. asterism

5. In this chapter, what is defined as projected direction of Earth’s rotation axis?

- A. celestial equator
- B. asterism
- C. celestial pole
- D. altitude

2. A learner describes angular height above the horizon. What is the correct term?

- A. zenith
- B. celestial pole
- C. constellation
- D. altitude

4. Which expression would a scientist use for imaginary sphere used to map the sky?

- A. celestial pole
- B. celestial sphere
- C. constellation
- D. cardinal directions

Exercises (continued)

6. Which statement best explains Horizon?

- A. altitude is measured from 0 degrees at the horizon to 90 degrees at the zenith
- B. the apparent daily rotation of the sky is organised around the celestial poles
- C. the horizon is an apparent circle, not a physical edge of Earth
- D. every direction around an observer meets the horizon; objects rise above it and later approach it again

8. Which statement correctly avoids a common misunderstanding about Celestial sphere?

- A. the celestial equator is not Earth's equator in the atmosphere; it is a projected reference circle
- B. stars are not attached to a shell and are not all equally far away
- C. one of the officially defined regions of the sky, traditionally associated with a star pattern
- D. astronomers project directions to stars and planets onto the sphere to organise coordinates and apparent paths

10. Which example most directly illustrates Constellation?

- A. the Big Dipper is an asterism within Ursa Major
- B. a star just above a distant building has low altitude
- C. a constellation is a viewing pattern, not a close cluster whose stars travel together
- D. Orion is recognised by a set of bright stars and also names a larger defined region

7. Which example most directly illustrates Altitude?

- A. an object halfway from horizon to overhead has an altitude near 45 degrees
- B. a star map represents part of this imaginary sphere on a flat page
- C. near the north celestial pole, stars trace small circles during a long exposure
- D. altitude is an angle, not the object's distance from Earth

9. Which statement best explains Celestial poles and equator?

- A. modern boundaries ensure every sky position belongs to one constellation region
- B. every direction around an observer meets the horizon; objects rise above it and later approach it again
- C. the apparent daily rotation of the sky is organised around the celestial poles
- D. the celestial equator is not Earth's equator in the atmosphere; it is a projected reference circle

B. Answer in one or two sentences

1. Define cardinal directions and altitude in your own words.
2. Explain how Altitude works. Use one cause-and-effect link.
3. Compare Celestial sphere and Celestial poles and equator. State one similarity or connection and one difference.
4. Use the example of Constellation to explain why context or measurement matters.

C. Answer in detail

1. Explain the main ideas of "Directions, Constellations and the Celestial Sphere" as a connected account. Include at least four important terms from the chapter.
2. Two stars appear one finger-width apart. Does this tell you their actual separation in space? Explain.

D. Apply and reason

1. At different latitudes, the celestial poles appear at different altitudes.
2. "All stars in a constellation are neighbours." Explain the line-of-sight error.

E. Activity or project

Design a simple evening star-map key showing horizon, zenith, direction labels and symbol sizes.

CHAPTER 3

Telescopes and Astronomical Events

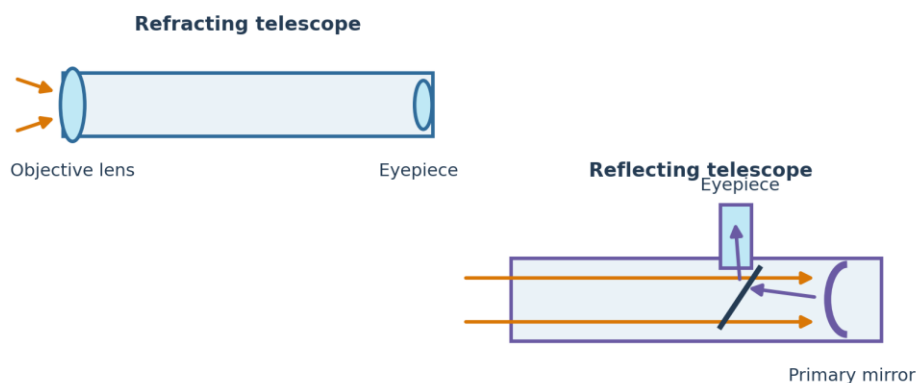
A telescope is mainly a light-collecting instrument. It can reveal faint objects and fine detail because its lens or mirror gathers far more light than a human pupil. Observers must also choose suitable times and safe methods. Some events, such as meteor showers or eclipses, are predictable, while sudden clouds or local light can still spoil the view.

Opening question Why can a larger telescope show a faint galaxy even when it does not magnify it very much?

Learning outcomes

- Compare refracting and reflecting telescopes.
- Explain light-gathering and magnification.
- Identify common observable events.
- Apply safe observing rules.

Two Common Optical Telescope Designs



Lenses bend light; mirrors reflect it. Both collect more light than the unaided eye.

Figure 3.1 How telescopes form images. Original illustration; not to scale where stated.

How telescopes form images

Optical telescopes use lenses, mirrors or both to collect and focus visible light.

Refracting telescope

The phrase Refracting telescope refers to a telescope whose main objective is a lens. Here, the curved objective bends incoming light toward a focus where an eyepiece forms a magnified view. Consider the following case: small school telescopes often use a long tube with a lens at the front. The careful interpretation is that a refractor does not work because the tube itself magnifies; the optical surfaces control light.

Reflecting telescope

A telescope whose main objective is a curved mirror. In practice, the primary mirror reflects incoming light toward a focus, often using a smaller secondary mirror to direct it. An example is many large research telescopes are reflectors because large mirrors can be supported from behind. However, a reflector is not a telescope that merely uses any flat mirror.

Check your understanding Which part of either design gathers most of the incoming light?

Light, detail and magnification

A useful telescope must balance aperture, optical quality, steadiness and magnification.

Aperture

The diameter of the main light-collecting lens or mirror. In practice, a larger aperture gathers more light and can separate finer detail under suitable conditions. An example is doubling the aperture can reveal objects too faint for a smaller instrument. However, extreme magnification cannot restore detail that the aperture, atmosphere or focus did not capture.

Magnification

To understand Magnification, begin with the apparent enlargement produced by the telescope and eyepiece combination. Next, note that changing eyepieces changes magnification, but the image may become dim and unsteady at excessive power. Moderate magnification often shows the Moon more clearly than a very high setting. This does not mean that magnification and aperture are related to different jobs: size of view versus collected light and resolution.

Check your understanding Why may a lower-power eyepiece produce a clearer view on a shaky night?

Observing events safely

Preparation is part of observation: know the event, location, timing, weather and safety limits.

Astronomical event

A noticeable occurrence involving celestial objects or their apparent positions. Evidence for the idea comes from situations such as a meteor shower is best observed over time from a dark site, not through a narrow telescope field. The explanation is events include conjunctions, eclipses, meteor showers, occultations and changing phases. Its boundary is equally important: not every bright streak is a meteor; aircraft and satellites may also cross the sky.

Solar observing safety

The use of certified methods that prevent dangerous sunlight from entering the eye. Proper front-mounted solar filters block almost all visible, infrared and ultraviolet radiation before it enters an instrument. For example, projecting the Sun through a suitable small telescope can be safe only with informed adult supervision and correct equipment. One limit must remain clear: sunglasses, smoked glass, exposed film and an unfiltered telescope are never safe solar viewers.

Check your understanding What is the single most important rule before pointing any optical instrument near the Sun?

Activity: Compare aperture using paper tubes

Aim	to model how opening size affects collected light
Materials	two cardboard tubes of different diameter, white paper and a dim indoor target
Steps	1. Stand the tubes side by side and look through each at the same dim target. 2. Keep eye distance and target distance similar. 3. Compare how much of the target and surrounding light each tube admits. 4. Explain which parts of this model represent aperture and field of view.
Observe	The wider opening admits rays from a larger area, although a paper tube does not focus them like a telescope.
Expected learning	Models can isolate one feature while leaving out others.
Safety	Use only indoors or away from the Sun; do not point tubes at bright lamps.

Chapter Review

Key ideas

- A telescope whose main objective is a lens.
- The diameter of the main light-collecting lens or mirror.
- A noticeable occurrence involving celestial objects or their apparent positions.
- “A telescope’s most important job is to make objects look huge.” Correct the statement.

Important terms

telescope: instrument that collects and focuses radiation	refractor: telescope using a main lens
reflector: telescope using a main mirror	aperture: diameter of the light-collecting opening
magnification: apparent enlargement of an image	resolution: ability to separate fine detail
meteor shower: period when many meteors seem to radiate from one sky region	solar filter: certified filter designed for safe Sun viewing

A. Choose the correct answer

1. Which term names instrument that collects and focuses radiation?

- A. refractor
- B. aperture
- C. resolution
- D. telescope

3. Choose the vocabulary item defined as telescope using a main mirror.

- A. aperture
- B. reflector
- C. resolution
- D. solar filter

5. In this chapter, what is defined as apparent enlargement of an image?

- A. resolution
- B. solar filter
- C. refractor
- D. magnification

2. A learner describes telescope using a main lens. What is the correct term?

- A. refractor
- B. reflector
- C. magnification
- D. meteor shower

4. Which expression would a scientist use for diameter of the light-collecting opening?

- A. magnification
- B. meteor shower
- C. aperture
- D. telescope

Exercises (continued)

6. Which statement best explains Refracting telescope?

- A. the curved objective bends incoming light toward a focus where an eyepiece forms a magnified view
- B. the primary mirror reflects incoming light toward a focus, often using a smaller secondary mirror to direct it
- C. changing eyepieces changes magnification, but the image may become dim and unsteady at excessive power
- D. a refractor does not work because the tube itself magnifies; the optical surfaces control light

8. Which statement correctly avoids a common misunderstanding about Aperture?

- A. magnification and aperture are related to different jobs: size of view versus collected light and resolution
- B. a noticeable occurrence involving celestial objects or their apparent positions
- C. extreme magnification cannot restore detail that the aperture, atmosphere or focus did not capture
- D. a larger aperture gathers more light and can separate finer detail under suitable conditions

10. Which example most directly illustrates Astronomical event?

- A. a meteor shower is best observed over time from a dark site, not through a narrow telescope field
- B. projecting the Sun through a suitable small telescope can be safe only with informed adult supervision and correct equipment
- C. small school telescopes often use a long tube with a lens at the front
- D. not every bright streak is a meteor; aircraft and satellites may also cross the sky

7. Which example most directly illustrates Reflecting telescope?

- A. doubling the aperture can reveal objects too faint for a smaller instrument
- B. many large research telescopes are reflectors because large mirrors can be supported from behind
- C. moderate magnification often shows the Moon more clearly than a very high setting
- D. a reflector is not a telescope that merely uses any flat mirror

9. Which statement best explains Magnification?

- A. events include conjunctions, eclipses, meteor showers, occultations and changing phases
- B. the curved objective bends incoming light toward a focus where an eyepiece forms a magnified view
- C. magnification and aperture are related to different jobs: size of view versus collected light and resolution
- D. changing eyepieces changes magnification, but the image may become dim and unsteady at excessive power

B. Answer in one or two sentences

1. Define telescope and refractor in your own words.
2. Explain how Reflecting telescope works. Use one cause-and-effect link.
3. Compare Aperture and Magnification. State one similarity or connection and one difference.
4. Use the example of Astronomical event to explain why context or measurement matters.

C. Answer in detail

1. Explain the main ideas of “Telescopes and Astronomical Events” as a connected account. Include at least four important terms from the chapter.
2. A shop advertises a tiny telescope mainly by “600× power.” What questions should a careful buyer ask?

D. Apply and reason

1. If aperture increases while optical quality remains good, faint objects and fine detail generally become easier to detect.
2. “A telescope’s most important job is to make objects look huge.” Correct the statement.

E. Activity or project

Prepare a safe observing plan for a lunar eclipse, including equipment, timing and weather alternatives.

UNIT 1 REVIEW

Fundamentals of Astronomy - Mixed Review

This review combines ideas from all chapters in the unit. Use diagrams, units and evidence where relevant. The questions are new combinations rather than copies of the chapter exercises.

A. Mixed multiple-choice review

1. Which option links a term with its correct meaning?

- A. celestial object - an explanation based on observations
- B. celestial object - natural object beyond most of Earth's atmosphere
- C. celestial object - artificial brightening of the night sky
- D. celestial object - point directly overhead

3. Only one pair is scientifically correct. Which is it?

- A. horizon - angular height above the horizon
- B. horizon - projected direction of Earth's rotation axis
- C. horizon - instrument that collects and focuses radiation
- D. horizon - apparent boundary between Earth and sky

5. Find the pair that uses the vocabulary correctly.

- A. celestial equator - recognisable informal star pattern
- B. celestial equator - projection of Earth's equator on the sky
- C. celestial equator - telescope using a main mirror
- D. celestial equator - period when many meteors seem to radiate from one sky region

7. Choose the correctly defined term from Unit 1.

- A. aperture - ability to separate fine detail
- B. aperture - scientific study of space and celestial objects
- C. aperture - motion as it seems from an observer's viewpoint
- D. aperture - diameter of the light-collecting opening

9. Which statement best explains Observation in Chapter 1?

- A. Earth's rotation makes most sky objects appear to travel from east to west during a night
- B. astronomers project directions to stars and planets onto the sphere to organise coordinates and apparent paths
- C. an observation reports what was detected; an inference explains what it may mean
- D. good observations include time, place, conditions and a description that another person could check

11. Which statement best explains Asterism in Chapter 2?

- A. the primary mirror reflects incoming light toward a focus, often using a smaller secondary mirror to direct it
- B. asterisms are informal but useful for finding other objects
- C. astronomers compare observations from different times, instruments and places to test explanations
- D. an asterism is not one of the official sky regions

2. Select the accurately matched term and definition.

- A. apparent motion - artificial brightening of the night sky
- B. apparent motion - angular height above the horizon
- C. apparent motion - motion as it seems from an observer's viewpoint
- D. apparent motion - projection of Earth's equator on the sky

4. Which term-definition pair would belong in the Unit 1 glossary?

- A. zenith - point directly overhead
- B. zenith - projected direction of Earth's rotation axis
- C. zenith - recognisable informal star pattern
- D. zenith - diameter of the light-collecting opening

6. Which pairing preserves the meaning used in this unit?

- A. telescope - telescope using a main mirror
- B. telescope - ability to separate fine detail
- C. telescope - instrument that collects and focuses radiation
- D. telescope - natural object beyond most of Earth's atmosphere

8. Which row could be copied into an accurate revision table?

- A. meteor shower - period when many meteors seem to radiate from one sky region
- B. meteor shower - scientific study of space and celestial objects
- C. meteor shower - an explanation based on observations
- D. meteor shower - apparent boundary between Earth and sky

10. Which statement best explains Horizon in Chapter 2?

- A. every direction around an observer meets the horizon; objects rise above it and later approach it again
- B. astronomers project directions to stars and planets onto the sphere to organise coordinates and apparent paths
- C. the primary mirror reflects incoming light toward a focus, often using a smaller secondary mirror to direct it
- D. the horizon is an apparent circle, not a physical edge of Earth

12. Which statement best explains Astronomical event in Chapter 3?

- A. astronomers compare observations from different times, instruments and places to test explanations
- B. scattered light reduces contrast, so faint stars and the Milky Way become harder to see
- C. events include conjunctions, eclipses, meteor showers, occultations and changing phases
- D. not every bright streak is a meteor; aircraft and satellites may also cross the sky

UNIT 1 REVIEW - OLYMPIAD CHALLENGE

B. Written and visual reasoning

1. Connect one idea from Chapter 1 with one idea from Chapter 3. Explain the relationship.
2. Choose two terms from different chapters in Unit 1 and compare their roles.
3. A student says, "That bright point must be the brightest star." Explain what additional observations would be needed.
4. If aperture increases while optical quality remains good, faint objects and fine detail generally become easier to detect.
5. Draw or describe one visual model that would help teach the central idea of Unit 1. State one limitation of the model.
6. Write one claim about Unit 1, give two pieces of evidence and state one uncertainty or boundary condition.

Olympiad Challenge

1. A student is reasoning about Apparent motion. Which statement is the most scientifically careful?

- A. a journal is not an art competition; labels, dates and consistency matter more than decoration
- B. apparent motion describes the view; true motion describes movement in space
- C. altitude is an angle, not the object's distance from Earth
- D. an asterism is not one of the official sky regions

3. When checking a claim about Solar observing safety, which statement sets the right boundary?

- A. astronomy uses evidence; astrology makes personal claims that are not tested in the same scientific way
- B. apparent motion describes the view; true motion describes movement in space
- C. altitude is an angle, not the object's distance from Earth
- D. sunglasses, smoked glass, exposed film and an unfiltered telescope are never safe solar viewers

5. Which conclusion about Reflecting telescope avoids overclaiming?

- A. extreme magnification cannot restore detail that the aperture, atmosphere or focus did not capture
- B. a reflector is not a telescope that merely uses any flat mirror
- C. sunglasses, smoked glass, exposed film and an unfiltered telescope are never safe solar viewers
- D. apparent motion describes the view; true motion describes movement in space

7. Which explanation of Celestial poles and equator is precise enough for a science report?

- A. a constellation is a viewing pattern, not a close cluster whose stars travel together
- B. a reflector is not a telescope that merely uses any flat mirror
- C. sunglasses, smoked glass, exposed film and an unfiltered telescope are never safe solar viewers
- D. the celestial equator is not Earth's equator in the atmosphere; it is a projected reference circle

2. Which situation is the strongest application of Constellation?

- A. the Big Dipper is an asterism within Ursa Major
- B. doubling the aperture can reveal objects too faint for a smaller instrument
- C. Orion is recognised by a set of bright stars and also names a larger defined region
- D. measuring how a planet changes position night after night

4. Which unfamiliar case is best explained by Horizon?

- A. a star just above a distant building has low altitude
- B. an object halfway from horizon to overhead has an altitude near 45 degrees
- C. Orion is recognised by a set of bright stars and also names a larger defined region
- D. doubling the aperture can reveal objects too faint for a smaller instrument

6. Where would Celestial object be most useful?

- A. a constellation rises in the east and later sets in the west although its stars have not circled Earth that night
- B. a star just above a distant building has low altitude
- C. the Sun is a star, the Moon is a natural satellite and Venus is a planet
- D. Orion is recognised by a set of bright stars and also names a larger defined region

8. Which example provides the clearest evidence of Astronomical event?

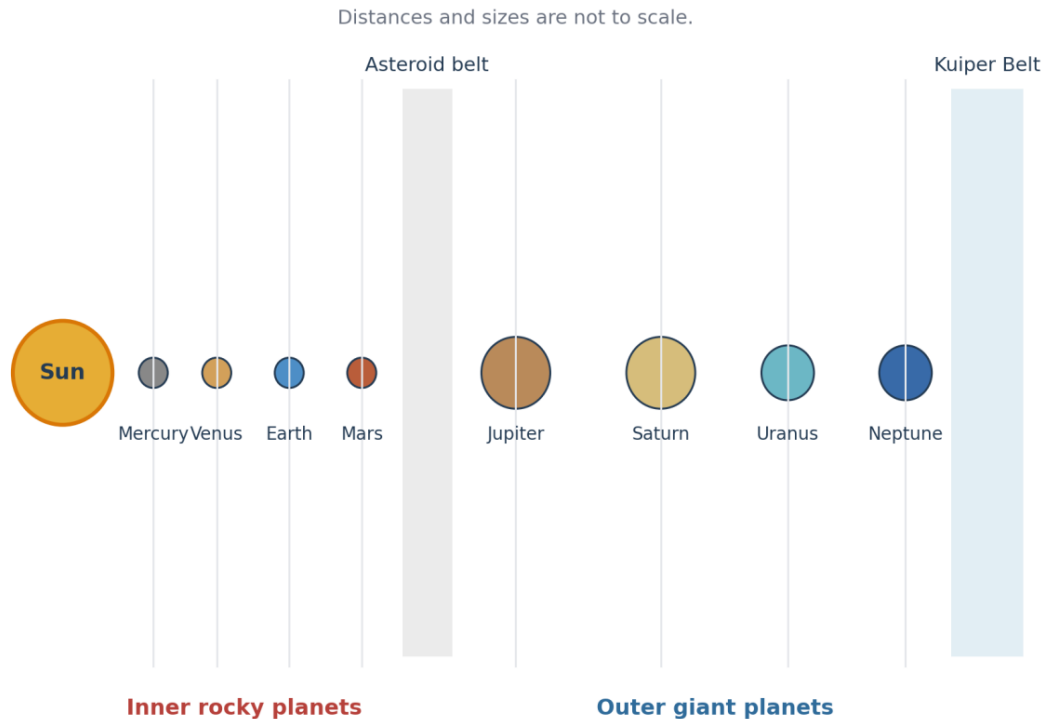
- A. a meteor shower is best observed over time from a dark site, not through a narrow telescope field
- B. projecting the Sun through a suitable small telescope can be safe only with informed adult supervision and correct equipment
- C. the Sun is a star, the Moon is a natural satellite and Venus is a planet
- D. a star just above a distant building has low altitude

UNIT 2

The Solar System

The Sun and the worlds that travel around it form a connected system shaped by gravity, motion and a common origin.

The Solar System: Orbits and Regions



Unit 2 opening visual - original illustration.

Major concepts

- Chapter 4: Formation and Architecture of the Solar System
- Chapter 5: The Sun and the Rocky Planets
- Chapter 6: Giant Planets, Dwarf Planets and Moons
- Chapter 7: Asteroids, Comets and Meteors

Expected unit outcomes

- Outline the nebular model.
- Describe the sun as a star.
- Compare gas and ice giants.
- Distinguish asteroid and comet.

CHAPTER 4

Formation and Architecture of the Solar System

The Solar System formed about 4.6 billion years ago from part of a cold cloud of gas and dust. Gravity caused the material to collapse; rotation helped flatten it into a disc. Most mass collected at the centre to form the young Sun, while grains and chunks in the disc collided and grew into planets, moons and smaller bodies. This model explains both shared patterns and important differences among the worlds.

Opening question Why do most planets orbit the Sun in nearly the same direction and roughly the same plane?

Learning outcomes

- Outline the nebular model.
- Explain accretion and differentiation.
- Identify main solar system regions.
- Use scale models cautiously.

A Simplified Solar System Formation Sequence

Gravity pulled material together while rotation flattened much of it into a disc.

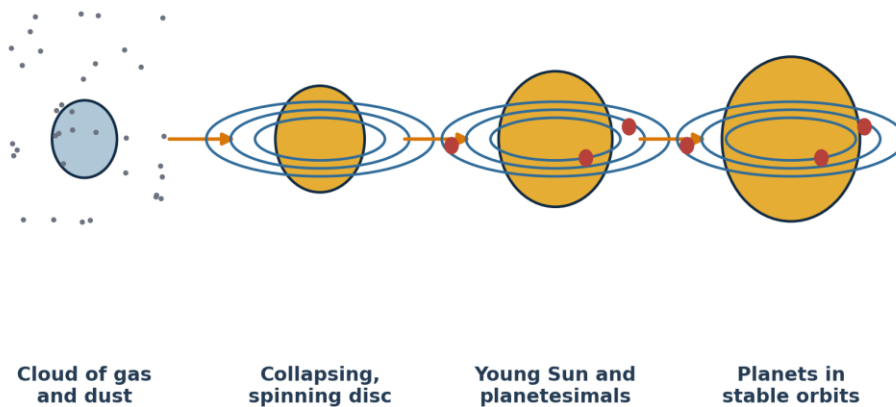


Figure 4.1 From cloud to spinning disc. Original illustration; not to scale where stated.

From cloud to spinning disc

Formation models connect present-day evidence with processes that operated long ago.

Solar nebula

The rotating cloud of gas and dust from which the Solar System formed. In practice, gravity amplified denser regions, collapse made rotation faster and collisions flattened much material into a disc. An example is a spinning pizza dough becomes flatter as it spreads, though the physics is only a rough analogy. However, the nebula was not an empty hole; it contained matter distributed through space.

Accretion

To understand Accretion, begin with growth by repeated sticking, collision and gravitational gathering of smaller pieces. Next, note that dust grains formed clumps, clumps became planetesimals and some planetesimals grew into protoplanets. Small snowballs gain material as they roll, but early planetary collisions were much faster and hotter. This does not mean that accretion includes destructive collisions as well as successful growth.

Check your understanding What evidence in planetary orbits supports formation from a disc?

Heating and separation

Growing worlds changed internally as impacts, compression and radioactive decay supplied heat.

Differentiation

To understand Differentiation, begin with separation of materials inside a body according to density and chemical properties. Next, note that when a young world partly melted, dense metals tended to sink while lighter rocky material rose. Earth's metallic core and rocky mantle record internal separation. This does not mean that differentiation is not the same as the planet splitting into separate bodies.

Frost line

Frost line describes a region in the early disc beyond which temperatures were low enough for water and other substances to freeze. Its behaviour follows from this relationship: more solid material was available beyond this line, helping large cores and giant planets form. The example of icy moons are common in the colder outer Solar System shows the idea in use. Remember that the frost line was a broad changing zone, not a painted circle fixed forever.

Check your understanding Why was solid material more abundant in the colder outer disc?

A structured system

The Solar System contains nested regions rather than a random mixture of bodies.

Inner and outer planets

Two broad planet groups separated by composition, size and location. The inner four are smaller and rocky; the outer four are much larger and rich in gases or ices. For example, Mars and Jupiter lie on opposite sides of the main asteroid belt. One limit must remain clear: “outer” describes position, not that the planets are outside the Solar System.

Small-body reservoirs

Small-body reservoirs means regions containing many asteroids, comets and dwarf planets. It works because the asteroid belt lies mainly between Mars and Jupiter, while the Kuiper Belt lies beyond Neptune and the distant Oort Cloud is inferred to surround the system. A concrete case is Ceres belongs to the asteroid belt and Pluto to the Kuiper Belt. By contrast, these regions are mostly empty space; spacecraft do not face a solid wall of rocks.

Check your understanding How are the asteroid belt and Kuiper Belt alike, and how are they different?

Activity: Build a distance strip

Aim	to represent the order and unequal spacing of planets
Materials	long paper strip, ruler, pencil and a supplied scale table
Steps	1. Mark the Sun at one end. 2. Using a chosen compressed scale, mark average orbital distances for all eight planets. 3. Write planet names without drawing planet sizes on the same scale. 4. Compare the crowded inner region with the widely spaced outer region.
Observe	No convenient classroom scale can show planet diameters and orbital distances accurately at the same time.
Expected learning	A model must state what is and is not to scale.
Safety	Keep long strips clear of walkways.

Chapter Review

Key ideas

- The rotating cloud of gas and dust from which the solar system formed.
- Separation of materials inside a body according to density and chemical properties.
- Two broad planet groups separated by composition, size and location.
- “The asteroid belt is the remains of one exploded planet.” Explain why the formation story is more complex.

Important terms

solar nebula: cloud from which the Solar System formed	accretion: growth by gathering smaller material
planetesimal: small early body that could grow into a planet	protoplanet: large developing planetary body
differentiation: internal separation by density and composition	frost line: early-disc region beyond which ices could condense
asteroid belt: small-body region between Mars and Jupiter	Kuiper Belt: icy-body region beyond Neptune

A. Choose the correct answer

1. Which term names cloud from which the Solar System formed?

- A. solar nebula
- B. accretion
- C. protoplanet
- D. frost line

3. Choose the vocabulary item defined as small early body that could grow into a planet.

- A. protoplanet
- B. frost line
- C. planetesimal
- D. Kuiper Belt

5. In this chapter, what is defined as internal separation by density and composition?

- A. differentiation
- B. frost line
- C. Kuiper Belt
- D. accretion

2. A learner describes growth by gathering smaller material. What is the correct term?

- A. planetesimal
- B. accretion
- C. differentiation
- D. asteroid belt

4. Which expression would a scientist use for large developing planetary body?

- A. differentiation
- B. asteroid belt
- C. solar nebula
- D. protoplanet

Exercises (continued)

6. Which statement best explains Solar nebula?

- A. dust grains formed clumps, clumps became planetesimals and some planetesimals grew into protoplanets
- B. gravity amplified denser regions, collapse made rotation faster and collisions flattened much material into a disc
- C. more solid material was available beyond this line, helping large cores and giant planets form
- D. the nebula was not an empty hole; it contained matter distributed through space

8. Which statement correctly avoids a common misunderstanding about Differentiation?

- A. the frost line was a broad changing zone, not a painted circle fixed forever
- B. two broad planet groups separated by composition, size and location
- C. when a young world partly melted, dense metals tended to sink while lighter rocky material rose
- D. differentiation is not the same as the planet splitting into separate bodies

10. Which example most directly illustrates Inner and outer planets?

- A. Ceres belongs to the asteroid belt and Pluto to the Kuiper Belt
- B. Mars and Jupiter lie on opposite sides of the main asteroid belt
- C. a spinning pizza dough becomes flatter as it spreads, though the physics is only a rough analogy
- D. “outer” describes position, not that the planets are outside the Solar System

7. Which example most directly illustrates Accretion?

- A. Earth’s metallic core and rocky mantle record internal separation
- B. icy moons are common in the colder outer Solar System
- C. small snowballs gain material as they roll, but early planetary collisions were much faster and hotter
- D. accretion includes destructive collisions as well as successful growth

9. Which statement best explains Frost line?

- A. more solid material was available beyond this line, helping large cores and giant planets form
- B. the inner four are smaller and rocky; the outer four are much larger and rich in gases or ices
- C. gravity amplified denser regions, collapse made rotation faster and collisions flattened much material into a disc
- D. the frost line was a broad changing zone, not a painted circle fixed forever

B. Answer in one or two sentences

1. Define solar nebula and accretion in your own words.
2. Explain how Accretion works. Use one cause-and-effect link.
3. Compare Differentiation and Frost line. State one similarity or connection and one difference.
4. Use the example of Inner and outer planets to explain why context or measurement matters.

C. Answer in detail

1. Explain the main ideas of “Formation and Architecture of the Solar System” as a connected account. Include at least four important terms from the chapter.
2. A diagram shows all planets equally spaced. What useful information does it show, and what false impression might it create?

D. Apply and reason

1. If the early disc had not been flattened, planetary orbital planes would likely be less closely aligned.
2. “The asteroid belt is the remains of one exploded planet.” Explain why the formation story is more complex.

E. Activity or project

Create a labelled flowchart from cloud to mature Solar System and include two sources of evidence.

CHAPTER 5

The Sun and the Rocky Planets

The Sun contains almost all the mass of the Solar System and supplies the energy that warms planets. Closest to it are four terrestrial, or rocky, planets: Mercury, Venus, Earth and Mars. They share solid surfaces and layered interiors, yet their atmospheres and histories have produced sharply different environments.

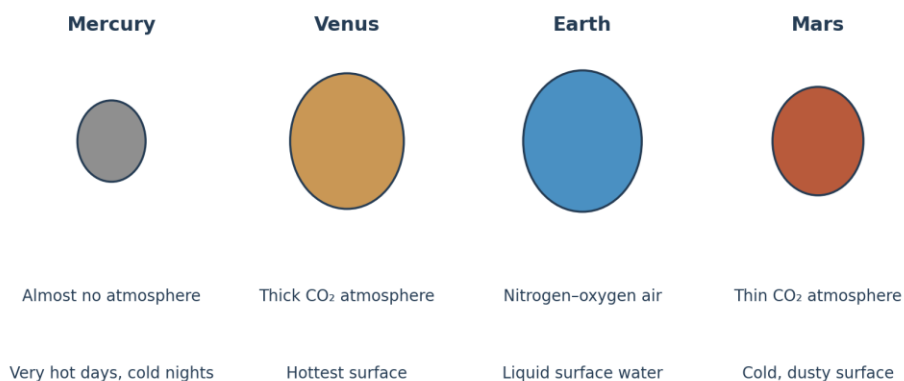
Opening question How can Venus be hotter than Mercury even though Mercury is closer to the Sun?

Learning outcomes

- Describe the sun as a star.
- Identify major solar layers.
- Compare rocky planets.
- Relate atmosphere to surface conditions.

Rocky Planets: A Comparison

All four have solid surfaces, but their atmospheres and temperatures differ greatly.



Planet sizes shown approximately relative to one another; distances are not represented.

Figure 5.1 Our star. Original illustration; not to scale where stated.

Our star

The Sun is a middle-aged star whose gravity holds the Solar System together.

Nuclear fusion

To understand Nuclear fusion, begin with the process in which light atomic nuclei combine and release energy. Next, note that in the solar core, hydrogen nuclei ultimately form helium while a small amount of mass is converted to energy. Sunlight and heat at Earth began as energy released in the core. This does not mean that fusion is not ordinary burning; it does not require oxygen and occurs under extreme pressure and temperature.

Solar layers

Solar layers describes regions of the Sun distinguished by how energy moves and what observers can detect. Its behaviour follows from this relationship: energy leaves the core through the radiative and convective zones, reaches the visible photosphere and influences the thin outer atmosphere. The example of granular patterns on the photosphere trace convection below shows the idea in use. Remember that the photosphere is called a surface for convenience even though the Sun is gas and plasma.

Check your understanding Where is solar energy produced, and from which layer does most visible light escape?

Mercury and Venus

Neighbouring planets can have very different conditions because distance is only one factor.

Mercury

Mercury describes the smallest planet and the closest to the Sun. Its behaviour follows from this relationship: its weak gravity and solar heating left it with only an extremely thin exosphere, so temperatures change greatly between day and night. The example of its cratered surface resembles the Moon in many images shows the idea in use. Remember that Mercury is not the hottest planet on average despite being nearest the Sun.

Venus

A rocky planet similar in size to Earth but covered by a very thick carbon-dioxide atmosphere. Because a strong greenhouse effect traps heat, and clouds conceal a hot, high-pressure surface, observers may find that radar instruments map Venus through its clouds. The conclusion should remain precise: Venus is not Earth's twin in present surface conditions; size alone does not determine habitability.

Check your understanding Which property of Venus explains its exceptionally high surface temperature?

Earth and Mars

Earth and Mars preserve evidence of water, atmosphere and geological change in different ways.

Earth

Earth means the rocky planet with abundant surface liquid water and known life. It works because its atmosphere, magnetic field, oceans and active geology interact in a complex climate system. A concrete case is water cycles through oceans, air, ice, soil and living things. By contrast, Earth is not protected by one single feature; habitability depends on interacting conditions.

Mars

A cold rocky planet with a thin carbon-dioxide atmosphere, polar caps, volcanoes and ancient water-carved features. The underlying relationship is that its small mass and changing environment allowed much atmosphere and surface water to be lost or locked away. This can be seen when rovers examine layered rocks and minerals for evidence of past environments. Mars is not simply a smaller Earth today, and no confirmed Martian life has been found.

Check your understanding What kinds of Martian evidence suggest that liquid water existed there in the past?

Activity: Compare planetary evidence cards

Aim	to infer conditions from selected evidence
Materials	teacher-prepared cards describing atmosphere, temperature, craters and water evidence
Steps	1. Work in groups and sort each card under Mercury, Venus, Earth or Mars. 2. Give one reason for every placement. 3. Identify a card that could fit more than one planet and state what extra evidence is needed. 4. Check the final classification with a reliable planet reference.
Observe	Good classifications use several clues rather than one familiar word.
Expected learning	Inference becomes stronger when independent evidence agrees.
Safety	No special safety requirement.

Chapter Review

Key ideas

- The process in which light atomic nuclei combine and release energy.
- The smallest planet and the closest to the sun.
- The rocky planet with abundant surface liquid water and known life.
- “The nearest planet to the Sun must be the hottest.” Use Mercury and Venus to correct this rule.

Important terms

fusion: combining light nuclei to release energy	photosphere: visible layer of the Sun
solar wind: stream of charged particles from the Sun	terrestrial planet: planet with a rocky solid surface
exosphere: extremely thin outer atmosphere	greenhouse effect: warming caused when an atmosphere slows heat loss
magnetic field: region in which magnetic forces act	habitability: ability of an environment to support life as defined

A. Choose the correct answer

1. Which term names combining light nuclei to release energy?

- A. photosphere
- B. fusion
- C. terrestrial planet
- D. greenhouse effect

3. Choose the vocabulary item defined as stream of charged particles from the Sun.

- A. terrestrial planet
- B. greenhouse effect
- C. habitability
- D. solar wind

5. In this chapter, what is defined as extremely thin outer atmosphere?

- A. greenhouse effect
- B. exosphere
- C. habitability
- D. photosphere

2. A learner describes visible layer of the Sun. What is the correct term?

- A. solar wind
- B. exosphere
- C. photosphere
- D. magnetic field

4. Which expression would a scientist use for planet with a rocky solid surface?

- A. terrestrial planet
- B. exosphere
- C. magnetic field
- D. fusion

Exercises (continued)

6. Which statement best explains Nuclear fusion?

- A. energy leaves the core through the radiative and convective zones, reaches the visible photosphere and influences the thin outer atmosphere
- B. a strong greenhouse effect traps heat, and clouds conceal a hot, high-pressure surface
- C. in the solar core, hydrogen nuclei ultimately form helium while a small amount of mass is converted to energy
- D. fusion is not ordinary burning; it does not require oxygen and occurs under extreme pressure and temperature

8. Which statement correctly avoids a common misunderstanding about Mercury?

- A. Mercury is not the hottest planet on average despite being nearest the Sun
- B. Venus is not Earth's twin in present surface conditions; size alone does not determine habitability
- C. the rocky planet with abundant surface liquid water and known life
- D. its weak gravity and solar heating left it with only an extremely thin exosphere, so temperatures change greatly between day and night

10. Which example most directly illustrates Earth?

- A. rovers examine layered rocks and minerals for evidence of past environments
- B. sunlight and heat at Earth began as energy released in the core
- C. water cycles through oceans, air, ice, soil and living things
- D. Earth is not protected by one single feature; habitability depends on interacting conditions

7. Which example most directly illustrates Solar layers?

- A. its cratered surface resembles the Moon in many images
- B. radar instruments map Venus through its clouds
- C. the photosphere is called a surface for convenience even though the Sun is gas and plasma
- D. granular patterns on the photosphere trace convection below

9. Which statement best explains Venus?

- A. its atmosphere, magnetic field, oceans and active geology interact in a complex climate system
- B. a strong greenhouse effect traps heat, and clouds conceal a hot, high-pressure surface
- C. in the solar core, hydrogen nuclei ultimately form helium while a small amount of mass is converted to energy
- D. Venus is not Earth's twin in present surface conditions; size alone does not determine habitability

B. Answer in one or two sentences

1. Define fusion and photosphere in your own words.
2. Explain how Solar layers works. Use one cause-and-effect link.
3. Compare Mercury and Venus. State one similarity or connection and one difference.
4. Use the example of Earth to explain why context or measurement matters.

C. Answer in detail

1. Explain the main ideas of "The Sun and the Rocky Planets" as a connected account. Include at least four important terms from the chapter.
2. A planet is Earth-sized and close to its star. Why is that not enough to claim it is Earth-like?

D. Apply and reason

1. Removing most of Venus's atmosphere would greatly change its heat balance, but the exact outcome would require a climate model.
2. "The nearest planet to the Sun must be the hottest." Use Mercury and Venus to correct this rule.

E. Activity or project

Prepare a four-column comparison of the rocky planets using only properties explained in this chapter.

CHAPTER 6

Giant Planets, Dwarf Planets and Moons

Beyond the asteroid belt are Jupiter, Saturn, Uranus and Neptune. These giant planets have deep atmospheres, many moons and ring systems. The Solar System also contains dwarf planets: nearly round bodies that orbit the Sun but have not cleared other objects from their orbital neighbourhood. Moons themselves are diverse worlds, some with atmospheres, oceans or active geology.

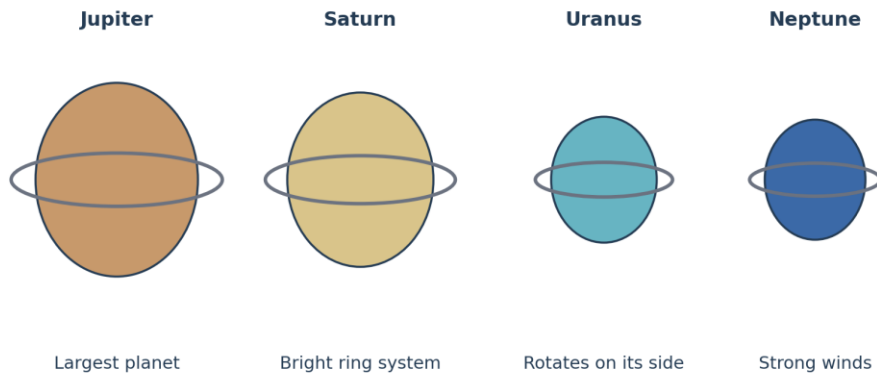
Opening question Why is being round and orbiting the Sun not enough for an object to be classified as a planet?

Learning outcomes

- Compare gas and ice giants.
- Explain the planet and dwarf-planet distinction.
- Describe the diversity of moons.
- Interpret ring systems correctly.

Giant Planets and Their Rings

The outer planets are large worlds dominated by gases or icy materials.



Simplified comparison; not to scale.

Figure 6.1 Four giant worlds. Original illustration; not to scale where stated.

Four giant worlds

The outer planets are not enlarged versions of one another.

Gas giants

Gas giants describes Jupiter and Saturn, large planets dominated by hydrogen and helium. Its behaviour follows from this relationship: their great mass compressed material deeply, producing thick atmospheres, powerful weather and no ordinary solid surface to stand on. The example of Jupiter's Great Red Spot is a long-lived storm system shows the idea in use. Remember that "gas giant" does not mean the planet is a loose cloud with no deeper dense layers.

Ice giants

Uranus and Neptune, giant planets with larger proportions of water-, ammonia- and methane-rich materials than Jupiter and Saturn. Because their different composition and internal heat influence atmosphere colour and weather, observers may find that methane absorbs red light and contributes to blue-green appearances. The conclusion should remain precise: the word ice refers to chemical materials under planetary conditions, not simply frozen cubes.

Check your understanding Name one compositional difference between gas giants and ice giants.

Rings and moons

Gravity organises particles and satellites around giant planets.

Planetary rings

Many small particles orbiting a planet in a flattened band. Because collisions, resonances and moon gravity shape gaps and narrow structures, observers may find that Saturn's bright rings contain ice-rich particles ranging from dust-sized grains to larger chunks. The conclusion should remain precise: rings are not single solid discs.

Natural satellite

In this chapter, Natural satellite is used for a natural body that orbits a planet, dwarf planet or small body. Moons can be captured, co-formed or produced by impacts, and tidal forces may heat their interiors. One useful example is Jupiter's moon Europa is studied because evidence suggests an ocean beneath ice. It should be separated from this different claim: not every moon is geologically dead or airless.

Check your understanding How can the gravity of a moon help shape a ring?

Dwarf planets and classification

Scientific categories are definitions chosen to organise evidence consistently.

Planet

In the Solar System, a body that orbits the Sun, is nearly round and has cleared its orbital neighbourhood. The underlying relationship is that clearing means its gravity dominates nearby objects over long periods. This can be seen when Earth fulfils all three conditions. Clearing does not mean the orbit is literally empty.

Dwarf planet

The phrase Dwarf planet refers to a nearly round Sun-orbiting body that has not cleared its orbital neighbourhood and is not a moon. Here, known examples occupy the asteroid belt or distant trans-Neptunian region. Consider the following case: Ceres and Pluto are officially recognised dwarf planets. The careful interpretation is that a dwarf planet is not merely a very small planet under the formal definition.

Check your understanding Which criterion separates a dwarf planet from a planet in the IAU definition?

Activity: Model a ring system with moving particles

Aim	to show why a ring is many orbiting pieces
Materials	large sheet, drawn planet, small counters and pencils
Steps	1. Draw several circular ring paths around the planet. 2. Place many counters on different paths instead of making one solid band. 3. Move counters a small distance in the same direction, with inner counters taking shorter paths. 4. Add a small “shepherd moon” near one ring edge and discuss possible effects.
Observe	Particles share an overall direction but occupy different orbits and do not behave as a rigid plate.
Expected learning	A system can appear continuous from far away while being made of separate parts.
Safety	Keep small counters away from very young children.

Chapter Review

Key ideas

- Jupiter and Saturn, large planets dominated by hydrogen and helium.
- Many small particles orbiting a planet in a flattened band.
- In the solar system, a body that orbits the sun, is nearly round and has cleared its orbital neighbourhood.
- “All moons are smaller copies of Earth’s Moon.” Give two counterexamples in property, not just name.

Important terms

gas giant: large planet dominated by hydrogen and helium	ice giant: giant rich in heavier volatile materials
ring: orbiting band of many particles	natural satellite: natural body orbiting another non-stellar body
tidal heating: internal heating caused by changing gravitational stretching	planet: round Sun-orbiting body that has cleared its neighbourhood
dwarf planet: round Sun-orbiting body that has not cleared its neighbourhood	orbital neighbourhood: region around an orbit influenced by a body’s gravity

A. Choose the correct answer

1. Which term names large planet dominated by hydrogen and helium?

- A. ice giant
- B. natural satellite
- C. gas giant
- D. planet

3. Choose the vocabulary item defined as orbiting band of many particles.

- A. ring
- B. natural satellite
- C. planet
- D. orbital neighbourhood

5. In this chapter, what is defined as internal heating caused by changing gravitational stretching?

- A. planet
- B. orbital neighbourhood
- C. tidal heating
- D. ice giant

2. A learner describes giant rich in heavier volatile materials. What is the correct term?

- A. ring
- B. tidal heating
- C. dwarf planet
- D. ice giant

4. Which expression would a scientist use for natural body orbiting another non-stellar body?

- A. tidal heating
- B. natural satellite
- C. dwarf planet
- D. gas giant

Exercises (continued)

6. Which statement best explains Gas giants?

- A. their different composition and internal heat influence atmosphere colour and weather
- B. moons can be captured, co-formed or produced by impacts, and tidal forces may heat their interiors
- C. “gas giant” does not mean the planet is a loose cloud with no deeper dense layers
- D. their great mass compressed material deeply, producing thick atmospheres, powerful weather and no ordinary solid surface to stand on

8. Which statement correctly avoids a common misunderstanding about Planetary rings?

- A. not every moon is geologically dead or airless
- B. rings are not single solid discs
- C. in the Solar System, a body that orbits the Sun, is nearly round and has cleared its orbital neighbourhood
- D. collisions, resonances and moon gravity shape gaps and narrow structures

10. Which example most directly illustrates Planet?

- A. Ceres and Pluto are officially recognised dwarf planets
- B. Jupiter’s Great Red Spot is a long-lived storm system
- C. clearing does not mean the orbit is literally empty
- D. Earth fulfils all three conditions

7. Which example most directly illustrates Ice giants?

- A. methane absorbs red light and contributes to blue-green appearances
- B. Saturn’s bright rings contain ice-rich particles ranging from dust-sized grains to larger chunks
- C. Jupiter’s moon Europa is studied because evidence suggests an ocean beneath ice
- D. the word ice refers to chemical materials under planetary conditions, not simply frozen cubes

9. Which statement best explains Natural satellite?

- A. clearing means its gravity dominates nearby objects over long periods
- B. their great mass compressed material deeply, producing thick atmospheres, powerful weather and no ordinary solid surface to stand on
- C. moons can be captured, co-formed or produced by impacts, and tidal forces may heat their interiors
- D. not every moon is geologically dead or airless

B. Answer in one or two sentences

1. Define gas giant and ice giant in your own words.
2. Explain how Ice giants works. Use one cause-and-effect link.
3. Compare Planetary rings and Natural satellite. State one similarity or connection and one difference.
4. Use the example of Planet to explain why context or measurement matters.

C. Answer in detail

1. Explain the main ideas of “Giant Planets, Dwarf Planets and Moons” as a connected account. Include at least four important terms from the chapter.
2. A newly found object is round and orbits the Sun beyond Neptune. What further evidence is needed for classification?

D. Apply and reason

1. If a ring’s particles suddenly stuck into one rigid sheet, their orbital behaviour would be physically very different.
2. “All moons are smaller copies of Earth’s Moon.” Give two counterexamples in property, not just name.

E. Activity or project

Create a classification key that separates planets, dwarf planets, moons and small bodies.

CHAPTER 7

Asteroids, Comets and Meteors

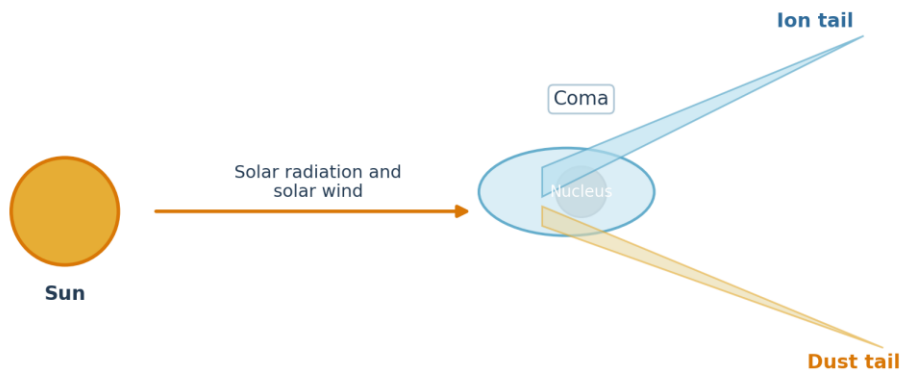
Small Solar System bodies preserve material from early formation and sometimes cross planetary orbits. Asteroids are mostly rocky or metallic; comets contain substantial volatile material that becomes active near the Sun. Meteoroids are smaller pieces in space. A meteor is the streak of light produced in an atmosphere, while a meteorite is material that reaches the ground.

Opening question Why are “meteor” and “meteorite” names for different stages rather than different kinds of rock?

Learning outcomes

- Distinguish asteroid and comet.
- Use meteoroid, meteor and meteorite correctly.
- Describe comet activity.
- Explain impact risk in proportion.

Comet Anatomy Near the Sun



A comet's tails point generally away from the Sun, not simply behind its motion.

Figure 7.1 Asteroids and their orbits. Original illustration; not to scale where stated.

Asteroids and their orbits

Most known asteroids follow stable paths, though some approach planets.

Asteroid

A small rocky or metallic body orbiting the Sun. Because asteroids are leftovers and fragments shaped by collisions and planetary gravity, observers may find that the main belt contains many asteroids but is spread over a vast volume. The conclusion should remain precise: an asteroid is not defined as any object that might hit Earth.

Near-Earth object

In this chapter, Near-Earth object is used for an asteroid or comet whose orbit brings it into the neighbourhood of Earth's orbit. Surveys repeatedly measure positions so future paths and uncertainties can be calculated. One useful example is a potentially hazardous asteroid receives careful monitoring, not an automatic prediction of impact. It should be separated from this different claim: "near-Earth" is an orbital category and can still involve very large distances.

Check your understanding Why are repeated position measurements needed before estimating impact risk?

Comets become active

A comet changes visibly as solar heating affects its volatile-rich nucleus.

Comet nucleus

In this chapter, Comet nucleus is used for a solid body containing ices, dust and rocky material. Near the Sun, warming releases gas and dust that create a coma and tails. One useful example is a periodic comet may brighten on each return to the inner Solar System. It should be separated from this different claim: the nucleus is much smaller than the enormous faint coma visible around it.

Comet tail

A stream of dust or ionised gas extending generally away from the Sun. The process can be followed as radiation pressure and solar wind shape different tails, so direction depends mainly on the Sun rather than the comet's path. For instance, a comet moving away from the Sun still has tails pointing generally away from the Sun. A common overstatement is avoided by recognising that a tail does not always trail directly behind like smoke from a vehicle.

Check your understanding Why can a comet have more than one tail?

From space rock to ground sample

Correct terms describe where the material is and what is happening.

Meteoroid, meteor, meteorite

The phrase Meteoroid, meteor, meteorite refers to a meteoroid is a small natural solid in space; a meteor is the luminous atmospheric event; a meteorite is surviving material on the ground. Here, rapid entry compresses and heats surrounding air, while ablation removes material from the object. Consider the following case: a brief “shooting star” is a meteor, not a star falling. The careful interpretation is that the glow is not produced mainly by friction rubbing against air in the ordinary everyday sense.

Impact crater

A roughly circular depression formed when a fast object transfers energy to a surface. In practice, shock waves excavate and deform material far beyond the object’s own size. An example is the Moon preserves many craters because erosion and active geology erase them slowly. However, crater size is not equal to the incoming object’s diameter.

Check your understanding Put meteoroid, meteor and meteorite in the order that one piece of material might experience.

Activity: Make an impact-evidence tray

Aim	to observe how speed and size affect a model crater
Materials	tray of flour topped with cocoa powder, small balls of different sizes, ruler and eye protection
Steps	1. Smooth the powder surface without compacting it. 2. Drop one ball vertically from a fixed height and measure the crater. 3. Repeat with a different ball or height, smoothing the tray each time. 4. Compare crater width, depth and ejected rays.
Observe	Larger or faster model impactors usually move more material, though the classroom model omits shock physics and high speed.
Expected learning	Crater form provides evidence about an impact event.
Safety	Wear eye protection, keep faces away from the tray and clean spilled powder promptly.

Chapter Review

Key ideas

- A small rocky or metallic body orbiting the sun.
- A solid body containing ices, dust and rocky material.
- A meteoroid is a small natural solid in space; a meteor is the luminous atmospheric event; a meteorite is surviving material on the ground.
- “A shooting star is a burning star.” Correct every part of the phrase scientifically.

Important terms

asteroid: small rocky or metallic Sun-orbiting body	near-Earth object: small body with an orbit approaching Earth’s orbital region
comet: volatile-rich body that can become active near the Sun	nucleus: solid central body of a comet
coma: cloud around an active comet nucleus	meteoroid: small natural solid travelling in space
meteor: visible atmospheric streak caused by an incoming body	meteorite: surviving material found on the ground

A. Choose the correct answer

1. Which term names small rocky or metallic Sun-orbiting body?

- A. near-Earth object
- B. nucleus
- C. meteoroid
- D. asteroid

3. Choose the vocabulary item defined as volatile-rich body that can become active near the Sun.

- A. nucleus
- B. comet
- C. meteoroid
- D. meteorite

5. In this chapter, what is defined as cloud around an active comet nucleus?

- A. meteoroid
- B. meteorite
- C. near-Earth object
- D. coma

2. A learner describes small body with an orbit approaching Earth’s orbital region. What is the correct term?

- A. near-Earth object
- B. comet
- C. coma
- D. meteor

4. Which expression would a scientist use for solid central body of a comet?

- A. coma
- B. meteor
- C. nucleus
- D. asteroid

Exercises (continued)

6. Which statement best explains Asteroid?

- A. asteroids are leftovers and fragments shaped by collisions and planetary gravity
- B. surveys repeatedly measure positions so future paths and uncertainties can be calculated
- C. radiation pressure and solar wind shape different tails, so direction depends mainly on the Sun rather than the comet's path
- D. an asteroid is not defined as any object that might hit Earth

8. Which statement correctly avoids a common misunderstanding about Comet nucleus?

- A. a tail does not always trail directly behind like smoke from a vehicle
- B. a meteoroid is a small natural solid in space; a meteor is the luminous atmospheric event; a meteorite is surviving material on the ground
- C. the nucleus is much smaller than the enormous faint coma visible around it
- D. near the Sun, warming releases gas and dust that create a coma and tails

10. Which example most directly illustrates Meteoroid, meteor, meteorite?

- A. a brief "shooting star" is a meteor, not a star falling
- B. the Moon preserves many craters because erosion and active geology erase them slowly
- C. the main belt contains many asteroids but is spread over a vast volume
- D. the glow is not produced mainly by friction rubbing against air in the ordinary everyday sense

7. Which example most directly illustrates Near-Earth object?

- A. a periodic comet may brighten on each return to the inner Solar System
- B. a potentially hazardous asteroid receives careful monitoring, not an automatic prediction of impact
- C. a comet moving away from the Sun still has tails pointing generally away from the Sun
- D. "near-Earth" is an orbital category and can still involve very large distances

9. Which statement best explains Comet tail?

- A. rapid entry compresses and heats surrounding air, while ablation removes material from the object
- B. asteroids are leftovers and fragments shaped by collisions and planetary gravity
- C. a tail does not always trail directly behind like smoke from a vehicle
- D. radiation pressure and solar wind shape different tails, so direction depends mainly on the Sun rather than the comet's path

B. Answer in one or two sentences

1. Define asteroid and near-Earth object in your own words.
2. Explain how Near-Earth object works. Use one cause-and-effect link.
3. Compare Comet nucleus and Comet tail. State one similarity or connection and one difference.
4. Use the example of Meteoroid, meteor, meteorite to explain why context or measurement matters.

C. Answer in detail

1. Explain the main ideas of "Asteroids, Comets and Meteors" as a connected account. Include at least four important terms from the chapter.
2. A news headline says an asteroid has a 0.1% impact probability. Explain why uncertainty and later observations matter.

D. Apply and reason

1. Increasing entry speed greatly increases kinetic energy because speed is squared in the energy formula.
2. "A shooting star is a burning star." Correct every part of the phrase scientifically.

E. Activity or project

Create a three-panel public-information poster that explains NEO monitoring without exaggerating danger.

UNIT 2 REVIEW

The Solar System - Mixed Review

This review combines ideas from all chapters in the unit. Use diagrams, units and evidence where relevant. The questions are new combinations rather than copies of the chapter exercises.

A. Mixed multiple-choice review

1. Which option links a term with its correct meaning?

- A. planetesimal - internal separation by density and composition
- B. planetesimal - icy-body region beyond Neptune
- C. planetesimal - small early body that could grow into a planet
- D. planetesimal - planet with a rocky solid surface

3. Only one pair is scientifically correct. Which is it?

- A. fusion - combining light nuclei to release energy
- B. fusion - stream of charged particles from the Sun
- C. fusion - warming caused when an atmosphere slows heat loss
- D. fusion - giant rich in heavier volatile materials

5. Find the pair that uses the vocabulary correctly.

- A. magnetic field - large planet dominated by hydrogen and helium
- B. magnetic field - natural body orbiting another non-stellar body
- C. magnetic field - region in which magnetic forces act
- D. magnetic field - region around an orbit influenced by a body's gravity

7. Choose the correctly defined term from Unit 2.

- A. tidal heating - internal heating caused by changing gravitational stretching
- B. tidal heating - round Sun-orbiting body that has not cleared its neighbourhood
- C. tidal heating - small body with an orbit approaching Earth's orbital region
- D. tidal heating - small natural solid travelling in space

9. Which statement best explains Differentiation in Chapter 4?

- A. when a young world partly melted, dense metals tended to sink while lighter rocky material rose
- B. the inner four are smaller and rocky; the outer four are much larger and rich in gases or ices
- C. a strong greenhouse effect traps heat, and clouds conceal a hot, high-pressure surface
- D. differentiation is not the same as the planet splitting into separate bodies

11. Which statement best explains Gas giants in Chapter 6?

- A. collisions, resonances and moon gravity shape gaps and narrow structures
- B. surveys repeatedly measure positions so future paths and uncertainties can be calculated
- C. their great mass compressed material deeply, producing thick atmospheres, powerful weather and no ordinary solid surface to stand on
- D. "gas giant" does not mean the planet is a loose cloud with no deeper dense layers

2. Select the accurately matched term and definition.

- A. frost line - icy-body region beyond Neptune
- B. frost line - stream of charged particles from the Sun
- C. frost line - region in which magnetic forces act
- D. frost line - early-disc region beyond which ices could condense

4. Which term-definition pair would belong in the Unit 2 glossary?

- A. terrestrial planet - warming caused when an atmosphere slows heat loss
- B. terrestrial planet - planet with a rocky solid surface
- C. terrestrial planet - large planet dominated by hydrogen and helium
- D. terrestrial planet - internal heating caused by changing gravitational stretching

6. Which pairing preserves the meaning used in this unit?

- A. ice giant - natural body orbiting another non-stellar body
- B. ice giant - round Sun-orbiting body that has not cleared its neighbourhood
- C. ice giant - volatile-rich body that can become active near the Sun
- D. ice giant - giant rich in heavier volatile materials

8. Which row could be copied into an accurate revision table?

- A. orbital neighbourhood - small body with an orbit approaching Earth's orbital region
- B. orbital neighbourhood - region around an orbit influenced by a body's gravity
- C. orbital neighbourhood - cloud around an active comet nucleus
- D. orbital neighbourhood - cloud from which the Solar System formed

10. Which statement best explains Solar layers in Chapter 5?

- A. a strong greenhouse effect traps heat, and clouds conceal a hot, high-pressure surface
- B. energy leaves the core through the radiative and convective zones, reaches the visible photosphere and influences the thin outer atmosphere
- C. collisions, resonances and moon gravity shape gaps and narrow structures
- D. the photosphere is called a surface for convenience even though the Sun is gas and plasma

12. Which statement best explains Dwarf planet in Chapter 6?

- A. surveys repeatedly measure positions so future paths and uncertainties can be calculated
- B. gravity amplified denser regions, collapse made rotation faster and collisions flattened much material into a disc
- C. a dwarf planet is not merely a very small planet under the formal definition
- D. known examples occupy the asteroid belt or distant trans-Neptunian region

UNIT 2 REVIEW - OLYMPIAD CHALLENGE

B. Written and visual reasoning

1. Connect one idea from Chapter 4 with one idea from Chapter 7. Explain the relationship.
2. Choose two terms from different chapters in Unit 2 and compare their roles.
3. A diagram shows all planets equally spaced. What useful information does it show, and what false impression might it create?
4. Increasing entry speed greatly increases kinetic energy because speed is squared in the energy formula.
5. Draw or describe one visual model that would help teach the central idea of Unit 2. State one limitation of the model.
6. Write one claim about Unit 2, give two pieces of evidence and state one uncertainty or boundary condition.

Olympiad Challenge

1. A student is reasoning about Nuclear fusion. Which statement is the most scientifically careful?

- A. the photosphere is called a surface for convenience even though the Sun is gas and plasma
- B. Earth is not protected by one single feature; habitability depends on interacting conditions
- C. fusion is not ordinary burning; it does not require oxygen and occurs under extreme pressure and temperature
- D. rings are not single solid discs

3. When checking a claim about Comet nucleus, which statement sets the right boundary?

- A. the nucleus is much smaller than the enormous faint coma visible around it
- B. a tail does not always trail directly behind like smoke from a vehicle
- C. the nebula was not an empty hole; it contained matter distributed through space
- D. "outer" describes position, not that the planets are outside the Solar System

5. Which conclusion about Earth avoids overclaiming?

- A. Mars is not simply a smaller Earth today, and no confirmed Martian life has been found
- B. rings are not single solid discs
- C. Earth is not protected by one single feature; habitability depends on interacting conditions
- D. an asteroid is not defined as any object that might hit Earth

7. Which explanation of Solar nebula is precise enough for a science report?

- A. the nebula was not an empty hole; it contained matter distributed through space
- B. accretion includes destructive collisions as well as successful growth
- C. "outer" describes position, not that the planets are outside the Solar System
- D. Mercury is not the hottest planet on average despite being nearest the Sun

2. Which situation is the strongest application of Ice giants?

- A. Saturn's bright rings contain ice-rich particles ranging from dust-sized grains to larger chunks
- B. Ceres and Pluto are officially recognised dwarf planets
- C. a comet moving away from the Sun still has tails pointing generally away from the Sun
- D. methane absorbs red light and contributes to blue-green appearances

4. Which unfamiliar case is best explained by Frost line?

- A. Mars and Jupiter lie on opposite sides of the main asteroid belt
- B. icy moons are common in the colder outer Solar System
- C. granular patterns on the photosphere trace convection below
- D. rovers examine layered rocks and minerals for evidence of past environments

6. Where would Dwarf planet be most useful?

- A. the main belt contains many asteroids but is spread over a vast volume
- B. a comet moving away from the Sun still has tails pointing generally away from the Sun
- C. small snowballs gain material as they roll, but early planetary collisions were much faster and hotter
- D. Ceres and Pluto are officially recognised dwarf planets

8. Which example provides the clearest evidence of Solar layers?

- A. its cratered surface resembles the Moon in many images
- B. granular patterns on the photosphere trace convection below
- C. rovers examine layered rocks and minerals for evidence of past environments
- D. Jupiter's moon Europa is studied because evidence suggests an ocean beneath ice

UNIT 3

Stars, Galaxies and the Universe

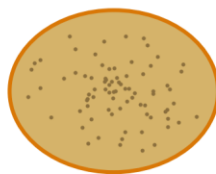
Stars are distant suns. In groups and galaxies, they reveal a universe far larger and older than the Solar System.

Three Broad Galaxy Shapes

Galaxies contain stars, gas, dust and dark matter held together by gravity.



Spiral



Elliptical



Irregular

Unit 3 opening visual - original illustration.

Major concepts

- Chapter 8: Stars: Properties and Life Cycles
- Chapter 9: Constellations, Star Clusters and Nebulae
- Chapter 10: Galaxies, the Milky Way and the Universe

Expected unit outcomes

- Relate colour to surface temperature.
- Distinguish constellation and cluster.
- Identify broad galaxy shapes.

CHAPTER 8

Stars: Properties and Life Cycles

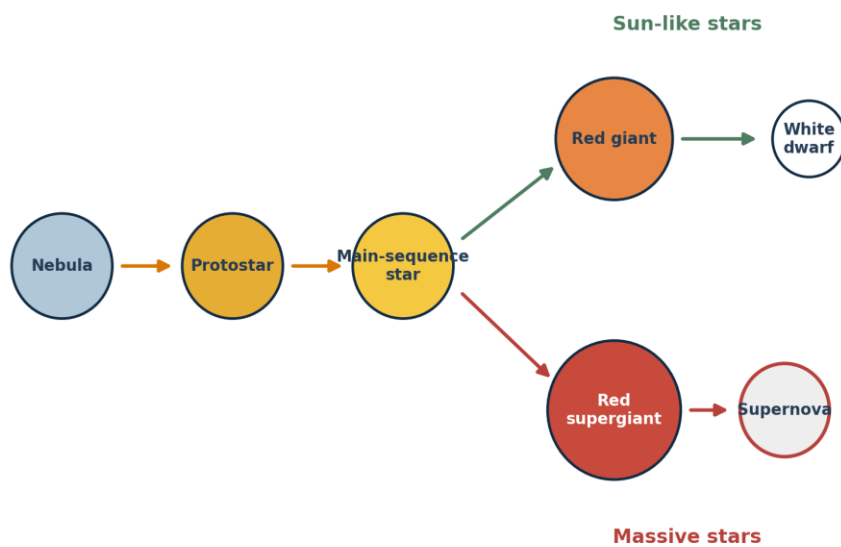
A star is a massive sphere of hot plasma held together by gravity and powered for most of its life by nuclear fusion. Stars differ in mass, surface temperature, colour, size and brightness. Their life cycles are not identical: mass determines how quickly fuel is used and which later stages are possible. Astronomers infer these properties from light rather than by visiting the stars.

Opening question Why can a dim-looking star actually release more energy than a bright-looking nearby star?

Learning outcomes

- Relate colour to surface temperature.
- Distinguish apparent and intrinsic brightness.
- Explain stellar energy and balance.
- Outline mass-dependent life cycles.

A Simplified Life Cycle of Stars



A star's mass strongly influences its later stages. Diagram is simplified.

Figure 8.1 Reading starlight. Original illustration; not to scale where stated.

Reading starlight

A point of light contains clues when it is measured carefully.

Stellar colour and temperature

In this chapter, Stellar colour and temperature is used for the connection between a star's visible colour and the temperature of its outer layers. Hotter stars tend to appear blue-white while cooler stars tend to appear orange-red, because temperature changes the distribution of emitted light. One useful example is Rigel appears blue-white and Betelgeuse appears reddish. It should be separated from this different claim: colour is a temperature clue, not a simple sign that a star is approaching or moving away.

Apparent and intrinsic brightness

Apparent brightness is how bright a star looks from Earth; luminosity is the total energy it emits each second. The process can be followed as distance spreads light over a larger area, so a luminous far star can appear faint. For instance, a nearby torch can look brighter than a distant floodlight even though it produces less light overall. A common overstatement is avoided by recognising that brightness in our sky cannot be compared fairly without accounting for distance.

Check your understanding What two main factors control how bright a star appears from Earth?

A stable main-sequence star

Gravity and pressure act in opposite directions inside a normal star.

Hydrostatic balance

The near-balance between inward gravity and outward pressure in a star. The process can be followed as gravity compresses material while hot gas and radiation pressure resist collapse; fusion helps maintain the temperature that supports pressure. For instance, the Sun remains roughly stable over human timescales because these effects balance. A common overstatement is avoided by recognising that balance does not mean nothing moves; energy and material continue to flow.

Main sequence

Main sequence concerns the long life stage during which a star fuses hydrogen in its core. The key cause-and-effect link is that higher-mass stars have more fuel but consume it much faster because their cores are hotter and their luminosities are greater. This appears in the Sun is a main-sequence star about halfway through this broad stage. At the same time, main sequence is not a row of stars in physical space; it is a pattern in a property diagram.

Check your understanding Why can a massive star have a shorter life despite containing more fuel?

Later stages

When core fuel changes, the balance and structure of a star change.

Red giant and white dwarf

Late stages expected for a Sun-like star: expansion into a red giant followed by a compact hot remnant. In practice, core changes make outer layers expand; after the outer gas is lost, the remaining core becomes a white dwarf and gradually cools. An example is planetary nebulae surround some newly exposed white dwarfs. However, a white dwarf is not a small main-sequence star and a planetary nebula does not form a planet.

Supergiant and supernova

To understand Supergiant and supernova, begin with late stages of a massive star, ending in a powerful stellar explosion. Next, note that massive cores build heavier elements until fusion can no longer support the core; collapse and rebound eject outer material. Supernova remnants enrich space with elements that can enter later stars and planets. This does not mean that not every star becomes a supernova; the outcome depends strongly on mass.

Check your understanding Which life-cycle path is expected for the Sun, and which is not?

Activity: Compare apparent brightness with lamps

Aim	to model why distance changes received light
Materials	two identical small lamps, a ruler and a dim room
Steps	1. Place one lamp twice as far from a wall as the other. 2. Compare the illuminated areas and brightness on the wall. 3. Move the lamps while keeping their power unchanged. 4. Relate the model to stars of equal luminosity at different distances.
Observe	Light from a more distant source spreads over a larger area and appears dimmer.
Expected learning	Distance must be considered when inferring a source's true output.
Safety	Use cool, low-voltage lamps and avoid trailing wires.

Chapter Review

Key ideas

- The connection between a star's visible colour and the temperature of its outer layers.
- The near-balance between inward gravity and outward pressure in a star.
- Late stages expected for a sun-like star: expansion into a red giant followed by a compact hot remnant.
- "Red stars are hotter because red means fire." Correct the colour-temperature reasoning.

Important terms

star: self-gravitating luminous sphere of hot plasma	plasma: ionised state of matter with free charged particles
luminosity: total energy a star emits per second	apparent brightness: brightness received by an observer
hydrostatic balance: balance of inward gravity and outward pressure	main sequence: core hydrogen-fusing life stage
red giant: expanded late stage of a lower-mass star	supernova: powerful explosion marking the death of some massive stars

A. Choose the correct answer

1. Which term names self-gravitating luminous sphere of hot plasma?

- A. star
- B. plasma
- C. apparent brightness
- D. main sequence

3. Choose the vocabulary item defined as total energy a star emits per second.

- A. apparent brightness
- B. main sequence
- C. luminosity
- D. supernova

5. In this chapter, what is defined as balance of inward gravity and outward pressure?

- A. hydrostatic balance
- B. main sequence
- C. supernova
- D. plasma

2. A learner describes ionised state of matter with free charged particles. What is the correct term?

- A. luminosity
- B. plasma
- C. hydrostatic balance
- D. red giant

4. Which expression would a scientist use for brightness received by an observer?

- A. hydrostatic balance
- B. red giant
- C. star
- D. apparent brightness

Exercises (continued)

6. Which statement best explains Stellar colour and temperature?

- A. distance spreads light over a larger area, so a luminous far star can appear faint
- B. hotter stars tend to appear blue-white while cooler stars tend to appear orange-red, because temperature changes the distribution of emitted light
- C. higher-mass stars have more fuel but consume it much faster because their cores are hotter and their luminosities are greater
- D. colour is a temperature clue, not a simple sign that a star is approaching or moving away

8. Which statement correctly avoids a common misunderstanding about Hydrostatic balance?

- A. main sequence is not a row of stars in physical space; it is a pattern in a property diagram
- B. late stages expected for a Sun-like star: expansion into a red giant followed by a compact hot remnant
- C. gravity compresses material while hot gas and radiation pressure resist collapse; fusion helps maintain the temperature that supports pressure
- D. balance does not mean nothing moves; energy and material continue to flow

10. Which example most directly illustrates Red giant and white dwarf?

- A. supernova remnants enrich space with elements that can enter later stars and planets
- B. planetary nebulae surround some newly exposed white dwarfs
- C. Rigel appears blue-white and Betelgeuse appears reddish
- D. a white dwarf is not a small main-sequence star and a planetary nebula does not form a planet

7. Which example most directly illustrates Apparent and intrinsic brightness?

- A. the Sun remains roughly stable over human timescales because these effects balance
- B. the Sun is a main-sequence star about halfway through this broad stage
- C. a nearby torch can look brighter than a distant floodlight even though it produces less light overall
- D. brightness in our sky cannot be compared fairly without accounting for distance

9. Which statement best explains Main sequence?

- A. higher-mass stars have more fuel but consume it much faster because their cores are hotter and their luminosities are greater
- B. core changes make outer layers expand; after the outer gas is lost, the remaining core becomes a white dwarf and gradually cools
- C. hotter stars tend to appear blue-white while cooler stars tend to appear orange-red, because temperature changes the distribution of emitted light
- D. main sequence is not a row of stars in physical space; it is a pattern in a property diagram

B. Answer in one or two sentences

1. Define star and plasma in your own words.
2. Explain how Apparent and intrinsic brightness works. Use one cause-and-effect link.
3. Compare Hydrostatic balance and Main sequence. State one similarity or connection and one difference.
4. Use the example of Red giant and white dwarf to explain why context or measurement matters.

C. Answer in detail

1. Explain the main ideas of “Stars: Properties and Life Cycles” as a connected account. Include at least four important terms from the chapter.
2. Two stars have the same apparent brightness, but one is ten times farther away. What can you infer qualitatively about luminosity?

D. Apply and reason

1. If core fusion stopped without another pressure source, gravity would compress the core.
2. “Red stars are hotter because red means fire.” Correct the colour-temperature reasoning.

E. Activity or project

Create a branching flowchart showing two simplified stellar life cycles and the role of mass.

CHAPTER 9

Constellations, Star Clusters and Nebulae

Stars may form apparent patterns, physical groups or glowing clouds. A constellation is a region viewed from Earth; a star cluster is a real group whose stars formed together; a nebula is a cloud of gas and dust that may form stars, reflect light, glow or mark a stellar death. Distinguishing these categories prevents a common mistake: treating everything that looks close in the sky as physically close.

Opening question How could measurements show that a set of stars is a real cluster rather than only a chance line-of-sight pattern?

Learning outcomes

- Distinguish constellation and cluster.
- Compare open and globular clusters.
- Classify broad nebula types.
- Explain how distance changes apparent patterns.

A Constellation Is a Pattern Seen from Earth

The stars can be at very different distances even though they form one apparent shape.

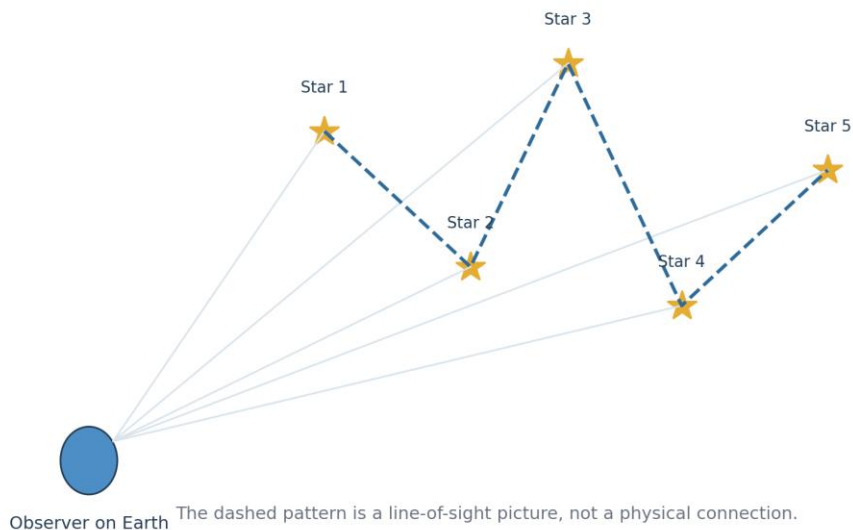


Figure 9.1 Patterns versus physical groups. Original illustration; not to scale where stated.

Patterns versus physical groups

The sky is a two-dimensional view of a three-dimensional universe.

Line-of-sight pattern

An apparent arrangement produced when objects at different distances lie in similar directions. The process can be followed as from Earth, depth is compressed into a sky picture; changing viewpoint would alter the shape. For instance, the familiar outline of Orion joins stars that are not all near one another. A common overstatement is avoided by recognising that drawing lines between stars does not prove a gravitational connection.

Common motion and distance

Common motion and distance concerns evidence that objects may form a real physical group. The key cause-and-effect link is that stars in a cluster often have similar distances, ages, chemical compositions and motions through space. This appears in measurements can show Pleiades stars moving together. At the same time, one shared property alone is not enough; several kinds of evidence strengthen the conclusion.

Check your understanding Which measurements would be most useful for testing whether stars form a real group?

Two broad cluster families

Clusters are natural laboratories because many member stars formed under similar conditions.

Open cluster

Open cluster concerns a relatively loose group of tens to thousands of generally young stars in a galactic disc. The key cause-and-effect link is that the group gradually disperses as internal motions and outside gravity pull members apart. This appears in the Pleiades is a nearby open cluster visible without a telescope. At the same time, “open” refers to a loose arrangement, not an opening in space.

Globular cluster

A dense, roughly spherical group containing many old stars. Evidence for the idea comes from situations such as a telescope resolves the edges of a globular cluster into many points. The explanation is strong mutual gravity keeps large numbers of stars together for very long times, often in a galaxy’s halo. Its boundary is equally important: a globular cluster is far smaller than a galaxy even though both contain many stars.

Check your understanding Why are cluster stars useful for comparing how stars change with age?

Nebulae: clouds with different stories

The word nebula describes appearance and material, but not one single process.

Star-forming nebula

To understand Star-forming nebula, begin with a cold or partly glowing cloud where dense regions can collapse to form stars. Next, note that gravity competes with gas pressure and turbulence; newly formed stars may illuminate or erode the surrounding cloud. The Orion Nebula contains active star formation. This does not mean that all nebulae are not nurseries; some are remains of dead stars.

Reflection, emission and remnant nebulae

Reflection, emission and remnant nebulae describes broad classes based on how a cloud is seen and how it formed. Its behaviour follows from this relationship: reflection nebulae scatter nearby starlight, emission nebulae glow when gas is energised and supernova remnants are expanding debris. The example of different wavelengths reveal dust, hot gas and shock fronts shows the idea in use. Remember that colourful images may combine filtered data and do not always match unaided-eye colour.

Check your understanding How does an emission nebula produce light differently from a reflection nebula?

Activity: Make a three-dimensional constellation

Aim	to show how one pattern can arise from different depths
Materials	cardboard base, five sticks of different lengths, clay and small paper stars
Steps	1. Attach paper stars to sticks of different lengths. 2. Place the sticks so the stars form a chosen pattern when viewed from one marked point. 3. Move to the side and observe how the pattern changes. 4. Label the model as a line-of-sight demonstration, not a scale model.
Observe	The intended pattern appears only from a particular viewpoint.
Expected learning	Apparent closeness on the sky does not prove physical closeness.
Safety	Use blunt sticks or drinking straws; handle scissors carefully.

Chapter Review

Key ideas

- An apparent arrangement produced when objects at different distances lie in similar directions.
- A relatively loose group of tens to thousands of generally young stars in a galactic disc.
- A cold or partly glowing cloud where dense regions can collapse to form stars.
- “A nebula is always where stars are being born.” Give two corrections.

Important terms

line of sight: direction from observer to an object	star cluster: physical group of stars
open cluster: loose generally young cluster	globular cluster: dense spherical generally old cluster
nebula: interstellar cloud of gas or dust	emission nebula: gas cloud that emits light after being energised
reflection nebula: dust cloud seen by scattered starlight	supernova remnant: expanding material from a stellar explosion

A. Choose the correct answer

1. Which term names direction from observer to an object?

- A. star cluster
- B. line of sight
- C. globular cluster
- D. emission nebula

3. Choose the vocabulary item defined as loose generally young cluster.

- A. globular cluster
- B. emission nebula
- C. supernova remnant
- D. open cluster

5. In this chapter, what is defined as interstellar cloud of gas or dust?

- A. emission nebula
- B. nebula
- C. supernova remnant
- D. star cluster

2. A learner describes physical group of stars. What is the correct term?

- A. open cluster
- B. nebula
- C. star cluster
- D. reflection nebula

4. Which expression would a scientist use for dense spherical generally old cluster?

- A. globular cluster
- B. nebula
- C. reflection nebula
- D. line of sight

Exercises (continued)

6. Which statement best explains Line-of-sight pattern?

- A. stars in a cluster often have similar distances, ages, chemical compositions and motions through space
- B. strong mutual gravity keeps large numbers of stars together for very long times, often in a galaxy's halo
- C. from Earth, depth is compressed into a sky picture; changing viewpoint would alter the shape
- D. drawing lines between stars does not prove a gravitational connection

8. Which statement correctly avoids a common misunderstanding about Open cluster?

- A. "open" refers to a loose arrangement, not an opening in space
- B. a globular cluster is far smaller than a galaxy even though both contain many stars
- C. a cold or partly glowing cloud where dense regions can collapse to form stars
- D. the group gradually disperses as internal motions and outside gravity pull members apart

10. Which example most directly illustrates Star-forming nebula?

- A. different wavelengths reveal dust, hot gas and shock fronts
- B. the familiar outline of Orion joins stars that are not all near one another
- C. the Orion Nebula contains active star formation
- D. all nebulae are not nurseries; some are remains of dead stars

7. Which example most directly illustrates Common motion and distance?

- A. the Pleiades is a nearby open cluster visible without a telescope
- B. a telescope resolves the edges of a globular cluster into many points
- C. one shared property alone is not enough; several kinds of evidence strengthen the conclusion
- D. measurements can show Pleiades stars moving together

9. Which statement best explains Globular cluster?

- A. gravity competes with gas pressure and turbulence; newly formed stars may illuminate or erode the surrounding cloud
- B. strong mutual gravity keeps large numbers of stars together for very long times, often in a galaxy's halo
- C. from Earth, depth is compressed into a sky picture; changing viewpoint would alter the shape
- D. a globular cluster is far smaller than a galaxy even though both contain many stars

B. Answer in one or two sentences

1. Define line of sight and star cluster in your own words.
2. Explain how Common motion and distance works. Use one cause-and-effect link.
3. Compare Open cluster and Globular cluster. State one similarity or connection and one difference.
4. Use the example of Star-forming nebula to explain why context or measurement matters.

C. Answer in detail

1. Explain the main ideas of "Constellations, Star Clusters and Nebulae" as a connected account. Include at least four important terms from the chapter.
2. A group of stars has the same age and motion but looks spread out. Could it still be a cluster? Explain.

D. Apply and reason

1. From a distant location in the Galaxy, familiar Earth constellations would look different.
2. "A nebula is always where stars are being born." Give two corrections.

E. Activity or project

Prepare an identification guide that compares constellation, open cluster, globular cluster and nebula.

CHAPTER 10

Galaxies, the Milky Way and the Universe

A galaxy is a vast system of stars, gas, dust and dark matter held together by gravity. The Sun belongs to the Milky Way, a barred spiral galaxy containing an enormous number of stars. Galaxies themselves form groups and clusters across the observable universe. At foundation level, the most important idea is scale: the Solar System is a tiny part of one galaxy, and the Milky Way is one among many galaxies.

Opening question Why does the Milky Way look like a band across a dark sky when we are located inside it?

Learning outcomes

- Identify broad galaxy shapes.
- Locate the solar system within the milky way.
- Explain universe and observable universe.
- Use light-year as a distance unit.

Three Broad Galaxy Shapes

Galaxies contain stars, gas, dust and dark matter held together by gravity.

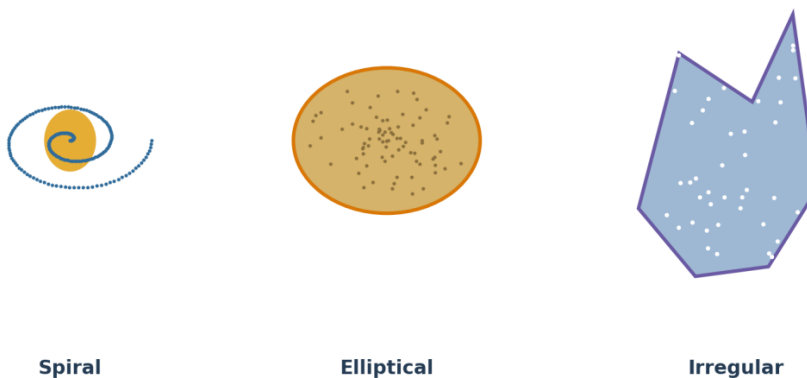


Figure 10.1 What makes a galaxy. Original illustration; not to scale where stated.

What makes a galaxy

Galaxies are organised gravitational systems, not random patches of stars.

Galaxy

Galaxy concerns a gravitationally bound system containing stars, stellar remnants, gas, dust and dark matter. The key cause-and-effect link is that stars orbit a common galactic mass distribution while gas can form new stars and feedback can reshape the galaxy. This appears in the Andromeda Galaxy is a large neighbouring spiral galaxy. At the same time, a galaxy is not the same as a Solar System; it contains many stellar systems.

Galaxy shapes

Broad observed forms such as spiral, elliptical and irregular. Evidence for the idea comes from situations such as spiral galaxies have discs and arms; ellipticals are smoother and irregulars lack a simple overall form. The explanation is shape reflects rotation, star formation, past interactions and mass distribution, though real galaxies show many intermediate structures. Its boundary is equally important: shape is a description, not a complete history by itself.

Check your understanding Which galaxy type usually shows a disc with spiral arms?

Our place in the Milky Way

Because we observe from inside the galactic disc, our view is crowded and partly blocked by dust.

Milky Way

The barred spiral galaxy that contains the Sun. Evidence for the idea comes from situations such as the pale night-sky band is the combined light of many distant stars along the disc. The explanation is the Solar System lies in the Orion Spur, away from the central bulge, and orbits the galactic centre over hundreds of millions of years. Its boundary is equally important: the Milky Way is not the entire universe and the Sun is not at its centre.

Light-year

The distance light travels in one year, about 9.46 trillion kilometres. Using a distance unit based on light avoids writing extremely long kilometre values for interstellar scales. For example, a star ten light-years away is seen by light that left about ten years earlier. One limit must remain clear: a light-year measures distance, not time, even though “year” appears in the name.

Check your understanding What do we mean when we say a star is 100 light-years away?

The larger universe

The universe includes space, time, matter, energy and the physical relationships among them.

Universe

Universe describes the totality of space-time, matter and energy described by physical laws. Its behaviour follows from this relationship: observations show galaxies at vast distances and evidence that the universe has changed over time. The example of telescopes that look farther away also see earlier periods because light takes time to arrive shows the idea in use. Remember that the universe is not known to have a simple edge like a room wall.

Observable universe

The region from which light or other signals have had time to reach us. Because the finite age of the universe and expansion limit what we can currently observe, observers may find that some regions may exist beyond the observable boundary even though their light has not reached Earth. The conclusion should remain precise: observable does not mean visible to human eyes; instruments detect many wavelengths and particles.

Check your understanding Why is looking far into space also looking back in time?

Activity: Create a scale ladder

Aim	to organise structures from local to cosmic scale
Materials	index cards and a large table or floor area
Steps	1. Write one structure on each card: Moon, Earth, Solar System, Milky Way, galaxy group and observable universe. 2. Arrange the cards from smaller to larger containing structures. 3. Add "is part of" arrows between neighbouring levels. 4. Explain why exact scale drawings are impractical.
Observe	The arrangement is nested but not perfectly simple; galaxies contain many systems and groups contain many galaxies.
Expected learning	Scale reasoning prevents confusion between objects and systems.
Safety	Keep floor layouts away from walkways.

Chapter Review

Key ideas

- A gravitationally bound system containing stars, stellar remnants, gas, dust and dark matter.
- The barred spiral galaxy that contains the sun.
- The totality of space-time, matter and energy described by physical laws.
- “The Milky Way is a milky cloud around the Solar System.” Correct the structure and viewpoint.

Important terms

galaxy: vast gravitational system of stars, gas and other matter	spiral galaxy: disc galaxy with spiral structure
elliptical galaxy: smooth rounded or oval galaxy	irregular galaxy: galaxy without a simple regular form
Milky Way: galaxy containing the Solar System	Orion Spur: small spiral feature containing the Sun
light-year: distance light travels in one year	observable universe: region whose signals can reach us

A. Choose the correct answer

1. Which term names vast gravitational system of stars, gas and other matter?

- A. spiral galaxy
- B. irregular galaxy
- C. galaxy
- D. Orion Spur

3. Choose the vocabulary item defined as smooth rounded or oval galaxy.

- A. elliptical galaxy
- B. irregular galaxy
- C. Orion Spur
- D. observable universe

5. In this chapter, what is defined as galaxy containing the Solar System?

- A. Orion Spur
- B. observable universe
- C. Milky Way
- D. spiral galaxy

2. A learner describes disc galaxy with spiral structure. What is the correct term?

- A. elliptical galaxy
- B. Milky Way
- C. light-year
- D. spiral galaxy

4. Which expression would a scientist use for galaxy without a simple regular form?

- A. Milky Way
- B. irregular galaxy
- C. light-year
- D. galaxy

Exercises (continued)

6. Which statement best explains Galaxy?

- A. shape reflects rotation, star formation, past interactions and mass distribution, though real galaxies show many intermediate structures
- B. using a distance unit based on light avoids writing extremely long kilometre values for interstellar scales
- C. a galaxy is not the same as a Solar System; it contains many stellar systems
- D. stars orbit a common galactic mass distribution while gas can form new stars and feedback can reshape the galaxy

8. Which statement correctly avoids a common misunderstanding about Milky Way?

- A. a light-year measures distance, not time, even though “year” appears in the name
- B. the Milky Way is not the entire universe and the Sun is not at its centre
- C. the totality of space-time, matter and energy described by physical laws
- D. the Solar System lies in the Orion Spur, away from the central bulge, and orbits the galactic centre over hundreds of millions of years

10. Which example most directly illustrates Universe?

- A. some regions may exist beyond the observable boundary even though their light has not reached Earth
- B. the Andromeda Galaxy is a large neighbouring spiral galaxy
- C. the universe is not known to have a simple edge like a room wall
- D. telescopes that look farther away also see earlier periods because light takes time to arrive

7. Which example most directly illustrates Galaxy shapes?

- A. spiral galaxies have discs and arms; ellipticals are smoother and irregulars lack a simple overall form
- B. the pale night-sky band is the combined light of many distant stars along the disc
- C. a star ten light-years away is seen by light that left about ten years earlier
- D. shape is a description, not a complete history by itself

9. Which statement best explains Light-year?

- A. observations show galaxies at vast distances and evidence that the universe has changed over time
- B. stars orbit a common galactic mass distribution while gas can form new stars and feedback can reshape the galaxy
- C. using a distance unit based on light avoids writing extremely long kilometre values for interstellar scales
- D. a light-year measures distance, not time, even though “year” appears in the name

B. Answer in one or two sentences

1. Define galaxy and spiral galaxy in your own words.
2. Explain how Galaxy shapes works. Use one cause-and-effect link.
3. Compare Milky Way and Light-year. State one similarity or connection and one difference.
4. Use the example of Universe to explain why context or measurement matters.

C. Answer in detail

1. Explain the main ideas of “Galaxies, the Milky Way and the Universe” as a connected account. Include at least four important terms from the chapter.
2. A student says the Andromeda Galaxy is “another Solar System.” What hierarchy mistake has been made?

D. Apply and reason

1. If light travelled instantly, distant observations would not show earlier states, but many other physical relationships would also change.
2. “The Milky Way is a milky cloud around the Solar System.” Correct the structure and viewpoint.

E. Activity or project

Make a fold-out scale ladder with one verified comparison at each level.

UNIT 3 REVIEW

Stars, Galaxies and the Universe - Mixed Review

This review combines ideas from all chapters in the unit. Use diagrams, units and evidence where relevant. The questions are new combinations rather than copies of the chapter exercises.

A. Mixed multiple-choice review

1. Which option links a term with its correct meaning?

- A. apparent brightness - core hydrogen-fusing life stage
- B. apparent brightness - direction from observer to an object
- C. apparent brightness - interstellar cloud of gas or dust
- D. apparent brightness - brightness received by an observer

3. Only one pair is scientifically correct. Which is it?

- A. star cluster - dense spherical generally old cluster
- B. star cluster - physical group of stars
- C. star cluster - dust cloud seen by scattered starlight
- D. star cluster - smooth rounded or oval galaxy

5. Find the pair that uses the vocabulary correctly.

- A. supernova remnant - disc galaxy with spiral structure
- B. supernova remnant - galaxy containing the Solar System
- C. supernova remnant - self-gravitating luminous sphere of hot plasma
- D. supernova remnant - expanding material from a stellar explosion

7. Choose the correctly defined term from Unit 3.

- A. Orion Spur - region whose signals can reach us
- B. Orion Spur - small spiral feature containing the Sun
- C. Orion Spur - total energy a star emits per second
- D. Orion Spur - expanded late stage of a lower-mass star

9. Which statement best explains Main sequence in Chapter 8?

- A. massive cores build heavier elements until fusion can no longer support the core; collapse and rebound eject outer material
- B. higher-mass stars have more fuel but consume it much faster because their cores are hotter and their luminosities are greater
- C. gravity competes with gas pressure and turbulence; newly formed stars may illuminate or erode the surrounding cloud
- D. main sequence is not a row of stars in physical space; it is a pattern in a property diagram

11. Which statement best explains Galaxy shapes in Chapter 10?

- A. using a distance unit based on light avoids writing extremely long kilometre values for interstellar scales
- B. gravity compresses material while hot gas and radiation pressure resist collapse; fusion helps maintain the temperature that supports pressure
- C. shape is a description, not a complete history by itself
- D. shape reflects rotation, star formation, past interactions and mass distribution, though real galaxies show many intermediate structures

2. Select the accurately matched term and definition.

- A. red giant - expanded late stage of a lower-mass star
- B. red giant - direction from observer to an object
- C. red giant - dense spherical generally old cluster
- D. red giant - expanding material from a stellar explosion

4. Which term-definition pair would belong in the Unit 3 glossary?

- A. nebula - dust cloud seen by scattered starlight
- B. nebula - disc galaxy with spiral structure
- C. nebula - interstellar cloud of gas or dust
- D. nebula - small spiral feature containing the Sun

6. Which pairing preserves the meaning used in this unit?

- A. elliptical galaxy - smooth rounded or oval galaxy
- B. elliptical galaxy - galaxy containing the Solar System
- C. elliptical galaxy - region whose signals can reach us
- D. elliptical galaxy - brightness received by an observer

8. Which row could be copied into an accurate revision table?

- A. star - total energy a star emits per second
- B. star - core hydrogen-fusing life stage
- C. star - self-gravitating luminous sphere of hot plasma
- D. star - physical group of stars

10. Which statement best explains Open cluster in Chapter 9?

- A. gravity competes with gas pressure and turbulence; newly formed stars may illuminate or erode the surrounding cloud
- B. using a distance unit based on light avoids writing extremely long kilometre values for interstellar scales
- C. the group gradually disperses as internal motions and outside gravity pull members apart
- D. "open" refers to a loose arrangement, not an opening in space

12. Which statement best explains Stellar colour and temperature in Chapter 8?

- A. hotter stars tend to appear blue-white while cooler stars tend to appear orange-red, because temperature changes the distribution of emitted light
- B. gravity compresses material while hot gas and radiation pressure resist collapse; fusion helps maintain the temperature that supports pressure
- C. stars in a cluster often have similar distances, ages, chemical compositions and motions through space
- D. colour is a temperature clue, not a simple sign that a star is approaching or moving away

UNIT 3 REVIEW - OLYMPIAD CHALLENGE

B. Written and visual reasoning

1. Connect one idea from Chapter 8 with one idea from Chapter 10. Explain the relationship.
2. Choose two terms from different chapters in Unit 3 and compare their roles.
3. Two stars have the same apparent brightness, but one is ten times farther away. What can you infer qualitatively about luminosity?
4. If light travelled instantly, distant observations would not show earlier states, but many other physical relationships would also change.
5. Draw or describe one visual model that would help teach the central idea of Unit 3. State one limitation of the model.
6. Write one claim about Unit 3, give two pieces of evidence and state one uncertainty or boundary condition.

Olympiad Challenge

1. A student is reasoning about Globular cluster. Which statement is the most scientifically careful?

- A. all nebulae are not nurseries; some are remains of dead stars
- B. shape is a description, not a complete history by itself
- C. observable does not mean visible to human eyes; instruments detect many wavelengths and particles
- D. a globular cluster is far smaller than a galaxy even though both contain many stars

3. When checking a claim about Supergiant and supernova, which statement sets the right boundary?

- A. drawing lines between stars does not prove a gravitational connection
- B. not every star becomes a supernova; the outcome depends strongly on mass
- C. a globular cluster is far smaller than a galaxy even though both contain many stars
- D. shape is a description, not a complete history by itself

5. Which conclusion about Apparent and intrinsic brightness avoids overclaiming?

- A. balance does not mean nothing moves; energy and material continue to flow
- B. not every star becomes a supernova; the outcome depends strongly on mass
- C. a globular cluster is far smaller than a galaxy even though both contain many stars
- D. brightness in our sky cannot be compared fairly without accounting for distance

7. Which explanation of Light-year is precise enough for a science report?

- A. the universe is not known to have a simple edge like a room wall
- B. a light-year measures distance, not time, even though "year" appears in the name
- C. brightness in our sky cannot be compared fairly without accounting for distance
- D. not every star becomes a supernova; the outcome depends strongly on mass

2. Which situation is the strongest application of Universe?

- A. telescopes that look farther away also see earlier periods because light takes time to arrive
- B. some regions may exist beyond the observable boundary even though their light has not reached Earth
- C. the Sun remains roughly stable over human timescales because these effects balance
- D. the familiar outline of Orion joins stars that are not all near one another

4. Which unfamiliar case is best explained by Galaxy?

- A. spiral galaxies have discs and arms; ellipticals are smoother and irregulars lack a simple overall form
- B. telescopes that look farther away also see earlier periods because light takes time to arrive
- C. the Andromeda Galaxy is a large neighbouring spiral galaxy
- D. the Sun remains roughly stable over human timescales because these effects balance

6. Where would Open cluster be most useful?

- A. the Pleiades is a nearby open cluster visible without a telescope
- B. a telescope resolves the edges of a globular cluster into many points
- C. the Andromeda Galaxy is a large neighbouring spiral galaxy
- D. telescopes that look farther away also see earlier periods because light takes time to arrive

8. Which example provides the clearest evidence of Red giant and white dwarf?

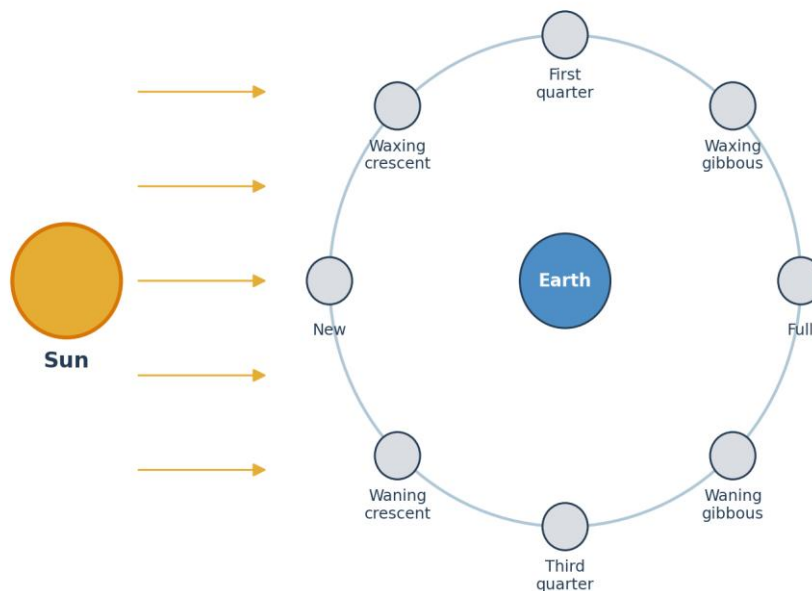
- A. supernova remnants enrich space with elements that can enter later stars and planets
- B. the Pleiades is a nearby open cluster visible without a telescope
- C. planetary nebulae surround some newly exposed white dwarfs
- D. the Andromeda Galaxy is a large neighbouring spiral galaxy

UNIT 4**Earth and Moon**

Earth's motions and its partnership with the Moon produce familiar cycles: day, year, seasons, phases, eclipses and tides.

Moon Phases: Sunlight and Our View

The Sun always illuminates half the Moon. The phase depends on viewing geometry.



Positions and sizes are not to scale.

Unit 4 opening visual - original illustration.

Major concepts

- Chapter 11: Earth's Structure and Atmosphere
- Chapter 12: Rotation, Revolution, Seasons and Time
- Chapter 13: The Moon: Phases, Eclipses and Tides

Expected unit outcomes

- Name major internal layers.
- Distinguish rotation and revolution.
- Sequence the lunar phases.

CHAPTER 11

Earth's Structure and Atmosphere

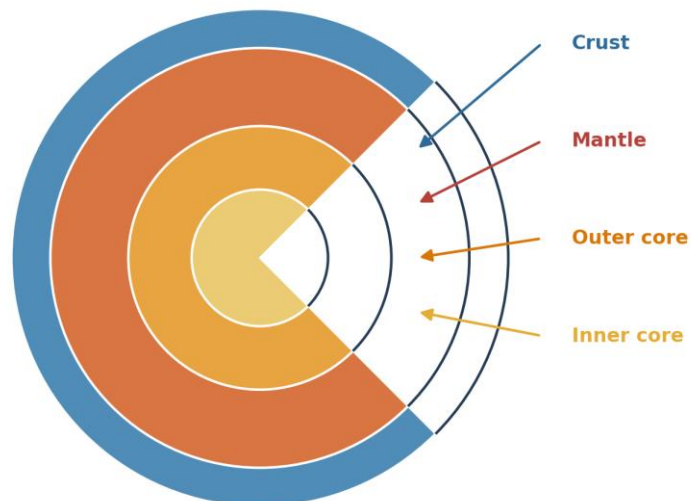
Earth is a layered rocky planet. Its solid outer crust, thick mantle, metallic core and surrounding atmosphere interact through flows of matter and energy. We cannot travel to the deep interior, so scientists combine earthquake waves, gravity, magnetism, laboratory experiments and meteorite evidence. The atmosphere is also layered, becoming thinner with height while supporting weather and shielding life.

Opening question How can scientists know about Earth's core without drilling anywhere near it?

Learning outcomes

- Name major internal layers.
- Explain evidence from seismic waves.
- Describe atmospheric layers.
- Connect atmosphere to weather and protection.

Earth's Internal Layers



Layer thicknesses are not shown to scale.

Figure 11.1 Inside a differentiated planet. Original illustration; not to scale where stated.

Inside a differentiated planet

Earth's layers can be described by composition or by physical behaviour.

Crust and mantle

The rocky outer regions of Earth, with a thin crust above a much thicker mantle. Evidence for the idea comes from situations such as volcanic rock brings material from upper regions toward the surface. The explanation is slow mantle flow transfers heat and helps move tectonic plates, while the crust forms continents and ocean floors. Its boundary is equally important: the mantle is mostly solid rock that can flow slowly; it is not a global ocean of liquid magma.

Core

Earth's mainly iron-nickel central region, divided into liquid outer core and solid inner core. Pressure keeps the inner core solid despite high temperature, while movement in the liquid outer core helps generate Earth's magnetic field. For example, seismic waves change speed and some types do not pass through liquids. One limit must remain clear: the core is not empty and is not made of ordinary fire.

Check your understanding Which evidence suggests that the outer core is liquid?

Reading earthquake waves

Earthquakes act like natural probes of the interior.

Seismic wave

A vibration that carries energy through Earth. Different wave types travel at different speeds and respond differently to solids and liquids, creating patterns detected around the planet. For example, a shadow zone shows where certain waves do not arrive directly. One limit must remain clear: scientists infer layers from many events and detectors, not from one earthquake alone.

Density and pressure

Density and pressure means properties that generally increase toward Earth's centre. It works because gravity pulled denser material inward during differentiation, and the weight of overlying layers produces enormous pressure. A concrete case is iron-rich meteorites provide clues to materials available during planet formation. By contrast, high pressure can change melting behaviour, so high temperature does not always mean liquid.

Check your understanding Why are many seismometer stations more useful than one?

A layered atmosphere

Atmospheric names mark broad regions with characteristic temperature behaviour and processes.

Troposphere and stratosphere

The lowest layers: the troposphere contains most weather; the stratosphere includes the ozone-rich region. Because solar heating of the surface drives tropospheric circulation, while ozone absorbs much ultraviolet radiation higher up, observers may find that commercial aircraft often cruise near the upper troposphere or lower stratosphere. The conclusion should remain precise: weather does not occur equally through all atmospheric layers.

Upper atmosphere

In this chapter, Upper atmosphere is used for the mesosphere, thermosphere and exosphere above the stratosphere. Meteors commonly glow in the mesosphere; auroras occur mainly in the thermosphere; the exosphere gradually merges with space. One useful example is some satellites orbit where extremely thin atmospheric particles still exist. It should be separated from this different claim: space does not begin at a perfectly sharp natural wall.

Check your understanding In which broad atmospheric layer does most everyday weather occur?

Activity: Model indirect evidence

Aim	to infer hidden interiors from measurements
Materials	three sealed identical boxes containing different safe objects, balance and magnet
Steps	1. Without opening the boxes, compare mass and response to gentle tilting. 2. Test whether a magnet affects the contents through the box. 3. Record observations before proposing what each box contains. 4. Open the boxes only after every group has stated evidence and uncertainty.
Observe	Different evidence narrows possibilities but may not identify one unique answer immediately.
Expected learning	Science often studies inaccessible regions through indirect effects.
Safety	Use sealed lightweight objects and keep magnets away from electronics.

Chapter Review

Key ideas

- The rocky outer regions of earth, with a thin crust above a much thicker mantle.
- A vibration that carries energy through earth.
- The lowest layers: the troposphere contains most weather; the stratosphere includes the ozone-rich region.
- “The mantle is liquid because volcanoes contain lava.” Explain the mistake.

Important terms

crust: thin rocky outer layer of Earth	mantle: thick rocky layer that flows slowly
outer core: liquid metallic layer	inner core: solid metallic centre
seismic wave: wave carrying earthquake energy through Earth	troposphere: lowest atmospheric layer and main weather region
stratosphere: layer above the troposphere containing much ozone	exosphere: outermost atmospheric region fading into space

A. Choose the correct answer

1. Which term names thin rocky outer layer of Earth?

- A. mantle
- B. inner core
- C. troposphere
- D. crust

3. Choose the vocabulary item defined as liquid metallic layer.

- A. inner core
- B. outer core
- C. troposphere
- D. exosphere

5. In this chapter, what is defined as wave carrying earthquake energy through Earth?

- A. troposphere
- B. exosphere
- C. mantle
- D. seismic wave

2. A learner describes thick rocky layer that flows slowly. What is the correct term?

- A. mantle
- B. outer core
- C. seismic wave
- D. stratosphere

4. Which expression would a scientist use for solid metallic centre?

- A. seismic wave
- B. stratosphere
- C. inner core
- D. crust

Exercises (continued)

6. Which statement best explains Crust and mantle?

- A. slow mantle flow transfers heat and helps move tectonic plates, while the crust forms continents and ocean floors
- B. pressure keeps the inner core solid despite high temperature, while movement in the liquid outer core helps generate Earth's magnetic field
- C. gravity pulled denser material inward during differentiation, and the weight of overlying layers produces enormous pressure
- D. the mantle is mostly solid rock that can flow slowly; it is not a global ocean of liquid magma

8. Which statement correctly avoids a common misunderstanding about Seismic wave?

- A. high pressure can change melting behaviour, so high temperature does not always mean liquid
- B. the lowest layers: the troposphere contains most weather; the stratosphere includes the ozone-rich region
- C. scientists infer layers from many events and detectors, not from one earthquake alone
- D. different wave types travel at different speeds and respond differently to solids and liquids, creating patterns detected around the planet

10. Which example most directly illustrates Troposphere and stratosphere?

- A. commercial aircraft often cruise near the upper troposphere or lower stratosphere
- B. some satellites orbit where extremely thin atmospheric particles still exist
- C. volcanic rock brings material from upper regions toward the surface
- D. weather does not occur equally through all atmospheric layers

7. Which example most directly illustrates Core?

- A. a shadow zone shows where certain waves do not arrive directly
- B. seismic waves change speed and some types do not pass through liquids
- C. iron-rich meteorites provide clues to materials available during planet formation
- D. the core is not empty and is not made of ordinary fire

9. Which statement best explains Density and pressure?

- A. solar heating of the surface drives tropospheric circulation, while ozone absorbs much ultraviolet radiation higher up
- B. slow mantle flow transfers heat and helps move tectonic plates, while the crust forms continents and ocean floors
- C. high pressure can change melting behaviour, so high temperature does not always mean liquid
- D. gravity pulled denser material inward during differentiation, and the weight of overlying layers produces enormous pressure

B. Answer in one or two sentences

1. Define crust and mantle in your own words.
2. Explain how Core works. Use one cause-and-effect link.
3. Compare Seismic wave and Density and pressure. State one similarity or connection and one difference.
4. Use the example of Troposphere and stratosphere to explain why context or measurement matters.

C. Answer in detail

1. Explain the main ideas of "Earth's Structure and Atmosphere" as a connected account. Include at least four important terms from the chapter.
2. If seismic waves suddenly travelled at one speed everywhere, what information about layers would be lost?

D. Apply and reason

1. A stronger atmosphere would alter surface pressure and climate, but outcomes would depend on composition and sunlight.
2. "The mantle is liquid because volcanoes contain lava." Explain the mistake.

E. Activity or project

Create a labelled Earth cross-section with evidence beside each inferred layer.

CHAPTER 12

Rotation, Revolution, Seasons and Time

Earth turns on its axis and travels around the Sun. Rotation produces the daily cycle of day and night and the apparent daily motion of the sky. Revolution, combined with a tilted axis, produces seasonal changes in sunlight angle and day length. Longitude and Earth's rotation also provide the basis for local time and time zones.

Opening question Why is summer not caused simply by Earth being much closer to the Sun?

Learning outcomes

- Distinguish rotation and revolution.
- Explain day and night.
- Explain seasons using axial tilt.
- Solve basic longitude-time problems.

Why Earth Has Seasons

Earth's axis keeps nearly the same tilt direction as Earth travels around the Sun.

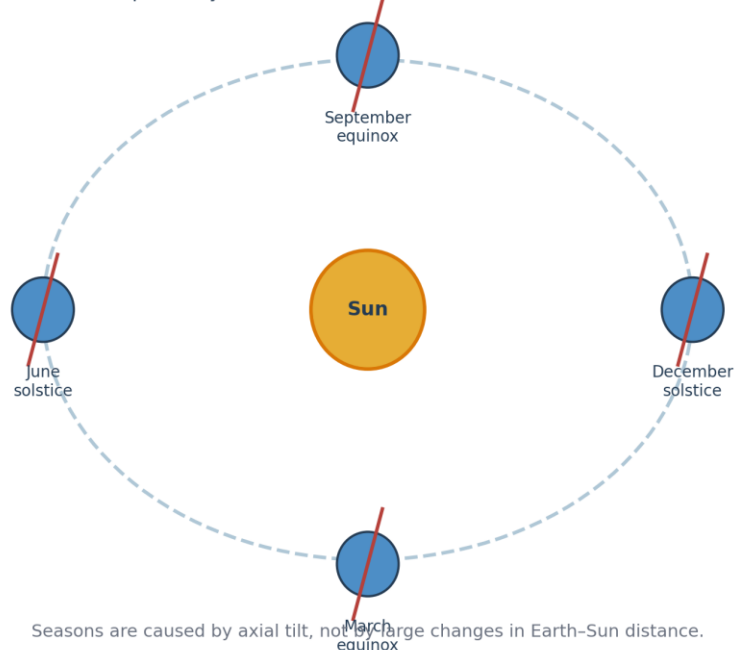


Figure 12.1 Two motions, two time scales. Original illustration; not to scale where stated.

Two motions, two time scales

The same planet can rotate and revolve at the same time.

Rotation

Spinning of Earth around its axis approximately once each day. Locations turn into and out of sunlight, creating day and night, while sky objects appear to move westward. For example, India faces the Sun at local daytime while another region may face away. One limit must remain clear: the Sun does not circle Earth once daily even though that is the apparent view.

Revolution

Revolution means Earth's orbital motion around the Sun once each year. It works because gravity continually bends Earth's forward motion into an orbit. A concrete case is the changing background of stars across the year reflects Earth's changing orbital position. By contrast, revolution is not the same as spinning around the axis.

Check your understanding Which motion mainly explains a 24-hour cycle, and which explains a year?

Sunlight and seasons

Earth's axis is tilted about 23.5 degrees relative to the perpendicular to its orbital plane.

Axial tilt

Axial tilt means the angle between Earth's rotation axis and the perpendicular to its orbital plane. It works because as Earth revolves, one hemisphere is tilted more toward the Sun and later more away, changing day length and sunlight angle. A concrete case is Northern Hemisphere summer occurs when the north is tilted toward the Sun. By contrast, the axis does not repeatedly lean toward and away from the Sun; it maintains nearly the same direction in space.

Season

A part of the year marked by regular changes in daylight and solar heating. The underlying relationship is that more direct sunlight concentrates energy over a smaller surface area and longer days increase heating time. This can be seen when December is summer in much of the Southern Hemisphere while it is winter in the Northern Hemisphere. Seasons are not simultaneous and identical everywhere, and local climate has additional influences.

Check your understanding How do sunlight angle and day length both contribute to seasonal heating?

Longitude and clock time

Earth's rotation links east-west position to the time when the Sun appears highest.

Local solar time

In this chapter, Local solar time is used for time based on the Sun's apparent daily position for a particular longitude. Places farther east turn toward the Sun earlier, so their local solar time is ahead of places to the west. One useful example is when it is noon near one longitude, a point 15 degrees west is about one hour earlier by solar time. It should be separated from this different claim: countries use standard times, so clock noon need not equal local solar noon.

Time zone and IST

A region using a common standard clock time; Indian Standard Time is based on 82.5 degrees east longitude. The process can be followed as Earth turns 360 degrees in about 24 hours, giving 15 degrees per hour; a standard meridian prevents every town using a different clock. For instance, IST is UTC plus 5 hours 30 minutes. A common overstatement is avoided by recognising that time-zone borders often follow political boundaries rather than exact longitude lines.

Check your understanding If one place lies 30 degrees east of another, about how much later is the western place by local solar time?

Activity: Test sunlight angle with graph paper

Aim	to compare energy spread at different angles
Materials	torch, graph paper, books and a dim room
Steps	1. Hold the torch at a fixed distance and shine it straight onto graph paper. 2. Trace or count the brightly lit squares. 3. Tilt the paper without changing torch distance and repeat. 4. Compare the illuminated area and brightness.
Observe	Oblique light spreads the same beam over a larger area and is less concentrated.
Expected learning	Sun angle changes the concentration of received energy.
Safety	Use a cool torch, not a hot lamp or the Sun.

Chapter Review

Key ideas

- Spinning of earth around its axis approximately once each day.
- The angle between earth's rotation axis and the perpendicular to its orbital plane.
- Time based on the sun's apparent daily position for a particular longitude.
- "Earth is closest to the Sun during summer everywhere." Correct the hemisphere and distance reasoning.

Important terms

rotation: spinning about an axis	axis: imaginary line about which a body rotates
revolution: orbital motion around another body	axial tilt: inclination of a rotation axis
season: yearly period with characteristic sunlight conditions	longitude: east-west angular coordinate on Earth
time zone: region using a common standard time	IST: Indian Standard Time, UTC+5:30

A. Choose the correct answer

1. Which term names spinning about an axis?

- A. rotation
- B. axis
- C. axial tilt
- D. longitude

3. Choose the vocabulary item defined as orbital motion around another body.

- A. axial tilt
- B. longitude
- C. revolution
- D. IST

5. In this chapter, what is defined as yearly period with characteristic sunlight conditions?

- A. season
- B. longitude
- C. IST
- D. axis

2. A learner describes imaginary line about which a body rotates.

What is the correct term?

- A. revolution
- B. axis
- C. season
- D. time zone

4. Which expression would a scientist use for inclination of a rotation axis?

- A. season
- B. time zone
- C. rotation
- D. axial tilt

Exercises (continued)

6. Which statement best explains Rotation?

- A. gravity continually bends Earth's forward motion into an orbit
- B. locations turn into and out of sunlight, creating day and night, while sky objects appear to move westward
- C. more direct sunlight concentrates energy over a smaller surface area and longer days increase heating time
- D. the Sun does not circle Earth once daily even though that is the apparent view

8. Which statement correctly avoids a common misunderstanding about Axial tilt?

- A. seasons are not simultaneous and identical everywhere, and local climate has additional influences
- B. time based on the Sun's apparent daily position for a particular longitude
- C. as Earth revolves, one hemisphere is tilted more toward the Sun and later more away, changing day length and sunlight angle
- D. the axis does not repeatedly lean toward and away from the Sun; it maintains nearly the same direction in space

10. Which example most directly illustrates Local solar time?

- A. IST is UTC plus 5 hours 30 minutes
- B. when it is noon near one longitude, a point 15 degrees west is about one hour earlier by solar time
- C. India faces the Sun at local daytime while another region may face away
- D. countries use standard times, so clock noon need not equal local solar noon

7. Which example most directly illustrates Revolution?

- A. Northern Hemisphere summer occurs when the north is tilted toward the Sun
- B. December is summer in much of the Southern Hemisphere while it is winter in the Northern Hemisphere
- C. the changing background of stars across the year reflects Earth's changing orbital position
- D. revolution is not the same as spinning around the axis

9. Which statement best explains Season?

- A. more direct sunlight concentrates energy over a smaller surface area and longer days increase heating time
- B. places farther east turn toward the Sun earlier, so their local solar time is ahead of places to the west
- C. locations turn into and out of sunlight, creating day and night, while sky objects appear to move westward
- D. seasons are not simultaneous and identical everywhere, and local climate has additional influences

B. Answer in one or two sentences

1. Define rotation and axis in your own words.
2. Explain how Revolution works. Use one cause-and-effect link.
3. Compare Axial tilt and Season. State one similarity or connection and one difference.
4. Use the example of Local solar time to explain why context or measurement matters.

C. Answer in detail

1. Explain the main ideas of "Rotation, Revolution, Seasons and Time" as a connected account. Include at least four important terms from the chapter.
2. Two cities have the same clock time but different longitudes. Why might local solar noon differ?

D. Apply and reason

1. Without axial tilt, seasonal day-length and sunlight-angle changes would be much smaller.
2. "Earth is closest to the Sun during summer everywhere." Correct the hemisphere and distance reasoning.

E. Activity or project

Construct four Earth–Sun diagrams and annotate the seasons in both hemispheres.

CHAPTER 13

The Moon: Phases, Eclipses and Tides

The Moon orbits Earth while reflecting sunlight. As its position changes, observers see different fractions of the illuminated half, producing phases. Eclipses occur only when the Sun, Earth and Moon align closely near the tilted orbit's crossing points. The Moon's gravity also contributes strongly to ocean tides, together with the Sun and the shape of ocean basins.

Opening question Why do Moon phases occur every month but eclipses do not?

Learning outcomes

- Sequence the lunar phases.
- Distinguish phases from shadows.
- Explain solar and lunar eclipses.
- Describe the basic tidal pattern.

Moon Phases: Sunlight and Our View

The Sun always illuminates half the Moon. The phase depends on viewing geometry.

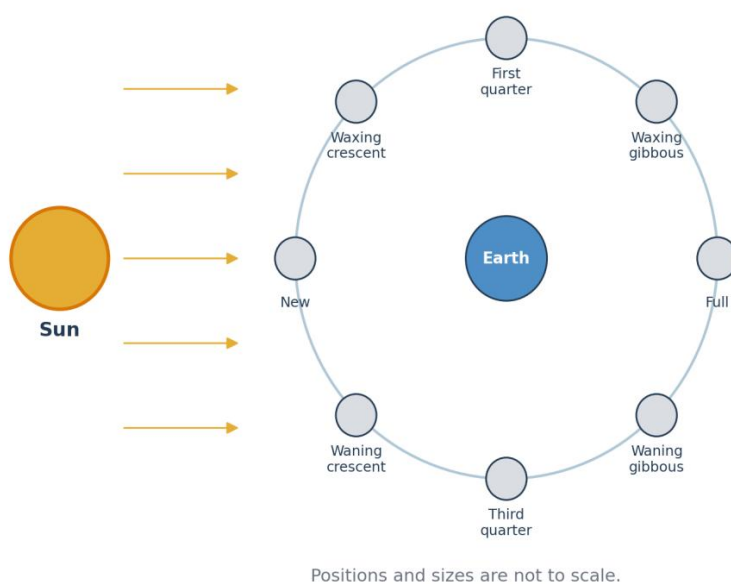


Figure 13.1 Phases are viewing geometry. Original illustration; not to scale where stated.

Phases are viewing geometry

Half of the Moon is sunlit at every ordinary phase.

Lunar phase

Lunar phase means the changing visible fraction of the Moon's sunlit half. It works because the Moon's position around Earth changes the angle between the Sun, Moon and observer. A concrete case is the sequence runs from new through waxing phases to full, then waning phases back to new. By contrast, Earth's shadow does not cause ordinary phases.

Waxing and waning

Waxing means the illuminated fraction seen from Earth is increasing; waning means it is decreasing. The underlying relationship is that the synodic phase cycle takes about 29.5 days because Earth also moves around the Sun while the Moon orbits Earth. This can be seen when a first-quarter Moon is half illuminated in appearance but about one quarter through the phase cycle. "quarter" names orbital progress, not the fraction of the disc that is bright.

Check your understanding At which phase is the Moon roughly opposite the Sun in the sky?

Eclipse alignments

The Moon's orbit is tilted by about five degrees to Earth's orbital plane.

Solar eclipse

An event in which the Moon passes between Earth and the Sun and its shadow falls on part of Earth. The underlying relationship is that only observers in particular shadow regions see totality, while a wider region may see a partial eclipse. This can be seen when a total solar eclipse reveals the Sun's corona for a short time. A solar eclipse is unsafe to view without proper protection except during the brief total phase at a location experiencing totality.

Lunar eclipse

The phrase Lunar eclipse refers to an event in which the full Moon passes through Earth's shadow. Here, sunlight filtered and bent through Earth's atmosphere can make the Moon appear copper-red during totality. Consider the following case: a lunar eclipse can be seen from much of Earth's night side. The careful interpretation is that the Moon does not produce red light; Earth's atmosphere changes the sunlight reaching it.

Check your understanding Which body lies in the middle during a lunar eclipse?

Tides and gravitational difference

Tides arise from differences in gravitational pull across Earth and from the Earth–Moon system’s motion.

Tide

A regular rise and fall of sea level caused mainly by the Moon’s gravity, with an important solar contribution. The process can be followed as ocean water forms broad bulges and Earth rotates through them, while coastlines and depth alter timing and height. For instance, many coasts experience about two high and two low tides each lunar day. A common overstatement is avoided by recognising that a high tide is not simply water being pulled only toward the Moon.

Spring and neap tides

Spring and neap tides concerns larger tidal ranges near new and full Moon; smaller ranges near first and third quarter. The key cause-and-effect link is that solar and lunar tidal effects reinforce when Sun, Earth and Moon are aligned and partly oppose when their directions are near right angles. This appears in spring tides occur twice each phase cycle in every season. At the same time, “spring” here means springing up, not the season.

Check your understanding At which lunar phases are spring tides expected?

Activity: Model Moon phases with a ball

Aim	to connect orbital position with illuminated appearance
Materials	white ball on a stick, one unshaded lamp and a darkened room
Steps	1. Place the lamp at the centre and hold the ball at arm’s length. 2. Turn slowly while keeping the ball near eye height. 3. Name the visible phase at eight positions. 4. Notice that the lamp always illuminates half the ball.
Observe	The viewed fraction changes even though the illuminated half remains.
Expected learning	Phases result from geometry, not changing light production.
Safety	Keep clear of furniture and use a cool lamp; do not use the Sun.

Chapter Review

Key ideas

- The changing visible fraction of the moon's sunlit half.
- An event in which the moon passes between earth and the sun and its shadow falls on part of earth.
- A regular rise and fall of sea level caused mainly by the moon's gravity, with an important solar contribution.
- "The Moon becomes dark because Earth blocks sunlight every month." Correct the claim.

Important terms

lunar phase: appearance of the Moon's illuminated portion	waxing: increasing visible illuminated fraction
waning: decreasing visible illuminated fraction	solar eclipse: Moon blocks the Sun from part of Earth
lunar eclipse: Moon passes through Earth's shadow	umbra: dark central part of a shadow
tide: periodic sea-level change driven mainly by gravity	spring tide: large tidal range near alignment

A. Choose the correct answer

1. Which term names appearance of the Moon's illuminated portion?

- A. waxing
- B. lunar phase
- C. solar eclipse
- D. umbra

3. Choose the vocabulary item defined as decreasing visible illuminated fraction.

- A. solar eclipse
- B. umbra
- C. spring tide
- D. waning

5. In this chapter, what is defined as Moon passes through Earth's shadow?

- A. umbra
- B. lunar eclipse
- C. spring tide
- D. waxing

2. A learner describes increasing visible illuminated fraction. What is the correct term?

- A. waning
- B. lunar eclipse
- C. waxing
- D. tide

4. Which expression would a scientist use for Moon blocks the Sun from part of Earth?

- A. solar eclipse
- B. lunar eclipse
- C. tide
- D. lunar phase

Exercises (continued)

6. Which statement best explains Lunar phase?

- A. the synodic phase cycle takes about 29.5 days because Earth also moves around the Sun while the Moon orbits Earth
- B. sunlight filtered and bent through Earth's atmosphere can make the Moon appear copper-red during totality
- C. the Moon's position around Earth changes the angle between the Sun, Moon and observer
- D. Earth's shadow does not cause ordinary phases

8. Which statement correctly avoids a common misunderstanding about Solar eclipse?

- A. a solar eclipse is unsafe to view without proper protection except during the brief total phase at a location experiencing totality
- B. the Moon does not produce red light; Earth's atmosphere changes the sunlight reaching it
- C. a regular rise and fall of sea level caused mainly by the Moon's gravity, with an important solar contribution
- D. only observers in particular shadow regions see totality, while a wider region may see a partial eclipse

10. Which example most directly illustrates Tide?

- A. spring tides occur twice each phase cycle in every season
- B. the sequence runs from new through waxing phases to full, then waning phases back to new
- C. many coasts experience about two high and two low tides each lunar day
- D. a high tide is not simply water being pulled only toward the Moon

7. Which example most directly illustrates Waxing and waning?

- A. a total solar eclipse reveals the Sun's corona for a short time
- B. a lunar eclipse can be seen from much of Earth's night side
- C. "quarter" names orbital progress, not the fraction of the disc that is bright
- D. a first-quarter Moon is half illuminated in appearance but about one quarter through the phase cycle

9. Which statement best explains Lunar eclipse?

- A. ocean water forms broad bulges and Earth rotates through them, while coastlines and depth alter timing and height
- B. sunlight filtered and bent through Earth's atmosphere can make the Moon appear copper-red during totality
- C. the Moon's position around Earth changes the angle between the Sun, Moon and observer
- D. the Moon does not produce red light; Earth's atmosphere changes the sunlight reaching it

B. Answer in one or two sentences

1. Define lunar phase and waxing in your own words.
2. Explain how Waxing and waning works. Use one cause-and-effect link.
3. Compare Solar eclipse and Lunar eclipse. State one similarity or connection and one difference.
4. Use the example of Tide to explain why context or measurement matters.

C. Answer in detail

1. Explain the main ideas of "The Moon: Phases, Eclipses and Tides" as a connected account. Include at least four important terms from the chapter.
2. During a full Moon, why is there usually no lunar eclipse?

D. Apply and reason

1. If the Moon's orbit were not tilted, solar and lunar eclipses would occur much more often.
2. "The Moon becomes dark because Earth blocks sunlight every month." Correct the claim.

E. Activity or project

Keep a one-month Moon record and compare observed dates with a reliable phase calendar.

UNIT 4 REVIEW

Earth and Moon - Mixed Review

This review combines ideas from all chapters in the unit. Use diagrams, units and evidence where relevant. The questions are new combinations rather than copies of the chapter exercises.

A. Mixed multiple-choice review

1. Which option links a term with its correct meaning?

- A. seismic wave - wave carrying earthquake energy through Earth
- B. seismic wave - layer above the troposphere containing much ozone
- C. seismic wave - imaginary line about which a body rotates
- D. seismic wave - east-west angular coordinate on Earth

3. Only one pair is scientifically correct. Which is it?

- A. revolution - yearly period with characteristic sunlight conditions
- B. revolution - Indian Standard Time, UTC+5:30
- C. revolution - orbital motion around another body
- D. revolution - Moon blocks the Sun from part of Earth

5. Find the pair that uses the vocabulary correctly.

- A. lunar phase - appearance of the Moon's illuminated portion
- B. lunar phase - decreasing visible illuminated fraction
- C. lunar phase - dark central part of a shadow
- D. lunar phase - thick rocky layer that flows slowly

7. Choose the correctly defined term from Unit 4.

- A. tide - thin rocky outer layer of Earth
- B. tide - solid metallic centre
- C. tide - periodic sea-level change driven mainly by gravity
- D. tide - outermost atmospheric region fading into space

9. Which statement best explains Troposphere and stratosphere in Chapter 11?

- A. locations turn into and out of sunlight, creating day and night, while sky objects appear to move westward
- B. Earth turns 360 degrees in about 24 hours, giving 15 degrees per hour; a standard meridian prevents every town using a different clock
- C. solar heating of the surface drives tropospheric circulation, while ozone absorbs much ultraviolet radiation higher up
- D. weather does not occur equally through all atmospheric layers

11. Which statement best explains Solar eclipse in Chapter 13?

- A. only observers in particular shadow regions see totality, while a wider region may see a partial eclipse
- B. ocean water forms broad bulges and Earth rotates through them, while coastlines and depth alter timing and height
- C. gravity pulled denser material inward during differentiation, and the weight of overlying layers produces enormous pressure
- D. a solar eclipse is unsafe to view without proper protection except during the brief total phase at a location experiencing totality

2. Select the accurately matched term and definition.

- A. exosphere - imaginary line about which a body rotates
- B. exosphere - outermost atmospheric region fading into space
- C. exosphere - yearly period with characteristic sunlight conditions
- D. exosphere - appearance of the Moon's illuminated portion

4. Which term-definition pair would belong in the Unit 4 glossary?

- A. longitude - Indian Standard Time, UTC+5:30
- B. longitude - decreasing visible illuminated fraction
- C. longitude - periodic sea-level change driven mainly by gravity
- D. longitude - east-west angular coordinate on Earth

6. Which pairing preserves the meaning used in this unit?

- A. solar eclipse - dark central part of a shadow
- B. solar eclipse - Moon blocks the Sun from part of Earth
- C. solar eclipse - thin rocky outer layer of Earth
- D. solar eclipse - wave carrying earthquake energy through Earth

8. Which row could be copied into an accurate revision table?

- A. mantle - solid metallic centre
- B. mantle - layer above the troposphere containing much ozone
- C. mantle - orbital motion around another body
- D. mantle - thick rocky layer that flows slowly

10. Which statement best explains Season in Chapter 12?

- A. Earth turns 360 degrees in about 24 hours, giving 15 degrees per hour; a standard meridian prevents every town using a different clock
- B. ocean water forms broad bulges and Earth rotates through them, while coastlines and depth alter timing and height
- C. seasons are not simultaneous and identical everywhere, and local climate has additional influences
- D. more direct sunlight concentrates energy over a smaller surface area and longer days increase heating time

12. Which statement best explains Core in Chapter 11?

- A. gravity pulled denser material inward during differentiation, and the weight of overlying layers produces enormous pressure
- B. pressure keeps the inner core solid despite high temperature, while movement in the liquid outer core helps generate Earth's magnetic field
- C. as Earth revolves, one hemisphere is tilted more toward the Sun and later more away, changing day length and sunlight angle
- D. the core is not empty and is not made of ordinary fire

UNIT 4 REVIEW - OLYMPIAD CHALLENGE

B. Written and visual reasoning

1. Connect one idea from Chapter 11 with one idea from Chapter 13. Explain the relationship.
2. Choose two terms from different chapters in Unit 4 and compare their roles.
3. If seismic waves suddenly travelled at one speed everywhere, what information about layers would be lost?
4. If the Moon's orbit were not tilted, solar and lunar eclipses would occur much more often.
5. Draw or describe one visual model that would help teach the central idea of Unit 4. State one limitation of the model.
6. Write one claim about Unit 4, give two pieces of evidence and state one uncertainty or boundary condition.

Olympiad Challenge

1. A student is reasoning about Lunar phase. Which statement is the most scientifically careful?

- A. Earth's shadow does not cause ordinary phases
- B. "quarter" names orbital progress, not the fraction of the disc that is bright
- C. a high tide is not simply water being pulled only toward the Moon
- D. scientists infer layers from many events and detectors, not from one earthquake alone

3. When checking a claim about Axial tilt, which statement sets the right boundary?

- A. seasons are not simultaneous and identical everywhere, and local climate has additional influences
- B. Earth's shadow does not cause ordinary phases
- C. the axis does not repeatedly lean toward and away from the Sun; it maintains nearly the same direction in space
- D. a high tide is not simply water being pulled only toward the Moon

5. Which conclusion about Troposphere and stratosphere avoids overclaiming?

- A. weather does not occur equally through all atmospheric layers
- B. space does not begin at a perfectly sharp natural wall
- C. the axis does not repeatedly lean toward and away from the Sun; it maintains nearly the same direction in space
- D. Earth's shadow does not cause ordinary phases

7. Which explanation of Crust and mantle is precise enough for a science report?

- A. the core is not empty and is not made of ordinary fire
- B. weather does not occur equally through all atmospheric layers
- C. the mantle is mostly solid rock that can flow slowly; it is not a global ocean of liquid magma
- D. the axis does not repeatedly lean toward and away from the Sun; it maintains nearly the same direction in space

2. Which situation is the strongest application of Core?

- A. a shadow zone shows where certain waves do not arrive directly
- B. seismic waves change speed and some types do not pass through liquids
- C. some satellites orbit where extremely thin atmospheric particles still exist
- D. December is summer in much of the Southern Hemisphere while it is winter in the Northern Hemisphere

4. Which unfamiliar case is best explained by Lunar eclipse?

- A. many coasts experience about two high and two low tides each lunar day
- B. seismic waves change speed and some types do not pass through liquids
- C. some satellites orbit where extremely thin atmospheric particles still exist
- D. a lunar eclipse can be seen from much of Earth's night side

6. Where would Time zone and IST be most useful?

- A. the sequence runs from new through waxing phases to full, then waning phases back to new
- B. IST is UTC plus 5 hours 30 minutes
- C. a lunar eclipse can be seen from much of Earth's night side
- D. seismic waves change speed and some types do not pass through liquids

8. Which example provides the clearest evidence of Revolution?

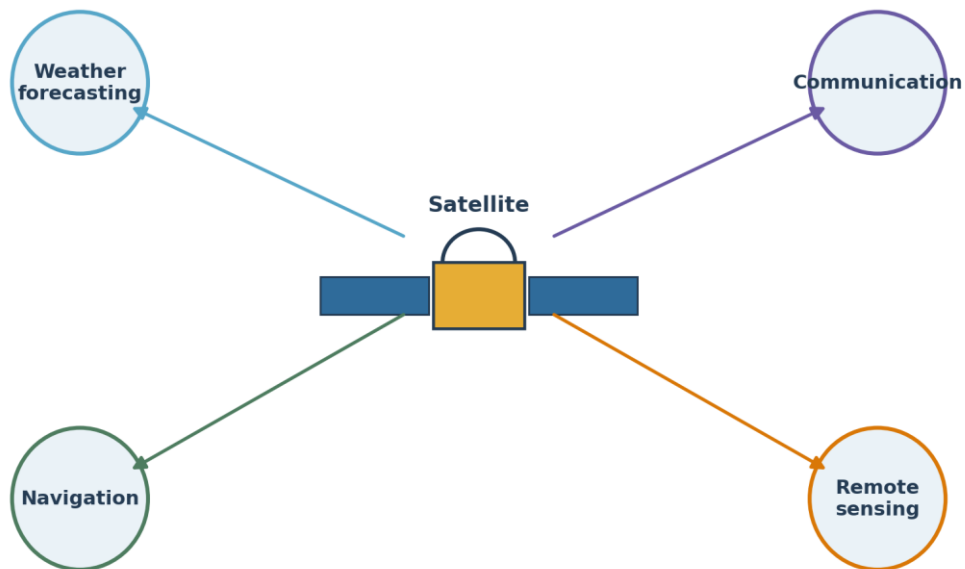
- A. Northern Hemisphere summer occurs when the north is tilted toward the Sun
- B. IST is UTC plus 5 hours 30 minutes
- C. a lunar eclipse can be seen from much of Earth's night side
- D. the changing background of stars across the year reflects Earth's changing orbital position

UNIT 5

Satellites and Space Technology

Satellites turn orbital motion into practical services for communication, navigation, weather study and observation of Earth.

Satellite Applications: From Orbit to Everyday Life



Satellites collect, relay or time information; ground systems turn it into services.

Unit 5 opening visual - original illustration.

Major concepts

- Chapter 14: Natural and Artificial Satellites
- Chapter 15: Communication, Navigation, Weather and Remote Sensing

Expected unit outcomes

- Define natural and artificial satellite.
- Explain signal relay.

CHAPTER 14

Natural and Artificial Satellites

A satellite is an object that moves in an orbit around another body. The Moon is Earth's natural satellite, while artificial satellites are machines placed into orbit for particular purposes. An orbit is continuous falling: gravity bends a spacecraft's forward motion so that it keeps missing the surface. Different altitudes and shapes suit different jobs.

Opening question How can a satellite be falling toward Earth continuously without crashing into it?

Learning outcomes

- Define natural and artificial satellite.
- Explain orbit as free fall.
- Compare basic orbit types.
- Relate altitude to orbital period and coverage.

Common Artificial-Satellite Orbits

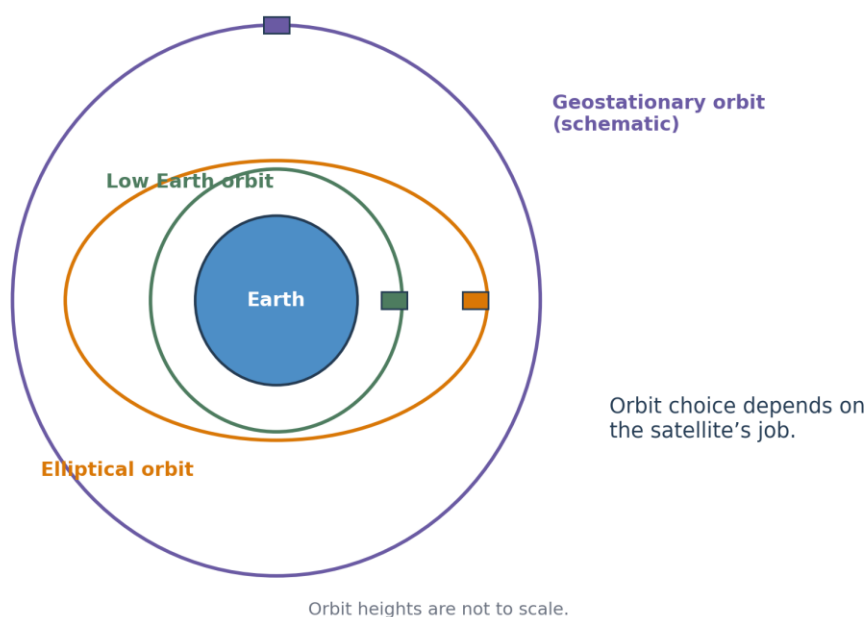


Figure 14.1 Orbiting under gravity. Original illustration; not to scale where stated.

Orbiting under gravity

A spacecraft needs sideways speed as well as height.

Orbit

A curved path produced when gravity continually changes an object's direction of motion. The underlying relationship is that a fast sideways-moving satellite falls toward Earth while the curved surface falls away beneath it. This can be seen when throwing a ball faster makes it travel farther before landing; orbit is an extreme version without air resistance. An orbit does not mean gravity has disappeared.

Free fall

The phrase Free fall refers to motion controlled mainly by gravity, with support forces absent or very small. Here, satellites and astronauts share nearly the same acceleration, so astronauts feel weightless relative to their spacecraft. Consider the following case: an orbiting crew floats even though Earth's gravity is still strong. The careful interpretation is that weightlessness in orbit is not caused by being beyond gravity.

Check your understanding What two conditions prevent an orbiting satellite from simply dropping vertically?

Choosing an orbit

Orbit altitude, inclination and shape control what a satellite can observe and how often.

Low Earth orbit

The phrase Low Earth orbit refers to an orbit relatively close to Earth, commonly used for imaging, science stations and many small satellites. Here, short orbital periods give repeated passes, but a single satellite sees only part of Earth at one time. Consider the following case: the International Space Station is in low Earth orbit. The careful interpretation is that low orbit does not remain fixed over one point unless special motion and latitude conditions are involved.

Geostationary orbit

A circular equatorial orbit whose period matches Earth's rotation, making a satellite appear fixed over one longitude. In practice, the altitude and direction must be precise; this position provides continuous view of a broad region. An example is communication and weather satellites can monitor the same area continuously. However, a geostationary satellite is not motionless in space; it orbits once per day.

Check your understanding Why must a geostationary orbit be above the equator?

Natural and artificial systems

The word satellite describes an orbital relationship rather than composition.

Natural satellite

Natural satellite concerns a naturally formed body orbiting a planet, dwarf planet or small body. The key cause-and-effect link is that its orbit records formation, capture and long-term gravitational interactions. This appears in Earth's Moon stabilises aspects of Earth's axial motion and drives strong tides. At the same time, a moon need not be spherical if its gravity is weak.

Artificial satellite

A human-made object intentionally placed in orbit. Evidence for the idea comes from situations such as a weather satellite carries sensors while solar panels supply electricity. The explanation is it combines payload, power, communication, thermal control, attitude control and structure. Its boundary is equally important: the spent upper stage near it may also orbit, but it is space debris rather than an operating satellite.

Check your understanding Which spacecraft parts are needed even when they are not the main mission instrument?

Activity: Swinging-orbit analogy

Aim	to separate inward force from forward motion
Materials	rubber stopper or soft ball tied securely to string, open area and eye protection
Steps	1. An adult or teacher checks every knot. 2. Swing the object gently in a horizontal circle. 3. Observe that the string pulls inward while the object's motion is sideways. 4. Discuss how gravity supplies an inward acceleration in an orbit, while noting limits of the model.
Observe	The object moves around because its direction changes continuously, not because it stops moving forward.
Expected learning	Circular motion requires inward acceleration.
Safety	Teacher demonstration only; use a soft object, secure knots and a clear area.

Chapter Review

Key ideas

- A curved path produced when gravity continually changes an object's direction of motion.
- An orbit relatively close to earth, commonly used for imaging, science stations and many small satellites.
- A naturally formed body orbiting a planet, dwarf planet or small body.
- "Astronauts float because there is no gravity in low Earth orbit." Correct the statement.

Important terms

satellite: object orbiting another body	orbit: gravity-curved path around a body
free fall: motion mainly under gravity	low Earth orbit: relatively low-altitude Earth orbit
geostationary orbit: equatorial daily orbit appearing fixed	inclination: tilt of an orbit to a reference plane
payload: mission equipment carried by a spacecraft	space debris: non-functional human-made material in orbit

A. Choose the correct answer

1. Which term names object orbiting another body?

- A. orbit
- B. low Earth orbit
- C. satellite
- D. inclination

3. Choose the vocabulary item defined as motion mainly under gravity.

- A. free fall
- B. low Earth orbit
- C. inclination
- D. space debris

5. In this chapter, what is defined as equatorial daily orbit appearing fixed?

- A. inclination
- B. space debris
- C. geostationary orbit
- D. orbit

2. A learner describes gravity-curved path around a body. What is the correct term?

- A. free fall
- B. geostationary orbit
- C. payload
- D. orbit

4. Which expression would a scientist use for relatively low-altitude Earth orbit?

- A. geostationary orbit
- B. low Earth orbit
- C. payload
- D. satellite

Exercises (continued)

6. Which statement best explains Orbit?

- A. satellites and astronauts share nearly the same acceleration, so astronauts feel weightless relative to their spacecraft
- B. the altitude and direction must be precise; this position provides continuous view of a broad region
- C. an orbit does not mean gravity has disappeared
- D. a fast sideways-moving satellite falls toward Earth while the curved surface falls away beneath it

8. Which statement correctly avoids a common misunderstanding about Low Earth orbit?

- A. a geostationary satellite is not motionless in space; it orbits once per day
- B. low orbit does not remain fixed over one point unless special motion and latitude conditions are involved
- C. a naturally formed body orbiting a planet, dwarf planet or small body
- D. short orbital periods give repeated passes, but a single satellite sees only part of Earth at one time

10. Which example most directly illustrates Natural satellite?

- A. a weather satellite carries sensors while solar panels supply electricity
- B. throwing a ball faster makes it travel farther before landing; orbit is an extreme version without air resistance
- C. a moon need not be spherical if its gravity is weak
- D. Earth's Moon stabilises aspects of Earth's axial motion and drives strong tides

7. Which example most directly illustrates Free fall?

- A. an orbiting crew floats even though Earth's gravity is still strong
- B. the International Space Station is in low Earth orbit
- C. communication and weather satellites can monitor the same area continuously
- D. weightlessness in orbit is not caused by being beyond gravity

9. Which statement best explains Geostationary orbit?

- A. its orbit records formation, capture and long-term gravitational interactions
- B. a fast sideways-moving satellite falls toward Earth while the curved surface falls away beneath it
- C. the altitude and direction must be precise; this position provides continuous view of a broad region
- D. a geostationary satellite is not motionless in space; it orbits once per day

B. Answer in one or two sentences

1. Define satellite and orbit in your own words.
2. Explain how Free fall works. Use one cause-and-effect link.
3. Compare Low Earth orbit and Geostationary orbit. State one similarity or connection and one difference.
4. Use the example of Natural satellite to explain why context or measurement matters.

C. Answer in detail

1. Explain the main ideas of "Natural and Artificial Satellites" as a connected account. Include at least four important terms from the chapter.
2. A satellite's engine switches off after reaching orbit. Why can it continue moving?

D. Apply and reason

1. If forward speed becomes too small at a given altitude, the path can intersect the atmosphere or surface.
2. "Astronauts float because there is no gravity in low Earth orbit." Correct the statement.

E. Activity or project

Design a satellite card naming orbit, payload, power source and intended service.

CHAPTER 15

Communication, Navigation, Weather and Remote Sensing

Satellite applications depend on instruments, orbit choice and ground systems. Communication satellites relay signals. Navigation constellations transmit precise timing information. Weather satellites observe clouds, temperature and moisture. Remote-sensing satellites measure reflected or emitted radiation from Earth to map land, water, crops, disasters and environmental change.

Opening question Why is a navigation satellite mainly a precise clock in orbit rather than a map that directly sees every user?

Learning outcomes

- Explain signal relay.
- Describe satellite navigation by timing.
- Identify weather observations.
- Interpret remote-sensing data cautiously.

Satellite Applications: From Orbit to Everyday Life

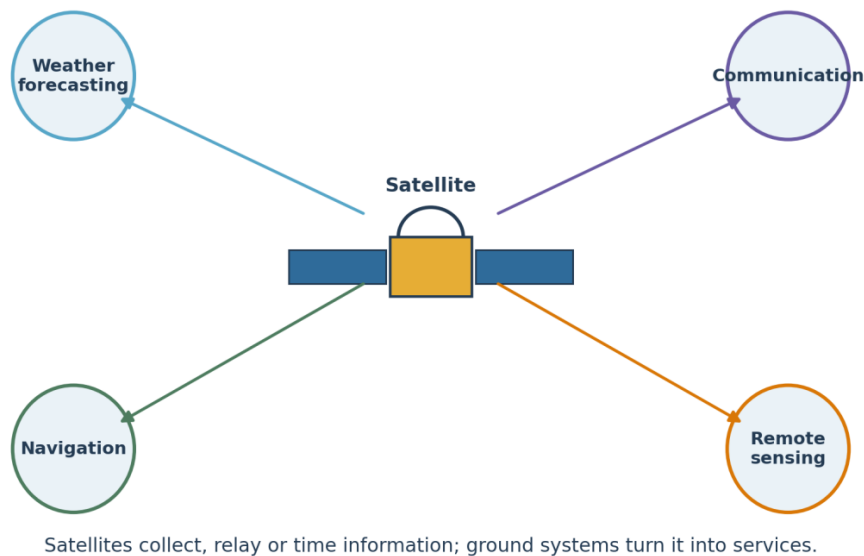


Figure 15.1 Relaying information. Original illustration; not to scale where stated.

Relaying information

A communication link includes a transmitter, satellite and receiver.

Communication satellite

The phrase Communication satellite refers to a satellite that receives, processes or redirects radio signals between locations. Here, an uplink carries signals to the satellite and a downlink returns them to a ground region; repeaters or digital processors manage channels. Consider the following case: television, telephony, broadband and emergency links can cross large distances. The careful interpretation is that the satellite does not create the programme content; it transports encoded signals.

Coverage and delay

The area served by a satellite beam and the travel time of its signal. In practice, higher satellites cover larger areas but signals travel farther, increasing delay and reducing received strength. An example is geostationary links suit broad continuous coverage, while low-orbit constellations can reduce delay. However, one orbit is not best for every communication need.

Check your understanding Why do low-orbit systems usually need many satellites for continuous global coverage?

Position from time

Navigation satellites allow a receiver to calculate distance from signal travel time.

Satellite navigation

Positioning based on timed radio signals from multiple satellites whose orbits and clock corrections are known. In practice, the receiver compares arrival times and solves for its three-dimensional position and clock error. An example is India's NavIC provides regional navigation services. However, a receiver usually does not send its position to the satellites in order to calculate it.

Triangulation and trilateration

To understand Triangulation and trilateration, begin with geometric methods for position; satellite navigation primarily uses distances rather than measured angles. Next, note that each measured range defines a sphere of possible positions, and several spheres intersect near one solution. Four or more satellites help solve position and receiver clock offset. This does not mean that the everyday word triangulation is often used loosely, but timing systems mainly perform trilateration.

Check your understanding Why is one satellite signal insufficient for a unique ground position?

Observing Earth

Sensors turn radiation into measurements that must be calibrated and interpreted.

Weather satellite

A satellite observing clouds, temperature, water vapour, storms and related atmospheric conditions. Evidence for the idea comes from situations such as rapid image sequences reveal cyclone development and movement. The explanation is geostationary satellites provide frequent views of the same region, while polar systems collect global detailed strips. Its boundary is equally important: a cloud image alone cannot provide a complete forecast; models and ground data are also needed.

Remote sensing

Collecting information about an object or area without direct contact, often through reflected or emitted radiation. Different wavelength bands reveal vegetation, water, heat, soil moisture or surface roughness. For example, radar can observe through cloud and at night because it sends and receives microwave signals. One limit must remain clear: false-colour imagery is not fake; colours may represent non-visible measurements.

Check your understanding Why might radar be selected instead of an ordinary camera during monsoon cloud cover?

Activity: Interpret a false-colour image key

Aim	to connect displayed colour with measured meaning
Materials	teacher-provided simplified image and legend
Steps	1. Read the legend before interpreting the image. 2. Identify two areas with different displayed colours. 3. State what measurement each colour represents. 4. Write one conclusion supported by the image and one question it cannot answer alone.
Observe	A data image is meaningful only when its bands, date, resolution and legend are known.
Expected learning	Visual evidence must be interpreted with metadata.
Safety	No special safety requirement.

Chapter Review

Key ideas

- A satellite that receives, processes or redirects radio signals between locations.
- Positioning based on timed radio signals from multiple satellites whose orbits and clock corrections are known.
- A satellite observing clouds, temperature, water vapour, storms and related atmospheric conditions.
- “Green in a satellite image always means vegetation.” Explain why the legend must be checked.

Important terms

uplink: signal sent from ground to satellite	downlink: signal sent from satellite to ground
coverage footprint: ground area reached by a satellite beam	navigation satellite: satellite broadcasting precise timing and orbit data
trilateration: position finding from measured distances	weather satellite: satellite observing atmospheric conditions
remote sensing: measurement without direct contact	false colour: display colours assigned to measured data bands

A. Choose the correct answer

1. Which term names signal sent from ground to satellite?

- A. downlink
- B. navigation satellite
- C. weather satellite
- D. uplink

3. Choose the vocabulary item defined as ground area reached by a satellite beam.

- A. navigation satellite
- B. coverage footprint
- C. weather satellite
- D. false colour

5. In this chapter, what is defined as position finding from measured distances?

- A. weather satellite
- B. false colour
- C. downlink
- D. trilateration

2. A learner describes signal sent from satellite to ground. What is the correct term?

- A. downlink
- B. coverage footprint
- C. trilateration
- D. remote sensing

4. Which expression would a scientist use for satellite broadcasting precise timing and orbit data?

- A. trilateration
- B. remote sensing
- C. navigation satellite
- D. uplink

Exercises (continued)

6. Which statement best explains Communication satellite?

- A. an uplink carries signals to the satellite and a downlink returns them to a ground region; repeaters or digital processors manage channels
- B. higher satellites cover larger areas but signals travel farther, increasing delay and reducing received strength
- C. each measured range defines a sphere of possible positions, and several spheres intersect near one solution
- D. the satellite does not create the programme content; it transports encoded signals

8. Which statement correctly avoids a common misunderstanding about Satellite navigation?

- A. the everyday word triangulation is often used loosely, but timing systems mainly perform trilateration
- B. a satellite observing clouds, temperature, water vapour, storms and related atmospheric conditions
- C. a receiver usually does not send its position to the satellites in order to calculate it
- D. the receiver compares arrival times and solves for its three-dimensional position and clock error

10. Which example most directly illustrates Weather satellite?

- A. rapid image sequences reveal cyclone development and movement
- B. radar can observe through cloud and at night because it sends and receives microwave signals
- C. television, telephony, broadband and emergency links can cross large distances
- D. a cloud image alone cannot provide a complete forecast; models and ground data are also needed

7. Which example most directly illustrates Coverage and delay?

- A. India's NavIC provides regional navigation services
- B. geostationary links suit broad continuous coverage, while low-orbit constellations can reduce delay
- C. four or more satellites help solve position and receiver clock offset
- D. one orbit is not best for every communication need

9. Which statement best explains Triangulation and trilateration?

- A. geostationary satellites provide frequent views of the same region, while polar systems collect global detailed strips
- B. an uplink carries signals to the satellite and a downlink returns them to a ground region; repeaters or digital processors manage channels
- C. the everyday word triangulation is often used loosely, but timing systems mainly perform trilateration
- D. each measured range defines a sphere of possible positions, and several spheres intersect near one solution

B. Answer in one or two sentences

1. Define uplink and downlink in your own words.
2. Explain how Coverage and delay works. Use one cause-and-effect link.
3. Compare Satellite navigation and Triangulation and trilateration. State one similarity or connection and one difference.
4. Use the example of Weather satellite to explain why context or measurement matters.

C. Answer in detail

1. Explain the main ideas of "Communication, Navigation, Weather and Remote Sensing" as a connected account. Include at least four important terms from the chapter.
2. A navigation app shows a wrong position near tall buildings. Name two reasons signals may be affected.

D. Apply and reason

1. Removing accurate clock information would make distance estimates unreliable even if satellite locations were known.
2. "Green in a satellite image always means vegetation." Explain why the legend must be checked.

E. Activity or project

Plan a four-satellite service system and justify which application needs each sensor and orbit.

UNIT 5 REVIEW

Satellites and Space Technology - Mixed Review

This review combines ideas from all chapters in the unit. Use diagrams, units and evidence where relevant. The questions are new combinations rather than copies of the chapter exercises.

A. Mixed multiple-choice review

1. Which option links a term with its correct meaning?

- A. inclination - non-functional human-made material in orbit
- B. inclination - tilt of an orbit to a reference plane
- C. inclination - ground area reached by a satellite beam
- D. inclination - measurement without direct contact

3. Only one pair is scientifically correct. Which is it?

- A. navigation satellite - satellite observing atmospheric conditions
- B. navigation satellite - object orbiting another body
- C. navigation satellite - equatorial daily orbit appearing fixed
- D. navigation satellite - satellite broadcasting precise timing and orbit data

5. Find the pair that uses the vocabulary correctly.

- A. orbit - relatively low-altitude Earth orbit
- B. orbit - gravity-curved path around a body
- C. orbit - mission equipment carried by a spacecraft
- D. orbit - ground area reached by a satellite beam

7. Choose the correctly defined term from Unit 5.

- A. space debris - signal sent from satellite to ground
- B. space debris - position finding from measured distances
- C. space debris - object orbiting another body
- D. space debris - non-functional human-made material in orbit

9. Which statement best explains Artificial satellite in Chapter 14?

- A. higher satellites cover larger areas but signals travel farther, increasing delay and reducing received strength
- B. a fast sideways-moving satellite falls toward Earth while the curved surface falls away beneath it
- C. the spent upper stage near it may also orbit, but it is space debris rather than an operating satellite
- D. it combines payload, power, communication, thermal control, attitude control and structure

11. Which statement best explains Geostationary orbit in Chapter 14?

- A. it combines payload, power, communication, thermal control, attitude control and structure
- B. the altitude and direction must be precise; this position provides continuous view of a broad region
- C. geostationary satellites provide frequent views of the same region, while polar systems collect global detailed strips
- D. a geostationary satellite is not motionless in space; it orbits once per day

2. Select the accurately matched term and definition.

- A. uplink - ground area reached by a satellite beam
- B. uplink - satellite observing atmospheric conditions
- C. uplink - signal sent from ground to satellite
- D. uplink - gravity-curved path around a body

4. Which term-definition pair would belong in the Unit 5 glossary?

- A. remote sensing - measurement without direct contact
- B. remote sensing - object orbiting another body
- C. remote sensing - relatively low-altitude Earth orbit
- D. remote sensing - non-functional human-made material in orbit

6. Which pairing preserves the meaning used in this unit?

- A. geostationary orbit - mission equipment carried by a spacecraft
- B. geostationary orbit - signal sent from satellite to ground
- C. geostationary orbit - equatorial daily orbit appearing fixed
- D. geostationary orbit - satellite observing atmospheric conditions

8. Which row could be copied into an accurate revision table?

- A. coverage footprint - ground area reached by a satellite beam
- B. coverage footprint - position finding from measured distances
- C. coverage footprint - display colours assigned to measured data bands
- D. coverage footprint - relatively low-altitude Earth orbit

10. Which statement best explains Weather satellite in Chapter 15?

- A. geostationary satellites provide frequent views of the same region, while polar systems collect global detailed strips
- B. a fast sideways-moving satellite falls toward Earth while the curved surface falls away beneath it
- C. it combines payload, power, communication, thermal control, attitude control and structure
- D. a cloud image alone cannot provide a complete forecast; models and ground data are also needed

12. Which statement best explains Satellite navigation in Chapter 15?

- A. geostationary satellites provide frequent views of the same region, while polar systems collect global detailed strips
- B. the altitude and direction must be precise; this position provides continuous view of a broad region
- C. the receiver compares arrival times and solves for its three-dimensional position and clock error
- D. a receiver usually does not send its position to the satellites in order to calculate it

UNIT 5 REVIEW - OLYMPIAD CHALLENGE

B. Written and visual reasoning

1. Connect one idea from Chapter 14 with one idea from Chapter 15. Explain the relationship.
2. Choose two terms from different chapters in Unit 5 and compare their roles.
3. A satellite's engine switches off after reaching orbit. Why can it continue moving?
4. Removing accurate clock information would make distance estimates unreliable even if satellite locations were known.
5. Draw or describe one visual model that would help teach the central idea of Unit 5. State one limitation of the model.
6. Write one claim about Unit 5, give two pieces of evidence and state one uncertainty or boundary condition.

Olympiad Challenge

1. A student is reasoning about Geostationary orbit. Which statement is the most scientifically careful?

- A. a moon need not be spherical if its gravity is weak
- B. a geostationary satellite is not motionless in space; it orbits once per day
- C. one orbit is not best for every communication need
- D. false-colour imagery is not fake; colours may represent non-visible measurements

3. When checking a claim about Artificial satellite, which statement sets the right boundary?

- A. the satellite does not create the programme content; it transports encoded signals
- B. the everyday word triangulation is often used loosely, but timing systems mainly perform trilateration
- C. weightlessness in orbit is not caused by being beyond gravity
- D. the spent upper stage near it may also orbit, but it is space debris rather than an operating satellite

5. Which conclusion about Coverage and delay avoids overclaiming?

- A. a receiver usually does not send its position to the satellites in order to calculate it
- B. one orbit is not best for every communication need
- C. false-colour imagery is not fake; colours may represent non-visible measurements
- D. a geostationary satellite is not motionless in space; it orbits once per day

7. Which explanation of Triangulation and trilateration is precise enough for a science report?

- A. a cloud image alone cannot provide a complete forecast; models and ground data are also needed
- B. weightlessness in orbit is not caused by being beyond gravity
- C. the spent upper stage near it may also orbit, but it is space debris rather than an operating satellite
- D. the everyday word triangulation is often used loosely, but timing systems mainly perform trilateration

2. Which situation is the strongest application of Weather satellite?

- A. radar can observe through cloud and at night because it sends and receives microwave signals
- B. the International Space Station is in low Earth orbit
- C. rapid image sequences reveal cyclone development and movement
- D. television, telephony, broadband and emergency links can cross large distances

4. Which unfamiliar case is best explained by Orbit?

- A. throwing a ball faster makes it travel farther before landing; orbit is an extreme version without air resistance
- B. an orbiting crew floats even though Earth's gravity is still strong
- C. Earth's Moon stabilises aspects of Earth's axial motion and drives strong tides
- D. India's NavIC provides regional navigation services

6. Where would Low Earth orbit be most useful?

- A. communication and weather satellites can monitor the same area continuously
- B. television, telephony, broadband and emergency links can cross large distances
- C. the International Space Station is in low Earth orbit
- D. rapid image sequences reveal cyclone development and movement

8. Which example provides the clearest evidence of Natural satellite?

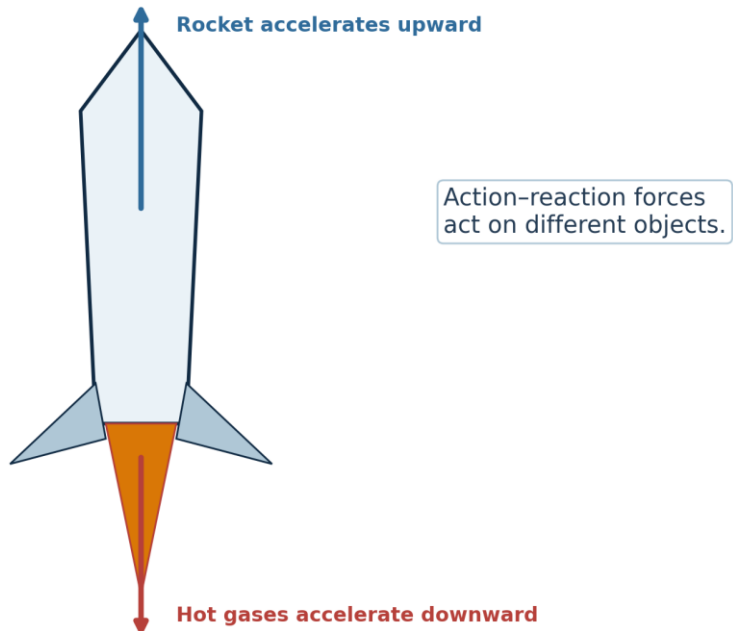
- A. Earth's Moon stabilises aspects of Earth's axial motion and drives strong tides
- B. a weather satellite carries sensors while solar panels supply electricity
- C. India's NavIC provides regional navigation services
- D. throwing a ball faster makes it travel farther before landing; orbit is an extreme version without air resistance

UNIT 6

Rockets and Launch Vehicles

A launch vehicle must overcome gravity, control its path and deliver a payload to the correct orbit.

Rocket Thrust and Newton's Third Law



The rocket does not need air to “push against”; it pushes exhaust gases.

Unit 6 opening visual - original illustration.

Major concepts

- Chapter 16: Rocket Principles and Flight
- Chapter 17: Stages, Payloads, Launch Sites and Reusable Rockets

Expected unit outcomes

- Apply newton's third law to thrust.
- Explain staging.

CHAPTER 16

Rocket Principles and Flight

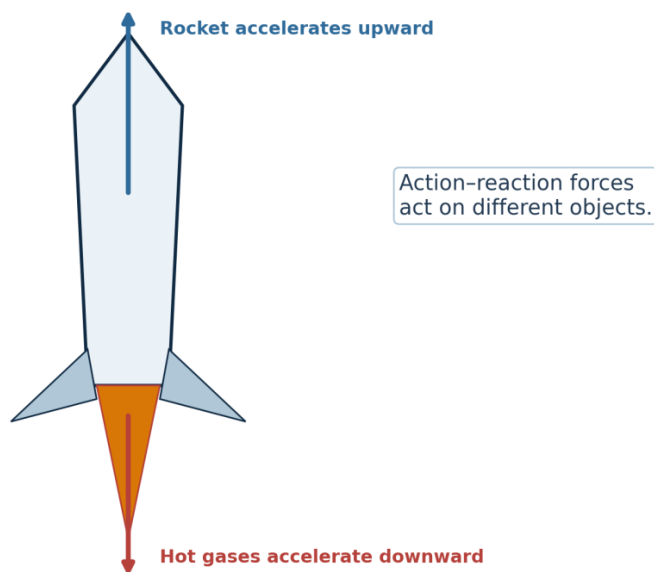
A rocket accelerates by expelling mass at high speed. The exhaust is pushed backward, and an equal and opposite force acts on the rocket. This works in vacuum because the forces are between the rocket and its exhaust, not between the rocket and surrounding air. A successful flight must also manage gravity, atmospheric drag, stability and guidance.

Opening question Why can a rocket accelerate in space where there is no air to push against?

Learning outcomes

- Apply Newton's third law to thrust.
- Distinguish mass and weight.
- Describe drag and gravity losses.
- Explain stability, guidance and control.

Rocket Thrust and Newton's Third Law



The rocket does not need air to "push against"; it pushes exhaust gases.

Figure 16.1 Producing thrust. Original illustration; not to scale where stated.

Producing thrust

A rocket engine converts stored energy into a directed high-speed exhaust.

Thrust

The forward force produced when a propulsion system accelerates exhaust backward. In practice, combustion or another energy source raises pressure; a nozzle directs expanding gas and increases exhaust speed. An example is a balloon released without tying moves opposite to escaping air. However, thrust is not a one-time push; it acts while mass is expelled.

Newton's third law

To understand Newton's third law, begin with when one object exerts a force on another, the second exerts an equal-magnitude opposite-direction force on the first. Next, note that the engine pushes exhaust backward and exhaust pushes the engine forward. A rocket and its exhaust form the action-reaction pair. This does not mean that the equal forces do not cancel because they act on different objects.

Check your understanding Identify the two objects in the rocket action-reaction force pair.

Forces during launch

The engine must produce enough net upward force to increase vertical speed.

Mass and weight

To understand Mass and weight, begin with mass measures amount of matter and resistance to acceleration; weight is gravitational force on that mass. Next, note that mass remains nearly the same from Earth to orbit, while weight depends on local gravity and support conditions. A 10-kilogram instrument has the same mass on the Moon but less weight. This does not mean that kilograms measure mass in SI usage; newtons measure force.

Drag and gravity loss

Drag and gravity loss describes drag opposes motion through atmosphere, while gravity continually pulls downward. Its behaviour follows from this relationship: launch trajectories and throttling balance atmospheric stress, efficiency and safety. The example of rockets often turn gradually rather than rising straight up to orbit shows the idea in use. Remember that reaching space height alone does not create orbit; sufficient sideways speed is essential.

Check your understanding Why does a launch vehicle gradually tilt toward the horizontal during ascent?

Controlling attitude and path

A rocket must point accurately while moving through changing conditions.

Stability

The tendency of a vehicle to resist or correct unwanted rotation or deviation. Shape, mass distribution, fins in air and active control systems influence stability. For example, a model rocket with a forward centre of mass is usually easier to stabilise. One limit must remain clear: stability alone does not guarantee the vehicle follows the desired route.

Guidance, navigation and control

Guidance, navigation and control means navigation estimates position and motion, guidance chooses the desired path and control commands actuators to follow it. It works because computers use sensors such as gyroscopes and accelerometers, then steer engines or small thrusters. A concrete case is gimballed engines tilt thrust to rotate and steer a launch vehicle. By contrast, guidance is not a ground operator manually pointing the rocket every second.

Check your understanding How do navigation, guidance and control differ from one another?

Activity: Balloon rocket investigation

Aim	to observe action-reaction motion
Materials	balloon, drinking straw, long string, tape and two supports
Steps	1. Thread the string through the straw and pull it tight horizontally. 2. Tape an inflated untied balloon to the straw with its opening pointing backward. 3. Release the opening and observe motion. 4. Repeat with different inflation levels while keeping other conditions similar.
Observe	More expelled air can change thrust duration and speed, but balloon shape and leaks also matter.
Expected learning	Expelling mass produces an opposite force on the vehicle.
Safety	Use latex-free balloons where needed and keep balloons away from small children.

Chapter Review

Key ideas

- The forward force produced when a propulsion system accelerates exhaust backward.
- Mass measures amount of matter and resistance to acceleration; weight is gravitational force on that mass.
- The tendency of a vehicle to resist or correct unwanted rotation or deviation.
- “A rocket rises because exhaust pushes on the launch pad and then on air.” Correct the force explanation.

Important terms

thrust: force produced by accelerating exhaust	nozzle: engine passage that directs expanding gas
mass: measure of matter and inertia	weight: gravitational force on an object
drag: force opposing motion through a fluid	stability: resistance or correction of unwanted motion
guidance: selection of desired path	control: actuation that changes motion to follow a command

A. Choose the correct answer

1. Which term names force produced by accelerating exhaust?

- A. thrust
- B. nozzle
- C. weight
- D. stability

3. Choose the vocabulary item defined as measure of matter and inertia.

- A. weight
- B. stability
- C. mass
- D. control

5. In this chapter, what is defined as force opposing motion through a fluid?

- A. drag
- B. stability
- C. control
- D. nozzle

2. A learner describes engine passage that directs expanding gas.

What is the correct term?

- A. mass
- B. nozzle
- C. drag
- D. guidance

4. Which expression would a scientist use for gravitational force on an object?

- A. drag
- B. guidance
- C. thrust
- D. weight

Exercises (continued)

6. Which statement best explains Thrust?

- A. the engine pushes exhaust backward and exhaust pushes the engine forward
- B. combustion or another energy source raises pressure; a nozzle directs expanding gas and increases exhaust speed
- C. launch trajectories and throttling balance atmospheric stress, efficiency and safety
- D. thrust is not a one-time push; it acts while mass is expelled

8. Which statement correctly avoids a common misunderstanding about Mass and weight?

- A. reaching space height alone does not create orbit; sufficient sideways speed is essential
- B. the tendency of a vehicle to resist or correct unwanted rotation or deviation
- C. mass remains nearly the same from Earth to orbit, while weight depends on local gravity and support conditions
- D. kilograms measure mass in SI usage; newtons measure force

10. Which example most directly illustrates Stability?

- A. gimballed engines tilt thrust to rotate and steer a launch vehicle
- B. a model rocket with a forward centre of mass is usually easier to stabilise
- C. a balloon released without tying moves opposite to escaping air
- D. stability alone does not guarantee the vehicle follows the desired route

7. Which example most directly illustrates Newton's third law?

- A. a 10-kilogram instrument has the same mass on the Moon but less weight
- B. rockets often turn gradually rather than rising straight up to orbit
- C. a rocket and its exhaust form the action-reaction pair
- D. the equal forces do not cancel because they act on different objects

9. Which statement best explains Drag and gravity loss?

- A. launch trajectories and throttling balance atmospheric stress, efficiency and safety
- B. shape, mass distribution, fins in air and active control systems influence stability
- C. combustion or another energy source raises pressure; a nozzle directs expanding gas and increases exhaust speed
- D. reaching space height alone does not create orbit; sufficient sideways speed is essential

B. Answer in one or two sentences

1. Define thrust and nozzle in your own words.
2. Explain how Newton's third law works. Use one cause-and-effect link.
3. Compare Mass and weight and Drag and gravity loss. State one similarity or connection and one difference.
4. Use the example of Stability to explain why context or measurement matters.

C. Answer in detail

1. Explain the main ideas of "Rocket Principles and Flight" as a connected account. Include at least four important terms from the chapter.
2. A rocket has upward thrust equal to its weight at lift-off. What happens to vertical acceleration initially?

D. Apply and reason

1. In a perfect vacuum drag disappears, but gravity and the need for orbital speed remain.
2. "A rocket rises because exhaust pushes on the launch pad and then on air." Correct the force explanation.

E. Activity or project

Create a force diagram for three moments: on the pad, just after lift-off and high in the atmosphere.

CHAPTER 17

Stages, Payloads, Launch Sites and Reusable Rockets

A launch vehicle carries propellant, tanks, engines, structure, computers and a payload. Empty tanks and engines become unnecessary mass, so many vehicles discard stages during flight. Launch sites are selected for safety, tracking, transport and useful launch directions. Reusable systems recover some hardware, but recovery adds mass, fuel and operational complexity.

Opening question Why can throwing away an empty stage make the remaining rocket more capable?

Learning outcomes

- Explain staging.
- Distinguish launch vehicle and payload.
- Identify launch-site factors.
- Evaluate basic benefits and costs of reusability.

Why Rockets Use Stages

Dropping empty stages reduces mass, allowing the remaining vehicle to accelerate more efficiently.

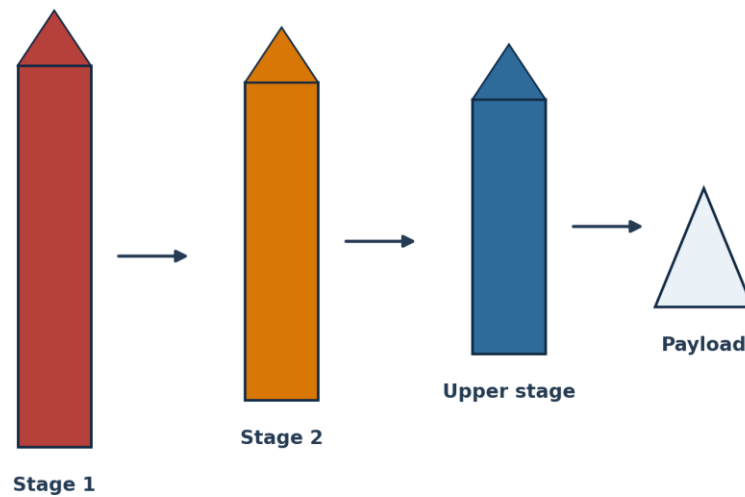


Figure 17.1 Dividing the vehicle. Original illustration; not to scale where stated.

Dividing the vehicle

Staging is a mass-management strategy.

Rocket stage

To understand Rocket stage, begin with a section containing engines, propellant and structure that operates during part of ascent. Next, note that after its propellant is used, separating the stage removes empty tanks and engines so later stages accelerate less mass. A first stage supplies strong initial thrust while an upper stage may place the payload precisely. This does not mean that stage separation must be timed and controlled; it is not simply pieces breaking off.

Payload and fairing

Payload and fairing describes the payload is the mission cargo; the fairing is a protective aerodynamic cover used during atmospheric flight. Its behaviour follows from this relationship: the fairing shields spacecraft from airflow, sound and heating, then is discarded when no longer needed. The example of a payload may be a satellite, probe, crew module or scientific instrument shows the idea in use. Remember that the payload is not the whole rocket and the fairing is not usually carried to the final orbit.

Check your understanding Why is the payload fairing normally released after leaving dense atmosphere?

Selecting a launch site

Geography affects both safety and performance.

Launch azimuth and downrange safety

Launch azimuth and downrange safety describes the initial flight direction and the region beneath later stages. Its behaviour follows from this relationship: launch paths are chosen so falling hardware and possible failures avoid dense population when possible. The example of coastal launch centres can direct selected flights over open water shows the idea in use. Remember that every mission cannot use the same direction because required orbital inclination differs.

Earth-rotation advantage

Eastward launch sites receive some speed from Earth's rotation, greatest near the equator. Because launching eastward to suitable orbits can reduce the velocity the rocket must supply, observers may find that Sriharikota's location supports valuable eastward launch corridors. The conclusion should remain precise: rotation advantage does not replace rocket thrust and is not useful for every orbit.

Check your understanding Why are polar-orbit launches often directed differently from equatorial launches?

Recovering hardware

Reusability is an engineering trade-off, not an automatic guarantee of lower cost.

Reusable launch system

Reusable launch system means a system designed to recover and fly again some launch hardware. It works because boosters may land using engines, wings, parachutes or combinations, then require inspection and refurbishment. A concrete case is recovery can reduce manufacturing of repeated hardware when flight rate and operations support it. By contrast, reusable does not mean no parts are discarded and no maintenance is needed.

Mission trade-off

A decision balancing performance, cost, risk, schedule and objectives. The underlying relationship is that landing propellant and hardware reduce mass available for payload, while reuse can offer operational savings and learning. This can be seen when a mission may choose expendable flight when maximum payload is more important than recovery. One design cannot optimise every mission at once.

Check your understanding Name one performance cost and one possible operational benefit of recovery.

Activity: Mass-fraction paper model

Aim	to show why empty structure matters
Materials	stack of index cards, paper clips and a spring scale or balance
Steps	1. Build a “rocket” from cards labelled payload, upper stage and first stage. 2. Measure total mass. 3. Remove the first-stage cards after an imagined burn and measure the remaining mass. 4. Discuss why the same upper-stage force produces more acceleration for less mass.
Observe	For a given force, reducing mass increases acceleration according to Newton’s second law.
Expected learning	Staging improves performance by discarding no-longer-needed mass.
Safety	Handle spring scales and paper clips carefully.

Chapter Review

Key ideas

- A section containing engines, propellant and structure that operates during part of ascent.
- The initial flight direction and the region beneath later stages.
- A system designed to recover and fly again some launch hardware.
- “Reusable rockets are free after the first flight.” Identify at least three continuing costs.

Important terms

stage: rocket section used during part of flight	stage separation: controlled release of a completed stage
payload: cargo carried for the mission	fairing: protective cover around payload
launch site: facility and region supporting launch operations	azimuth: horizontal direction of a launch path
reusable launcher: vehicle with recoverable flight hardware	trade-off: balance in which improving one factor may cost another

A. Choose the correct answer

1. Which term names rocket section used during part of flight?

- A. stage separation
- B. stage
- C. fairing
- D. azimuth

3. Choose the vocabulary item defined as cargo carried for the mission.

- A. fairing
- B. azimuth
- C. trade-off
- D. payload

5. In this chapter, what is defined as facility and region supporting launch operations?

- A. azimuth
- B. launch site
- C. trade-off
- D. stage separation

2. A learner describes controlled release of a completed stage. What is the correct term?

- A. payload
- B. launch site
- C. stage separation
- D. reusable launcher

4. Which expression would a scientist use for protective cover around payload?

- A. fairing
- B. launch site
- C. reusable launcher
- D. stage

Exercises (continued)

6. Which statement best explains Rocket stage?

- A. the fairing shields spacecraft from airflow, sound and heating, then is discarded when no longer needed
- B. launching eastward to suitable orbits can reduce the velocity the rocket must supply
- C. after its propellant is used, separating the stage removes empty tanks and engines so later stages accelerate less mass
- D. stage separation must be timed and controlled; it is not simply pieces breaking off

8. Which statement correctly avoids a common misunderstanding about Launch azimuth and downrange safety?

- A. every mission cannot use the same direction because required orbital inclination differs
- B. rotation advantage does not replace rocket thrust and is not useful for every orbit
- C. a system designed to recover and fly again some launch hardware
- D. launch paths are chosen so falling hardware and possible failures avoid dense population when possible

10. Which example most directly illustrates Reusable launch system?

- A. a mission may choose expendable flight when maximum payload is more important than recovery
- B. a first stage supplies strong initial thrust while an upper stage may place the payload precisely
- C. recovery can reduce manufacturing of repeated hardware when flight rate and operations support it
- D. reusable does not mean no parts are discarded and no maintenance is needed

7. Which example most directly illustrates Payload and fairing?

- A. coastal launch centres can direct selected flights over open water
- B. Sriharikota's location supports valuable eastward launch corridors
- C. the payload is not the whole rocket and the fairing is not usually carried to the final orbit
- D. a payload may be a satellite, probe, crew module or scientific instrument

9. Which statement best explains Earth-rotation advantage?

- A. boosters may land using engines, wings, parachutes or combinations, then require inspection and refurbishment
- B. launching eastward to suitable orbits can reduce the velocity the rocket must supply
- C. after its propellant is used, separating the stage removes empty tanks and engines so later stages accelerate less mass
- D. rotation advantage does not replace rocket thrust and is not useful for every orbit

B. Answer in one or two sentences

1. Define stage and stage separation in your own words.
2. Explain how Payload and fairing works. Use one cause-and-effect link.
3. Compare Launch azimuth and downrange safety and Earth-rotation advantage. State one similarity or connection and one difference.
4. Use the example of Reusable launch system to explain why context or measurement matters.

C. Answer in detail

1. Explain the main ideas of "Stages, Payloads, Launch Sites and Reusable Rockets" as a connected account. Include at least four important terms from the chapter.
2. A reusable booster reserves fuel for landing. How can this affect payload capacity?

D. Apply and reason

1. A site farther east is not automatically better; orbit direction, weather, safety and infrastructure also matter.
2. "Reusable rockets are free after the first flight." Identify at least three continuing costs.

E. Activity or project

Compare two imaginary launch plans and recommend one using a weighted decision table.

UNIT 6 REVIEW

Rockets and Launch Vehicles - Mixed Review

This review combines ideas from all chapters in the unit. Use diagrams, units and evidence where relevant. The questions are new combinations rather than copies of the chapter exercises.

A. Mixed multiple-choice review

1. Which option links a term with its correct meaning?

- A. guidance - rocket section used during part of flight
- B. guidance - protective cover around payload
- C. guidance - selection of desired path
- D. guidance - balance in which improving one factor may cost another

3. Only one pair is scientifically correct. Which is it?

- A. launch site - facility and region supporting launch operations
- B. launch site - vehicle with recoverable flight hardware
- C. launch site - engine passage that directs expanding gas
- D. launch site - resistance or correction of unwanted motion

5. Find the pair that uses the vocabulary correctly.

- A. mass - force opposing motion through a fluid
- B. mass - actuation that changes motion to follow a command
- C. mass - measure of matter and inertia
- D. mass - protective cover around payload

7. Choose the correctly defined term from Unit 6.

- A. stage - rocket section used during part of flight
- B. stage - cargo carried for the mission
- C. stage - horizontal direction of a launch path
- D. stage - engine passage that directs expanding gas

9. Which statement best explains Rocket stage in Chapter 17?

- A. after its propellant is used, separating the stage removes empty tanks and engines so later stages accelerate less mass
- B. launch paths are chosen so falling hardware and possible failures avoid dense population when possible
- C. the engine pushes exhaust backward and exhaust pushes the engine forward
- D. stage separation must be timed and controlled; it is not simply pieces breaking off

11. Which statement best explains Stability in Chapter 16?

- A. after its propellant is used, separating the stage removes empty tanks and engines so later stages accelerate less mass
- B. landing propellant and hardware reduce mass available for payload, while reuse can offer operational savings and learning
- C. shape, mass distribution, fins in air and active control systems influence stability
- D. stability alone does not guarantee the vehicle follows the desired route

2. Select the accurately matched term and definition.

- A. stage separation - protective cover around payload
- B. stage separation - vehicle with recoverable flight hardware
- C. stage separation - measure of matter and inertia
- D. stage separation - controlled release of a completed stage

4. Which term-definition pair would belong in the Unit 6 glossary?

- A. trade-off - engine passage that directs expanding gas
- B. trade-off - balance in which improving one factor may cost another
- C. trade-off - force opposing motion through a fluid
- D. trade-off - rocket section used during part of flight

6. Which pairing preserves the meaning used in this unit?

- A. stability - actuation that changes motion to follow a command
- B. stability - cargo carried for the mission
- C. stability - vehicle with recoverable flight hardware
- D. stability - resistance or correction of unwanted motion

8. Which row could be copied into an accurate revision table?

- A. fairing - horizontal direction of a launch path
- B. fairing - protective cover around payload
- C. fairing - force produced by accelerating exhaust
- D. fairing - force opposing motion through a fluid

10. Which statement best explains Mission trade-off in Chapter 17?

- A. the engine pushes exhaust backward and exhaust pushes the engine forward
- B. landing propellant and hardware reduce mass available for payload, while reuse can offer operational savings and learning
- C. after its propellant is used, separating the stage removes empty tanks and engines so later stages accelerate less mass
- D. one design cannot optimise every mission at once

12. Which statement best explains Earth-rotation advantage in Chapter 17?

- A. landing propellant and hardware reduce mass available for payload, while reuse can offer operational savings and learning
- B. shape, mass distribution, fins in air and active control systems influence stability
- C. rotation advantage does not replace rocket thrust and is not useful for every orbit
- D. launching eastward to suitable orbits can reduce the velocity the rocket must supply

UNIT 6 REVIEW - OLYMPIAD CHALLENGE

B. Written and visual reasoning

1. Connect one idea from Chapter 16 with one idea from Chapter 17. Explain the relationship.
2. Choose two terms from different chapters in Unit 6 and compare their roles.
3. A rocket has upward thrust equal to its weight at lift-off. What happens to vertical acceleration initially?
4. A site farther east is not automatically better; orbit direction, weather, safety and infrastructure also matter.
5. Draw or describe one visual model that would help teach the central idea of Unit 6. State one limitation of the model.
6. Write one claim about Unit 6, give two pieces of evidence and state one uncertainty or boundary condition.

Olympiad Challenge

1. A student is reasoning about Rocket stage. Which statement is the most scientifically careful?

- A. the payload is not the whole rocket and the fairing is not usually carried to the final orbit
- B. reusable does not mean no parts are discarded and no maintenance is needed
- C. stage separation must be timed and controlled; it is not simply pieces breaking off
- D. kilograms measure mass in SI usage; newtons measure force

3. When checking a claim about Launch azimuth and downrange safety, which statement sets the right boundary?

- A. every mission cannot use the same direction because required orbital inclination differs
- B. rotation advantage does not replace rocket thrust and is not useful for every orbit
- C. thrust is not a one-time push; it acts while mass is expelled
- D. stability alone does not guarantee the vehicle follows the desired route

5. Which conclusion about Reusable launch system avoids overclaiming?

- A. one design cannot optimise every mission at once
- B. kilograms measure mass in SI usage; newtons measure force
- C. reusable does not mean no parts are discarded and no maintenance is needed
- D. stage separation must be timed and controlled; it is not simply pieces breaking off

7. Which explanation of Thrust is precise enough for a science report?

- A. thrust is not a one-time push; it acts while mass is expelled
- B. the equal forces do not cancel because they act on different objects
- C. stability alone does not guarantee the vehicle follows the desired route
- D. every mission cannot use the same direction because required orbital inclination differs

2. Which situation is the strongest application of Newton's third law?

- A. a 10-kilogram instrument has the same mass on the Moon but less weight
- B. gimballed engines tilt thrust to rotate and steer a launch vehicle
- C. Sriharikota's location supports valuable eastward launch corridors
- D. a rocket and its exhaust form the action-reaction pair

4. Which unfamiliar case is best explained by Drag and gravity loss?

- A. a model rocket with a forward centre of mass is usually easier to stabilise
- B. rockets often turn gradually rather than rising straight up to orbit
- C. a payload may be a satellite, probe, crew module or scientific instrument
- D. a mission may choose expendable flight when maximum payload is more important than recovery

6. Where would Guidance, navigation and control be most useful?

- A. a first stage supplies strong initial thrust while an upper stage may place the payload precisely
- B. Sriharikota's location supports valuable eastward launch corridors
- C. a rocket and its exhaust form the action-reaction pair
- D. gimballed engines tilt thrust to rotate and steer a launch vehicle

8. Which example provides the clearest evidence of Payload and fairing?

- A. coastal launch centres can direct selected flights over open water
- B. a payload may be a satellite, probe, crew module or scientific instrument
- C. a mission may choose expendable flight when maximum payload is more important than recovery
- D. rockets often turn gradually rather than rising straight up to orbit

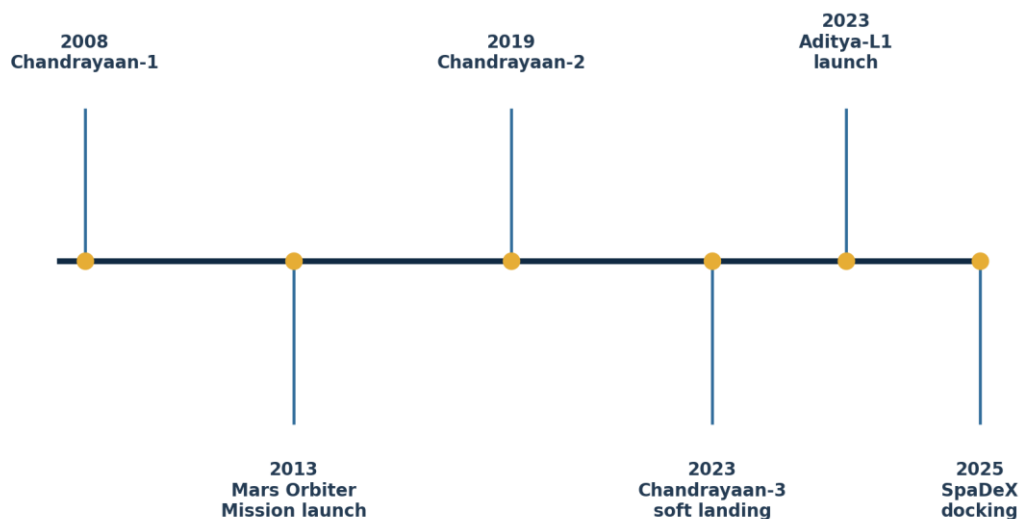
UNIT 7

ISRO and India's Space Missions

India's space programme combines launch systems, science missions and applications that serve society.

Selected Indian Space-Science Milestones

Timeline shows selected milestones, not every Indian mission.



Unit 7 opening visual - original illustration.

Major concepts

- Chapter 18: ISRO, Major Centres, PSLV and GSLV
- Chapter 19: Chandrayaan, Mangalyaan and Aditya-L1
- Chapter 20: Gaganyaan and India's Recent Space Achievements

Expected unit outcomes

- State isro's broad roles.
- Summarise chandrayaan mission goals.
- State gaganyaan's objective.

CHAPTER 18

ISRO, Major Centres, PSLV and GSLV

The Indian Space Research Organisation develops launch vehicles, satellites, science missions and applications under the Department of Space. Its work is distributed across specialised centres for launch-vehicle development, spacecraft, propulsion, applications, remote sensing, tracking and launch operations. PSLV, GSLV and LVM3 are different launch systems chosen according to mission needs.

Opening question Why would one space organisation divide work among several specialised centres instead of building everything at one location?

Learning outcomes

- State isro's broad roles.
- Locate selected major centres by function.
- Compare pslv, gslv and lvm3 at foundation level.
- Explain mission-vehicle matching.

Selected ISRO Centres: A Schematic Locator

Locations are approximate and the outline is not a political map.

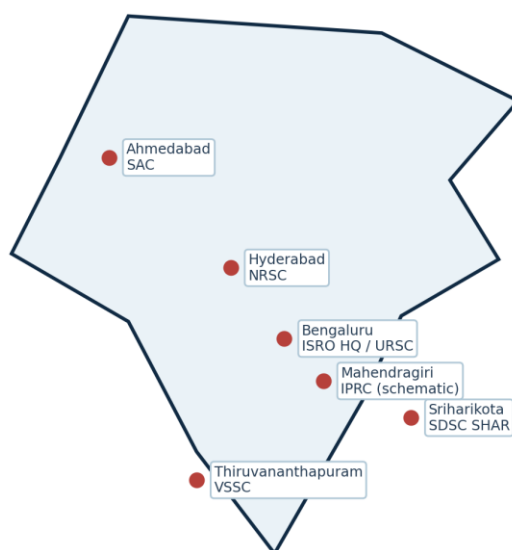


Figure 18.1 A national space system. Original illustration; not to scale where stated.

A national space system

Space missions require connected institutions rather than one laboratory.

ISRO

ISRO describes India's national space agency responsible for space research, technology and applications. Its behaviour follows from this relationship: teams design spacecraft and launchers, operate missions, analyse data and work with users in weather, communication, navigation and Earth observation. The example of an Earth-observation mission links engineers, launch teams, tracking networks, scientists and public-service agencies shows the idea in use. Remember that ISRO is not only a rocket-launching organisation.

Space applications

Use of space systems and data to meet scientific, economic or public needs. Because satellite services support communication, cyclone warning, mapping, resource monitoring, navigation and disaster response, observers may find that remote-sensing maps can help plan flood relief routes. The conclusion should remain precise: an application is useful only when space data are processed, delivered and understood by ground users.

Check your understanding Name two activities that occur after a satellite has been launched.

Selected centres and functions

Names are easier to remember when linked to the work performed there.

VSSC, URSC and SAC

VSSC leads major launch-vehicle development; URSC develops satellites; SAC develops many payloads and applications systems. Because complex projects share designs and test results across centres, observers may find that a launch vehicle may be developed by VSSC while its satellite payload includes work from SAC and its spacecraft bus is led by URSC. The conclusion should remain precise: centres collaborate; their functions are not completely isolated boxes.

SDSC SHAR, IPRC and NRSC

In this chapter, SDSC SHAR, IPRC and NRSC is used for SDSC SHAR is the principal launch centre at Sriharikota; IPRC supports propulsion testing and integration; NRSC acquires and processes remote-sensing data. Launch preparation, engine qualification and data services occur at different specialised facilities. One useful example is a remote-sensing satellite may launch from Sriharikota and later supply data processed through NRSC systems. It should be separated from this different claim: a launch centre is not the place where every satellite is designed.

Check your understanding Which centre-function pair is most directly connected with launching from Sriharikota?

Launch vehicles for different tasks

Vehicle families evolve through variants and mission experience.

PSLV

Polar Satellite Launch Vehicle, a versatile multi-stage Indian launcher widely used for Earth-orbit and interplanetary mission injections. The underlying relationship is that solid and liquid stages work in sequence, and variants are selected according to payload and trajectory. This can be seen when PSLV launched Chandrayaan-1, the Mars Orbiter Mission and Aditya-L1. The word polar does not mean PSLV can launch only polar satellites.

GSLV and LVM3

The phrase GSLV and LVM3 refers to launch-vehicle families developed for heavier payloads and missions requiring high-energy upper stages; LVM3 is India's largest operational launcher family at this level. Here, cryogenic upper-stage technology improves performance for demanding transfers, while LVM3 also supports crewed-programme development and commercial missions. Consider the following case: GSLV-F16 launched NISAR in July 2025, and LVM3 has launched Chandrayaan-2 and Chandrayaan-3. The careful interpretation is that GSLV and LVM3 are related to geosynchronous missions but are not limited to one exact orbit.

Check your understanding Why is a launch vehicle selected by payload mass and target trajectory rather than by mission name alone?

Activity: Match centres to mission tasks

Aim	to understand specialisation and collaboration
Materials	task cards and centre cards prepared from the chapter
Steps	1. Spread the mission-task cards on a table. 2. Match each task to the centre most directly associated with it. 3. Draw arrows where more than one centre must cooperate. 4. Explain one complete path from development to launch to data use.
Observe	Most realistic missions require several linked centres rather than one centre completing every task.
Expected learning	Large engineering programmes are systems of specialised teams.
Safety	No special safety requirement.

Chapter Review

Key ideas

- India's national space agency responsible for space research, technology and applications.
- Vssc leads major launch-vehicle development; ursc develops satellites; sac develops many payloads and applications systems.
- Polar satellite launch vehicle, a versatile multi-stage indian launcher widely used for earth-orbit and interplanetary mission injections.
- "PSLV is only for polar missions." Use mission examples to correct the label-based assumption.

Important terms

ISRO: Indian Space Research Organisation	VSSC: centre leading launch-vehicle development
URSC: centre leading spacecraft development	SAC: centre developing payloads and applications
SDSC SHAR: launch centre at Sriharikota	PSLV: Polar Satellite Launch Vehicle
GSLV: Geosynchronous Satellite Launch Vehicle family	LVM3: heavy-lift Indian launch vehicle family

A. Choose the correct answer

1. Which term names Indian Space Research Organisation?

- A. VSSC
- B. SAC
- C. ISRO
- D. PSLV

3. Choose the vocabulary item defined as centre leading spacecraft development.

- A. URSC
- B. SAC
- C. PSLV
- D. LVM3

5. In this chapter, what is defined as launch centre at Sriharikota?

- A. PSLV
- B. LVM3
- C. SDSC SHAR
- D. VSSC

2. A learner describes centre leading launch-vehicle development.

What is the correct term?

- A. URSC
- B. SDSC SHAR
- C. GSLV
- D. VSSC

4. Which expression would a scientist use for centre developing payloads and applications?

- A. SDSC SHAR
- B. SAC
- C. GSLV
- D. ISRO

Exercises (continued)

6. Which statement best explains ISRO?

- A. satellite services support communication, cyclone warning, mapping, resource monitoring, navigation and disaster response
- B. launch preparation, engine qualification and data services occur at different specialised facilities
- C. ISRO is not only a rocket-launching organisation
- D. teams design spacecraft and launchers, operate missions, analyse data and work with users in weather, communication, navigation and Earth observation

8. Which statement correctly avoids a common misunderstanding about VSSC, URSC and SAC?

- A. a launch centre is not the place where every satellite is designed
- B. centres collaborate; their functions are not completely isolated boxes
- C. Polar Satellite Launch Vehicle, a versatile multi-stage Indian launcher widely used for Earth-orbit and interplanetary mission injections
- D. complex projects share designs and test results across centres

10. Which example most directly illustrates PSLV?

- A. GSLV-F16 launched NISAR in July 2025, and LVM3 has launched Chandrayaan-2 and Chandrayaan-3
- B. an Earth-observation mission links engineers, launch teams, tracking networks, scientists and public-service agencies
- C. the word polar does not mean PSLV can launch only polar satellites
- D. PSLV launched Chandrayaan-1, the Mars Orbiter Mission and Aditya-L1

7. Which example most directly illustrates Space applications?

- A. remote-sensing maps can help plan flood relief routes
- B. a launch vehicle may be developed by VSSC while its satellite payload includes work from SAC and its spacecraft bus is led by URSC
- C. a remote-sensing satellite may launch from Sriharikota and later supply data processed through NRSC systems
- D. an application is useful only when space data are processed, delivered and understood by ground users

9. Which statement best explains SDSC SHAR, IPRC and NRSC?

- A. solid and liquid stages work in sequence, and variants are selected according to payload and trajectory
- B. teams design spacecraft and launchers, operate missions, analyse data and work with users in weather, communication, navigation and Earth observation
- C. launch preparation, engine qualification and data services occur at different specialised facilities
- D. a launch centre is not the place where every satellite is designed

B. Answer in one or two sentences

1. Define ISRO and VSSC in your own words.
2. Explain how Space applications works. Use one cause-and-effect link.
3. Compare VSSC, URSC and SAC and SDSC SHAR, IPRC and NRSC. State one similarity or connection and one difference.
4. Use the example of PSLV to explain why context or measurement matters.

C. Answer in detail

1. Explain the main ideas of “ISRO, Major Centres, PSLV and GSLV” as a connected account. Include at least four important terms from the chapter.
2. A weather satellite has reached orbit but users receive no images. Which parts of the space system should be checked?

D. Apply and reason

1. If every facility duplicated every capability, coordination might be simpler locally but cost and expertise would be used inefficiently.
2. “PSLV is only for polar missions.” Use mission examples to correct the label-based assumption.

E. Activity or project

Prepare a schematic map and workflow for one Indian satellite mission from payload design to public service.

CHAPTER 19

Chandrayaan, Mangalyaan and Aditya-L1

India's planetary and solar missions answer different scientific questions and require different trajectories. Chandrayaan missions study the Moon; the Mars Orbiter Mission demonstrated interplanetary capability and studied Mars; Aditya-L1 observes the Sun from a halo orbit near the Sun–Earth L1 region. Their success is not measured only by launch: navigation, instrument operation, data quality and mission learning also matter.

Opening question Why is a spacecraft sent near the Sun–Earth L1 point instead of simply placing it in an ordinary low Earth orbit to study the Sun?

Learning outcomes

- Summarise chandrayaan mission goals.
- Describe the mars orbiter mission.
- Explain aditya-l1's observing position.
- Distinguish orbiter, lander and rover roles.

Selected Indian Space-Science Milestones

Timeline shows selected milestones, not every Indian mission.

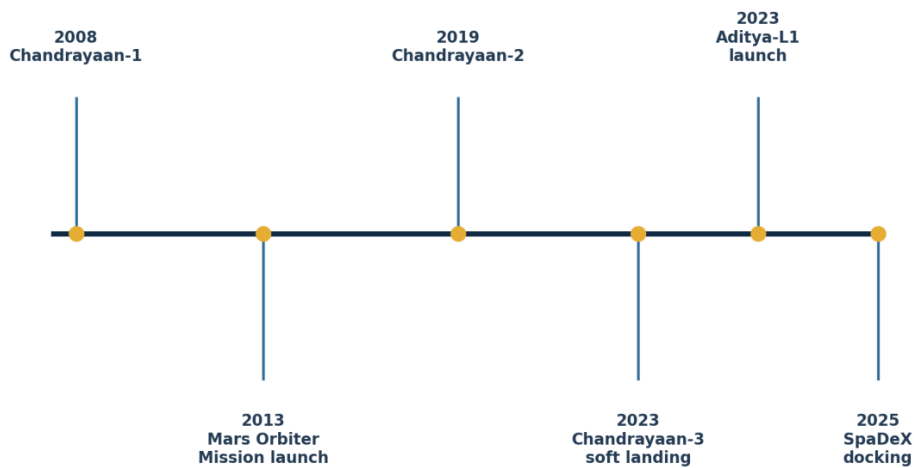


Figure 19.1 Exploring the Moon. Original illustration; not to scale where stated.

Exploring the Moon

Each Chandrayaan mission combined scientific goals with new engineering capabilities.

Chandrayaan-1 and Chandrayaan-2

Chandrayaan-1 orbited the Moon and contributed important evidence about lunar water-related molecules; Chandrayaan-2 placed an orbiter successfully though its Vikram lander did not soft-land. Because orbiters map wide areas repeatedly, while landing attempts test descent, hazard avoidance and surface operations, observers may find that the Chandrayaan-2 orbiter continued scientific observations after the landing attempt. The conclusion should remain precise: a mission can return valuable science and engineering lessons even when one element does not meet its intended goal.

Chandrayaan-3

In this chapter, Chandrayaan-3 is used for India's mission that achieved a soft landing near the lunar south polar region on 23 August 2023 with the Vikram lander and Pragyan rover. The propulsion module delivered the lander system to lunar orbit, the lander controlled descent and the rover made local measurements. One useful example is surface instruments studied temperature, seismic activity, plasma and elemental composition during the operating period. It should be separated from this different claim: "south polar region" does not mean the spacecraft landed exactly at the geographic south pole.

Check your understanding How do an orbiter, lander and rover gather different kinds of evidence?

The Mars Orbiter Mission

Interplanetary navigation requires precise timing, communication and independent spacecraft operation.

Mars Orbiter Mission

In this chapter, Mars Orbiter Mission is used for India's first Mars mission, launched in 2013 and inserted into Mars orbit in 2014. The spacecraft used Earth-orbit-raising manoeuvres, a trans-Mars injection and autonomous operations during long signal delays. One useful example is the mission imaged Mars and studied its atmosphere and surface while demonstrating deep-space technologies. It should be separated from this different claim: an orbiter does not land, and a technology demonstration can also produce scientific observations.

Interplanetary trajectory

A path transferring a spacecraft between the gravitational regions of planets. The process can be followed as launch timing uses the moving positions of Earth and the target so the spacecraft and planet arrive at the same region. For instance, a spacecraft aims for where Mars will be, not where it appears at launch. A common overstatement is avoided by recognising that the shortest straight-line path is not normally the most fuel-efficient or physically possible path.

Check your understanding Why are Mars launch opportunities separated by long intervals rather than available every day?

Watching the Sun from L1

A stable observing geometry can be as valuable as being close to the target.

Aditya-L1

The phrase Aditya-L1 refers to India's first space-based solar observatory, carrying seven payloads to study the photosphere, chromosphere, corona, solar wind and related phenomena. Here, it operates in a halo orbit around the Sun–Earth L1 region, allowing a nearly continuous view of the Sun without regular Earth eclipses. Consider the following case: its instruments combine imaging and in-situ measurements of particles and magnetic fields. The careful interpretation is that Aditya-L1 does not sit motionless exactly on the mathematical L1 point.

Space weather

Conditions in near-Earth space influenced by solar radiation, charged particles and magnetic activity. In practice, solar eruptions can disturb satellites, radio communication, navigation and power systems, so early observation improves understanding and preparedness. An example is coronal mass ejections can produce geomagnetic storms after travelling outward from the Sun. However, space weather is not ordinary atmospheric weather, though both can affect technology.

Check your understanding What advantage does the L1 observing region give a solar mission?

Activity: Plan a mission architecture

Aim	to select spacecraft elements for a science question
Materials	mission cards listing Moon questions, budget tokens and orbiter/lander/rover options
Steps	1. Choose one scientific question about the Moon. 2. Select an orbiter, lander, rover or combination using the available tokens. 3. State what each chosen element measures and how data reach Earth. 4. Identify one risk and one backup observation.
Observe	More complex architectures can answer local and global questions but add cost and failure points.
Expected learning	Mission design begins with questions, not with adding every possible machine.
Safety	No special safety requirement.

Chapter Review

Key ideas

- Chandrayaan-1 orbited the moon and contributed important evidence about lunar water-related molecules; Chandrayaan-2 placed an orbiter successfully though its Vikram lander did not soft-land.
- India's first Mars mission, launched in 2013 and inserted into Mars orbit in 2014.
- India's first space-based solar observatory, carrying seven payloads to study the photosphere, chromosphere, corona, solar wind and related phenomena.
- "Aditya-L1 is parked halfway between Earth and the Sun." Correct the distance and motion.

Important terms

orbiter: spacecraft operating around a world	lander: spacecraft designed to reach and work on a surface
rover: mobile surface exploration vehicle	soft landing: controlled landing at survivable speed
interplanetary trajectory: path between planetary regions	Aditya-L1: Indian solar-observatory mission
L1 region: Sun–Earth gravitational region useful for solar viewing	space weather: space-environment conditions influenced by solar activity

A. Choose the correct answer

1. Which term names spacecraft operating around a world?

- A. lander
- B. soft landing
- C. Aditya-L1
- D. orbiter

3. Choose the vocabulary item defined as mobile surface exploration vehicle.

- A. soft landing
- B. rover
- C. Aditya-L1
- D. space weather

5. In this chapter, what is defined as path between planetary regions?

- A. Aditya-L1
- B. space weather
- C. lander
- D. interplanetary trajectory

2. A learner describes spacecraft designed to reach and work on a surface. What is the correct term?

- A. lander
- B. rover
- C. interplanetary trajectory
- D. L1 region

4. Which expression would a scientist use for controlled landing at survivable speed?

- A. interplanetary trajectory
- B. L1 region
- C. soft landing
- D. orbiter

Exercises (continued)

6. Which statement best explains Chandrayaan-1 and Chandrayaan-2?

- A. orbiters map wide areas repeatedly, while landing attempts test descent, hazard avoidance and surface operations
- B. the propulsion module delivered the lander system to lunar orbit, the lander controlled descent and the rover made local measurements
- C. launch timing uses the moving positions of Earth and the target so the spacecraft and planet arrive at the same region
- D. a mission can return valuable science and engineering lessons even when one element does not meet its intended goal

8. Which statement correctly avoids a common misunderstanding about Mars Orbiter Mission?

- A. the shortest straight-line path is not normally the most fuel-efficient or physically possible path
- B. India's first space-based solar observatory, carrying seven payloads to study the photosphere, chromosphere, corona, solar wind and related phenomena
- C. an orbiter does not land, and a technology demonstration can also produce scientific observations
- D. the spacecraft used Earth-orbit-raising manoeuvres, a trans-Mars injection and autonomous operations during long signal delays

10. Which example most directly illustrates Aditya-L1?

- A. its instruments combine imaging and in-situ measurements of particles and magnetic fields
- B. coronal mass ejections can produce geomagnetic storms after travelling outward from the Sun
- C. the Chandrayaan-2 orbiter continued scientific observations after the landing attempt
- D. Aditya-L1 does not sit motionless exactly on the mathematical L1 point

7. Which example most directly illustrates Chandrayaan-3?

- A. the mission imaged Mars and studied its atmosphere and surface while demonstrating deep-space technologies
- B. surface instruments studied temperature, seismic activity, plasma and elemental composition during the operating period
- C. a spacecraft aims for where Mars will be, not where it appears at launch
- D. "south polar region" does not mean the spacecraft landed exactly at the geographic south pole

9. Which statement best explains Interplanetary trajectory?

- A. it operates in a halo orbit around the Sun–Earth L1 region, allowing a nearly continuous view of the Sun without regular Earth eclipses
- B. orbiters map wide areas repeatedly, while landing attempts test descent, hazard avoidance and surface operations
- C. the shortest straight-line path is not normally the most fuel-efficient or physically possible path
- D. launch timing uses the moving positions of Earth and the target so the spacecraft and planet arrive at the same region

B. Answer in one or two sentences

1. Define orbiter and lander in your own words.
2. Explain how Chandrayaan-3 works. Use one cause-and-effect link.
3. Compare Mars Orbiter Mission and Interplanetary trajectory. State one similarity or connection and one difference.
4. Use the example of Aditya-L1 to explain why context or measurement matters.

C. Answer in detail

1. Explain the main ideas of "Chandrayaan, Mangalyaan and Aditya-L1" as a connected account. Include at least four important terms from the chapter.
2. A lander works perfectly but its communication relay fails. How could mission architecture reduce this risk?

D. Apply and reason

1. If Earth and Mars did not move in orbits, transfer-window planning would be completely different.
2. "Aditya-L1 is parked halfway between Earth and the Sun." Correct the distance and motion.

E. Activity or project

Create a comparison table for Chandrayaan-3, Mars Orbiter Mission and Aditya-L1 using target, spacecraft type, launch vehicle and key objective.

CHAPTER 20

Gaganyaan and India's Recent Space Achievements

Gaganyaan is India's programme to demonstrate indigenous human spaceflight capability to low Earth orbit. Human-rated systems require much stricter reliability, escape, life-support, recovery and medical planning than an uncrewed satellite. Related achievements such as SpaDeX docking, NISAR Earth observation and Indian participation in Axiom Mission 4 develop experience useful for future science and human-spaceflight goals. Status must always be stated with a date.

Opening question Why must a crewed programme test emergency systems even when engineers expect the main launch to work normally?

Learning outcomes

- State gaganyaan's objective.
- Identify major crew-safety systems.
- Describe the significance of docking.
- Summarise selected achievements through June 2026.

Human Spaceflight: Launch, Orbit and Recovery

Crew safety depends on many systems working in sequence and on extensive testing.

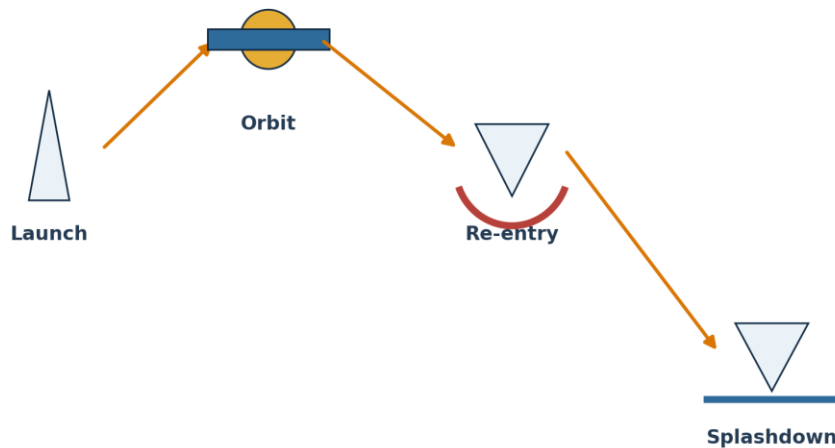


Figure 20.1 Designing for people. Original illustration; not to scale where stated.

Designing for people

A human-rated mission treats the crew as both operators and vulnerable living payloads.

Crew module and service module

In this chapter, Crew module and service module is used for the crew module provides a protected living and re-entry space; the service module supports propulsion, power and other functions before separation. The crew module must survive ascent environments, maintain life support, orient for re-entry, slow with parachutes and be recovered from the sea. One useful example is uncrewed test flights can carry instruments and humanoid systems before people fly. It should be separated from this different claim: a crew capsule is not simply a satellite with seats added.

Crew escape system

A rapid system intended to pull or propel the crew module away from a failing launch vehicle during critical early phases. The process can be followed as sensors and commands must act quickly, and the escape path must lead to safe descent and recovery. For instance, test-vehicle missions examine abort sequences under controlled conditions. A common overstatement is avoided by recognising that testing escape does not imply a launch failure is expected; it prepares for a low-probability high-consequence event.

Check your understanding Which mission phase is the crew escape system designed to protect most directly?

Docking and operations in orbit

Future stations and complex missions require spacecraft to meet and connect safely.

Rendezvous and docking

Rendezvous brings spacecraft to matching position and velocity; docking creates a controlled mechanical connection. The process can be followed as navigation sensors, communication, relative-motion control and capture mechanisms must work at very low closing speeds. For instance, SpaDeX demonstrated docking in January 2025, later undocking and a second docking with power transfer. A common overstatement is avoided by recognising that docking is not a high-speed collision and matching position alone is insufficient without matching velocity.

Power transfer and formation flight

Power transfer and formation flight concerns power transfer shares electrical energy across a docked interface; formation flight coordinates separate spacecraft. The key cause-and-effect link is that these capabilities support future servicing, station assembly and multi-spacecraft observations. This appears in SpaDeX's demonstrated power transfer tested more than physical attachment. At the same time, two satellites moving near each other are not automatically a controlled formation.

Check your understanding Why must two spacecraft match velocity before docking?

A dated achievement record

Current-awareness claims require an information cut-off and a clear status label.

NISAR and Earth science

NISAR is a joint NASA–ISRO radar mission launched by GSLV-F16 on 30 July 2025 and reported in science phase by late November 2025. In practice, its L-band and S-band radar observations are designed to measure changes in land, ice, vegetation and natural hazards with repeated global coverage. An example is radar's long wavelengths and active illumination provide capabilities unavailable to ordinary visible-light images. However, a successful launch is not the same as completion of calibration or start of routine science.

Human-spaceflight preparation by June 2026

To understand Human-spaceflight preparation by June 2026, begin with the programme had continued propulsion, recovery and integrated tests, including the second Integrated Air Drop Test reported on 10 April 2026; Indian astronaut Shubhanshu Shukla completed Axiom Mission 4 in 2025. Next, note that air-drop tests examine parachute deployment and sea recovery of a crew-module-like article, while an ISS mission provides operational and microgravity experience. Experience from international flight can inform training and experiments without being the same as the Gaganyaan orbital mission. This does not mean that planned Gaganyaan flights must not be described as completed before official confirmation.

Check your understanding Why does a current-affairs textbook need to distinguish completed, ongoing and planned activities?

Activity: Build a crew-safety fault tree

Aim	to see how backup systems respond to hazards
Materials	large paper, sticky notes and mission-phase cards
Steps	1. Place launch, orbit, re-entry and recovery in sequence. 2. Under each phase, add one possible hazard. 3. Add a prevention, detection and response measure for each hazard. 4. Mark which measures are independent backups.
Observe	Crew safety uses layers: prevent failure, detect it, respond and survive if prevention fails.
Expected learning	Redundancy and testing reduce risk but cannot make complex systems risk-free.
Safety	No special safety requirement.

Chapter Review

Key ideas

- The crew module provides a protected living and re-entry space; the service module supports propulsion, power and other functions before separation.
- Rendezvous brings spacecraft to matching position and velocity; docking creates a controlled mechanical connection.
- Nisar is a joint nasa–isro radar mission launched by gslv-f16 on 30 july 2025 and reported in science phase by late november 2025.
- “SpaDeX and Gaganyaan are the same mission.” Explain the technology relationship without merging their objectives.

Important terms

Gaganyaan: Indian human-spaceflight programme	human-rated: designed and verified to protect crew with high reliability
crew module: re-entry spacecraft containing crew	crew escape system: emergency system separating crew from a failing launcher
rendezvous: controlled approach with matched relative motion	docking: mechanical connection of spacecraft
SpaDeX: Indian Space Docking Experiment	NISAR: NASA–ISRO Synthetic Aperture Radar mission

A. Choose the correct answer

1. Which term names Indian human-spaceflight programme?

- A. Gaganyaan
- B. human-rated
- C. crew escape system
- D. docking

3. Choose the vocabulary item defined as re-entry spacecraft containing crew.

- A. crew escape system
- B. docking
- C. crew module
- D. NISAR

5. In this chapter, what is defined as controlled approach with matched relative motion?

- A. rendezvous
- B. docking
- C. NISAR
- D. human-rated

2. A learner describes designed and verified to protect crew with high reliability. What is the correct term?

- A. crew module
- B. human-rated
- C. rendezvous
- D. SpaDeX

4. Which expression would a scientist use for emergency system separating crew from a failing launcher?

- A. rendezvous
- B. SpaDeX
- C. Gaganyaan
- D. crew escape system

Exercises (continued)

6. Which statement best explains Crew module and service module?

- A. sensors and commands must act quickly, and the escape path must lead to safe descent and recovery
- B. the crew module must survive ascent environments, maintain life support, orient for re-entry, slow with parachutes and be recovered from the sea
- C. these capabilities support future servicing, station assembly and multi-spacecraft observations
- D. a crew capsule is not simply a satellite with seats added

8. Which statement correctly avoids a common misunderstanding about Rendezvous and docking?

- A. two satellites moving near each other are not automatically a controlled formation
- B. NISAR is a joint NASA–ISRO radar mission launched by GSLV-F16 on 30 July 2025 and reported in science phase by late November 2025
- C. navigation sensors, communication, relative-motion control and capture mechanisms must work at very low closing speeds
- D. docking is not a high-speed collision and matching position alone is insufficient without matching velocity

10. Which example most directly illustrates NISAR and Earth science?

- A. experience from international flight can inform training and experiments without being the same as the Gaganyaan orbital mission
- B. radar's long wavelengths and active illumination provide capabilities unavailable to ordinary visible-light images
- C. uncrewed test flights can carry instruments and humanoid systems before people fly
- D. a successful launch is not the same as completion of calibration or start of routine science

7. Which example most directly illustrates Crew escape system?

- A. SpaDeX demonstrated docking in January 2025, later undocking and a second docking with power transfer
- B. SpaDeX's demonstrated power transfer tested more than physical attachment
- C. test-vehicle missions examine abort sequences under controlled conditions
- D. testing escape does not imply a launch failure is expected; it prepares for a low-probability high-consequence event

9. Which statement best explains Power transfer and formation flight?

- A. these capabilities support future servicing, station assembly and multi-spacecraft observations
- B. its L-band and S-band radar observations are designed to measure changes in land, ice, vegetation and natural hazards with repeated global coverage
- C. the crew module must survive ascent environments, maintain life support, orient for re-entry, slow with parachutes and be recovered from the sea
- D. two satellites moving near each other are not automatically a controlled formation

B. Answer in one or two sentences

1. Define Gaganyaan and human-rated in your own words.
2. Explain how Crew escape system works. Use one cause-and-effect link.
3. Compare Rendezvous and docking and Power transfer and formation flight. State one similarity or connection and one difference.
4. Use the example of NISAR and Earth science to explain why context or measurement matters.

C. Answer in detail

1. Explain the main ideas of “Gaganyaan and India’s Recent Space Achievements” as a connected account. Include at least four important terms from the chapter.
2. A parachute test succeeds once. Why is one success not enough for crew certification?

D. Apply and reason

1. Without docking capability, station assembly and exchange of crew or supplies would be far more limited.
2. “SpaDeX and Gaganyaan are the same mission.” Explain the technology relationship without merging their objectives.

E. Activity or project

Write a dated mission-status bulletin using only official sources and status labels.

UNIT 7 REVIEW

ISRO and India's Space Missions - Mixed Review

This review combines ideas from all chapters in the unit. Use diagrams, units and evidence where relevant. The questions are new combinations rather than copies of the chapter exercises.

A. Mixed multiple-choice review

1. Which option links a term with its correct meaning?

- A. LVM3 - spacecraft designed to reach and work on a surface
- B. LVM3 - path between planetary regions
- C. LVM3 - Indian human-spaceflight programme
- D. LVM3 - heavy-lift Indian launch vehicle family

3. Only one pair is scientifically correct. Which is it?

- A. Aditya-L1 - space-environment conditions influenced by solar activity
- B. Aditya-L1 - Indian solar-observatory mission
- C. Aditya-L1 - re-entry spacecraft containing crew
- D. Aditya-L1 - Indian Space Docking Experiment

5. Find the pair that uses the vocabulary correctly.

- A. crew escape system - mechanical connection of spacecraft
- B. crew escape system - Indian Space Research Organisation
- C. crew escape system - launch centre at Sriharikota
- D. crew escape system - emergency system separating crew from a failing launcher

7. Choose the correctly defined term from Unit 7.

- A. VSSC - centre developing payloads and applications
- B. VSSC - centre leading launch-vehicle development
- C. VSSC - Geosynchronous Satellite Launch Vehicle family
- D. VSSC - mobile surface exploration vehicle

9. Which statement best explains Chandrayaan-3 in Chapter 19?

- A. launch timing uses the moving positions of Earth and the target so the spacecraft and planet arrive at the same region
- B. the propulsion module delivered the lander system to lunar orbit, the lander controlled descent and the rover made local measurements
- C. navigation sensors, communication, relative-motion control and capture mechanisms must work at very low closing speeds
- D. "south polar region" does not mean the spacecraft landed exactly at the geographic south pole

11. Which statement best explains Human-spaceflight preparation by June 2026 in Chapter 20?

- A. satellite services support communication, cyclone warning, mapping, resource monitoring, navigation and disaster response
- B. orbiters map wide areas repeatedly, while landing attempts test descent, hazard avoidance and surface operations
- C. planned Gaganyaan flights must not be described as completed before official confirmation
- D. air-drop tests examine parachute deployment and sea recovery of a crew-module-like article, while an ISS mission provides operational and microgravity experience

2. Select the accurately matched term and definition.

- A. rover - mobile surface exploration vehicle
- B. rover - path between planetary regions
- C. rover - space-environment conditions influenced by solar activity
- D. rover - emergency system separating crew from a failing launcher

4. Which term-definition pair would belong in the Unit 7 glossary?

- A. Gaganyaan - re-entry spacecraft containing crew
- B. Gaganyaan - mechanical connection of spacecraft
- C. Gaganyaan - Indian human-spaceflight programme
- D. Gaganyaan - centre leading launch-vehicle development

6. Which pairing preserves the meaning used in this unit?

- A. SpaDeX - Indian Space Docking Experiment
- B. SpaDeX - Indian Space Research Organisation
- C. SpaDeX - centre developing payloads and applications
- D. SpaDeX - heavy-lift Indian launch vehicle family

8. Which row could be copied into an accurate revision table?

- A. SDSC SHAR - Geosynchronous Satellite Launch Vehicle family
- B. SDSC SHAR - spacecraft designed to reach and work on a surface
- C. SDSC SHAR - launch centre at Sriharikota
- D. SDSC SHAR - Indian solar-observatory mission

10. Which statement best explains Crew module and service module in Chapter 20?

- A. navigation sensors, communication, relative-motion control and capture mechanisms must work at very low closing speeds
- B. satellite services support communication, cyclone warning, mapping, resource monitoring, navigation and disaster response
- C. the crew module must survive ascent environments, maintain life support, orient for re-entry, slow with parachutes and be recovered from the sea
- D. a crew capsule is not simply a satellite with seats added

12. Which statement best explains PSLV in Chapter 18?

- A. solid and liquid stages work in sequence, and variants are selected according to payload and trajectory
- B. orbiters map wide areas repeatedly, while landing attempts test descent, hazard avoidance and surface operations
- C. solar eruptions can disturb satellites, radio communication, navigation and power systems, so early observation improves understanding and preparedness
- D. the word polar does not mean PSLV can launch only polar satellites

UNIT 7 REVIEW - OLYMPIAD CHALLENGE

B. Written and visual reasoning

1. Connect one idea from Chapter 18 with one idea from Chapter 20. Explain the relationship.
2. Choose two terms from different chapters in Unit 7 and compare their roles.
3. A weather satellite has reached orbit but users receive no images. Which parts of the space system should be checked?
4. Without docking capability, station assembly and exchange of crew or supplies would be far more limited.
5. Draw or describe one visual model that would help teach the central idea of Unit 7. State one limitation of the model.
6. Write one claim about Unit 7, give two pieces of evidence and state one uncertainty or boundary condition.

Olympiad Challenge

1. A student is reasoning about SDSC SHAR, IPRC and NRSC. Which statement is the most scientifically careful?

- A. the word polar does not mean PSLV can launch only polar satellites
- B. "south polar region" does not mean the spacecraft landed exactly at the geographic south pole
- C. space weather is not ordinary atmospheric weather, though both can affect technology
- D. a launch centre is not the place where every satellite is designed

3. When checking a claim about Human-spaceflight preparation by June 2026, which statement sets the right boundary?

- A. ISRO is not only a rocket-launching organisation
- B. planned Gaganyaan flights must not be described as completed before official confirmation
- C. a launch centre is not the place where every satellite is designed
- D. "south polar region" does not mean the spacecraft landed exactly at the geographic south pole

5. Which conclusion about Crew escape system avoids overclaiming?

- A. docking is not a high-speed collision and matching position alone is insufficient without matching velocity
- B. planned Gaganyaan flights must not be described as completed before official confirmation
- C. a launch centre is not the place where every satellite is designed
- D. testing escape does not imply a launch failure is expected; it prepares for a low-probability high-consequence event

7. Which explanation of Interplanetary trajectory is precise enough for a science report?

- A. Aditya-L1 does not sit motionless exactly on the mathematical L1 point
- B. the shortest straight-line path is not normally the most fuel-efficient or physically possible path
- C. testing escape does not imply a launch failure is expected; it prepares for a low-probability high-consequence event
- D. planned Gaganyaan flights must not be described as completed before official confirmation

2. Which situation is the strongest application of Aditya-L1?

- A. its instruments combine imaging and in-situ measurements of particles and magnetic fields
- B. coronal mass ejections can produce geomagnetic storms after travelling outward from the Sun
- C. SpaDeX demonstrated docking in January 2025, later undocking and a second docking with power transfer
- D. an Earth-observation mission links engineers, launch teams, tracking networks, scientists and public-service agencies

4. Which unfamiliar case is best explained by Chandrayaan-1 and Chandrayaan-2?

- A. surface instruments studied temperature, seismic activity, plasma and elemental composition during the operating period
- B. its instruments combine imaging and in-situ measurements of particles and magnetic fields
- C. the Chandrayaan-2 orbiter continued scientific observations after the landing attempt
- D. SpaDeX demonstrated docking in January 2025, later undocking and a second docking with power transfer

6. Where would VSSC, URSC and SAC be most useful?

- A. a launch vehicle may be developed by VSSC while its satellite payload includes work from SAC and its spacecraft bus is led by URSC
- B. a remote-sensing satellite may launch from Sriharikota and later supply data processed through NRSC systems
- C. the Chandrayaan-2 orbiter continued scientific observations after the landing attempt
- D. its instruments combine imaging and in-situ measurements of particles and magnetic fields

8. Which example provides the clearest evidence of NISAR and Earth science?

- A. experience from international flight can inform training and experiments without being the same as the Gaganyaan orbital mission
- B. a launch vehicle may be developed by VSSC while its satellite payload includes work from SAC and its spacecraft bus is led by URSC
- C. radar's long wavelengths and active illumination provide capabilities unavailable to ordinary visible-light images
- D. the Chandrayaan-2 orbiter continued scientific observations after the landing attempt

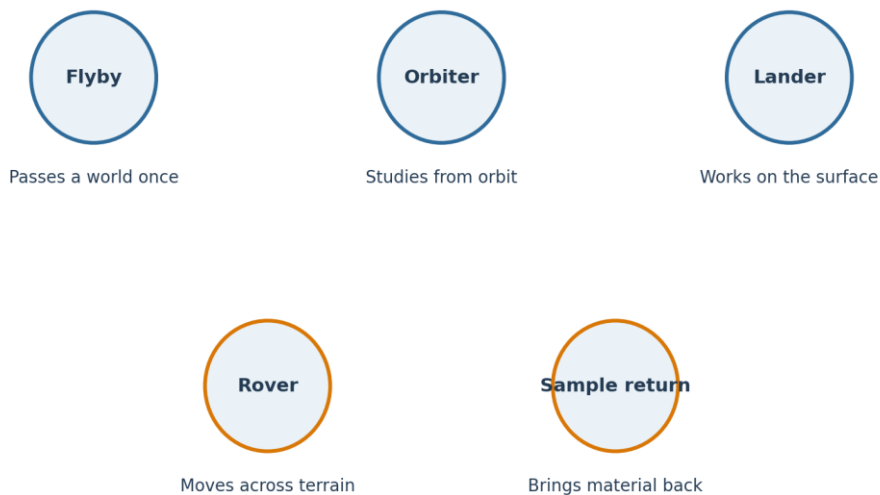
UNIT 8

Space Exploration

Robotic spacecraft and human crews explore places that cannot be studied fully from Earth.

Robotic Exploration: Different Missions, Different Jobs

Mission design matches the scientific question, distance, risk and available technology.



Unit 8 opening visual - original illustration.

Major concepts

- Chapter 21: Astronauts and Life in Space
- Chapter 22: Space Stations, Moon Missions and Mars Missions
- Chapter 23: Major Space Agencies and International Cooperation

Expected unit outcomes

- Explain microgravity.
- Describe functions of a space station.
- Identify selected major agencies.

CHAPTER 21

Astronauts and Life in Space

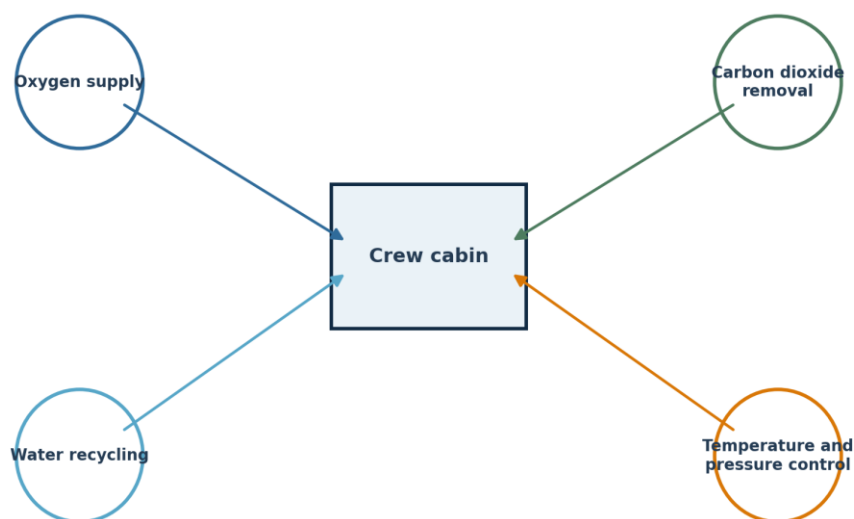
Astronauts are trained crew members who operate spacecraft, conduct experiments, maintain systems and respond to emergencies. In orbit they experience microgravity because they and the vehicle are in continuous free fall. Living safely requires controlled air, water, temperature, waste, food, exercise, medical monitoring and psychological support. Every routine task is redesigned for a moving, enclosed environment.

Opening question Why do astronauts need daily exercise when moving objects feels easier in microgravity?

Learning outcomes

- Explain microgravity.
- Identify life-support functions.
- Describe astronaut work and training.
- Explain common effects on the human body.

Life Support on a Space Station



Closed systems reduce resupply needs, but no real station is perfectly closed.

Figure 21.1 Working while falling. Original illustration; not to scale where stated.

Working while falling

Microgravity changes the effects of weight but not mass or inertia.

Microgravity

A condition in which objects experience very small apparent weight because they share free fall. The process can be followed as the spacecraft, crew and loose objects all accelerate around Earth together, so support forces are greatly reduced. For instance, a gently released tool floats beside an astronaut instead of dropping to the floor. A common overstatement is avoided by recognising that microgravity does not mean zero gravity; Earth's pull remains strong in low orbit.

Inertia in orbit

Inertia in orbit concerns the resistance of mass to a change in motion remains even when weight is small. The key cause-and-effect link is that a massive object may float but still requires force to start, stop or turn, and can injure a crew member if moving. This appears in astronauts use handrails and restraints to control their own motion. At the same time, "easy to lift" does not mean "easy to control".

Check your understanding Why can a heavy floating equipment box still be difficult to stop?

Keeping a small world habitable

A crewed spacecraft is a managed environment with limited supplies.

Life-support system

Life-support system concerns equipment that controls air pressure and composition, removes carbon dioxide, manages water and temperature and monitors contaminants. The key cause-and-effect link is that fans move air because warm air does not rise normally in microgravity, and recycling reduces the amount carried from Earth. This appears in water recovered from cabin humidity and waste can be purified for reuse. At the same time, life support is not only an oxygen tank.

Food, hygiene and waste

Systems that provide nutrition and prevent loose liquids, crumbs and microbes from harming equipment or crew. Evidence for the idea comes from situations such as surface tension makes liquids form floating blobs. The explanation is food packages, airflow, wipes, toilets and sealed storage are designed for the cabin environment. Its boundary is equally important: ordinary showering and open cups are impractical in microgravity.

Check your understanding Why are fans essential even when a cabin contains enough oxygen overall?

Human adaptation

The body changes when it no longer supports weight in the usual way.

Spaceflight deconditioning

To understand Spaceflight deconditioning, begin with loss of bone mineral, muscle strength and cardiovascular capacity during prolonged microgravity exposure. Next, note that less mechanical loading reduces the signals that maintain bone and muscle, while body fluids shift toward the head. Resistance and aerobic exercise help limit these changes. This does not mean that exercise reduces risk but does not make the space environment identical to Earth.

Astronaut training

Astronaut training describes preparation in spacecraft systems, science, robotics, survival, teamwork, language and emergency response. Its behaviour follows from this relationship: simulators and underwater training allow repeated practice of complex procedures before flight. The example of neutral-buoyancy pools help rehearse body positioning for spacewalk tasks, though water is not microgravity shows the idea in use. Remember that astronaut selection is not based only on physical bravery; technical skill and cooperation are central.

Check your understanding Which parts of underwater training model spacewalking well, and which do not?

Activity: Move and work with limited anchors

Aim	to model the need for restraints during tasks
Materials	rolling chair, lightweight objects and a clear supervised space
Steps	1. Sit on a stable rolling chair with feet raised slightly under supervision. 2. Try to push a light object without holding an anchor. 3. Repeat while holding a fixed support. 4. Relate chair motion to action-reaction and the need for foot restraints.
Observe	Without an anchor, the worker moves opposite to the push, making precise work harder.
Expected learning	Microgravity work depends on managing reaction forces.
Safety	Adult supervision; use a low stable chair, light objects and no sudden pushes.

Chapter Review

Key ideas

- A condition in which objects experience very small apparent weight because they share free fall.
- Equipment that controls air pressure and composition, removes carbon dioxide, manages water and temperature and monitors contaminants.
- Loss of bone mineral, muscle strength and cardiovascular capacity during prolonged microgravity exposure.
- “Astronauts can breathe anywhere in a space station because oxygen floats evenly by itself.” Explain the airflow problem.

Important terms

astronaut: trained person who travels or works in space	microgravity: very small apparent weight during shared free fall
inertia: resistance to change in motion	life support: systems maintaining a safe crew environment
carbon-dioxide scrubbing: removal of exhaled carbon dioxide	deconditioning: loss of physical capacity during reduced loading
spacewalk: crew activity outside a spacecraft	simulator: training system reproducing selected mission conditions

A. Choose the correct answer

1. Which term names trained person who travels or works in space?

- A. microgravity
- B. astronaut
- C. life support
- D. deconditioning

3. Choose the vocabulary item defined as resistance to change in motion.

- A. life support
- B. deconditioning
- C. simulator
- D. inertia

5. In this chapter, what is defined as removal of exhaled carbon dioxide?

- A. deconditioning
- B. carbon-dioxide scrubbing
- C. simulator
- D. microgravity

2. A learner describes very small apparent weight during shared free fall. What is the correct term?

- A. inertia
- B. carbon-dioxide scrubbing
- C. microgravity
- D. spacewalk

4. Which expression would a scientist use for systems maintaining a safe crew environment?

- A. life support
- B. carbon-dioxide scrubbing
- C. spacewalk
- D. astronaut

Exercises (continued)

6. Which statement best explains Microgravity?

- A. a massive object may float but still requires force to start, stop or turn, and can injure a crew member if moving
- B. food packages, airflow, wipes, toilets and sealed storage are designed for the cabin environment
- C. the spacecraft, crew and loose objects all accelerate around Earth together, so support forces are greatly reduced
- D. microgravity does not mean zero gravity; Earth's pull remains strong in low orbit

8. Which statement correctly avoids a common misunderstanding about Life-support system?

- A. life support is not only an oxygen tank
- B. ordinary showering and open cups are impractical in microgravity
- C. loss of bone mineral, muscle strength and cardiovascular capacity during prolonged microgravity exposure
- D. fans move air because warm air does not rise normally in microgravity, and recycling reduces the amount carried from Earth

10. Which example most directly illustrates Spaceflight deconditioning?

- A. neutral-buoyancy pools help rehearse body positioning for spacewalk tasks, though water is not microgravity
- B. a gently released tool floats beside an astronaut instead of dropping to the floor
- C. resistance and aerobic exercise help limit these changes
- D. exercise reduces risk but does not make the space environment identical to Earth

7. Which example most directly illustrates Inertia in orbit?

- A. water recovered from cabin humidity and waste can be purified for reuse
- B. surface tension makes liquids form floating blobs
- C. "easy to lift" does not mean "easy to control"
- D. astronauts use handrails and restraints to control their own motion

9. Which statement best explains Food, hygiene and waste?

- A. less mechanical loading reduces the signals that maintain bone and muscle, while body fluids shift toward the head
- B. food packages, airflow, wipes, toilets and sealed storage are designed for the cabin environment
- C. the spacecraft, crew and loose objects all accelerate around Earth together, so support forces are greatly reduced
- D. ordinary showering and open cups are impractical in microgravity

B. Answer in one or two sentences

1. Define astronaut and microgravity in your own words.
2. Explain how Inertia in orbit works. Use one cause-and-effect link.
3. Compare Life-support system and Food, hygiene and waste. State one similarity or connection and one difference.
4. Use the example of Spaceflight deconditioning to explain why context or measurement matters.

C. Answer in detail

1. Explain the main ideas of "Astronauts and Life in Space" as a connected account. Include at least four important terms from the chapter.
2. An astronaut's mass is unchanged in orbit. Which measurements would show this even when a scale reads almost zero?

D. Apply and reason

1. Without exercise, bone and muscle loss would generally increase during long missions.
2. "Astronauts can breathe anywhere in a space station because oxygen floats evenly by itself." Explain the airflow problem.

E. Activity or project

Design a one-day astronaut schedule balancing science, maintenance, exercise, meals and sleep.

CHAPTER 22

Space Stations, Moon Missions and Mars Missions

A space station supports repeated human activity in orbit, while lunar and Mars missions travel farther and face longer communication delays, greater radiation exposure and more difficult supply and recovery. Robotic missions usually go first because they tolerate risk and can operate for years. Human and robotic exploration answer overlapping but not identical questions.

Opening question Why can a Mars rover receive daily plans from Earth but not be driven like a remote-control car in real time?

Learning outcomes

- Describe functions of a space station.
- Compare robotic mission types.
- Explain communication delay.
- Compare moon and mars mission challenges.

Robotic Exploration: Different Missions, Different Jobs

Mission design matches the scientific question, distance, risk and available technology.

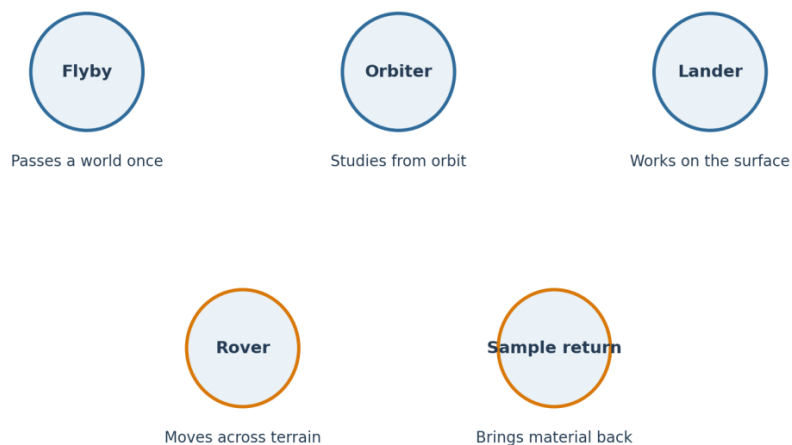


Figure 22.1 An orbital laboratory. Original illustration; not to scale where stated.

An orbital laboratory

A space station is both a spacecraft and a long-duration workplace.

Space station

Space station concerns a crewed orbital facility designed for extended habitation, research and visiting spacecraft. The key cause-and-effect link is that modular systems provide laboratories, power, thermal control, docking ports and living space, while regular vehicles carry crew and supplies. This appears in the International Space Station has supported research in biology, materials, Earth observation and technology. At the same time, a station does not float at one fixed point; it circles Earth repeatedly.

Microgravity research

Experiments that study processes when buoyancy, settling and weight-driven convection are greatly reduced. Evidence for the idea comes from situations such as protein crystals may grow differently in orbit than on Earth. The explanation is researchers can separate gravitational effects from other influences in flames, fluids, crystals and organisms. Its boundary is equally important: results from one space experiment do not automatically apply to every condition on Earth.

Check your understanding What makes an orbiting station useful for repeated experiments rather than a short spacecraft flight?

Robotic explorers

Mission type determines how closely a spacecraft can study one place.

Flyby, orbiter and lander

A flyby passes once, an orbiter circles repeatedly and a lander works on or near a surface. Evidence for the idea comes from situations such as Voyager performed flybys, Mars orbiters map globally and landers measure local conditions. The explanation is each step toward the surface usually adds navigation difficulty, mass and risk while enabling more local measurements. Its boundary is equally important: a flyby can still return major discoveries despite a short encounter.

Rover and sample return

A rover moves instruments across terrain; a sample-return mission aims to bring selected material to Earth. Mobility compares multiple sites, while returned samples can be studied with large laboratories for decades. For example, Mars rovers choose paths cautiously and perform analyses over many days. One limit must remain clear: a rover cannot travel wherever desired instantly because terrain, power and communication constrain it.

Check your understanding Which mission type is best for global mapping, and which for close study of several surface sites?

Moon and Mars are different targets

Distance changes almost every system requirement.

Communication delay

Communication delay describes the travel time of radio signals, limited by the speed of light. Its behaviour follows from this relationship: Moon communication takes roughly seconds for a round trip, while Mars delay varies from several to tens of minutes one way. The example of Mars rovers use onboard autonomy to avoid immediate hazards shows the idea in use. Remember that more powerful radio cannot eliminate light-travel time.

Surface environment and logistics

The combined conditions of gravity, atmosphere, temperature, dust, radiation and available resources. Because the Moon has no substantial atmosphere and extreme day-night cycles; Mars has a thin atmosphere, longer distance and widespread dust, observers may find that a lunar mission can return to Earth much faster than a Mars mission. The conclusion should remain precise: a technique suitable for the Moon may fail on Mars because entry, landing and power conditions differ.

Check your understanding Why does Mars's thin atmosphere make landing both helpful and difficult?

Activity: Choose the right mission type

Aim	to match scientific questions with spacecraft architecture
Materials	question cards and mission-type cards
Steps	1. Draw one science question card, such as “map all major canyons” or “measure soil chemistry at three sites.” 2. Select the mission type that answers it most efficiently. 3. State one limitation of your choice. 4. Compare answers and revise when another design is better supported.
Observe	The most complex mission is not always the most suitable; design follows the question.
Expected learning	Engineering choices are justified by evidence and objectives.
Safety	No special safety requirement.

Chapter Review

Key ideas

- A crewed orbital facility designed for extended habitation, research and visiting spacecraft.
- A flyby passes once, an orbiter circles repeatedly and a lander works on or near a surface.
- The travel time of radio signals, limited by the speed of light.
- “A space station is outside Earth’s gravity.” Use orbital motion to correct the claim.

Important terms

space station: long-duration crewed orbital facility	microgravity research: study under greatly reduced apparent weight
flyby: mission passing a target without orbiting	orbiter: spacecraft repeatedly circling a target
lander: stationary surface spacecraft	rover: mobile surface spacecraft
sample return: mission returning extraterrestrial material	communication delay: signal travel time between locations

A. Choose the correct answer

1. Which term names long-duration crewed orbital facility?

- A. microgravity research
- B. orbiter
- C. space station
- D. rover

3. Choose the vocabulary item defined as mission passing a target without orbiting.

- A. flyby
- B. orbiter
- C. rover
- D. communication delay

5. In this chapter, what is defined as stationary surface spacecraft?

- A. rover
- B. communication delay
- C. lander
- D. microgravity research

2. A learner describes study under greatly reduced apparent weight. What is the correct term?

- A. flyby
- B. lander
- C. sample return
- D. microgravity research

4. Which expression would a scientist use for spacecraft repeatedly circling a target?

- A. lander
- B. orbiter
- C. sample return
- D. space station

Exercises (continued)

6. Which statement best explains Space station?

- A. researchers can separate gravitational effects from other influences in flames, fluids, crystals and organisms
- B. mobility compares multiple sites, while returned samples can be studied with large laboratories for decades
- C. a station does not float at one fixed point; it circles Earth repeatedly
- D. modular systems provide laboratories, power, thermal control, docking ports and living space, while regular vehicles carry crew and supplies

8. Which statement correctly avoids a common misunderstanding about Flyby, orbiter and lander?

- A. a rover cannot travel wherever desired instantly because terrain, power and communication constrain it
- B. a flyby can still return major discoveries despite a short encounter
- C. the travel time of radio signals, limited by the speed of light
- D. each step toward the surface usually adds navigation difficulty, mass and risk while enabling more local measurements

10. Which example most directly illustrates Communication delay?

- A. a lunar mission can return to Earth much faster than a Mars mission
- B. the International Space Station has supported research in biology, materials, Earth observation and technology
- C. more powerful radio cannot eliminate light-travel time
- D. Mars rovers use onboard autonomy to avoid immediate hazards

7. Which example most directly illustrates Microgravity research?

- A. protein crystals may grow differently in orbit than on Earth
- B. Voyager performed flybys, Mars orbiters map globally and landers measure local conditions
- C. Mars rovers choose paths cautiously and perform analyses over many days
- D. results from one space experiment do not automatically apply to every condition on Earth

9. Which statement best explains Rover and sample return?

- A. Moon communication takes roughly seconds for a round trip, while Mars delay varies from several to tens of minutes one way
- B. modular systems provide laboratories, power, thermal control, docking ports and living space, while regular vehicles carry crew and supplies
- C. mobility compares multiple sites, while returned samples can be studied with large laboratories for decades
- D. a rover cannot travel wherever desired instantly because terrain, power and communication constrain it

B. Answer in one or two sentences

1. Define space station and microgravity research in your own words.
2. Explain how Microgravity research works. Use one cause-and-effect link.
3. Compare Flyby, orbiter and lander and Rover and sample return. State one similarity or connection and one difference.
4. Use the example of Communication delay to explain why context or measurement matters.

C. Answer in detail

1. Explain the main ideas of “Space Stations, Moon Missions and Mars Missions” as a connected account. Include at least four important terms from the chapter.
2. A Mars rover sees a hazard. Why must it sometimes decide locally rather than wait for Earth?

D. Apply and reason

1. If Mars were as close as the Moon, communication and emergency response would change, but its atmosphere and gravity would remain different.
2. “A space station is outside Earth’s gravity.” Use orbital motion to correct the claim.

E. Activity or project

Write a mission proposal choosing one target, one question, one mission type and four required subsystems.

CHAPTER 23

Major Space Agencies and International Cooperation

Space programmes are organised by national agencies, international partnerships, research institutions and companies. ISRO, NASA, ESA, JAXA, CNSA and Roscosmos are among the major agencies associated with launch, science, exploration and applications. Cooperation allows countries to share instruments, expertise, cost, tracking networks and data, but it also requires agreed standards, schedules and responsibilities.

Opening question Why might two agencies build different instruments for one spacecraft instead of each launching a complete separate mission?

Learning outcomes

- Identify selected major agencies.
- Describe forms of international cooperation.
- Explain shared standards and data.
- Evaluate benefits and challenges of partnership.

Satellite Applications: From Orbit to Everyday Life

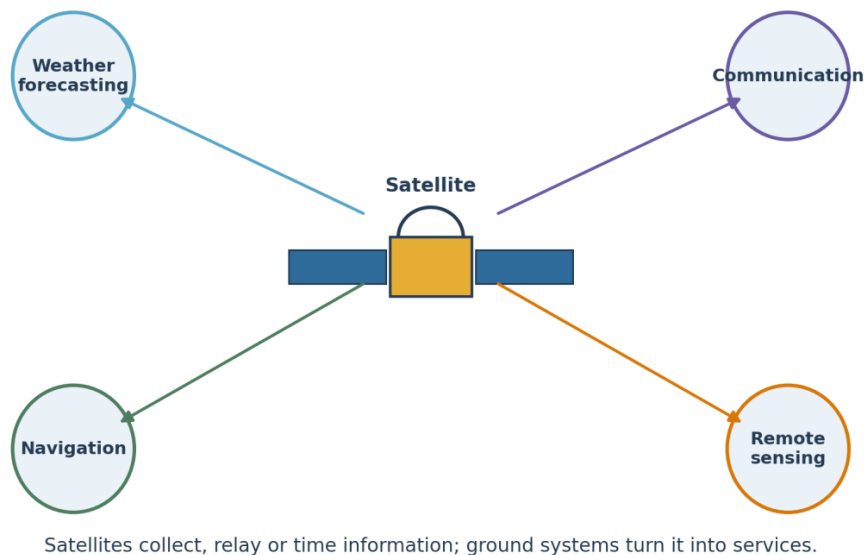


Figure 23.1 Different organisations, overlapping goals. Original illustration; not to scale where stated.

Different organisations, overlapping goals

Agencies develop capabilities according to national priorities, history and resources.

National space agency

A government organisation responsible for some combination of civil space research, technology and services. Evidence for the idea comes from situations such as NASA leads many United States civil missions, JAXA leads Japanese civil space activities and ISRO leads India's programme. The explanation is agencies may build missions, fund universities, operate launch sites, purchase services and cooperate internationally. Its boundary is equally important: one agency does not necessarily control all military, commercial and academic space work in its country.

ESA

The European Space Agency, an intergovernmental organisation through which participating European states cooperate on space programmes. Member contributions support shared launch, science, Earth-observation and exploration missions. For example, ESA instruments and astronauts may work with NASA, JAXA and other partners. One limit must remain clear: ESA is not a single country's agency.

Check your understanding How is an intergovernmental agency different from one national agency?

Ways missions cooperate

Partnership can occur at many levels short of building an entire spacecraft together.

Joint mission

A mission in which partners share major development, launch, operation or science responsibilities. Interfaces and schedules must be agreed early so hardware, software and data systems work together. For example, NISAR combines NASA's L-band radar contribution and ISRO's S-band radar and spacecraft/launch contributions. One limit must remain clear: joint does not mean every partner performs the same task.

Hosted instrument and tracking support

Hosted instrument and tracking support means one organisation may place an instrument on another's spacecraft or provide ground communication and navigation services. It works because sharing reduces duplication and increases global coverage of tracking networks. A concrete case is deep-space missions often communicate through stations located in different countries. By contrast, support agreements still require security, technical compatibility and clear data policies.

Check your understanding Why is a worldwide tracking network useful for a spacecraft far from Earth?

Standards, data and responsibility

Cooperation works when technical and scientific expectations are explicit.

Interface standard

An agreed physical, electrical, software or communication rule that lets systems connect. Because standard connectors, data formats and docking mechanisms reduce incompatibility and support future reuse, observers may find that spacecraft components exchange commands only when voltage levels and message formats match. The conclusion should remain precise: a standard does not guarantee quality; it guarantees a defined way of interacting.

Open scientific data

In this chapter, Open scientific data is used for observations released so researchers and the public can analyse them under stated terms. Broader access allows independent checking, new discoveries and educational use after calibration and documentation. One useful example is public mission archives may support studies never planned by the original team. It should be separated from this different claim: raw data without metadata and calibration are not automatically easy to interpret.

Check your understanding What information must accompany a dataset so that another scientist can use it correctly?

Activity: Negotiate a joint mission

Aim	to practise assigning responsibilities and interfaces
Materials	role cards for two agencies and a mission worksheet
Steps	1. Each team receives different strengths, such as launcher, radar instrument, ground network or spacecraft bus. 2. Agree on one mission objective and divide responsibilities. 3. List three interfaces that require common standards. 4. Decide how calibrated science data will be shared.
Observe	Cooperation gains capability but increases coordination and dependency.
Expected learning	Partnership design includes technical, legal and scientific planning.
Safety	No special safety requirement.

Chapter Review

Key ideas

- A government organisation responsible for some combination of civil space research, technology and services.
- A mission in which partners share major development, launch, operation or science responsibilities.
- An agreed physical, electrical, software or communication rule that lets systems connect.
- “International cooperation means no country has its own objectives.” Explain how shared and national goals can coexist.

Important terms

space agency: organisation managing public space activities	NASA: United States civil space agency
ESA: European intergovernmental space agency	JAXA: Japanese space agency
CNSA: Chinese national space administration	Roscosmos: Russian state space corporation
joint mission: mission with shared partner responsibilities	interface standard: agreed rule for connecting systems

A. Choose the correct answer

1. Which term names organisation managing public space activities?

- A. NASA
- B. JAXA
- C. Roscosmos
- D. space agency

3. Choose the vocabulary item defined as European intergovernmental space agency.

- A. JAXA
- B. ESA
- C. Roscosmos
- D. interface standard

5. In this chapter, what is defined as Chinese national space administration?

- A. Roscosmos
- B. interface standard
- C. NASA
- D. CNSA

2. A learner describes United States civil space agency. What is the correct term?

- A. NASA
- B. ESA
- C. CNSA
- D. joint mission

4. Which expression would a scientist use for Japanese space agency?

- A. CNSA
- B. joint mission
- C. JAXA
- D. space agency

Exercises (continued)

6. Which statement best explains National space agency?

- A. agencies may build missions, fund universities, operate launch sites, purchase services and cooperate internationally
- B. member contributions support shared launch, science, Earth-observation and exploration missions
- C. sharing reduces duplication and increases global coverage of tracking networks
- D. one agency does not necessarily control all military, commercial and academic space work in its country

8. Which statement correctly avoids a common misunderstanding about Joint mission?

- A. support agreements still require security, technical compatibility and clear data policies
- B. an agreed physical, electrical, software or communication rule that lets systems connect
- C. joint does not mean every partner performs the same task
- D. interfaces and schedules must be agreed early so hardware, software and data systems work together

10. Which example most directly illustrates Interface standard?

- A. spacecraft components exchange commands only when voltage levels and message formats match
- B. public mission archives may support studies never planned by the original team
- C. NASA leads many United States civil missions, JAXA leads Japanese civil space activities and ISRO leads India's programme
- D. a standard does not guarantee quality; it guarantees a defined way of interacting

7. Which example most directly illustrates ESA?

- A. NISAR combines NASA's L-band radar contribution and ISRO's S-band radar and spacecraft/launch contributions
- B. ESA instruments and astronauts may work with NASA, JAXA and other partners
- C. deep-space missions often communicate through stations located in different countries
- D. ESA is not a single country's agency

9. Which statement best explains Hosted instrument and tracking support?

- A. standard connectors, data formats and docking mechanisms reduce incompatibility and support future reuse
- B. agencies may build missions, fund universities, operate launch sites, purchase services and cooperate internationally
- C. support agreements still require security, technical compatibility and clear data policies
- D. sharing reduces duplication and increases global coverage of tracking networks

B. Answer in one or two sentences

1. Define space agency and NASA in your own words.
2. Explain how ESA works. Use one cause-and-effect link.
3. Compare Joint mission and Hosted instrument and tracking support. State one similarity or connection and one difference.
4. Use the example of Interface standard to explain why context or measurement matters.

C. Answer in detail

1. Explain the main ideas of "Major Space Agencies and International Cooperation" as a connected account. Include at least four important terms from the chapter.
2. One partner's instrument is delayed by six months. What schedule and risk options could the mission consider?

D. Apply and reason

1. Without shared communication standards, even excellent hardware may be unable to exchange commands or data.
2. "International cooperation means no country has its own objectives." Explain how shared and national goals can coexist.

E. Activity or project

Research one verified joint mission and create a responsibility chart using official sources.

UNIT 8 REVIEW

Space Exploration - Mixed Review

This review combines ideas from all chapters in the unit. Use diagrams, units and evidence where relevant. The questions are new combinations rather than copies of the chapter exercises.

A. Mixed multiple-choice review

1. Which option links a term with its correct meaning?

- A. space station - long-duration crewed orbital facility
- B. space station - mission passing a target without orbiting
- C. space station - mobile surface spacecraft
- D. space station - United States civil space agency

3. Only one pair is scientifically correct. Which is it?

- A. sample return - organisation managing public space activities
- B. sample return - Japanese space agency
- C. sample return - mission returning extraterrestrial material
- D. sample return - agreed rule for connecting systems

5. Find the pair that uses the vocabulary correctly.

- A. CNSA - Chinese national space administration
- B. CNSA - mission with shared partner responsibilities
- C. CNSA - very small apparent weight during shared free fall
- D. CNSA - loss of physical capacity during reduced loading

7. Choose the correctly defined term from Unit 8.

- A. inertia - removal of exhaled carbon dioxide
- B. inertia - training system reproducing selected mission conditions
- C. inertia - resistance to change in motion
- D. inertia - spacecraft repeatedly circling a target

9. Which statement best explains Flyby, orbiter and lander in Chapter 22?

- A. Moon communication takes roughly seconds for a round trip, while Mars delay varies from several to tens of minutes one way
- B. sharing reduces duplication and increases global coverage of tracking networks
- C. each step toward the surface usually adds navigation difficulty, mass and risk while enabling more local measurements
- D. a flyby can still return major discoveries despite a short encounter

11. Which statement best explains Microgravity in Chapter 21?

- A. the spacecraft, crew and loose objects all accelerate around Earth together, so support forces are greatly reduced
- B. fans move air because warm air does not rise normally in microgravity, and recycling reduces the amount carried from Earth
- C. researchers can separate gravitational effects from other influences in flames, fluids, crystals and organisms
- D. microgravity does not mean zero gravity; Earth's pull remains strong in low orbit

2. Select the accurately matched term and definition.

- A. orbiter - mobile surface spacecraft
- B. orbiter - spacecraft repeatedly circling a target
- C. orbiter - organisation managing public space activities
- D. orbiter - Chinese national space administration

4. Which term-definition pair would belong in the Unit 8 glossary?

- A. NASA - Japanese space agency
- B. NASA - mission with shared partner responsibilities
- C. NASA - resistance to change in motion
- D. NASA - United States civil space agency

6. Which pairing preserves the meaning used in this unit?

- A. interface standard - very small apparent weight during shared free fall
- B. interface standard - agreed rule for connecting systems
- C. interface standard - removal of exhaled carbon dioxide
- D. interface standard - long-duration crewed orbital facility

8. Which row could be copied into an accurate revision table?

- A. deconditioning - training system reproducing selected mission conditions
- B. deconditioning - mission passing a target without orbiting
- C. deconditioning - mission returning extraterrestrial material
- D. deconditioning - loss of physical capacity during reduced loading

10. Which statement best explains ESA in Chapter 23?

- A. sharing reduces duplication and increases global coverage of tracking networks
- B. fans move air because warm air does not rise normally in microgravity, and recycling reduces the amount carried from Earth
- C. ESA is not a single country's agency
- D. member contributions support shared launch, science, Earth-observation and exploration missions

12. Which statement best explains Astronaut training in Chapter 21?

- A. researchers can separate gravitational effects from other influences in flames, fluids, crystals and organisms
- B. simulators and underwater training allow repeated practice of complex procedures before flight
- C. agencies may build missions, fund universities, operate launch sites, purchase services and cooperate internationally
- D. astronaut selection is not based only on physical bravery; technical skill and cooperation are central

UNIT 8 REVIEW - OLYMPIAD CHALLENGE

B. Written and visual reasoning

1. Connect one idea from Chapter 21 with one idea from Chapter 23. Explain the relationship.
2. Choose two terms from different chapters in Unit 8 and compare their roles.
3. An astronaut's mass is unchanged in orbit. Which measurements would show this even when a scale reads almost zero?
4. Without shared communication standards, even excellent hardware may be unable to exchange commands or data.
5. Draw or describe one visual model that would help teach the central idea of Unit 8. State one limitation of the model.
6. Write one claim about Unit 8, give two pieces of evidence and state one uncertainty or boundary condition.

Olympiad Challenge

1. A student is reasoning about Space station. Which statement is the most scientifically careful?

- A. a station does not float at one fixed point; it circles Earth repeatedly
- B. results from one space experiment do not automatically apply to every condition on Earth
- C. more powerful radio cannot eliminate light-travel time
- D. joint does not mean every partner performs the same task

3. When checking a claim about Life-support system, which statement sets the right boundary?

- A. ordinary showering and open cups are impractical in microgravity
- B. a station does not float at one fixed point; it circles Earth repeatedly
- C. life support is not only an oxygen tank
- D. more powerful radio cannot eliminate light-travel time

5. Which conclusion about Interface standard avoids overclaiming?

- A. a standard does not guarantee quality; it guarantees a defined way of interacting
- B. raw data without metadata and calibration are not automatically easy to interpret
- C. life support is not only an oxygen tank
- D. a station does not float at one fixed point; it circles Earth repeatedly

7. Which explanation of National space agency is precise enough for a science report?

- A. ESA is not a single country's agency
- B. a standard does not guarantee quality; it guarantees a defined way of interacting
- C. one agency does not necessarily control all military, commercial and academic space work in its country
- D. life support is not only an oxygen tank

2. Which situation is the strongest application of ESA?

- A. NISAR combines NASA's L-band radar contribution and ISRO's S-band radar and spacecraft/launch contributions
- B. ESA instruments and astronauts may work with NASA, JAXA and other partners
- C. public mission archives may support studies never planned by the original team
- D. surface tension makes liquids form floating blobs

4. Which unfamiliar case is best explained by Rover and sample return?

- A. Mars rovers use onboard autonomy to avoid immediate hazards
- B. ESA instruments and astronauts may work with NASA, JAXA and other partners
- C. public mission archives may support studies never planned by the original team
- D. Mars rovers choose paths cautiously and perform analyses over many days

6. Where would Astronaut training be most useful?

- A. the International Space Station has supported research in biology, materials, Earth observation and technology
- B. neutral-buoyancy pools help rehearse body positioning for spacewalk tasks, though water is not microgravity
- C. Mars rovers choose paths cautiously and perform analyses over many days
- D. ESA instruments and astronauts may work with NASA, JAXA and other partners

8. Which example provides the clearest evidence of Inertia in orbit?

- A. water recovered from cabin humidity and waste can be purified for reuse
- B. neutral-buoyancy pools help rehearse body positioning for spacewalk tasks, though water is not microgravity
- C. Mars rovers choose paths cautiously and perform analyses over many days
- D. astronauts use handrails and restraints to control their own motion

UNIT 9

Basic Physics for Space Science

Forces, energy, waves, electricity and magnetism explain how spacecraft move, communicate and operate.

The Electromagnetic Spectrum



All bands are electromagnetic waves; instruments detect different bands to reveal different features.

Unit 9 opening visual - original illustration.

Major concepts

- Chapter 24: Force, Motion, Gravity and Newton's Laws
- Chapter 25: Work, Energy, Heat, Light and Sound
- Chapter 26: Electricity and Magnetism in Space Technology

Expected unit outcomes

- Distinguish speed, velocity and acceleration.
- Calculate simple work and kinetic-energy ideas qualitatively.
- Describe current, voltage and circuits.

CHAPTER 24

Force, Motion, Gravity and Newton's Laws

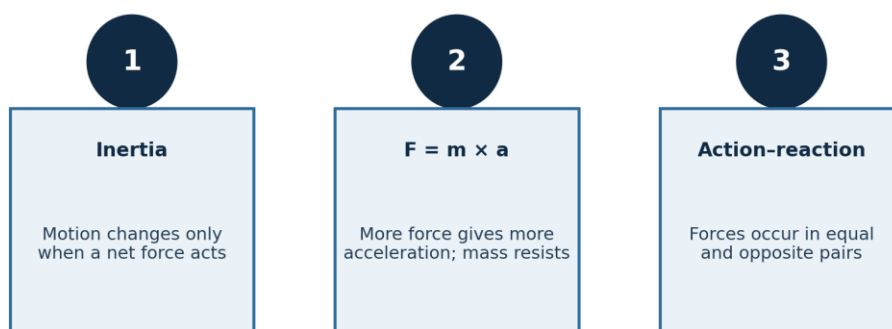
Physics describes motion using measurable quantities and explains change through forces. Speed tells how fast distance changes; velocity also includes direction; acceleration describes change in velocity. Newton's laws connect net force, mass and acceleration. Gravity is the attraction between masses and supplies the inward acceleration for planets, moons and satellites.

Opening question Why is an object moving in a circle accelerating even when its speed stays constant?

Learning outcomes

- Distinguish speed, velocity and acceleration.
- Apply all three newton laws.
- Calculate force in simple cases.
- Explain gravity and orbital motion.

Newton's Laws in Everyday and Space Motion



The laws describe how forces and motion are related; they apply on Earth and in space.

Figure 24.1 Describing motion. Original illustration; not to scale where stated.

Describing motion

A complete description depends on a reference point and consistent units.

Speed and velocity

Speed is distance travelled per time; velocity is speed in a stated direction. An object can keep constant speed while changing velocity by turning. For example, a satellite travelling east at 7.5 kilometres per second has a velocity, not only a speed. One limit must remain clear: speed cannot be negative in elementary treatment, while a chosen one-dimensional velocity can use signs for direction.

Acceleration

Acceleration means rate at which velocity changes. It works because speeding up, slowing down or changing direction all count as acceleration. A concrete case is a planet moving at nearly constant speed in a curved orbit accelerates toward the Sun. By contrast, acceleration is not another word for high speed.

Check your understanding Give an example of acceleration without an increase in speed.

Newton's three laws

The laws form a connected description of how forces change motion.

First and second laws

First and second laws means without net force, velocity remains constant; with net force, acceleration equals net force divided by mass. It works because greater force gives greater acceleration for the same mass, while greater mass gives smaller acceleration for the same force. A concrete case is a thruster changes a satellite's velocity, and the same thrust changes a small satellite faster than a massive one. By contrast, motion does not require a continuing force in empty space; changing motion does.

Third law

Forces between interacting objects occur in equal-magnitude opposite-direction pairs. The underlying relationship is that each force acts on a different object, so the pair does not cancel on either object's individual force diagram. This can be seen when Earth pulls an astronaut downward while the astronaut pulls Earth upward with equal force. Equal forces can produce very different accelerations when masses differ.

Check your understanding Why does Earth's acceleration toward a falling apple go unnoticed?

Gravity and orbits

Gravity weakens with distance but acts over enormous ranges.

Gravitational force

In this chapter, Gravitational force is used for mutual attraction between masses, stronger for larger masses and weaker at greater separation. Near Earth it gives objects downward acceleration; in space it shapes trajectories and holds systems together. One useful example is the Moon falls around Earth instead of moving away in a straight line. It should be separated from this different claim: gravity is not caused by Earth's atmosphere or rotation.

Circular-orbit acceleration

Inward acceleration required to continually change a moving object's direction. The process can be followed as gravity supplies this centripetal acceleration for an ideal orbit, while forward inertia carries the object sideways. For instance, a spacecraft can coast for long periods while gravity bends its path. A common overstatement is avoided by recognising that centripetal force is not an extra new force; it is the name for the net inward role played by real forces.

Check your understanding What real force usually provides centripetal acceleration for an orbiting satellite?

Activity: Measure force and acceleration qualitatively

Aim	to compare motion for different forces and masses
Materials	toy cart, elastic band, small masses and a marked floor
Steps	1. Pull the same cart with two different elastic-band stretches over the same distance. 2. Compare how quickly its speed changes. 3. Add mass to the cart and repeat with the same stretch. 4. Describe the results using force, mass and acceleration.
Observe	Greater pull generally gives greater acceleration; greater mass gives less acceleration for the same pull.
Expected learning	Newton's second law predicts patterns before exact calculation.
Safety	Keep the path clear and do not overstretch the elastic band.

Chapter Review

Key ideas

- Speed is distance travelled per time; velocity is speed in a stated direction.
- Without net force, velocity remains constant; with net force, acceleration equals net force divided by mass.
- Mutual attraction between masses, stronger for larger masses and weaker at greater separation.
- “An object moving at constant speed has zero acceleration.” Give the missing condition.

Important terms

speed: distance travelled per unit time	velocity: speed with direction
acceleration: rate of change of velocity	net force: combined effect of all forces
inertia: resistance to change in motion	Newton: SI unit of force
gravity: attraction between masses	centripetal acceleration: inward acceleration in curved motion

A. Choose the correct answer

1. Which term names distance travelled per unit time?

- A. speed
- B. velocity
- C. net force
- D. Newton

3. Choose the vocabulary item defined as rate of change of velocity.

- A. net force
- B. Newton
- C. acceleration
- D. centripetal acceleration

5. In this chapter, what is defined as resistance to change in motion?

- A. inertia
- B. Newton
- C. centripetal acceleration
- D. velocity

2. A learner describes speed with direction. What is the correct term?

- A. acceleration
- B. velocity
- C. inertia
- D. gravity

4. Which expression would a scientist use for combined effect of all forces?

- A. inertia
- B. gravity
- C. speed
- D. net force

Exercises (continued)

6. Which statement best explains Speed and velocity?

- A. speeding up, slowing down or changing direction all count as acceleration
- B. an object can keep constant speed while changing velocity by turning
- C. each force acts on a different object, so the pair does not cancel on either object's individual force diagram
- D. speed cannot be negative in elementary treatment, while a chosen one-dimensional velocity can use signs for direction

8. Which statement correctly avoids a common misunderstanding about First and second laws?

- A. equal forces can produce very different accelerations when masses differ
- B. mutual attraction between masses, stronger for larger masses and weaker at greater separation
- C. greater force gives greater acceleration for the same mass, while greater mass gives smaller acceleration for the same force
- D. motion does not require a continuing force in empty space; changing motion does

10. Which example most directly illustrates Gravitational force?

- A. a spacecraft can coast for long periods while gravity bends its path
- B. the Moon falls around Earth instead of moving away in a straight line
- C. a satellite travelling east at 7.5 kilometres per second has a velocity, not only a speed
- D. gravity is not caused by Earth's atmosphere or rotation

7. Which example most directly illustrates Acceleration?

- A. a thruster changes a satellite's velocity, and the same thrust changes a small satellite faster than a massive one
- B. Earth pulls an astronaut downward while the astronaut pulls Earth upward with equal force
- C. a planet moving at nearly constant speed in a curved orbit accelerates toward the Sun
- D. acceleration is not another word for high speed

9. Which statement best explains Third law?

- A. each force acts on a different object, so the pair does not cancel on either object's individual force diagram
- B. near Earth it gives objects downward acceleration; in space it shapes trajectories and holds systems together
- C. an object can keep constant speed while changing velocity by turning
- D. equal forces can produce very different accelerations when masses differ

B. Answer in one or two sentences

1. Define speed and velocity in your own words.
2. Explain how Acceleration works. Use one cause-and-effect link.
3. Compare First and second laws and Third law. State one similarity or connection and one difference.
4. Use the example of Gravitational force to explain why context or measurement matters.

C. Answer in detail

1. Explain the main ideas of "Force, Motion, Gravity and Newton's Laws" as a connected account. Include at least four important terms from the chapter.
2. A probe fires a thruster sideways while moving forward. Sketch how its velocity changes.

D. Apply and reason

1. If gravity vanished suddenly, an orbiting body would move approximately along the tangent instead of continuing the circle.
2. "An object moving at constant speed has zero acceleration." Give the missing condition.

E. Activity or project

Create six force diagrams for everyday and space situations, labelling only forces acting on the chosen object.

CHAPTER 25

Work, Energy, Heat, Light and Sound

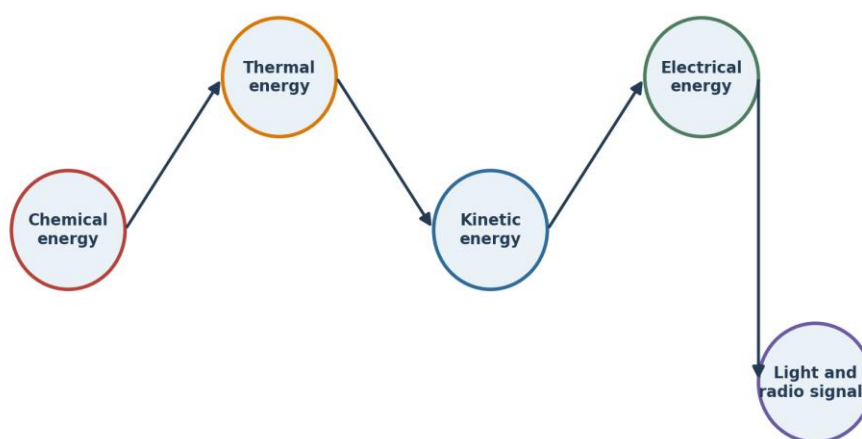
Energy is a measurable property that can be transferred and transformed. Work transfers energy when a force causes displacement. Moving objects have kinetic energy; separated or raised objects can have potential energy. Heat is energy transferred because of temperature difference. Light travels through vacuum as electromagnetic radiation, while sound requires matter and therefore cannot cross empty space.

Opening question Why can astronauts see an explosion in space but not hear it directly through vacuum?

Learning outcomes

- Calculate simple work and kinetic-energy ideas qualitatively.
- Distinguish heat and temperature.
- Explain light reflection and absorption.
- Explain why sound needs a medium.

Energy Changes Form



A rocket mission uses many transformations; energy is transferred and spread, not “used up” into nothing.

Figure 25.1 Work and mechanical energy. Original illustration; not to scale where stated.

Work and mechanical energy

In physics, work has a narrower meaning than everyday effort.

Work

Work means energy transferred when a force acts through a displacement in its direction. It works because for a constant aligned force, work equals force multiplied by distance. A concrete case is lifting a 5-newton object through 2 metres transfers about 10 joules of energy. By contrast, holding a heavy object still can be tiring but does no mechanical work on the object because displacement is zero.

Kinetic and potential energy

Kinetic energy is energy of motion; potential energy is stored through position or configuration. The underlying relationship is that a falling object converts gravitational potential energy mainly into kinetic energy, while engines transform chemical energy into vehicle motion and heat. This can be seen when a raised rocket has gravitational potential energy relative to a chosen reference level. Potential energy belongs to an interacting system, not to one isolated object in every description.

Check your understanding How does doubling speed affect kinetic energy qualitatively?

Temperature and heat transfer

Temperature and heat are related but not interchangeable.

Temperature

A measure related to the average random kinetic energy of particles. The underlying relationship is that objects at different temperatures exchange energy until they approach thermal equilibrium. This can be seen when a small spark can have high temperature but contain less total thermal energy than a warm swimming pool. Temperature does not measure the total amount of energy in an object.

Conduction, convection and radiation

The phrase Conduction, convection and radiation refers to three mechanisms of thermal energy transfer: particle contact, bulk fluid motion and electromagnetic waves. Here, spacecraft use conduction through structures and radiation to space; ordinary convection inside liquids or gases depends on buoyancy and gravity conditions. Consider the following case: a radiator panel emits infrared radiation to remove waste heat. The careful interpretation is that vacuum prevents conduction and convection across empty space but not radiation.

Check your understanding Which heat-transfer method allows the Sun to warm Earth across space?

Light and sound as waves

Different waves require different conditions for propagation.

Light

Electromagnetic radiation that can travel through vacuum and interacts by reflection, refraction, absorption and scattering. The process can be followed as telescopes form images by controlling light paths, while surfaces and atmospheres change which wavelengths reach a detector. For instance, a mirror reflects a beam and a lens refracts it. A common overstatement is avoided by recognising that visible light is only a small part of the electromagnetic spectrum.

Sound

Sound concerns a mechanical vibration transmitted through particles in a solid, liquid or gas. The key cause-and-effect link is that a vibrating source compresses and rarefies the medium, carrying energy without transporting the same particles all the way to the listener. This appears in radio can carry encoded sound across space, but the radio wave is electromagnetic; a speaker recreates sound at the receiver. At the same time, sound cannot propagate through a perfect vacuum.

Check your understanding What actually crosses space when mission control speaks to an astronaut by radio?

Activity: Compare thermal transfer paths

Aim	to identify conduction, convection and radiation in safe observations
Materials	metal and wooden spoons, warm water, lamp and thermometer if available
Steps	1. Place spoon ends in warm water and compare handle warming carefully. 2. Observe air movement above warm water using a light paper spiral at a safe distance. 3. Feel warmth from the side of a lamp without touching it. 4. Classify each observation and discuss where the model is imperfect.
Observe	Metal conducts better than wood; moving warm air shows convection; radiant warmth can cross an air gap.
Expected learning	Different mechanisms may operate together.
Safety	Use warm, not boiling, water; keep paper away from hot bulbs and use adult supervision.

Chapter Review

Key ideas

- Energy transferred when a force acts through a displacement in its direction.
- A measure related to the average random kinetic energy of particles.
- Electromagnetic radiation that can travel through vacuum and interacts by reflection, refraction, absorption and scattering.
- “Heat is the amount of hotness stored inside an object.” Replace the statement with correct terms.

Important terms

work: energy transfer by force through displacement	joule: SI unit of energy and work
kinetic energy: energy associated with motion	potential energy: energy associated with position or configuration
temperature: measure related to average particle motion	heat: energy transferred due to temperature difference
radiation: energy transfer by electromagnetic waves	sound: mechanical wave requiring a medium

A. Choose the correct answer

1. Which term names energy transfer by force through displacement?

- A. joule
- B. work
- C. potential energy
- D. heat

3. Choose the vocabulary item defined as energy associated with motion.

- A. potential energy
- B. heat
- C. sound
- D. kinetic energy

5. In this chapter, what is defined as measure related to average particle motion?

- A. heat
- B. temperature
- C. sound
- D. joule

2. A learner describes SI unit of energy and work. What is the correct term?

- A. kinetic energy
- B. temperature
- C. joule
- D. radiation

4. Which expression would a scientist use for energy associated with position or configuration?

- A. potential energy
- B. temperature
- C. radiation
- D. work

Exercises (continued)

6. Which statement best explains Work?

- A. a falling object converts gravitational potential energy mainly into kinetic energy, while engines transform chemical energy into vehicle motion and heat
- B. spacecraft use conduction through structures and radiation to space; ordinary convection inside liquids or gases depends on buoyancy and gravity conditions
- C. for a constant aligned force, work equals force multiplied by distance
- D. holding a heavy object still can be tiring but does no mechanical work on the object because displacement is zero

8. Which statement correctly avoids a common misunderstanding about Temperature?

- A. temperature does not measure the total amount of energy in an object
- B. vacuum prevents conduction and convection across empty space but not radiation
- C. electromagnetic radiation that can travel through vacuum and interacts by reflection, refraction, absorption and scattering
- D. objects at different temperatures exchange energy until they approach thermal equilibrium

10. Which example most directly illustrates Light?

- A. radio can carry encoded sound across space, but the radio wave is electromagnetic; a speaker recreates sound at the receiver
- B. lifting a 5-newton object through 2 metres transfers about 10 joules of energy
- C. a mirror reflects a beam and a lens refracts it
- D. visible light is only a small part of the electromagnetic spectrum

7. Which example most directly illustrates Kinetic and potential energy?

- A. a small spark can have high temperature but contain less total thermal energy than a warm swimming pool
- B. a radiator panel emits infrared radiation to remove waste heat
- C. potential energy belongs to an interacting system, not to one isolated object in every description
- D. a raised rocket has gravitational potential energy relative to a chosen reference level

9. Which statement best explains Conduction, convection and radiation?

- A. telescopes form images by controlling light paths, while surfaces and atmospheres change which wavelengths reach a detector
- B. spacecraft use conduction through structures and radiation to space; ordinary convection inside liquids or gases depends on buoyancy and gravity conditions
- C. for a constant aligned force, work equals force multiplied by distance
- D. vacuum prevents conduction and convection across empty space but not radiation

B. Answer in one or two sentences

1. Define work and joule in your own words.
2. Explain how Kinetic and potential energy works. Use one cause-and-effect link.
3. Compare Temperature and Conduction, convection and radiation. State one similarity or connection and one difference.
4. Use the example of Light to explain why context or measurement matters.

C. Answer in detail

1. Explain the main ideas of “Work, Energy, Heat, Light and Sound” as a connected account. Include at least four important terms from the chapter.
2. A spacecraft in sunlight overheats even though space is cold. Explain using radiation and limited heat loss.

D. Apply and reason

1. If sound travelled through vacuum, space communication and many physical processes would differ, but radio would still be electromagnetic.
2. “Heat is the amount of hotness stored inside an object.” Replace the statement with correct terms.

E. Activity or project

Trace at least six energy transformations from rocket propellant to a picture received on a ground computer.

CHAPTER 26

Electricity and Magnetism in Space Technology

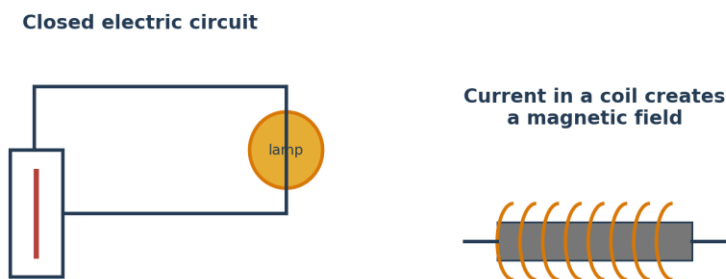
Spacecraft depend on electric charge, current, voltage, circuits, magnets and electromagnetic waves. Solar cells convert light energy to electrical energy; batteries store chemical energy; wires and electronic components distribute and control current. Magnetic fields support sensors, motors and attitude control, while radio waves carry commands and data across space.

Opening question Why can a spacecraft lose communication even when its computer and scientific instruments are still working?

Learning outcomes

- Describe current, voltage and circuits.
- Explain spacecraft power systems.
- Describe basic magnetic effects.
- Connect radio communication to electromagnetism.

Electric Circuits and Electromagnets



Spacecraft use electrical circuits, sensors, motors and magnetic devices in many subsystems.

Figure 26.1 Electric circuits. Original illustration; not to scale where stated.

Electric circuits

Charge moves through a complete conducting path when a source provides electrical potential difference.

Current and voltage

Current is the rate of electric-charge flow; voltage is energy transferred per unit charge between points. The underlying relationship is that a battery or solar array maintains a potential difference that drives current through suitable circuits. This can be seen when increased resistance can reduce current for a fixed voltage in a simple circuit. Voltage is not “the amount of current stored”.

Series and parallel paths

The phrase Series and parallel paths refers to components in series share one path; parallel branches provide more than one path. Here, spacecraft distribute power through carefully protected buses so one fault does not always disable every subsystem. Consider the following case: separate fused branches can isolate a damaged instrument. The careful interpretation is that parallel does not mean unlimited current; sources and wires have safe limits.

Check your understanding Why are protected parallel branches useful in a spacecraft?

Producing and storing power

Power availability changes with orientation, orbit and mission distance.

Solar array and battery

The phrase Solar array and battery refers to solar cells convert light to electricity; rechargeable batteries supply power during eclipse or peak demand. Here, power controllers regulate voltage, charge batteries and protect circuits. Consider the following case: an Earth satellite may charge in sunlight and use batteries while passing through Earth’s shadow. The careful interpretation is that a solar panel does not store energy by itself unless storage equipment is connected.

Power budget

A plan comparing expected generation, storage and consumption for every mission mode. In practice, engineers prioritise essential loads and schedule high-power instruments when enough energy is available. An example is a rover may pause driving to recharge or warm batteries. However, average power alone is insufficient if short peak demands exceed system capability.

Check your understanding What happens to a solar-powered Earth satellite during part of an orbital eclipse?

Magnetism and communication

Electric currents create magnetic fields, and changing electric and magnetic fields form electromagnetic waves.

Magnetic field and electromagnet

Magnetic field and electromagnet concerns a region in which magnetic forces act; a current-carrying coil can create a controllable magnetic field. The key cause-and-effect link is that spacecraft use motors, relays, magnetic sensors and magnetorquers that interact with Earth's field. This appears in a magnetorquer can slowly rotate a small satellite without expelling propellant. At the same time, magnetism is not blocked simply by vacuum.

Radio communication

Transmission of information by modulating electromagnetic waves. Evidence for the idea comes from situations such as a deep-space antenna uses a large dish and careful coding to detect weak data. The explanation is antennas convert electrical signals to waves and received waves back to electrical signals; pointing, frequency, power and noise affect the link. Its boundary is equally important: radio waves are not sound waves, even when they carry speech.

Check your understanding What conversion occurs at a transmitting antenna and at a receiving antenna?

Activity: Build a safe electromagnet

Aim	to observe a magnetic field produced by current
Materials	iron nail, insulated wire, 1.5 V cell, paper clips and adult supervision
Steps	1. Wrap insulated wire around the nail, leaving two free ends. 2. Briefly connect the wire ends to the cell. 3. Test whether the nail picks up paper clips. 4. Disconnect promptly and compare different coil turns without prolonged connection.
Observe	More coil turns can strengthen the effect within safe limits; the nail loses much of its magnetism after disconnection.
Expected learning	Electric current can create a controllable magnetic field.
Safety	Use only one low-voltage cell; disconnect after a few seconds because wire and cell can heat.

Chapter Review

Key ideas

- Current is the rate of electric-charge flow; voltage is energy transferred per unit charge between points.
- Solar cells convert light to electricity; rechargeable batteries supply power during eclipse or peak demand.
- A region in which magnetic forces act; a current-carrying coil can create a controllable magnetic field.
- “Radio works because sound travels through space to the antenna.” Correct both wave types and conversions.

Important terms

electric current: rate of charge flow	voltage: energy transferred per unit charge
circuit: complete path for electric current	solar cell: device converting light to electricity
battery: device storing chemical energy for electrical use	magnetic field: region of magnetic influence
electromagnet: magnet produced by electric current	radio wave: electromagnetic wave used for communication

A. Choose the correct answer

1. Which term names rate of charge flow?

- A. voltage
- B. solar cell
- C. electric current
- D. magnetic field

3. Choose the vocabulary item defined as complete path for electric current.

- A. circuit
- B. solar cell
- C. magnetic field
- D. radio wave

5. In this chapter, what is defined as device storing chemical energy for electrical use?

- A. magnetic field
- B. radio wave
- C. battery
- D. voltage

2. A learner describes energy transferred per unit charge. What is the correct term?

- A. circuit
- B. battery
- C. electromagnet
- D. voltage

4. Which expression would a scientist use for device converting light to electricity?

- A. battery
- B. solar cell
- C. electromagnet
- D. electric current

Exercises (continued)

6. Which statement best explains Current and voltage?

- A. spacecraft distribute power through carefully protected buses so one fault does not always disable every subsystem
- B. engineers prioritise essential loads and schedule high-power instruments when enough energy is available
- C. voltage is not “the amount of current stored”
- D. a battery or solar array maintains a potential difference that drives current through suitable circuits

8. Which statement correctly avoids a common misunderstanding about Solar array and battery?

- A. average power alone is insufficient if short peak demands exceed system capability
- B. a solar panel does not store energy by itself unless storage equipment is connected
- C. a region in which magnetic forces act; a current-carrying coil can create a controllable magnetic field
- D. power controllers regulate voltage, charge batteries and protect circuits

10. Which example most directly illustrates Magnetic field and electromagnet?

- A. a deep-space antenna uses a large dish and careful coding to detect weak data
- B. increased resistance can reduce current for a fixed voltage in a simple circuit
- C. magnetism is not blocked simply by vacuum
- D. a magnetorquer can slowly rotate a small satellite without expelling propellant

7. Which example most directly illustrates Series and parallel paths?

- A. separate fused branches can isolate a damaged instrument
- B. an Earth satellite may charge in sunlight and use batteries while passing through Earth’s shadow
- C. a rover may pause driving to recharge or warm batteries
- D. parallel does not mean unlimited current; sources and wires have safe limits

9. Which statement best explains Power budget?

- A. spacecraft use motors, relays, magnetic sensors and magnetorquers that interact with Earth’s field
- B. a battery or solar array maintains a potential difference that drives current through suitable circuits
- C. engineers prioritise essential loads and schedule high-power instruments when enough energy is available
- D. average power alone is insufficient if short peak demands exceed system capability

B. Answer in one or two sentences

1. Define electric current and voltage in your own words.
2. Explain how Series and parallel paths works. Use one cause-and-effect link.
3. Compare Solar array and battery and Power budget. State one similarity or connection and one difference.
4. Use the example of Magnetic field and electromagnet to explain why context or measurement matters.

C. Answer in detail

1. Explain the main ideas of “Electricity and Magnetism in Space Technology” as a connected account. Include at least four important terms from the chapter.
2. A spacecraft’s antenna points away from Earth. Which parts of the communication link are affected?

D. Apply and reason

1. Without energy storage, many solar-powered satellites would shut down during eclipses or darkness.
2. “Radio works because sound travels through space to the antenna.” Correct both wave types and conversions.

E. Activity or project

Prepare a spacecraft power budget with normal, science, safe and eclipse modes.

UNIT 9 REVIEW

Basic Physics for Space Science - Mixed Review

This review combines ideas from all chapters in the unit. Use diagrams, units and evidence where relevant. The questions are new combinations rather than copies of the chapter exercises.

A. Mixed multiple-choice review

1. Which option links a term with its correct meaning?

- A. joule - energy associated with position or configuration
- B. joule - SI unit of energy and work
- C. joule - energy transfer by electromagnetic waves
- D. joule - complete path for electric current

3. Only one pair is scientifically correct. Which is it?

- A. sound - energy transferred per unit charge
- B. sound - device storing chemical energy for electrical use
- C. sound - distance travelled per unit time
- D. sound - mechanical wave requiring a medium

5. Find the pair that uses the vocabulary correctly.

- A. magnetic field - electromagnetic wave used for communication
- B. magnetic field - region of magnetic influence
- C. magnetic field - rate of change of velocity
- D. magnetic field - attraction between masses

7. Choose the correctly defined term from Unit 9.

- A. net force - SI unit of force
- B. net force - energy transfer by force through displacement
- C. net force - measure related to average particle motion
- D. net force - combined effect of all forces

9. Which statement best explains Conduction, convection and radiation in Chapter 25?

- A. a vibrating source compresses and rarefies the medium, carrying energy without transporting the same particles all the way to the listener
- B. spacecraft use motors, relays, magnetic sensors and magnetorquers that interact with Earth's field
- C. vacuum prevents conduction and convection across empty space but not radiation
- D. spacecraft use conduction through structures and radiation to space; ordinary convection inside liquids or gases depends on buoyancy and gravity conditions

11. Which statement best explains Acceleration in Chapter 24?

- A. each force acts on a different object, so the pair does not cancel on either object's individual force diagram
- B. speeding up, slowing down or changing direction all count as acceleration
- C. objects at different temperatures exchange energy until they approach thermal equilibrium
- D. acceleration is not another word for high speed

2. Select the accurately matched term and definition.

- A. temperature - energy transfer by electromagnetic waves
- B. temperature - energy transferred per unit charge
- C. temperature - measure related to average particle motion
- D. temperature - region of magnetic influence

4. Which term-definition pair would belong in the Unit 9 glossary?

- A. circuit - complete path for electric current
- B. circuit - device storing chemical energy for electrical use
- C. circuit - electromagnetic wave used for communication
- D. circuit - combined effect of all forces

6. Which pairing preserves the meaning used in this unit?

- A. speed - rate of change of velocity
- B. speed - SI unit of force
- C. speed - distance travelled per unit time
- D. speed - SI unit of energy and work

8. Which row could be copied into an accurate revision table?

- A. gravity - attraction between masses
- B. gravity - energy transfer by force through displacement
- C. gravity - energy associated with position or configuration
- D. gravity - mechanical wave requiring a medium

10. Which statement best explains Solar array and battery in Chapter 26?

- A. power controllers regulate voltage, charge batteries and protect circuits
- B. spacecraft use motors, relays, magnetic sensors and magnetorquers that interact with Earth's field
- C. each force acts on a different object, so the pair does not cancel on either object's individual force diagram
- D. a solar panel does not store energy by itself unless storage equipment is connected

12. Which statement best explains Work in Chapter 25?

- A. objects at different temperatures exchange energy until they approach thermal equilibrium
- B. spacecraft distribute power through carefully protected buses so one fault does not always disable every subsystem
- C. for a constant aligned force, work equals force multiplied by distance
- D. holding a heavy object still can be tiring but does no mechanical work on the object because displacement is zero

UNIT 9 REVIEW - OLYMPIAD CHALLENGE

B. Written and visual reasoning

1. Connect one idea from Chapter 24 with one idea from Chapter 26. Explain the relationship.
2. Choose two terms from different chapters in Unit 9 and compare their roles.
3. A probe fires a thruster sideways while moving forward. Sketch how its velocity changes.
4. Without energy storage, many solar-powered satellites would shut down during eclipses or darkness.
5. Draw or describe one visual model that would help teach the central idea of Unit 9. State one limitation of the model.
6. Write one claim about Unit 9, give two pieces of evidence and state one uncertainty or boundary condition.

Olympiad Challenge

1. A student is reasoning about Conduction, convection and radiation. Which statement is the most scientifically careful?

- A. visible light is only a small part of the electromagnetic spectrum
- B. vacuum prevents conduction and convection across empty space but not radiation
- C. parallel does not mean unlimited current; sources and wires have safe limits
- D. radio waves are not sound waves, even when they carry speech

3. When checking a claim about Circular-orbit acceleration, which statement sets the right boundary?

- A. holding a heavy object still can be tiring but does no mechanical work on the object because displacement is zero
- B. vacuum prevents conduction and convection across empty space but not radiation
- C. parallel does not mean unlimited current; sources and wires have safe limits
- D. centripetal force is not an extra new force; it is the name for the net inward role played by real forces

5. Which conclusion about Acceleration avoids overclaiming?

- A. motion does not require a continuing force in empty space; changing motion does
- B. acceleration is not another word for high speed
- C. centripetal force is not an extra new force; it is the name for the net inward role played by real forces
- D. vacuum prevents conduction and convection across empty space but not radiation

7. Which explanation of Power budget is precise enough for a science report?

- A. magnetism is not blocked simply by vacuum
- B. acceleration is not another word for high speed
- C. centripetal force is not an extra new force; it is the name for the net inward role played by real forces
- D. average power alone is insufficient if short peak demands exceed system capability

2. Which situation is the strongest application of Magnetic field and electromagnet?

- A. a deep-space antenna uses a large dish and careful coding to detect weak data
- B. a thruster changes a satellite's velocity, and the same thrust changes a small satellite faster than a massive one
- C. a magnetorquer can slowly rotate a small satellite without expelling propellant
- D. lifting a 5-newton object through 2 metres transfers about 10 joules of energy

4. Which unfamiliar case is best explained by Current and voltage?

- A. increased resistance can reduce current for a fixed voltage in a simple circuit
- B. separate fused branches can isolate a damaged instrument
- C. a magnetorquer can slowly rotate a small satellite without expelling propellant
- D. a thruster changes a satellite's velocity, and the same thrust changes a small satellite faster than a massive one

6. Where would Temperature be most useful?

- A. a radiator panel emits infrared radiation to remove waste heat
- B. increased resistance can reduce current for a fixed voltage in a simple circuit
- C. a small spark can have high temperature but contain less total thermal energy than a warm swimming pool
- D. a magnetorquer can slowly rotate a small satellite without expelling propellant

8. Which example provides the clearest evidence of Gravitational force?

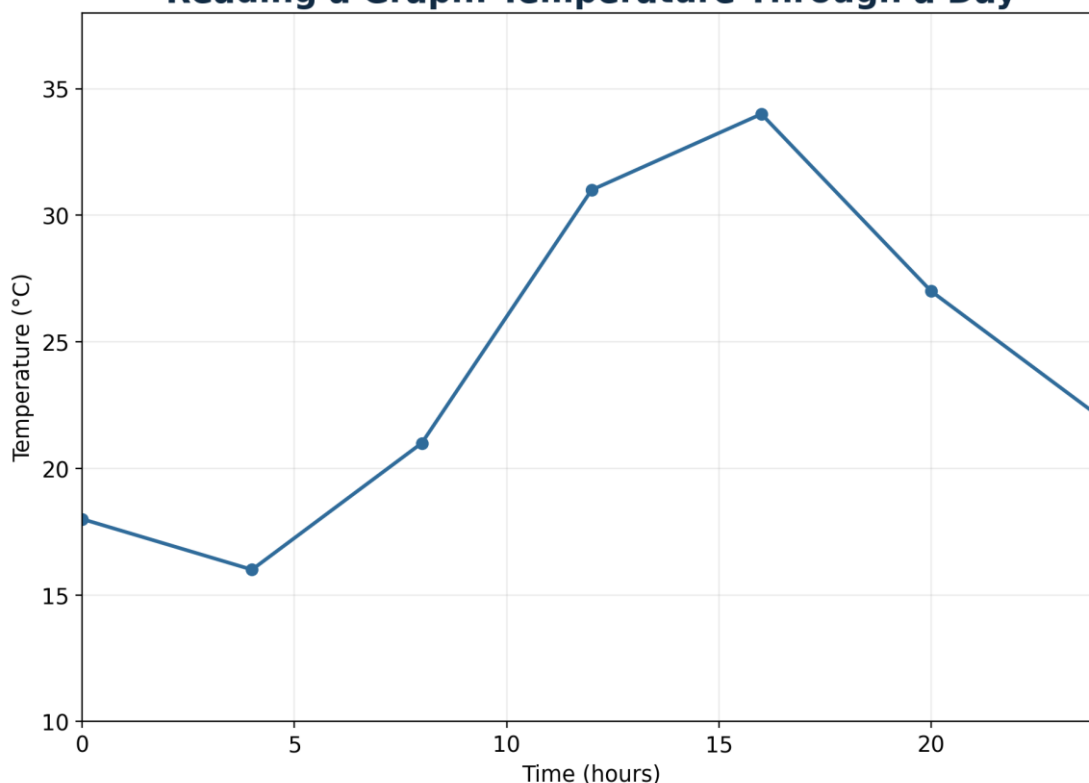
- A. the Moon falls around Earth instead of moving away in a straight line
- B. a spacecraft can coast for long periods while gravity bends its path
- C. a small spark can have high temperature but contain less total thermal energy than a warm swimming pool
- D. increased resistance can reduce current for a fixed voltage in a simple circuit

UNIT 10

Scientific Reasoning and Logical Aptitude

Olympiad success depends on observing evidence, recognising patterns and explaining how a conclusion follows.

Reading a Graph: Temperature Through a Day



Unit 10 opening visual - original illustration.

Major concepts

- Chapter 27: Scientific Observation, Measurement and Evidence
- Chapter 28: Patterns, Data Interpretation and Numerical Aptitude
- Chapter 29: Analytical Reasoning and Problem Solving

Expected unit outcomes

- Separate observation and inference.
- Read tables and graphs.
- Translate statements into constraints.

CHAPTER 27

Scientific Observation, Measurement and Evidence

Scientific reasoning starts by separating observation from interpretation. Measurements add numbers and units, but they also include uncertainty. A fair test changes one factor while controlling others. Repetition, careful records and independent checking help distinguish a real pattern from accident, bias or measurement error.

Opening question Why can two careful students obtain slightly different measurements without either being dishonest or careless?

Learning outcomes

- Separate observation and inference.
- Use units and estimate uncertainty.
- Design a fair test.
- Identify sources of error and bias.

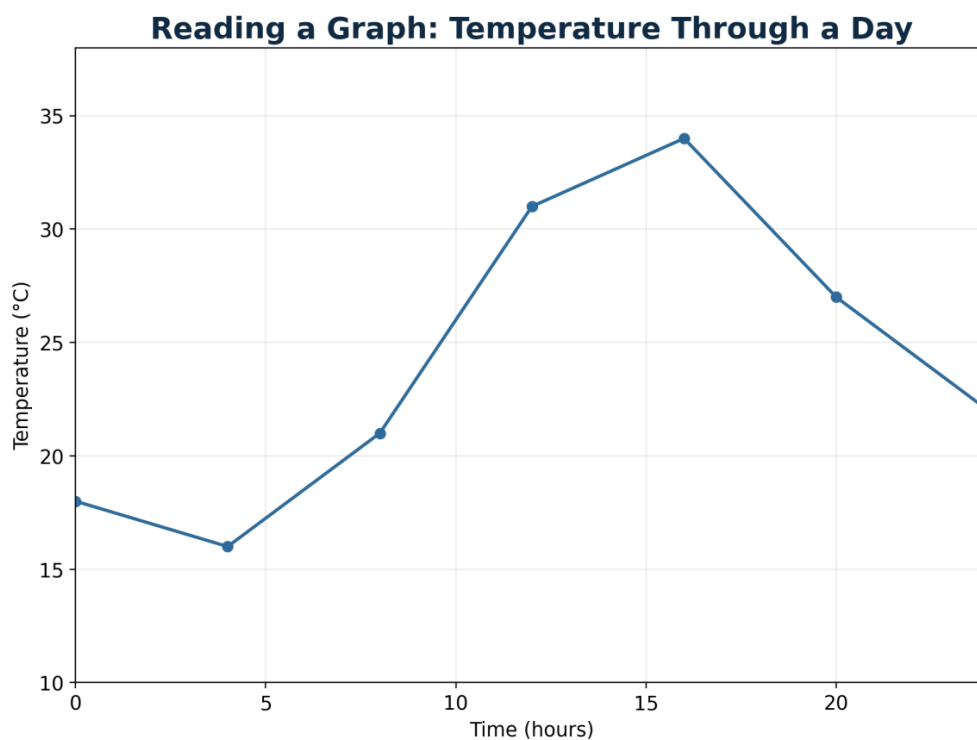


Figure 27.1 Claims and evidence. Original illustration; not to scale where stated.

Claims and evidence

A conclusion should be no stronger than the evidence supporting it.

Observation and inference

The phrase Observation and inference refers to an observation reports detected information; an inference is a reasoned explanation based on observations. Here, scientists state which details were measured and which ideas were concluded from them. Consider the following case: “the shadow length is 42 cm” is observation; “the Sun is higher than earlier” is an inference using a model. The careful interpretation is that an inference can be reasonable without being directly seen.

Hypothesis and prediction

A hypothesis is a testable proposed explanation; a prediction states an expected result under specified conditions. In practice, predictions link a hypothesis to evidence that could support or challenge it. An example is if higher Sun altitude shortens a vertical stick’s shadow, measurements near noon should be shorter than in early morning. However, a prediction made after seeing the result is weaker evidence than one recorded beforehand.

Check your understanding Which statement is more directly observable: “the object accelerated” or “its speed increased from 2 to 5 m/s”?

Measurement and uncertainty

Every measuring device has limited resolution and every procedure has variation.

Accuracy and precision

Accuracy is closeness to an accepted or true value; precision is closeness among repeated measurements. In practice, a set can be tightly grouped but shifted by calibration error, or spread out around the correct value. An example is a clock that always reads 3 seconds fast may be precise but inaccurate. However, more decimal places do not automatically make a result accurate.

Uncertainty

To understand Uncertainty, begin with a stated range or estimate showing limits of a measurement. Next, note that reading scale divisions, reaction time and changing conditions contribute to uncertainty; repetition helps estimate spread. A length recorded as about 12.4 ± 0.1 cm communicates more than “12.4000 cm” from a coarse ruler. This does not mean that uncertainty does not mean the measurement is useless.

Check your understanding How can repeating a measurement reveal random variation?

Fair tests and variables

A test is interpretable only when relevant conditions are controlled or recorded.

Independent and dependent variables

The independent variable is deliberately changed; the dependent variable is measured as the response. Evidence for the idea comes from situations such as when testing how ramp height affects cart travel distance, ramp height is changed and distance is measured. The explanation is all other influential factors should be kept as constant as reasonably possible. Its boundary is equally important: the variable that changes first in time is not automatically the independent variable.

Control and bias

A control provides comparison; bias is a systematic influence that favours a result or interpretation. Random assignment, blind procedures, standard instruments and predefined rules reduce some biases. For example, using the same stopwatch operator for all trials controls one source but may retain that operator's reaction bias. One limit must remain clear: bias is not only deliberate cheating; it can enter unnoticed through design.

Check your understanding Why should a fair test change one main factor at a time?

Activity: Time a repeated event

Aim	to estimate variation and improve a procedure
Materials	pendulum or bouncing ball, stopwatch and recording table
Steps	1. Choose a safe event that repeats, such as ten pendulum swings. 2. Predict the time, then measure it five times with the same method. 3. Calculate the range and discuss reaction-time effects. 4. Improve one part of the procedure and repeat.
Observe	Repeated values vary; timing several cycles reduces the relative effect of starting and stopping reaction.
Expected learning	Methods can be improved by identifying dominant uncertainty.
Safety	Secure the pendulum and keep the swing away from faces.

Chapter Review

Key ideas

- An observation reports detected information; an inference is a reasoned explanation based on observations.
- Accuracy is closeness to an accepted or true value; precision is closeness among repeated measurements.
- The independent variable is deliberately changed; the dependent variable is measured as the response.
- “One successful trial proves the hypothesis.” Explain why alternatives and repetition matter.

Important terms

observation: recorded information detected directly or by instrument	inference: explanation drawn from evidence
hypothesis: testable proposed explanation	prediction: expected result under stated conditions
accuracy: closeness to a correct value	precision: closeness among repeated values
uncertainty: estimated limit of a measurement	variable: factor that can change

A. Choose the correct answer

1. Which term names recorded information detected directly or by instrument?

- A. inference
- B. prediction
- C. precision
- D. observation

3. Choose the vocabulary item defined as testable proposed explanation.

- A. prediction
- B. hypothesis
- C. precision
- D. variable

5. In this chapter, what is defined as closeness to a correct value?

- A. precision
- B. variable
- C. inference
- D. accuracy

2. A learner describes explanation drawn from evidence. What is the correct term?

- A. inference
- B. hypothesis
- C. accuracy
- D. uncertainty

4. Which expression would a scientist use for expected result under stated conditions?

- A. accuracy
- B. uncertainty
- C. prediction
- D. observation

Exercises (continued)

6. Which statement best explains Observation and inference?

- A. scientists state which details were measured and which ideas were concluded from them
- B. predictions link a hypothesis to evidence that could support or challenge it
- C. reading scale divisions, reaction time and changing conditions contribute to uncertainty; repetition helps estimate spread
- D. an inference can be reasonable without being directly seen

8. Which statement correctly avoids a common misunderstanding about Accuracy and precision?

- A. uncertainty does not mean the measurement is useless
- B. the independent variable is deliberately changed; the dependent variable is measured as the response
- C. more decimal places do not automatically make a result accurate
- D. a set can be tightly grouped but shifted by calibration error, or spread out around the correct value

10. Which example most directly illustrates Independent and dependent variables?

- A. when testing how ramp height affects cart travel distance, ramp height is changed and distance is measured
- B. using the same stopwatch operator for all trials controls one source but may retain that operator's reaction bias
- C. "the shadow length is 42 cm" is observation; "the Sun is higher than earlier" is an inference using a model
- D. the variable that changes first in time is not automatically the independent variable

7. Which example most directly illustrates Hypothesis and prediction?

- A. a clock that always reads 3 seconds fast may be precise but inaccurate
- B. if higher Sun altitude shortens a vertical stick's shadow, measurements near noon should be shorter than in early morning
- C. a length recorded as about 12.4 ± 0.1 cm communicates more than "12.4000 cm" from a coarse ruler
- D. a prediction made after seeing the result is weaker evidence than one recorded beforehand

9. Which statement best explains Uncertainty?

- A. all other influential factors should be kept as constant as reasonably possible
- B. scientists state which details were measured and which ideas were concluded from them
- C. uncertainty does not mean the measurement is useless
- D. reading scale divisions, reaction time and changing conditions contribute to uncertainty; repetition helps estimate spread

B. Answer in one or two sentences

1. Define observation and inference in your own words.
2. Explain how Hypothesis and prediction works. Use one cause-and-effect link.
3. Compare Accuracy and precision and Uncertainty. State one similarity or connection and one difference.
4. Use the example of Independent and dependent variables to explain why context or measurement matters.

C. Answer in detail

1. Explain the main ideas of "Scientific Observation, Measurement and Evidence" as a connected account. Include at least four important terms from the chapter.
2. A digital display shows many decimals. What further information is needed before trusting them?

D. Apply and reason

1. If all measurement error were random, averaging many trials would help; systematic calibration error could remain.
2. "One successful trial proves the hypothesis." Explain why alternatives and repetition matter.

E. Activity or project

Write a fair-test plan for one safe question from this book, naming variables, controls and expected evidence.

CHAPTER 28

Patterns, Data Interpretation and Numerical Aptitude

Olympiad questions often present a table, graph, sequence or unfamiliar diagram. The task is not to guess what looks familiar; it is to identify the variables, units and rule, then test the rule against every given item. Numerical aptitude includes estimation, ratios, unit conversion and multi-step calculations written in a checkable order.

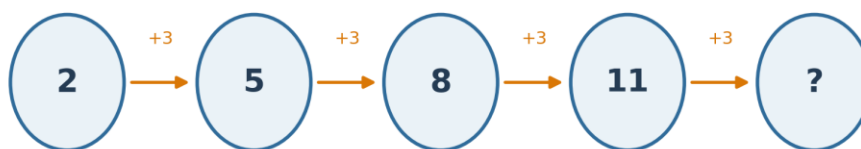
Opening question Why should you estimate an answer before entering exact numbers into a calculation?

Learning outcomes

- Read tables and graphs.
- Identify reliable numerical patterns.
- Use ratios and unit conversion.
- Check calculations by estimation and units.

Reasoning from a Pattern

Describe the rule before choosing the missing value.



A pattern is reliable only when the same relationship fits every given step.

Figure 28.1 Reading data displays. Original illustration; not to scale where stated.

Reading data displays

A graph is an argument about variables, not just a picture.

Axes and scale

The labelled directions and numerical intervals that define a graph. In practice, before reading points, identify the independent variable, dependent variable, units, origin and whether intervals are equal. An example is a graph beginning at 90 rather than zero can make a small change look dramatic. However, the visual steepness of a line depends on axis scale as well as data.

Trend and anomaly

To understand Trend and anomaly, begin with a trend is an overall relationship; an anomaly is a point that differs markedly from the pattern. Next, note that anomalies may reveal error, unusual conditions or new phenomena and should be investigated rather than deleted automatically. Temperature may rise through the morning with one lower point during cloud cover. This does not mean that one outlier does not by itself disprove every trend.

Check your understanding Why must you inspect axis labels before comparing two graphs?

Finding and testing patterns

A rule is accepted only if it fits all stated terms and conditions.

Sequence rule

To understand Sequence rule, begin with a relationship generating ordered values or shapes. Next, note that test differences, ratios, alternation, grouping and position-based rules, then use the simplest rule consistent with the evidence. 2, 5, 8, 11 increases by 3 each step. This does not mean that many rules can fit a very short sequence, so context and answer choices matter.

Classification pattern

Classification pattern describes a rule that groups items by shared properties. Its behaviour follows from this relationship: use more than appearance when possible and state both inclusion and exclusion conditions. The example of classifying planets by solid surface differs from classifying by distance from the Sun shows the idea in use. Remember that a useful category must be applied consistently to every item.

Check your understanding Why is “it looks similar” a weak classification rule?

Quantitative tools

Units and proportional reasoning prevent many avoidable mistakes.

Ratio and proportion

A ratio compares quantities; a proportion states that two ratios are equal. Scale drawings, speed, density and map calculations use multiplicative relationships. For example, if a model uses 1 cm for 1,000 km, 4.5 cm represents 4,500 km. One limit must remain clear: adding the same amount is different from multiplying by the same factor.

Dimensional check and estimation

Dimensional check and estimation means units are followed through a calculation and the result is compared with a sensible approximate range. It works because a speed calculation should end in distance per time, and an orbit period cannot reasonably be a negative number. A concrete case is estimating 298×51 as about 300×50 gives about 15,000 before exact work. By contrast, a calculator cannot identify a wrong formula or mistyped unit automatically.

Check your understanding What unit should result from dividing kilometres by hours?

Activity: Create and challenge a data graph

Aim	to practise complete graph communication
Materials	graph paper, ruler and a small supplied dataset
Steps	1. Choose an appropriate graph type. 2. Label axes with quantity and unit and select an even scale. 3. Plot all data, including any anomaly. 4. Exchange graphs and ask a partner to write two supported conclusions and one unsupported claim.
Observe	A readable graph allows another person to reproduce value readings and distinguish evidence from interpretation.
Expected learning	Data communication requires labels, scale and honest display.
Safety	Use rulers carefully.

Chapter Review

Key ideas

- The labelled directions and numerical intervals that define a graph.
- A relationship generating ordered values or shapes.
- A ratio compares quantities; a proportion states that two ratios are equal.
- “The calculator answer has eight digits, so it is exact.” Explain input, model and measurement limits.

Important terms

axis: reference line defining a graph variable	scale: relationship between display intervals and values
trend: overall pattern in data	anomaly: value differing strongly from a pattern
sequence: ordered set following a rule	ratio: comparison by division
proportion: equality of two ratios	estimation: approximate calculation used for checking

A. Choose the correct answer

1. Which term names reference line defining a graph variable?

- A. axis
- B. scale
- C. anomaly
- D. ratio

3. Choose the vocabulary item defined as overall pattern in data.

- A. anomaly
- B. ratio
- C. trend
- D. estimation

5. In this chapter, what is defined as ordered set following a rule?

- A. sequence
- B. ratio
- C. estimation
- D. scale

2. A learner describes relationship between display intervals and values. What is the correct term?

- A. trend
- B. scale
- C. sequence
- D. proportion

4. Which expression would a scientist use for value differing strongly from a pattern?

- A. sequence
- B. proportion
- C. axis
- D. anomaly

Exercises (continued)

6. Which statement best explains Axes and scale?

- A. anomalies may reveal error, unusual conditions or new phenomena and should be investigated rather than deleted automatically
- B. before reading points, identify the independent variable, dependent variable, units, origin and whether intervals are equal
- C. use more than appearance when possible and state both inclusion and exclusion conditions
- D. the visual steepness of a line depends on axis scale as well as data

8. Which statement correctly avoids a common misunderstanding about Sequence rule?

- A. a useful category must be applied consistently to every item
- B. a ratio compares quantities; a proportion states that two ratios are equal
- C. test differences, ratios, alternation, grouping and position-based rules, then use the simplest rule consistent with the evidence
- D. many rules can fit a very short sequence, so context and answer choices matter

10. Which example most directly illustrates Ratio and proportion?

- A. estimating 298×51 as about 300×50 gives about 15,000 before exact work
- B. if a model uses 1 cm for 1,000 km, 4.5 cm represents 4,500 km
- C. a graph beginning at 90 rather than zero can make a small change look dramatic
- D. adding the same amount is different from multiplying by the same factor

7. Which example most directly illustrates Trend and anomaly?

- A. 2, 5, 8, 11 increases by 3 each step
- B. classifying planets by solid surface differs from classifying by distance from the Sun
- C. temperature may rise through the morning with one lower point during cloud cover
- D. one outlier does not by itself disprove every trend

9. Which statement best explains Classification pattern?

- A. use more than appearance when possible and state both inclusion and exclusion conditions
- B. scale drawings, speed, density and map calculations use multiplicative relationships
- C. before reading points, identify the independent variable, dependent variable, units, origin and whether intervals are equal
- D. a useful category must be applied consistently to every item

B. Answer in one or two sentences

1. Define axis and scale in your own words.
2. Explain how Trend and anomaly works. Use one cause-and-effect link.
3. Compare Sequence rule and Classification pattern. State one similarity or connection and one difference.
4. Use the example of Ratio and proportion to explain why context or measurement matters.

C. Answer in detail

1. Explain the main ideas of “Patterns, Data Interpretation and Numerical Aptitude” as a connected account. Include at least four important terms from the chapter.
2. A graph’s vertical axis is truncated. How could this change the viewer’s impression without changing the data?

D. Apply and reason

1. If a pattern alternates two rules, testing only consecutive differences may hide its structure.
2. “The calculator answer has eight digits, so it is exact.” Explain input, model and measurement limits.

E. Activity or project

Build a ten-question mini-quiz containing one table, graph, sequence, ratio and unit-conversion problem.

CHAPTER 29

Analytical Reasoning and Problem Solving

Analytical reasoning turns a complicated question into a sequence of smaller decisions. Read conditions literally, represent them with a table or diagram, eliminate impossible choices and check the final answer against every condition. In science problems, the explanation is often more important than fast guessing because a correct option chosen for the wrong reason will fail on the next question.

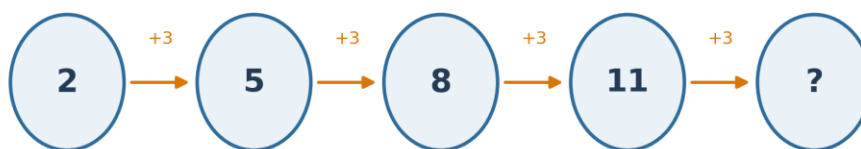
Opening question How can drawing a simple table make a word problem easier even when the table contains no new information?

Learning outcomes

- Translate statements into constraints.
- Use elimination and cases.
- Solve multi-step quantitative problems.
- Check conclusions for hidden assumptions.

Reasoning from a Pattern

Describe the rule before choosing the missing value.



A pattern is reliable only when the same relationship fits every given step.

Figure 29.1 Representing constraints. Original illustration; not to scale where stated.

Representing constraints

External representations reduce the load on memory.

Constraint

To understand Constraint, begin with a rule that limits possible arrangements or answers. Next, note that translate each sentence into a symbol, table entry, order relation or exclusion mark before combining conditions. “A launches before B” and “C is not on Tuesday” can be placed in a schedule grid. This does not mean that a condition applies only as worded; do not add everyday assumptions.

Case analysis

Case analysis describes division of a problem into a small number of possible situations. Its behaviour follows from this relationship: choose a condition with few possibilities, explore each case and reject cases that violate another rule. The example of if a satellite can use only orbit X or Y, test each branch against payload and coverage requirements shows the idea in use. Remember that case analysis should be complete; ignoring an inconvenient branch can miss the answer.

Check your understanding Why is the most restrictive condition often a good place to begin?

Elimination with evidence

Wrong options should be rejected for a stated reason.

Necessary and sufficient conditions

Necessary and sufficient conditions describes a necessary condition must be present; a sufficient condition guarantees the result. Its behaviour follows from this relationship: a planet must orbit the Sun to meet the Solar System planet definition, but orbiting the Sun alone is not sufficient. The example of having an atmosphere is not necessary for being a planet and is not sufficient for habitability shows the idea in use. Remember that confusing necessary with sufficient produces overconfident conclusions.

Counterexample

One valid example that disproves a universal statement. Because test words such as all, never and must by searching for an exception, observers may find that Venus is a counterexample to “the closest planet to the Sun is always hottest” if the statement is interpreted across Solar System planets. The conclusion should remain precise: a counterexample must meet the stated conditions, not a different problem.

Check your understanding What kind of claim can be disproved by one counterexample?

Multi-step problem solving

A clear written chain helps locate mistakes and earns partial credit.

Known–wanted–plan

Known–wanted–plan means a structure listing given information, required result and intended operations before calculating. It works because convert units early, select relationships and keep intermediate values labelled. A concrete case is for a satellite speed question, list distance, time, unit conversion and the speed formula. By contrast, starting arithmetic immediately can hide missing information or incompatible units.

Verification

Checking the result against conditions, units, magnitude and an alternative method where possible. The underlying relationship is that substitute an answer back into the problem and test every rule, not only the final one used. This can be seen when a timetable answer is valid only if all ordering and exclusion constraints hold. Reaching an answer choice does not prove the reasoning was complete.

Check your understanding List four checks that can be made before accepting a final numerical answer.

Activity: Solve a launch-schedule grid

Aim	to practise constraints and verification
Materials	teacher-prepared 4×4 schedule grid and six clues
Steps	1. Underline every limiting word in the clues. 2. Enter direct facts and exclusions in the grid. 3. Choose a clue that creates the fewest cases and test each case. 4. Verify the completed schedule against every clue.
Observe	A grid makes conflicts visible and prevents repeated mental guessing.
Expected learning	Reasoning quality is shown by a complete chain, not only the final arrangement.
Safety	No special safety requirement.

Chapter Review

Key ideas

- A rule that limits possible arrangements or answers.
- A necessary condition must be present; a sufficient condition guarantees the result.
- A structure listing given information, required result and intended operations before calculating.
- “One confirming example proves an all-statement.” Explain the asymmetry between proof and counterexample.

Important terms

constraint: rule limiting possible solutions	case analysis: systematic testing of possible situations
elimination: rejection of options using evidence	necessary condition: condition required for a result
sufficient condition: condition that guarantees a result	counterexample: example disproving a universal claim
assumption: unstated idea taken as true	verification: checking a result against evidence and rules

A. Choose the correct answer

1. Which term names rule limiting possible solutions?

- A. case analysis
- B. constraint
- C. necessary condition
- D. counterexample

3. Choose the vocabulary item defined as rejection of options using evidence.

- A. necessary condition
- B. counterexample
- C. verification
- D. elimination

5. In this chapter, what is defined as condition that guarantees a result?

- A. counterexample
- B. sufficient condition
- C. verification
- D. case analysis

2. A learner describes systematic testing of possible situations. What is the correct term?

- A. elimination
- B. sufficient condition
- C. case analysis
- D. assumption

4. Which expression would a scientist use for condition required for a result?

- A. necessary condition
- B. sufficient condition
- C. assumption
- D. constraint

Exercises (continued)

6. Which statement best explains Constraint?

- A. choose a condition with few possibilities, explore each case and reject cases that violate another rule
- B. test words such as all, never and must by searching for an exception
- C. translate each sentence into a symbol, table entry, order relation or exclusion mark before combining conditions
- D. a condition applies only as worded; do not add everyday assumptions

8. Which statement correctly avoids a common misunderstanding about Necessary and sufficient conditions?

- A. confusing necessary with sufficient produces overconfident conclusions
- B. a counterexample must meet the stated conditions, not a different problem
- C. a structure listing given information, required result and intended operations before calculating
- D. a planet must orbit the Sun to meet the Solar System planet definition, but orbiting the Sun alone is not sufficient

10. Which example most directly illustrates Known–wanted–plan?

- A. a timetable answer is valid only if all ordering and exclusion constraints hold
- B. “A launches before B” and “C is not on Tuesday” can be placed in a schedule grid
- C. for a satellite speed question, list distance, time, unit conversion and the speed formula
- D. starting arithmetic immediately can hide missing information or incompatible units

7. Which example most directly illustrates Case analysis?

- A. having an atmosphere is not necessary for being a planet and is not sufficient for habitability
- B. Venus is a counterexample to “the closest planet to the Sun is always hottest” if the statement is interpreted across Solar System planets
- C. case analysis should be complete; ignoring an inconvenient branch can miss the answer
- D. if a satellite can use only orbit X or Y, test each branch against payload and coverage requirements

9. Which statement best explains Counterexample?

- A. convert units early, select relationships and keep intermediate values labelled
- B. test words such as all, never and must by searching for an exception
- C. translate each sentence into a symbol, table entry, order relation or exclusion mark before combining conditions
- D. a counterexample must meet the stated conditions, not a different problem

B. Answer in one or two sentences

1. Define constraint and case analysis in your own words.
2. Explain how Case analysis works. Use one cause-and-effect link.
3. Compare Necessary and sufficient conditions and Counterexample. State one similarity or connection and one difference.
4. Use the example of Known–wanted–plan to explain why context or measurement matters.

C. Answer in detail

1. Explain the main ideas of “Analytical Reasoning and Problem Solving” as a connected account. Include at least four important terms from the chapter.
2. An option satisfies four of five conditions. Why is it still invalid?

D. Apply and reason

1. If a necessary condition is absent, the result is impossible; if it is present, further conditions may still be needed.
2. “One confirming example proves an all-statement.” Explain the asymmetry between proof and counterexample.

E. Activity or project

Create a two-stage space-logistics puzzle with a unique solution and provide a verification table.

UNIT 10 REVIEW

Scientific Reasoning and Logical Aptitude - Mixed Review

This review combines ideas from all chapters in the unit. Use diagrams, units and evidence where relevant. The questions are new combinations rather than copies of the chapter exercises.

A. Mixed multiple-choice review

1. Which option links a term with its correct meaning?

- A. trend - ordered set following a rule
- B. trend - approximate calculation used for checking
- C. trend - overall pattern in data
- D. trend - condition required for a result

3. Only one pair is scientifically correct. Which is it?

- A. constraint - rule limiting possible solutions
- B. constraint - rejection of options using evidence
- C. constraint - example disproving a universal claim
- D. constraint - explanation drawn from evidence

5. Find the pair that uses the vocabulary correctly.

- A. assumption - recorded information detected directly or by instrument
- B. assumption - expected result under stated conditions
- C. assumption - unstated idea taken as true
- D. assumption - factor that can change

7. Choose the correctly defined term from Unit 10.

- A. accuracy - closeness to a correct value
- B. accuracy - estimated limit of a measurement
- C. accuracy - relationship between display intervals and values
- D. accuracy - comparison by division

9. Which statement best explains Ratio and proportion in Chapter 28?

- A. scale drawings, speed, density and map calculations use multiplicative relationships
- B. translate each sentence into a symbol, table entry, order relation or exclusion mark before combining conditions
- C. substitute an answer back into the problem and test every rule, not only the final one used
- D. adding the same amount is different from multiplying by the same factor

11. Which statement best explains Accuracy and precision in Chapter 27?

- A. all other influential factors should be kept as constant as reasonably possible
- B. use more than appearance when possible and state both inclusion and exclusion conditions
- C. a set can be tightly grouped but shifted by calibration error, or spread out around the correct value
- D. more decimal places do not automatically make a result accurate

2. Select the accurately matched term and definition.

- A. ratio - approximate calculation used for checking
- B. ratio - rejection of options using evidence
- C. ratio - unstated idea taken as true
- D. ratio - comparison by division

4. Which term-definition pair would belong in the Unit 10 glossary?

- A. necessary condition - example disproving a universal claim
- B. necessary condition - condition required for a result
- C. necessary condition - recorded information detected directly or by instrument
- D. necessary condition - closeness to a correct value

6. Which pairing preserves the meaning used in this unit?

- A. inference - expected result under stated conditions
- B. inference - estimated limit of a measurement
- C. inference - overall pattern in data
- D. inference - explanation drawn from evidence

8. Which row could be copied into an accurate revision table?

- A. variable - relationship between display intervals and values
- B. variable - factor that can change
- C. variable - ordered set following a rule
- D. variable - rule limiting possible solutions

10. Which statement best explains Counterexample in Chapter 29?

- A. substitute an answer back into the problem and test every rule, not only the final one used
- B. test words such as all, never and must by searching for an exception
- C. all other influential factors should be kept as constant as reasonably possible
- D. a counterexample must meet the stated conditions, not a different problem

12. Which statement best explains Trend and anomaly in Chapter 28?

- A. use more than appearance when possible and state both inclusion and exclusion conditions
- B. a planet must orbit the Sun to meet the Solar System planet definition, but orbiting the Sun alone is not sufficient
- C. one outlier does not by itself disprove every trend
- D. anomalies may reveal error, unusual conditions or new phenomena and should be investigated rather than deleted automatically

UNIT 10 REVIEW - OLYMPIAD CHALLENGE

B. Written and visual reasoning

1. Connect one idea from Chapter 27 with one idea from Chapter 29. Explain the relationship.
2. Choose two terms from different chapters in Unit 10 and compare their roles.
3. A digital display shows many decimals. What further information is needed before trusting them?
4. If a necessary condition is absent, the result is impossible; if it is present, further conditions may still be needed.
5. Draw or describe one visual model that would help teach the central idea of Unit 10. State one limitation of the model.
6. Write one claim about Unit 10, give two pieces of evidence and state one uncertainty or boundary condition.

Olympiad Challenge

1. A student is reasoning about Constraint. Which statement is the most scientifically careful?

- A. case analysis should be complete; ignoring an inconvenient branch can miss the answer
- B. starting arithmetic immediately can hide missing information or incompatible units
- C. a condition applies only as worded; do not add everyday assumptions
- D. more decimal places do not automatically make a result accurate

3. When checking a claim about Sequence rule, which statement sets the right boundary?

- A. many rules can fit a very short sequence, so context and answer choices matter
- B. a useful category must be applied consistently to every item
- C. a condition applies only as worded; do not add everyday assumptions
- D. starting arithmetic immediately can hide missing information or incompatible units

5. Which conclusion about Independent and dependent variables avoids overclaiming?

- A. bias is not only deliberate cheating; it can enter unnoticed through design
- B. many rules can fit a very short sequence, so context and answer choices matter
- C. the variable that changes first in time is not automatically the independent variable
- D. a condition applies only as worded; do not add everyday assumptions

7. Which explanation of Observation and inference is precise enough for a science report?

- A. an inference can be reasonable without being directly seen
- B. a prediction made after seeing the result is weaker evidence than one recorded beforehand
- C. the variable that changes first in time is not automatically the independent variable
- D. many rules can fit a very short sequence, so context and answer choices matter

2. Which situation is the strongest application of Hypothesis and prediction?

- A. a clock that always reads 3 seconds fast may be precise but inaccurate
- B. using the same stopwatch operator for all trials controls one source but may retain that operator's reaction bias
- C. classifying planets by solid surface differs from classifying by distance from the Sun
- D. if higher Sun altitude shortens a vertical stick's shadow, measurements near noon should be shorter than in early morning

4. Which unfamiliar case is best explained by Counterexample?

- A. for a satellite speed question, list distance, time, unit conversion and the speed formula
- B. Venus is a counterexample to "the closest planet to the Sun is always hottest" if the statement is interpreted across Solar System planets
- C. if higher Sun altitude shortens a vertical stick's shadow, measurements near noon should be shorter than in early morning
- D. using the same stopwatch operator for all trials controls one source but may retain that operator's reaction bias

6. Where would Dimensional check and estimation be most useful?

- A. "A launches before B" and "C is not on Tuesday" can be placed in a schedule grid
- B. Venus is a counterexample to "the closest planet to the Sun is always hottest" if the statement is interpreted across Solar System planets
- C. if higher Sun altitude shortens a vertical stick's shadow, measurements near noon should be shorter than in early morning
- D. estimating 298×51 as about 300×50 gives about 15,000 before exact work

8. Which example provides the clearest evidence of Trend and anomaly?

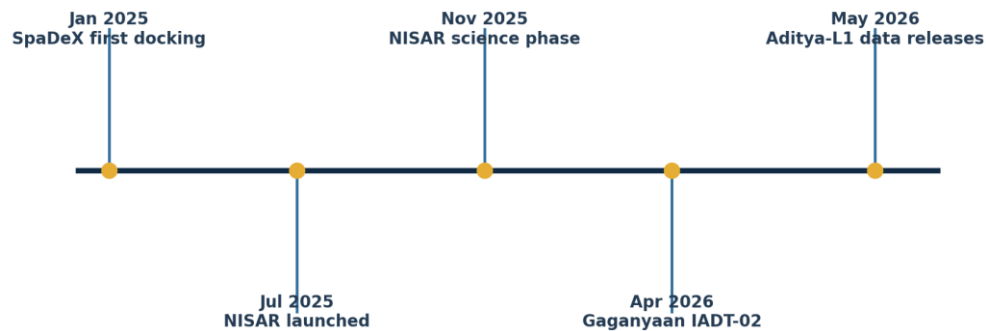
- A. 2, 5, 8, 11 increases by 3 each step
- B. temperature may rise through the morning with one lower point during cloud cover
- C. estimating 298×51 as about 300×50 gives about 15,000 before exact work
- D. Venus is a counterexample to "the closest planet to the Sun is always hottest" if the statement is interpreted across Solar System planets

UNIT 11**General Space Awareness**

Space science changes rapidly, so mission claims must always be connected to reliable sources and dates.

Space Awareness Update: Selected Events to June 2026

Mission status must always be read with a date; plans and outcomes can change.



Information cut-off: 29 June 2026.

Unit 11 opening visual - original illustration.

Major concepts

- Chapter 30: Space Awareness: Missions, Discoveries and People

Expected unit outcomes

- Apply status labels to mission news.

CHAPTER 30

Space Awareness: Missions, Discoveries and People

General space awareness combines durable knowledge with dated current information. A careful learner checks official mission pages, distinguishes announcement from accomplishment and records when a claim was verified. Through 29 June 2026, important topics include Aditya-L1 science operations, NISAR's science phase, SpaDeX docking demonstrations, Gaganyaan preparation and recent human-spaceflight experience. Historical scientists and astronauts should be remembered for specific contributions rather than slogans.

Opening question What words in a news report would tell you whether a mission is proposed, launched, operational or completed?

Learning outcomes

- Apply status labels to mission news.
- Summarise selected developments to June 2026.
- Recognise contributions of selected scientists and astronauts.
- Evaluate sources for space claims.

Space Awareness Update: Selected Events to June 2026

Mission status must always be read with a date; plans and outcomes can change.



Information cut-off: 29 June 2026.

Figure 30.1 Reading mission status correctly. Original illustration; not to scale where stated.

Reading mission status correctly

Dates and verbs prevent planned events from being reported as history.

Mission status

Mission status describes a clearly dated description such as proposed, approved, in development, launched, commissioning, operational, completed or unsuccessful. Its behaviour follows from this relationship: status changes after tests, launch and review, so a textbook should provide an information cut-off. The example of NISAR launched on 30 July 2025 and ISRO reported it had entered science phase on 28 November 2025 shows the idea in use. Remember that “will launch” in an old article must not be copied after the date without checking the result.

Primary source

An official mission page, technical report, dataset or statement produced by the responsible organisation or research team. Because primary sources give direct status and methods, though they still require careful reading and may later be updated, observers may find that ISRO mission releases are appropriate for confirming an Indian launch result. The conclusion should remain precise: a repost without date or link is weaker evidence than the original record.

Check your understanding Why should the publication date and last-updated date both be checked?

Selected developments to the cut-off

Current awareness should be concise, verified and honest about what remains in progress.

Active and recent Indian missions

Aditya-L1 continues solar observations and public data release; NISAR entered science phase after its 2025 launch; SpaDeX demonstrated docking, undocking, redocking and power transfer. Because these achievements concern different capabilities: solar science, Earth radar observation and orbital operations, observers may find that a May 2026 Aditya-L1 opportunity notice reported more than 27 terabytes of public data and peer-reviewed results. The conclusion should remain precise: data volume alone does not measure scientific importance; calibration, analysis and questions matter.

Gaganyaan preparation

In this chapter, Gaganyaan preparation is used for human-spaceflight systems were still being tested and developed at the cut-off, including recovery and air-drop work. A second integrated air-drop test in April 2026 used a simulated crew module released over sea to examine parachute and recovery performance. One useful example is ground and drop tests can validate subsystems before an uncrewed orbital flight. It should be separated from this different claim: testing progress must not be rewritten as a completed crewed Gaganyaan mission.

Check your understanding Which listed activity demonstrates orbital docking rather than Earth observation or solar science?

People and contributions

A contribution is described by work, context and evidence rather than fame alone.

Scientists and programme builders

Figures such as Aryabhata, Galileo Galilei, Isaac Newton, Meghnad Saha and Vikram Sarabhai contributed different ideas or institutions across centuries. The underlying relationship is that their work includes mathematical astronomy, telescopic observation, laws of motion and gravity, stellar ionisation theory and building India's modern space programme. This can be seen when Sarabhai argued for advanced space technology serving national development. Historical contributions should not be compressed into invented quotations or claims that one person "discovered space".

Astronauts connected with India

The phrase Astronauts connected with India refers to Rakesh Sharma became the first Indian citizen in space in 1984; Kalpana Chawla was the first woman of Indian origin in space; Sunita Williams has completed long-duration missions; Shubhanshu Shukla flew to the ISS on Axiom Mission 4 in 2025. Here, their careers illustrate military aviation, engineering, scientific operations, international crews and long training. Consider the following case: Shukla's mission experience is connected with but not identical to the Gaganyaan mission. The careful interpretation is that nationality, citizenship and heritage should be stated accurately rather than used interchangeably.

Check your understanding How can a biographical note avoid both hero worship and unfairly reducing a team achievement to one person?

Activity: Verify a space-news claim

Aim	to practise source tracing and status language
Materials	one recent headline, internet access supervised by a teacher and a verification worksheet
Steps	1. Copy the exact claim and publication date. 2. Trace it to an official agency or original research source. 3. Record mission status, event date and any uncertainty. 4. Rewrite the claim in one accurate sentence with an information cut-off.
Observe	Headlines often omit conditions or use future language; official records clarify dates and outcomes.
Expected learning	Current awareness is a method of verification, not memorising headlines.
Safety	Use school-approved sites and do not share personal information online.

Chapter Review

Key ideas

- A clearly dated description such as proposed, approved, in development, launched, commissioning, operational, completed or unsuccessful.
- Aditya-11 continues solar observations and public data release; nisar entered science phase after its 2025 launch; spadex demonstrated docking, undocking, redocking and power transfer.
- Figures such as aryabhata, galileo galilei, isaac newton, meghnad saha and vikram sarabhai contributed different ideas or institutions across centuries.
- “A successful launch means the mission has completed all scientific goals.” Correct the sequence of mission phases.

Important terms

mission status: dated stage of a mission or programme	information cut-off: latest date covered by a publication
primary source: original official or research record	commissioning: testing and calibrating a spacecraft after launch
operational: performing intended routine work	peer review: expert evaluation before scholarly publication
astronaut: trained human spaceflight crew member	space awareness: current and historical knowledge checked against reliable evidence

A. Choose the correct answer

1. Which term names dated stage of a mission or programme?

- A. information cut-off
- B. commissioning
- C. mission status
- D. peer review

3. Choose the vocabulary item defined as original official or research record.

- A. primary source
- B. commissioning
- C. peer review
- D. space awareness

5. In this chapter, what is defined as performing intended routine work?

- A. peer review
- B. space awareness
- C. operational
- D. information cut-off

2. A learner describes latest date covered by a publication. What is the correct term?

- A. primary source
- B. operational
- C. astronaut
- D. information cut-off

4. Which expression would a scientist use for testing and calibrating a spacecraft after launch?

- A. operational
- B. commissioning
- C. astronaut
- D. mission status

Exercises (continued)

6. Which statement best explains Mission status?

- A. primary sources give direct status and methods, though they still require careful reading and may later be updated
- B. a second integrated air-drop test in April 2026 used a simulated crew module released over sea to examine parachute and recovery performance
- C. “will launch” in an old article must not be copied after the date without checking the result
- D. status changes after tests, launch and review, so a textbook should provide an information cut-off

8. Which statement correctly avoids a common misunderstanding about Active and recent Indian missions?

- A. testing progress must not be rewritten as a completed crewed Gaganyaan mission
- B. data volume alone does not measure scientific importance; calibration, analysis and questions matter
- C. figures such as Aryabhata, Galileo Galilei, Isaac Newton, Meghnad Saha and Vikram Sarabhai contributed different ideas or institutions across centuries
- D. these achievements concern different capabilities: solar science, Earth radar observation and orbital operations

10. Which example most directly illustrates Scientists and programme builders?

- A. Shukla’s mission experience is connected with but not identical to the Gaganyaan mission
- B. NISAR launched on 30 July 2025 and ISRO reported it had entered science phase on 28 November 2025
- C. historical contributions should not be compressed into invented quotations or claims that one person “discovered space”
- D. Sarabhai argued for advanced space technology serving national development

7. Which example most directly illustrates Primary source?

- A. ISRO mission releases are appropriate for confirming an Indian launch result
- B. a May 2026 Aditya-L1 opportunity notice reported more than 27 terabytes of public data and peer-reviewed results
- C. ground and drop tests can validate subsystems before an uncrewed orbital flight
- D. a repost without date or link is weaker evidence than the original record

9. Which statement best explains Gaganyaan preparation?

- A. their work includes mathematical astronomy, telescopic observation, laws of motion and gravity, stellar ionisation theory and building India’s modern space programme
- B. status changes after tests, launch and review, so a textbook should provide an information cut-off
- C. a second integrated air-drop test in April 2026 used a simulated crew module released over sea to examine parachute and recovery performance
- D. testing progress must not be rewritten as a completed crewed Gaganyaan mission

B. Answer in one or two sentences

1. Define mission status and information cut-off in your own words.
2. Explain how Primary source works. Use one cause-and-effect link.
3. Compare Active and recent Indian missions and Gaganyaan preparation. State one similarity or connection and one difference.
4. Use the example of Scientists and programme builders to explain why context or measurement matters.

C. Answer in detail

1. Explain the main ideas of “Space Awareness: Missions, Discoveries and People” as a connected account. Include at least four important terms from the chapter.
2. An article from 2024 says a mission “is expected to launch in 2025.” What steps are needed before using it in a 2026 answer?

D. Apply and reason

1. If official sources disagree temporarily, state the conflict and wait for clarified evidence rather than invent certainty.
2. “A successful launch means the mission has completed all scientific goals.” Correct the sequence of mission phases.

E. Activity or project

Create a one-page dated space bulletin with three verified updates, one historical profile and a source list.

UNIT 11 REVIEW

General Space Awareness - Mixed Review

This review combines ideas from all chapters in the unit. Use diagrams, units and evidence where relevant. The questions are new combinations rather than copies of the chapter exercises.

A. Mixed multiple-choice review

1. Which option links a term with its correct meaning?

- A. commissioning - expert evaluation before scholarly publication
- B. commissioning - dated stage of a mission or programme
- C. commissioning - performing intended routine work
- D. commissioning - testing and calibrating a spacecraft after launch

3. Only one pair is scientifically correct. Which is it?

- A. information cut-off - testing and calibrating a spacecraft after launch
- B. information cut-off - latest date covered by a publication
- C. information cut-off - trained human spaceflight crew member
- D. information cut-off - original official or research record

5. Find the pair that uses the vocabulary correctly.

- A. space awareness - latest date covered by a publication
- B. space awareness - performing intended routine work
- C. space awareness - dated stage of a mission or programme
- D. space awareness - current and historical knowledge checked against reliable evidence

7. Choose the correctly defined term from Unit 11.

- A. peer review - current and historical knowledge checked against reliable evidence
- B. peer review - expert evaluation before scholarly publication
- C. peer review - original official or research record
- D. peer review - trained human spaceflight crew member

9. Which statement best explains Astronauts connected with India in Chapter 30?

- A. primary sources give direct status and methods, though they still require careful reading and may later be updated
- B. their careers illustrate military aviation, engineering, scientific operations, international crews and long training
- C. status changes after tests, launch and review, so a textbook should provide an information cut-off
- D. nationality, citizenship and heritage should be stated accurately rather than used interchangeably

11. Which statement best explains Gaganyaan preparation in Chapter 30?

- A. their careers illustrate military aviation, engineering, scientific operations, international crews and long training
- B. their work includes mathematical astronomy, telescopic observation, laws of motion and gravity, stellar ionisation theory and building India's modern space programme
- C. testing progress must not be rewritten as a completed crewed Gaganyaan mission
- D. a second integrated air-drop test in April 2026 used a simulated crew module released over sea to examine parachute and recovery performance

2. Select the accurately matched term and definition.

- A. astronaut - trained human spaceflight crew member
- B. astronaut - dated stage of a mission or programme
- C. astronaut - testing and calibrating a spacecraft after launch
- D. astronaut - current and historical knowledge checked against reliable evidence

4. Which term-definition pair would belong in the Unit 11 glossary?

- A. operational - trained human spaceflight crew member
- B. operational - latest date covered by a publication
- C. operational - performing intended routine work
- D. operational - expert evaluation before scholarly publication

6. Which pairing preserves the meaning used in this unit?

- A. primary source - original official or research record
- B. primary source - performing intended routine work
- C. primary source - current and historical knowledge checked against reliable evidence
- D. primary source - testing and calibrating a spacecraft after launch

8. Which row could be copied into an accurate revision table?

- A. mission status - original official or research record
- B. mission status - expert evaluation before scholarly publication
- C. mission status - dated stage of a mission or programme
- D. mission status - latest date covered by a publication

10. Which statement best explains Scientists and programme builders in Chapter 30?

- A. status changes after tests, launch and review, so a textbook should provide an information cut-off
- B. their careers illustrate military aviation, engineering, scientific operations, international crews and long training
- C. their work includes mathematical astronomy, telescopic observation, laws of motion and gravity, stellar ionisation theory and building India's modern space programme
- D. historical contributions should not be compressed into invented quotations or claims that one person "discovered space"

12. Which statement best explains Active and recent Indian missions in Chapter 30?

- A. these achievements concern different capabilities: solar science, Earth radar observation and orbital operations
- B. their work includes mathematical astronomy, telescopic observation, laws of motion and gravity, stellar ionisation theory and building India's modern space programme
- C. a second integrated air-drop test in April 2026 used a simulated crew module released over sea to examine parachute and recovery performance
- D. data volume alone does not measure scientific importance; calibration, analysis and questions matter

UNIT 11 REVIEW - OLYMPIAD CHALLENGE

B. Written and visual reasoning

1. Connect one idea from Chapter 30 with one idea from Chapter 30. Explain the relationship.
2. Choose two terms from different chapters in Unit 11 and compare their roles.
3. An article from 2024 says a mission “is expected to launch in 2025.” What steps are needed before using it in a 2026 answer?
4. If official sources disagree temporarily, state the conflict and wait for clarified evidence rather than invent certainty.
5. Draw or describe one visual model that would help teach the central idea of Unit 11. State one limitation of the model.
6. Write one claim about Unit 11, give two pieces of evidence and state one uncertainty or boundary condition.

Olympiad Challenge

1. A student is reasoning about Gaganyaan preparation. Which statement is the most scientifically careful?

- A. historical contributions should not be compressed into invented quotations or claims that one person “discovered space”
- B. a repost without date or link is weaker evidence than the original record
- C. nationality, citizenship and heritage should be stated accurately rather than used interchangeably
- D. testing progress must not be rewritten as a completed crewed Gaganyaan mission

3. When checking a claim about Astronauts connected with India, which statement sets the right boundary?

- A. “will launch” in an old article must not be copied after the date without checking the result
- B. nationality, citizenship and heritage should be stated accurately rather than used interchangeably
- C. testing progress must not be rewritten as a completed crewed Gaganyaan mission
- D. a repost without date or link is weaker evidence than the original record

5. Which conclusion about Primary source avoids overclaiming?

- A. data volume alone does not measure scientific importance; calibration, analysis and questions matter
- B. nationality, citizenship and heritage should be stated accurately rather than used interchangeably
- C. testing progress must not be rewritten as a completed crewed Gaganyaan mission
- D. a repost without date or link is weaker evidence than the original record

7. Which explanation of Gaganyaan preparation is precise enough for a science report?

- A. historical contributions should not be compressed into invented quotations or claims that one person “discovered space”
- B. testing progress must not be rewritten as a completed crewed Gaganyaan mission
- C. a repost without date or link is weaker evidence than the original record
- D. nationality, citizenship and heritage should be stated accurately rather than used interchangeably

2. Which situation is the strongest application of Scientists and programme builders?

- A. Sarabhai argued for advanced space technology serving national development
- B. Shukla’s mission experience is connected with but not identical to the Gaganyaan mission
- C. a May 2026 Aditya-L1 opportunity notice reported more than 27 terabytes of public data and peer-reviewed results
- D. NISAR launched on 30 July 2025 and ISRO reported it had entered science phase on 28 November 2025

4. Which unfamiliar case is best explained by Mission status?

- A. ISRO mission releases are appropriate for confirming an Indian launch result
- B. Sarabhai argued for advanced space technology serving national development
- C. NISAR launched on 30 July 2025 and ISRO reported it had entered science phase on 28 November 2025
- D. a May 2026 Aditya-L1 opportunity notice reported more than 27 terabytes of public data and peer-reviewed results

6. Where would Active and recent Indian missions be most useful?

- A. a May 2026 Aditya-L1 opportunity notice reported more than 27 terabytes of public data and peer-reviewed results
- B. ground and drop tests can validate subsystems before an uncrewed orbital flight
- C. NISAR launched on 30 July 2025 and ISRO reported it had entered science phase on 28 November 2025
- D. Sarabhai argued for advanced space technology serving national development

8. Which example provides the clearest evidence of Scientists and programme builders?

- A. Shukla’s mission experience is connected with but not identical to the Gaganyaan mission
- B. a May 2026 Aditya-L1 opportunity notice reported more than 27 terabytes of public data and peer-reviewed results
- C. Sarabhai argued for advanced space technology serving national development
- D. NISAR launched on 30 July 2025 and ISRO reported it had entered science phase on 28 November 2025

Cumulative Revision

Use this section after completing all eleven units. It deliberately mixes astronomy, technology, physics and reasoning so that the chapter source is not obvious.

Mixed objective questions

1. Which option links a term with its correct meaning?

- A. celestial object - an explanation based on observations
- B. celestial object - natural object beyond most of Earth's atmosphere
- C. celestial object - artificial brightening of the night sky
- D. celestial object - point directly overhead

3. Which option links a term with its correct meaning?

- A. planetesimal - internal separation by density and composition
- B. planetesimal - icy-body region beyond Neptune
- C. planetesimal - small early body that could grow into a planet
- D. planetesimal - planet with a rocky solid surface

5. Which option links a term with its correct meaning?

- A. apparent brightness - core hydrogen-fusing life stage
- B. apparent brightness - direction from observer to an object
- C. apparent brightness - interstellar cloud of gas or dust
- D. apparent brightness - brightness received by an observer

7. Which option links a term with its correct meaning?

- A. seismic wave - wave carrying earthquake energy through Earth
- B. seismic wave - layer above the troposphere containing much ozone
- C. seismic wave - imaginary line about which a body rotates
- D. seismic wave - east-west angular coordinate on Earth

9. Which option links a term with its correct meaning?

- A. inclination - non-functional human-made material in orbit
- B. inclination - tilt of an orbit to a reference plane
- C. inclination - ground area reached by a satellite beam
- D. inclination - measurement without direct contact

2. Select the accurately matched term and definition.

- A. apparent motion - artificial brightening of the night sky
- B. apparent motion - angular height above the horizon
- C. apparent motion - motion as it seems from an observer's viewpoint
- D. apparent motion - projection of Earth's equator on the sky

4. Select the accurately matched term and definition.

- A. frost line - icy-body region beyond Neptune
- B. frost line - stream of charged particles from the Sun
- C. frost line - region in which magnetic forces act
- D. frost line - early-disc region beyond which ices could condense

6. Select the accurately matched term and definition.

- A. red giant - expanded late stage of a lower-mass star
- B. red giant - direction from observer to an object
- C. red giant - dense spherical generally old cluster
- D. red giant - expanding material from a stellar explosion

8. Select the accurately matched term and definition.

- A. exosphere - imaginary line about which a body rotates
- B. exosphere - outermost atmospheric region fading into space
- C. exosphere - yearly period with characteristic sunlight conditions
- D. exosphere - appearance of the Moon's illuminated portion

10. Select the accurately matched term and definition.

- A. uplink - ground area reached by a satellite beam
- B. uplink - satellite observing atmospheric conditions
- C. uplink - signal sent from ground to satellite
- D. uplink - gravity-curved path around a body

Cumulative Revision (continued)

Mixed objective questions

11. Which option links a term with its correct meaning?

- A. guidance - rocket section used during part of flight
- B. guidance - protective cover around payload
- C. guidance - selection of desired path
- D. guidance - balance in which improving one factor may cost another

13. Which option links a term with its correct meaning?

- A. LVM3 - spacecraft designed to reach and work on a surface
- B. LVM3 - path between planetary regions
- C. LVM3 - Indian human-spaceflight programme
- D. LVM3 - heavy-lift Indian launch vehicle family

15. Which option links a term with its correct meaning?

- A. space station - long-duration crewed orbital facility
- B. space station - mission passing a target without orbiting
- C. space station - mobile surface spacecraft
- D. space station - United States civil space agency

17. Which option links a term with its correct meaning?

- A. joule - energy associated with position or configuration
- B. joule - SI unit of energy and work
- C. joule - energy transfer by electromagnetic waves
- D. joule - complete path for electric current

19. Which option links a term with its correct meaning?

- A. trend - ordered set following a rule
- B. trend - approximate calculation used for checking
- C. trend - overall pattern in data
- D. trend - condition required for a result

12. Select the accurately matched term and definition.

- A. stage separation - protective cover around payload
- B. stage separation - vehicle with recoverable flight hardware
- C. stage separation - measure of matter and inertia
- D. stage separation - controlled release of a completed stage

14. Select the accurately matched term and definition.

- A. rover - mobile surface exploration vehicle
- B. rover - path between planetary regions
- C. rover - space-environment conditions influenced by solar activity
- D. rover - emergency system separating crew from a failing launcher

16. Select the accurately matched term and definition.

- A. orbiter - mobile surface spacecraft
- B. orbiter - spacecraft repeatedly circling a target
- C. orbiter - organisation managing public space activities
- D. orbiter - Chinese national space administration

18. Select the accurately matched term and definition.

- A. temperature - energy transfer by electromagnetic waves
- B. temperature - energy transferred per unit charge
- C. temperature - measure related to average particle motion
- D. temperature - region of magnetic influence

20. Select the accurately matched term and definition.

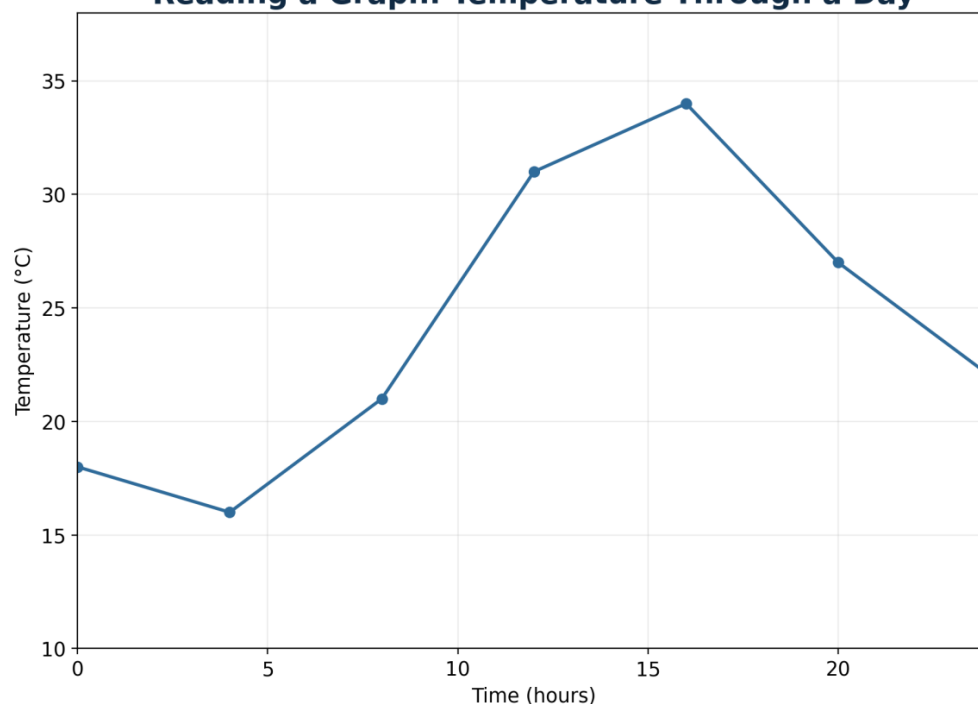
- A. ratio - approximate calculation used for checking
- B. ratio - rejection of options using evidence
- C. ratio - unstated idea taken as true
- D. ratio - comparison by division

Cumulative Revision (continued)

Data and calculation tasks

1. A satellite completes 15 orbits in 24 hours. Estimate the period of one orbit in minutes.
2. A model uses 1 cm for 10,000 km. What model length represents 150,000 km?
3. A graph shows temperature rising from 18 °C at 06:00 to 30 °C at 12:00. Calculate the average change per hour over this interval.
4. A launch vehicle has 6,000 kN upward thrust and 5,400 kN weight at one instant. Find the net upward force.
5. A radio signal takes 1.3 seconds to travel from Earth to the Moon. Estimate the round-trip delay.
6. Classify each as observation or inference: “the line moved 4 mm”; “the object is heating”; “the detector count doubled”; “a solar event caused the change”.
7. A table gives cloud cover, wind speed and rainfall for five days. State one trend and identify one value that should be checked as a possible anomaly.
8. A sequence of orbit numbers is 3, 6, 12, 24, __. State the rule and next value.

Reading a Graph: Temperature Through a Day



Use the graph to write two observations and one inference.

Cumulative Revision (continued)

Extended reasoning tasks

1. Explain why an unfiltered telescope makes Sun viewing more dangerous rather than safer.
2. Design a satellite mission for flood monitoring. Justify orbit, sensor, communication path and data users.
3. Compare a Moon rover and a Mars rover in communication delay, energy, atmosphere and emergency response.
4. A headline says “India has completed Gaganyaan” after a parachute test. Rewrite the headline accurately and explain the status error.
5. Use Newton’s laws to describe launch, coasting in orbit and a docking manoeuvre.
6. Create a concept map linking the Sun, solar wind, space weather, satellites and communication effects.

Self-check For every response, verify terminology, evidence, units, scale and mission date.

Practice Model Paper 1

Inferred practice pattern: 60 objective questions | 60 marks | Suggested time: 90 minutes. This is a practice model, not an official examination paper.

Instructions: Choose one best answer for each question. Use rough work for calculations. Do not assume that the longest option is correct.

1. Complete the definition: _____ means systems maintaining a safe crew environment.

- A. carbon-dioxide scrubbing
- B. spacewalk
- C. life support
- D. microgravity

3. Which conclusion follows most directly from this example: if a satellite can use only orbit X or Y, test each branch against payload and coverage requirements?

- A. choose a condition with few possibilities, explore each case and reject cases that violate another rule
- B. a planet must orbit the Sun to meet the Solar System planet definition, but orbiting the Sun alone is not sufficient
- C. a counterexample must meet the stated conditions, not a different problem
- D. checking the result against conditions, units, magnitude and an alternative method where possible

5. Which conclusion follows most directly from this example: projecting the Sun through a suitable small telescope can be safe only with informed adult supervision and correct equipment?

- A. the curved objective bends incoming light toward a focus where an eyepiece forms a magnified view
- B. a reflector is not a telescope that merely uses any flat mirror
- C. proper front-mounted solar filters block almost all visible, infrared and ultraviolet radiation before it enters an instrument
- D. the apparent enlargement produced by the telescope and eyepiece combination

7. Which conclusion follows most directly from this example: a spacecraft aims for where Mars will be, not where it appears at launch?

- A. it operates in a halo orbit around the Sun–Earth L1 region, allowing a nearly continuous view of the Sun without regular Earth eclipses
- B. space weather is not ordinary atmospheric weather, though both can affect technology
- C. launch timing uses the moving positions of Earth and the target so the spacecraft and planet arrive at the same region
- D. India's mission that achieved a soft landing near the lunar south polar region on 23 August 2023 with the Vikram lander and Pragyan rover

9. Which conclusion follows most directly from this example: Saturn's bright rings contain ice-rich particles ranging from dust-sized grains to larger chunks?

- A. moons can be captured, co-formed or produced by impacts, and tidal forces may heat their interiors
- B. collisions, resonances and moon gravity shape gaps and narrow structures
- C. clearing does not mean the orbit is literally empty
- D. Jupiter and Saturn, large planets dominated by hydrogen and helium

11. Select the correct vocabulary item for tilt of an orbit to a reference plane.

- A. payload
- B. satellite
- C. low Earth orbit
- D. inclination

2. Which chapter term is used for growth by gathering smaller material?

- A. planetesimal
- B. accretion
- C. differentiation
- D. Kuiper Belt

4. Which conclusion follows most directly from this example: a May 2026 Aditya-L1 opportunity notice reported more than 27 terabytes of public data and peer-reviewed results?

- A. a second integrated air-drop test in April 2026 used a simulated crew module released over sea to examine parachute and recovery performance
- B. these achievements concern different capabilities: solar science, Earth radar observation and orbital operations
- C. historical contributions should not be compressed into invented quotations or claims that one person "discovered space"
- D. a clearly dated description such as proposed, approved, in development, launched, commissioning, operational, completed or unsuccessful

6. Which chapter term is used for engine passage that directs expanding gas?

- A. mass
- B. nozzle
- C. drag
- D. control

8. Select the correct vocabulary item for designed and verified to protect crew with high reliability.

- A. crew module
- B. human-rated
- C. rendezvous
- D. NISAR

10. Which term matches this definition: protective cover around payload?

- A. launch site
- B. reusable launcher
- C. fairing
- D. stage separation

12. Which conclusion follows most directly from this example: the Andromeda Galaxy is a large neighbouring spiral galaxy?

- A. shape reflects rotation, star formation, past interactions and mass distribution, though real galaxies show many intermediate structures
- B. stars orbit a common galactic mass distribution while gas can form new stars and feedback can reshape the galaxy
- C. the Milky Way is not the entire universe and the Sun is not at its centre
- D. the totality of space-time, matter and energy described by physical laws

Practice Model Paper 1 (continued)

13. Which conclusion follows most directly from this example: a thruster changes a satellite's velocity, and the same thrust changes a small satellite faster than a massive one?

- A. each force acts on a different object, so the pair does not cancel on either object's individual force diagram
- B. gravity is not caused by Earth's atmosphere or rotation
- C. speed is distance travelled per time; velocity is speed in a stated direction
- D. greater force gives greater acceleration for the same mass, while greater mass gives smaller acceleration for the same force

15. Which conclusion follows most directly from this example: seismic waves change speed and some types do not pass through liquids?

- A. different wave types travel at different speeds and respond differently to solids and liquids, creating patterns detected around the planet
- B. high pressure can change melting behaviour, so high temperature does not always mean liquid
- C. pressure keeps the inner core solid despite high temperature, while movement in the liquid outer core helps generate Earth's magnetic field
- D. the mesosphere, thermosphere and exosphere above the stratosphere

17. Which conclusion follows most directly from this example: the International Space Station has supported research in biology, materials, Earth observation and technology?

- A. researchers can separate gravitational effects from other influences in flames, fluids, crystals and organisms
- B. modular systems provide laboratories, power, thermal control, docking ports and living space, while regular vehicles carry crew and supplies
- C. a flyby can still return major discoveries despite a short encounter
- D. the travel time of radio signals, limited by the speed of light

19. Which conclusion follows most directly from this example: a lunar eclipse can be seen from much of Earth's night side?

- A. sunlight filtered and bent through Earth's atmosphere can make the Moon appear copper-red during totality
- B. ocean water forms broad bulges and Earth rotates through them, while coastlines and depth alter timing and height
- C. "spring" here means springing up, not the season
- D. waxing means the illuminated fraction seen from Earth is increasing; waning means it is decreasing

21. Which conclusion follows most directly from this example: Northern Hemisphere summer occurs when the north is tilted toward the Sun?

- A. more direct sunlight concentrates energy over a smaller surface area and longer days increase heating time
- B. countries use standard times, so clock noon need not equal local solar noon
- C. spinning of Earth around its axis approximately once each day
- D. as Earth revolves, one hemisphere is tilted more toward the Sun and later more away, changing day length and sunlight angle

23. Which term matches this definition: planet with a rocky solid surface?

- A. exosphere
- B. magnetic field
- C. terrestrial planet
- D. photosphere

14. Which conclusion follows most directly from this example: Orion is recognised by a set of bright stars and also names a larger defined region?

- A. asterisms are informal but useful for finding other objects
- B. modern boundaries ensure every sky position belongs to one constellation region
- C. the horizon is an apparent circle, not a physical edge of Earth
- D. an imaginary sphere centred on the observer or Earth on which sky positions are pictured

16. Which conclusion follows most directly from this example: a payload may be a satellite, probe, crew module or scientific instrument?

- A. the fairing shields spacecraft from airflow, sound and heating, then is discarded when no longer needed
- B. launch paths are chosen so falling hardware and possible failures avoid dense population when possible
- C. rotation advantage does not replace rocket thrust and is not useful for every orbit
- D. a decision balancing performance, cost, risk, schedule and objectives

18. Name the concept described here: round Sun-orbiting body that has cleared its neighbourhood.

- A. dwarf planet
- B. gas giant
- C. natural satellite
- D. planet

20. Which conclusion follows most directly from this example: a spinning pizza dough becomes flatter as it spreads, though the physics is only a rough analogy?

- A. dust grains formed clumps, clumps became planetesimals and some planetesimals grew into protoplanets
- B. differentiation is not the same as the planet splitting into separate bodies
- C. two broad planet groups separated by composition, size and location
- D. gravity amplified denser regions, collapse made rotation faster and collisions flattened much material into a disc

22. This description refers to which term: surviving material found on the ground?

- A. meteorite
- B. asteroid
- C. comet
- D. meteoroid

24. Name the concept described here: Polar Satellite Launch Vehicle.

- A. GSLV
- B. ISRO
- C. SAC
- D. PSLV

Practice Model Paper 1 (continued)

25. Which conclusion follows most directly from this example: a launch vehicle may be developed by VSSC while its satellite payload includes work from SAC and its spacecraft bus is led by URSC?

- A. launch preparation, engine qualification and data services occur at different specialised facilities
- B. complex projects share designs and test results across centres
- C. the word polar does not mean PSLV can launch only polar satellites
- D. India's national space agency responsible for space research, technology and applications

27. Which conclusion follows most directly from this example: a constellation rises in the east and later sets in the west although its stars have not circled Earth that night?

- A. Earth's rotation makes most sky objects appear to travel from east to west during a night
- B. using the same location and similar time reduces unnecessary differences and makes patterns easier to compare
- C. clouds can make city glow worse, while darkness alone cannot overcome thick cloud or haze
- D. a careful, purposeful use of senses or instruments to gather information

29. Which conclusion follows most directly from this example: different wavelengths reveal dust, hot gas and shock fronts?

- A. reflection nebulae scatter nearby starlight, emission nebulae glow when gas is energised and supernova remnants are expanding debris
- B. from Earth, depth is compressed into a sky picture; changing viewpoint would alter the shape
- C. one shared property alone is not enough; several kinds of evidence strengthen the conclusion
- D. a dense, roughly spherical group containing many old stars

31. Which term matches this definition: agreed rule for connecting systems?

- A. interface standard
- B. space agency
- C. ESA
- D. Roscosmos

33. Which chapter term is used for mobile surface spacecraft?

- A. sample return
- B. space station
- C. orbiter
- D. rover

35. This description refers to which term: Moon blocks the Sun from part of Earth?

- A. lunar eclipse
- B. tide
- C. solar eclipse
- D. waxing

26. Which conclusion follows most directly from this example: planetary nebulae surround some newly exposed white dwarfs?

- A. massive cores build heavier elements until fusion can no longer support the core; collapse and rebound eject outer material
- B. colour is a temperature clue, not a simple sign that a star is approaching or moving away
- C. the near-balance between inward gravity and outward pressure in a star
- D. core changes make outer layers expand; after the outer gas is lost, the remaining core becomes a white dwarf and gradually cools

28. Which conclusion follows most directly from this example: ESA instruments and astronauts may work with NASA, JAXA and other partners?

- A. interfaces and schedules must be agreed early so hardware, software and data systems work together
- B. support agreements still require security, technical compatibility and clear data policies
- C. member contributions support shared launch, science, Earth-observation and exploration missions
- D. observations released so researchers and the public can analyse them under stated terms

30. Which conclusion follows most directly from this example: radar's long wavelengths and active illumination provide capabilities unavailable to ordinary visible-light images?

- A. air-drop tests examine parachute deployment and sea recovery of a crew-module-like article, while an ISS mission provides operational and microgravity experience
- B. a crew capsule is not simply a satellite with seats added
- C. rendezvous brings spacecraft to matching position and velocity; docking creates a controlled mechanical connection
- D. its L-band and S-band radar observations are designed to measure changes in land, ice, vegetation and natural hazards with repeated global coverage

32. Which conclusion follows most directly from this example: a balloon released without tying moves opposite to escaping air?

- A. the engine pushes exhaust backward and exhaust pushes the engine forward
- B. kilograms measure mass in SI usage; newtons measure force
- C. the tendency of a vehicle to resist or correct unwanted rotation or deviation
- D. combustion or another energy source raises pressure; a nozzle directs expanding gas and increases exhaust speed

34. Which conclusion follows most directly from this example: Earth's Moon stabilises aspects of Earth's axial motion and drives strong tides?

- A. it combines payload, power, communication, thermal control, attitude control and structure
- B. its orbit records formation, capture and long-term gravitational interactions
- C. an orbit does not mean gravity has disappeared
- D. an orbit relatively close to Earth, commonly used for imaging, science stations and many small satellites

36. Select the correct vocabulary item for ionised state of matter with free charged particles.

- A. luminosity
- B. plasma
- C. hydrostatic balance
- D. supernova

Practice Model Paper 1 (continued)

37. Which conclusion follows most directly from this example: granular patterns on the photosphere trace convection below?

- A. energy leaves the core through the radiative and convective zones, reaches the visible photosphere and influences the thin outer atmosphere
- B. its weak gravity and solar heating left it with only an extremely thin exosphere, so temperatures change greatly between day and night
- C. Venus is not Earth's twin in present surface conditions; size alone does not determine habitability
- D. a cold rocky planet with a thin carbon-dioxide atmosphere, polar caps, volcanoes and ancient water-carved features

39. Name the concept described here: imaginary line about which a body rotates.

- A. revolution
- B. axis
- C. season
- D. IST

41. Select the correct vocabulary item for region of magnetic influence.

- A. electromagnet
- B. electric current
- C. solar cell
- D. magnetic field

43. Which chapter term is used for relationship between display intervals and values?

- A. trend
- B. scale
- C. sequence
- D. estimation

45. Name the concept described here: speed with direction.

- A. acceleration
- B. velocity
- C. inertia
- D. centripetal acceleration

47. This description refers to which term: space-environment conditions influenced by solar activity?

- A. space weather
- B. orbiter
- C. rover
- D. Aditya-L1

38. Which conclusion follows most directly from this example: a radiator panel emits infrared radiation to remove waste heat?

- A. spacecraft use conduction through structures and radiation to space; ordinary convection inside liquids or gases depends on buoyancy and gravity conditions
- B. telescopes form images by controlling light paths, while surfaces and atmospheres change which wavelengths reach a detector
- C. sound cannot propagate through a perfect vacuum
- D. kinetic energy is energy of motion; potential energy is stored through position or configuration

40. Complete the definition: ____ means dense spherical generally old cluster.

- A. nebula
- B. reflection nebula
- C. globular cluster
- D. star cluster

42. Select the correct vocabulary item for projection of Earth's equator on the sky.

- A. constellation
- B. cardinal directions
- C. celestial sphere
- D. celestial equator

44. Which term matches this definition: outermost atmospheric region fading into space?

- A. exosphere
- B. crust
- C. outer core
- D. troposphere

46. Which conclusion follows most directly from this example: a magnetorquer can slowly rotate a small satellite without expelling propellant?

- A. antennas convert electrical signals to waves and received waves back to electrical signals; pointing, frequency, power and noise affect the link
- B. spacecraft use motors, relays, magnetic sensors and magnetorquers that interact with Earth's field
- C. voltage is not "the amount of current stored"
- D. solar cells convert light to electricity; rechargeable batteries supply power during eclipse or peak demand

48. Complete the definition: ____ means display colours assigned to measured data bands.

- A. false colour
- B. uplink
- C. coverage footprint
- D. weather satellite

Practice Model Paper 1 (continued)

49. Name the concept described here: expert evaluation before scholarly publication.

- A. astronaut
- B. mission status
- C. commissioning
- D. peer review

51. Which conclusion follows most directly from this example: a comet moving away from the Sun still has tails pointing generally away from the Sun?

- A. rapid entry compresses and heats surrounding air, while ablation removes material from the object
- B. crater size is not equal to the incoming object's diameter
- C. radiation pressure and solar wind shape different tails, so direction depends mainly on the Sun rather than the comet's path
- D. an asteroid or comet whose orbit brings it into the neighbourhood of Earth's orbit

53. Which conclusion follows most directly from this example: radar can observe through cloud and at night because it sends and receives microwave signals?

- A. an uplink carries signals to the satellite and a downlink returns them to a ground region; repeaters or digital processors manage channels
- B. one orbit is not best for every communication need
- C. different wavelength bands reveal vegetation, water, heat, soil moisture or surface roughness
- D. geometric methods for position; satellite navigation primarily uses distances rather than measured angles

55. Which chapter term is used for small spiral feature containing the Sun?

- A. light-year
- B. galaxy
- C. irregular galaxy
- D. Orion Spur

57. Which conclusion follows most directly from this example: neutral-buoyancy pools help rehearse body positioning for spacewalk tasks, though water is not microgravity?

- A. simulators and underwater training allow repeated practice of complex procedures before flight
- B. the spacecraft, crew and loose objects all accelerate around Earth together, so support forces are greatly reduced
- C. "easy to lift" does not mean "easy to control"
- D. systems that provide nutrition and prevent loose liquids, crumbs and microbes from harming equipment or crew

59. This description refers to which term: an explanation based on observations?

- A. apparent motion
- B. light pollution
- C. inference
- D. celestial object

50. Which conclusion follows most directly from this example: using the same stopwatch operator for all trials controls one source but may retain that operator's reaction bias?

- A. scientists state which details were measured and which ideas were concluded from them
- B. a prediction made after seeing the result is weaker evidence than one recorded beforehand
- C. random assignment, blind procedures, standard instruments and predefined rules reduce some biases
- D. a stated range or estimate showing limits of a measurement

52. Which conclusion follows most directly from this example: a graph beginning at 90 rather than zero can make a small change look dramatic?

- A. anomalies may reveal error, unusual conditions or new phenomena and should be investigated rather than deleted automatically
- B. many rules can fit a very short sequence, so context and answer choices matter
- C. a ratio compares quantities; a proportion states that two ratios are equal
- D. before reading points, identify the independent variable, dependent variable, units, origin and whether intervals are equal

54. This description refers to which term: energy associated with position or configuration?

- A. temperature
- B. radiation
- C. potential energy
- D. joule

56. Complete the definition: ____ means factor that can change.

- A. variable
- B. observation
- C. hypothesis
- D. precision

58. Complete the definition: ____ means certified filter designed for safe Sun viewing.

- A. solar filter
- B. telescope
- C. reflector
- D. resolution

60. Which term matches this definition: condition required for a result?

- A. sufficient condition
- B. assumption
- C. necessary condition
- D. case analysis

Practice Model Paper 2

Inferred practice pattern: 60 objective questions | 60 marks | Suggested time: 90 minutes. This is a practice model, not an official examination paper.

Instructions: Choose one best answer for each question. Use rough work for calculations. Do not assume that the longest option is correct.

1. Which statement is the best correction when discussing Planet?

- A. a dwarf planet is not merely a very small planet under the formal definition
- B. their different composition and internal heat influence atmosphere colour and weather
- C. clearing does not mean the orbit is literally empty
- D. Jupiter's moon Europa is studied because evidence suggests an ocean beneath ice

3. Complete the definition: ____ means mission equipment carried by a spacecraft.

- A. payload
- B. space debris
- C. orbit
- D. geostationary orbit

5. Complete the definition: ____ means re-entry spacecraft containing crew.

- A. crew escape system
- B. docking
- C. crew module
- D. Gaganyaan

7. Which chapter term is used for signal sent from ground to satellite?

- A. downlink
- B. uplink
- C. navigation satellite
- D. remote sensing

9. Which statement is the best correction when discussing Earth-rotation advantage?

- A. reusable does not mean no parts are discarded and no maintenance is needed
- B. rotation advantage does not replace rocket thrust and is not useful for every orbit
- C. after its propellant is used, separating the stage removes empty tanks and engines so later stages accelerate less mass
- D. coastal launch centres can direct selected flights over open water

11. Which statement is the best correction when discussing Reflecting telescope?

- A. extreme magnification cannot restore detail that the aperture, atmosphere or focus did not capture
- B. events include conjunctions, eclipses, meteor showers, occultations and changing phases
- C. small school telescopes often use a long tube with a lens at the front
- D. a reflector is not a telescope that merely uses any flat mirror

2. Which chapter term is used for removal of exhaled carbon dioxide?

- A. deconditioning
- B. simulator
- C. inertia
- D. carbon-dioxide scrubbing

4. This description refers to which term: round Sun-orbiting body that has not cleared its neighbourhood?

- A. dwarf planet
- B. orbital neighbourhood
- C. ice giant
- D. tidal heating

6. Which statement is the best correction when discussing Venus?

- A. Earth is not protected by one single feature; habitability depends on interacting conditions
- B. Venus is not Earth's twin in present surface conditions; size alone does not determine habitability
- C. in the solar core, hydrogen nuclei ultimately form helium while a small amount of mass is converted to energy
- D. its cratered surface resembles the Moon in many images

8. Which statement is the best correction when discussing Spring and neap tides?

- A. Earth's shadow does not cause ordinary phases
- B. "spring" here means springing up, not the season
- C. only observers in particular shadow regions see totality, while a wider region may see a partial eclipse
- D. many coasts experience about two high and two low tides each lunar day

10. Which term matches this definition: distance light travels in one year?

- A. light-year
- B. observable universe
- C. spiral galaxy
- D. Milky Way

12. Which statement is the best correction when discussing Differentiation?

- A. differentiation is not the same as the planet splitting into separate bodies
- B. the frost line was a broad changing zone, not a painted circle fixed forever
- C. the asteroid belt lies mainly between Mars and Jupiter, while the Kuiper Belt lies beyond Neptune and the distant Oort Cloud is inferred to surround the system
- D. small snowballs gain material as they roll, but early planetary collisions were much faster and hotter

Practice Model Paper 2 (continued)

13. Which statement is the best correction when discussing Horizon?

- A. altitude is an angle, not the object's distance from Earth
- B. the apparent daily rotation of the sky is organised around the celestial poles
- C. the horizon is an apparent circle, not a physical edge of Earth
- D. the Big Dipper is an asterism within Ursa Major

15. Which statement is the best correction when discussing Stellar colour and temperature?

- A. colour is a temperature clue, not a simple sign that a star is approaching or moving away
- B. brightness in our sky cannot be compared fairly without accounting for distance
- C. higher-mass stars have more fuel but consume it much faster because their cores are hotter and their luminosities are greater
- D. supernova remnants enrich space with elements that can enter later stars and planets

17. Which statement is the best correction when discussing Space weather?

- A. a mission can return valuable science and engineering lessons even when one element does not meet its intended goal
- B. the spacecraft used Earth-orbit-raising manoeuvres, a trans-Mars injection and autonomous operations during long signal delays
- C. its instruments combine imaging and in-situ measurements of particles and magnetic fields
- D. space weather is not ordinary atmospheric weather, though both can affect technology

19. Which term matches this definition: small early body that could grow into a planet?

- A. protoplanet
- B. frost line
- C. planetesimal
- D. solar nebula

21. Which statement is the best correction when discussing Gravitational force?

- A. gravity is not caused by Earth's atmosphere or rotation
- B. centripetal force is not an extra new force; it is the name for the net inward role played by real forces
- C. speeding up, slowing down or changing direction all count as acceleration
- D. Earth pulls an astronaut downward while the astronaut pulls Earth upward with equal force

23. Name the concept described here: thin rocky outer layer of Earth.

- A. mantle
- B. crust
- C. inner core
- D. stratosphere

14. This description refers to which term: Geosynchronous Satellite Launch Vehicle family?

- A. GSLV
- B. LVM3
- C. VSSC
- D. SDSC SHAR

16. Which term matches this definition: overall pattern in data?

- A. anomaly
- B. ratio
- C. trend
- D. axis

18. Select the correct vocabulary item for spacecraft operating around a world.

- A. lander
- B. orbiter
- C. soft landing
- D. L1 region

20. Which chapter term is used for recorded information detected directly or by instrument?

- A. inference
- B. observation
- C. prediction
- D. uncertainty

22. Name the concept described here: facility and region supporting launch operations.

- A. azimuth
- B. trade-off
- C. payload
- D. launch site

24. Complete the definition: ____ means total energy a star emits per second.

- A. apparent brightness
- B. main sequence
- C. luminosity
- D. star

Practice Model Paper 2 (continued)

25. Which statement is the best correction when discussing Inertia in orbit?

- A. life support is not only an oxygen tank
- B. “easy to lift” does not mean “easy to control”
- C. less mechanical loading reduces the signals that maintain bone and muscle, while body fluids shift toward the head
- D. a gently released tool floats beside an astronaut instead of dropping to the floor

27. Which statement is the best correction when discussing Scientists and programme builders?

- A. nationality, citizenship and heritage should be stated accurately rather than used interchangeably
- B. primary sources give direct status and methods, though they still require careful reading and may later be updated
- C. historical contributions should not be compressed into invented quotations or claims that one person “discovered space”
- D. ground and drop tests can validate subsystems before an uncrewed orbital flight

29. Which statement is the best correction when discussing Mass and weight?

- A. kilograms measure mass in SI usage; newtons measure force
- B. reaching space height alone does not create orbit; sufficient sideways speed is essential
- C. computers use sensors such as gyroscopes and accelerometers, then steer engines or small thrusters
- D. a rocket and its exhaust form the action-reaction pair

31. Select the correct vocabulary item for measure related to average particle motion.

- A. heat
- B. sound
- C. kinetic energy
- D. temperature

33. Complete the definition: ____ means officially defined region of the sky.

- A. constellation
- B. asterism
- C. altitude
- D. celestial pole

35. Select the correct vocabulary item for small rocky or metallic Sun-orbiting body.

- A. near-Earth object
- B. asteroid
- C. nucleus
- D. meteor

26. This description refers to which term: rate of change of velocity?

- A. net force
- B. Newton
- C. acceleration
- D. speed

28. Which statement is the best correction when discussing Coverage and delay?

- A. a receiver usually does not send its position to the satellites in order to calculate it
- B. geostationary satellites provide frequent views of the same region, while polar systems collect global detailed strips
- C. television, telephony, broadband and emergency links can cross large distances
- D. one orbit is not best for every communication need

30. Which statement is the best correction when discussing Flyby, orbiter and lander?

- A. a rover cannot travel wherever desired instantly because terrain, power and communication constrain it
- B. the Moon has no substantial atmosphere and extreme day-night cycles; Mars has a thin atmosphere, longer distance and widespread dust
- C. a flyby can still return major discoveries despite a short encounter
- D. protein crystals may grow differently in orbit than on Earth

32. Which chapter term is used for instrument that collects and focuses radiation?

- A. refractor
- B. telescope
- C. aperture
- D. meteor shower

34. Which statement is the best correction when discussing Crew module and service module?

- A. a crew capsule is not simply a satellite with seats added
- B. testing escape does not imply a launch failure is expected; it prepares for a low-probability high-consequence event
- C. these capabilities support future servicing, station assembly and multi-spacecraft observations
- D. experience from international flight can inform training and experiments without being the same as the Gaganyaan orbital mission

36. Select the correct vocabulary item for Moon passes through Earth’s shadow.

- A. umbra
- B. spring tide
- C. waning
- D. lunar eclipse

Practice Model Paper 2 (continued)

37. Which statement is the best correction when discussing Light pollution?

- A. astronomy uses evidence; astrology makes personal claims that are not tested in the same scientific way
- B. clouds can make city glow worse, while darkness alone cannot overcome thick cloud or haze
- C. objects may produce light, reflect light or block light, and their apparent motion depends on both their own movement and Earth's movement
- D. drawing the Moon's shape and position every evening for a month

39. Which statement is the best correction when discussing PSLV?

- A. GSLV and LVM3 are related to geosynchronous missions but are not limited to one exact orbit
- B. satellite services support communication, cyclone warning, mapping, resource monitoring, navigation and disaster response
- C. the word polar does not mean PSLV can launch only polar satellites
- D. a remote-sensing satellite may launch from Sriharikota and later supply data processed through NRSC systems

41. Which statement is the best correction when discussing Common motion and distance?

- A. "open" refers to a loose arrangement, not an opening in space
- B. one shared property alone is not enough; several kinds of evidence strengthen the conclusion
- C. gravity competes with gas pressure and turbulence; newly formed stars may illuminate or erode the surrounding cloud
- D. the familiar outline of Orion joins stars that are not all near one another

43. Complete the definition: ____ means magnet produced by electric current.

- A. electromagnet
- B. radio wave
- C. voltage
- D. battery

45. Which statement is the best correction when discussing Counterexample?

- A. starting arithmetic immediately can hide missing information or incompatible units
- B. a counterexample must meet the stated conditions, not a different problem
- C. translate each sentence into a symbol, table entry, order relation or exclusion mark before combining conditions
- D. having an atmosphere is not necessary for being a planet and is not sufficient for habitability

47. Name the concept described here: organisation managing public space activities.

- A. NASA
- B. space agency
- C. JAXA
- D. joint mission

38. Name the concept described here: condition that guarantees a result.

- A. counterexample
- B. verification
- C. elimination
- D. sufficient condition

40. This description refers to which term: orbital motion around another body?

- A. axial tilt
- B. longitude
- C. revolution
- D. rotation

42. Which statement is the best correction when discussing Current and voltage?

- A. parallel does not mean unlimited current; sources and wires have safe limits
- B. engineers prioritise essential loads and schedule high-power instruments when enough energy is available
- C. voltage is not "the amount of current stored"
- D. a deep-space antenna uses a large dish and careful coding to detect weak data

44. Which statement is the best correction when discussing Orbit?

- A. weightlessness in orbit is not caused by being beyond gravity
- B. the altitude and direction must be precise; this position provides continuous view of a broad region
- C. an orbit does not mean gravity has disappeared
- D. a weather satellite carries sensors while solar panels supply electricity

46. Which statement is the best correction when discussing Local solar time?

- A. countries use standard times, so clock noon need not equal local solar noon
- B. time-zone borders often follow political boundaries rather than exact longitude lines
- C. gravity continually bends Earth's forward motion into an orbit
- D. December is summer in much of the Southern Hemisphere while it is winter in the Northern Hemisphere

48. Which statement is the best correction when discussing Sequence rule?

- A. many rules can fit a very short sequence, so context and answer choices matter
- B. a useful category must be applied consistently to every item
- C. a speed calculation should end in distance per time, and an orbit period cannot reasonably be a negative number
- D. temperature may rise through the morning with one lower point during cloud cover

Practice Model Paper 2 (continued)

49. Which chapter term is used for interstellar cloud of gas or dust?

- A. emission nebula
- B. supernova remnant
- C. open cluster
- D. nebula

51. This description refers to which term: trained human spaceflight crew member?

- A. astronaut
- B. space awareness
- C. information cut-off
- D. operational

53. Which term matches this definition: mission returning extraterrestrial material?

- A. sample return
- B. communication delay
- C. microgravity research
- D. lander

55. Which statement is the best correction when discussing Hosted instrument and tracking support?

- A. a standard does not guarantee quality; it guarantees a defined way of interacting
- B. agencies may build missions, fund universities, operate launch sites, purchase services and cooperate internationally
- C. NISAR combines NASA's L-band radar contribution and ISRO's S-band radar and spacecraft/launch contributions
- D. support agreements still require security, technical compatibility and clear data policies

57. Which statement is the best correction when discussing Impact crater?

- A. an asteroid is not defined as any object that might hit Earth
- B. near the Sun, warming releases gas and dust that create a coma and tails
- C. a brief "shooting star" is a meteor, not a star falling
- D. crater size is not equal to the incoming object's diameter

59. Which statement is the best correction when discussing Density and pressure?

- A. weather does not occur equally through all atmospheric layers
- B. slow mantle flow transfers heat and helps move tectonic plates, while the crust forms continents and ocean floors
- C. a shadow zone shows where certain waves do not arrive directly
- D. high pressure can change melting behaviour, so high temperature does not always mean liquid

50. Which statement is the best correction when discussing Milky Way?

- A. a light-year measures distance, not time, even though "year" appears in the name
- B. the finite age of the universe and expansion limit what we can currently observe
- C. the Milky Way is not the entire universe and the Sun is not at its centre
- D. spiral galaxies have discs and arms; ellipticals are smoother and irregulars lack a simple overall form

52. Which statement is the best correction when discussing Sound?

- A. holding a heavy object still can be tiring but does no mechanical work on the object because displacement is zero
- B. sound cannot propagate through a perfect vacuum
- C. objects at different temperatures exchange energy until they approach thermal equilibrium
- D. a mirror reflects a beam and a lens refracts it

54. Which statement is the best correction when discussing Hypothesis and prediction?

- A. more decimal places do not automatically make a result accurate
- B. all other influential factors should be kept as constant as reasonably possible
- C. "the shadow length is 42 cm" is observation; "the Sun is higher than earlier" is an inference using a model
- D. a prediction made after seeing the result is weaker evidence than one recorded beforehand

56. Name the concept described here: extremely thin outer atmosphere.

- A. greenhouse effect
- B. habitability
- C. solar wind
- D. exosphere

58. Select the correct vocabulary item for motion as it seems from an observer's viewpoint.

- A. sky journal
- B. horizon
- C. observation
- D. apparent motion

60. Which term matches this definition: measure of matter and inertia?

- A. weight
- B. stability
- C. mass
- D. thrust

Answer Key and Guidance

Chapter 1: Astronomy and the Night Sky

MCQ: 1-B 2-C 3-D 4-A 5-B 6-C 7-D 8-A 9-B 10-C

8. A bright point is not automatically a star; planets can also look star-like
9. Earth's rotation makes most sky objects appear to travel from east to west during a night
10. Drawing the Moon's shape and position every evening for a month

Short-answer guidance

1. Astronomy: scientific study of space and celestial objects. Celestial object: natural object beyond most of Earth's atmosphere.
2. Expected points: a careful, purposeful use of senses or instruments to gather information; good observations include time, place, conditions and a description that another person could check
3. Expected points should use the definitions of Celestial object and Apparent motion, then give one accurate link and one distinction.
4. Use this evidence: drawing the Moon's shape and position every evening for a month Explain what it shows and what it does not prove.

Detailed and higher-order guidance

1. A complete response should define the chapter's central system, explain at least two processes, use one accurate example and correct one misconception.
2. Answers should identify the missing evidence or condition, apply the relevant chapter principle and state any remaining uncertainty.
3. Prediction responses must describe the changed condition, the expected consequence and the scientific reason linking them.
4. Correct the statement explicitly, then replace it with an accurate explanation using chapter terminology.

Chapter 2: Directions, Constellations and the Celestial Sphere

MCQ: 1-C 2-D 3-A 4-B 5-C 6-D 7-A 8-B 9-C 10-D

8. Stars are not attached to a shell and are not all equally far away
9. The apparent daily rotation of the sky is organised around the celestial poles
10. Orion is recognised by a set of bright stars and also names a larger defined region

Short-answer guidance

1. Cardinal directions: north, south, east and west. Altitude: angular height above the horizon.
2. Expected points: the angle of a celestial object above the horizon; altitude is measured from 0 degrees at the horizon to 90 degrees at the zenith
3. Expected points should use the definitions of Celestial sphere and Celestial poles and equator, then give one accurate link and one distinction.
4. Use this evidence: Orion is recognised by a set of bright stars and also names a larger defined region Explain what it shows and what it does not prove.

Detailed and higher-order guidance

1. A complete response should define the chapter's central system, explain at least two processes, use one accurate example and correct one misconception.
2. Answers should identify the missing evidence or condition, apply the relevant chapter principle and state any remaining uncertainty.
3. Prediction responses must describe the changed condition, the expected consequence and the scientific reason linking them.
4. Correct the statement explicitly, then replace it with an accurate explanation using chapter terminology.

Answer Key and Guidance (continued)

Chapter 3: Telescopes and Astronomical Events

MCQ: 1-D 2-A 3-B 4-C 5-D 6-A 7-B 8-C 9-D 10-A

8. Extreme magnification cannot restore detail that the aperture, atmosphere or focus did not capture
9. Changing eyepieces changes magnification, but the image may become dim and unsteady at excessive power
10. A meteor shower is best observed over time from a dark site, not through a narrow telescope field

Short-answer guidance

1. Telescope: instrument that collects and focuses radiation. Refractor: telescope using a main lens.
2. Expected points: a telescope whose main objective is a curved mirror; the primary mirror reflects incoming light toward a focus, often using a smaller secondary mirror to direct it
3. Expected points should use the definitions of Aperture and Magnification, then give one accurate link and one distinction.
4. Use this evidence: a meteor shower is best observed over time from a dark site, not through a narrow telescope field Explain what it shows and what it does not prove.

Detailed and higher-order guidance

1. A complete response should define the chapter's central system, explain at least two processes, use one accurate example and correct one misconception.
2. Answers should identify the missing evidence or condition, apply the relevant chapter principle and state any remaining uncertainty.
3. Prediction responses must describe the changed condition, the expected consequence and the scientific reason linking them.
4. Correct the statement explicitly, then replace it with an accurate explanation using chapter terminology.

Chapter 4: Formation and Architecture of the Solar System

MCQ: 1-A 2-B 3-C 4-D 5-A 6-B 7-C 8-D 9-A 10-B

8. Differentiation is not the same as the planet splitting into separate bodies
9. More solid material was available beyond this line, helping large cores and giant planets form
10. Mars and Jupiter lie on opposite sides of the main asteroid belt

Short-answer guidance

1. Solar nebula: cloud from which the Solar System formed. Accretion: growth by gathering smaller material.
2. Expected points: growth by repeated sticking, collision and gravitational gathering of smaller pieces; dust grains formed clumps, clumps became planetesimals and some planetesimals grew into protoplanets
3. Expected points should use the definitions of Differentiation and Frost line, then give one accurate link and one distinction.
4. Use this evidence: Mars and Jupiter lie on opposite sides of the main asteroid belt Explain what it shows and what it does not prove.

Detailed and higher-order guidance

1. A complete response should define the chapter's central system, explain at least two processes, use one accurate example and correct one misconception.
2. Answers should identify the missing evidence or condition, apply the relevant chapter principle and state any remaining uncertainty.
3. Prediction responses must describe the changed condition, the expected consequence and the scientific reason linking them.
4. Correct the statement explicitly, then replace it with an accurate explanation using chapter terminology.

Answer Key and Guidance (continued)

Chapter 5: The Sun and the Rocky Planets

MCQ: 1-B 2-C 3-D 4-A 5-B 6-C 7-D 8-A 9-B 10-C

8. Mercury is not the hottest planet on average despite being nearest the Sun
9. A strong greenhouse effect traps heat, and clouds conceal a hot, high-pressure surface
10. Water cycles through oceans, air, ice, soil and living things

Short-answer guidance

1. Fusion: combining light nuclei to release energy. Photosphere: visible layer of the Sun.
2. Expected points: regions of the Sun distinguished by how energy moves and what observers can detect; energy leaves the core through the radiative and convective zones, reaches the visible photosphere and influences the thin outer atmosphere
3. Expected points should use the definitions of Mercury and Venus, then give one accurate link and one distinction.
4. Use this evidence: water cycles through oceans, air, ice, soil and living things Explain what it shows and what it does not prove.

Detailed and higher-order guidance

1. A complete response should define the chapter's central system, explain at least two processes, use one accurate example and correct one misconception.
2. Answers should identify the missing evidence or condition, apply the relevant chapter principle and state any remaining uncertainty.
3. Prediction responses must describe the changed condition, the expected consequence and the scientific reason linking them.
4. Correct the statement explicitly, then replace it with an accurate explanation using chapter terminology.

Chapter 6: Giant Planets, Dwarf Planets and Moons

MCQ: 1-C 2-D 3-A 4-B 5-C 6-D 7-A 8-B 9-C 10-D

8. Rings are not single solid discs
9. Moons can be captured, co-formed or produced by impacts, and tidal forces may heat their interiors
10. Earth fulfils all three conditions

Short-answer guidance

1. Gas giant: large planet dominated by hydrogen and helium. Ice giant: giant rich in heavier volatile materials.
2. Expected points: Uranus and Neptune, giant planets with larger proportions of water-, ammonia- and methane-rich materials than Jupiter and Saturn; their different composition and internal heat influence atmosphere colour and weather
3. Expected points should use the definitions of Planetary rings and Natural satellite, then give one accurate link and one distinction.
4. Use this evidence: Earth fulfils all three conditions Explain what it shows and what it does not prove.

Detailed and higher-order guidance

1. A complete response should define the chapter's central system, explain at least two processes, use one accurate example and correct one misconception.
2. Answers should identify the missing evidence or condition, apply the relevant chapter principle and state any remaining uncertainty.
3. Prediction responses must describe the changed condition, the expected consequence and the scientific reason linking them.
4. Correct the statement explicitly, then replace it with an accurate explanation using chapter terminology.

Answer Key and Guidance (continued)

Chapter 7: Asteroids, Comets and Meteors

MCQ: 1-D 2-A 3-B 4-C 5-D 6-A 7-B 8-C 9-D 10-A

8. The nucleus is much smaller than the enormous faint coma visible around it
9. Radiation pressure and solar wind shape different tails, so direction depends mainly on the Sun rather than the comet's path
10. A brief "shooting star" is a meteor, not a star falling

Short-answer guidance

1. Asteroid: small rocky or metallic Sun-orbiting body. Near-earth object: small body with an orbit approaching Earth's orbital region.
2. Expected points: an asteroid or comet whose orbit brings it into the neighbourhood of Earth's orbit; surveys repeatedly measure positions so future paths and uncertainties can be calculated
3. Expected points should use the definitions of Comet nucleus and Comet tail, then give one accurate link and one distinction.
4. Use this evidence: a brief "shooting star" is a meteor, not a star falling Explain what it shows and what it does not prove.

Detailed and higher-order guidance

1. A complete response should define the chapter's central system, explain at least two processes, use one accurate example and correct one misconception.
2. Answers should identify the missing evidence or condition, apply the relevant chapter principle and state any remaining uncertainty.
3. Prediction responses must describe the changed condition, the expected consequence and the scientific reason linking them.
4. Correct the statement explicitly, then replace it with an accurate explanation using chapter terminology.

Chapter 8: Stars: Properties and Life Cycles

MCQ: 1-A 2-B 3-C 4-D 5-A 6-B 7-C 8-D 9-A 10-B

8. Balance does not mean nothing moves; energy and material continue to flow
9. Higher-mass stars have more fuel but consume it much faster because their cores are hotter and their luminosities are greater
10. Planetary nebulae surround some newly exposed white dwarfs

Short-answer guidance

1. Star: self-gravitating luminous sphere of hot plasma. Plasma: ionised state of matter with free charged particles.
2. Expected points: apparent brightness is how bright a star looks from Earth; luminosity is the total energy it emits each second; distance spreads light over a larger area, so a luminous far star can appear faint
3. Expected points should use the definitions of Hydrostatic balance and Main sequence, then give one accurate link and one distinction.
4. Use this evidence: planetary nebulae surround some newly exposed white dwarfs Explain what it shows and what it does not prove.

Detailed and higher-order guidance

1. A complete response should define the chapter's central system, explain at least two processes, use one accurate example and correct one misconception.
2. Answers should identify the missing evidence or condition, apply the relevant chapter principle and state any remaining uncertainty.
3. Prediction responses must describe the changed condition, the expected consequence and the scientific reason linking them.
4. Correct the statement explicitly, then replace it with an accurate explanation using chapter terminology.

Answer Key and Guidance (continued)

Chapter 9: Constellations, Star Clusters and Nebulae

MCQ: 1-B 2-C 3-D 4-A 5-B 6-C 7-D 8-A 9-B 10-C

8. “open” refers to a loose arrangement, not an opening in space
9. Strong mutual gravity keeps large numbers of stars together for very long times, often in a galaxy’s halo
10. The Orion Nebula contains active star formation

Short-answer guidance

1. Line of sight: direction from observer to an object. Star cluster: physical group of stars.
2. Expected points: evidence that objects may form a real physical group; stars in a cluster often have similar distances, ages, chemical compositions and motions through space
3. Expected points should use the definitions of Open cluster and Globular cluster, then give one accurate link and one distinction.
4. Use this evidence: the Orion Nebula contains active star formation Explain what it shows and what it does not prove.

Detailed and higher-order guidance

1. A complete response should define the chapter’s central system, explain at least two processes, use one accurate example and correct one misconception.
2. Answers should identify the missing evidence or condition, apply the relevant chapter principle and state any remaining uncertainty.
3. Prediction responses must describe the changed condition, the expected consequence and the scientific reason linking them.
4. Correct the statement explicitly, then replace it with an accurate explanation using chapter terminology.

Chapter 10: Galaxies, the Milky Way and the Universe

MCQ: 1-C 2-D 3-A 4-B 5-C 6-D 7-A 8-B 9-C 10-D

8. The Milky Way is not the entire universe and the Sun is not at its centre
9. Using a distance unit based on light avoids writing extremely long kilometre values for interstellar scales
10. Telescopes that look farther away also see earlier periods because light takes time to arrive

Short-answer guidance

1. Galaxy: vast gravitational system of stars, gas and other matter. Spiral galaxy: disc galaxy with spiral structure.
2. Expected points: broad observed forms such as spiral, elliptical and irregular; shape reflects rotation, star formation, past interactions and mass distribution, though real galaxies show many intermediate structures
3. Expected points should use the definitions of Milky Way and Light-year, then give one accurate link and one distinction.
4. Use this evidence: telescopes that look farther away also see earlier periods because light takes time to arrive Explain what it shows and what it does not prove.

Detailed and higher-order guidance

1. A complete response should define the chapter’s central system, explain at least two processes, use one accurate example and correct one misconception.
2. Answers should identify the missing evidence or condition, apply the relevant chapter principle and state any remaining uncertainty.
3. Prediction responses must describe the changed condition, the expected consequence and the scientific reason linking them.
4. Correct the statement explicitly, then replace it with an accurate explanation using chapter terminology.

Answer Key and Guidance (continued)

Chapter 11: Earth's Structure and Atmosphere

MCQ: 1-D 2-A 3-B 4-C 5-D 6-A 7-B 8-C 9-D 10-A

8. Scientists infer layers from many events and detectors, not from one earthquake alone
9. Gravity pulled denser material inward during differentiation, and the weight of overlying layers produces enormous pressure
10. Commercial aircraft often cruise near the upper troposphere or lower stratosphere

Short-answer guidance

1. Crust: thin rocky outer layer of Earth. Mantle: thick rocky layer that flows slowly.
2. Expected points: Earth's mainly iron-nickel central region, divided into liquid outer core and solid inner core; pressure keeps the inner core solid despite high temperature, while movement in the liquid outer core helps generate Earth's magnetic field
3. Expected points should use the definitions of Seismic wave and Density and pressure, then give one accurate link and one distinction.
4. Use this evidence: commercial aircraft often cruise near the upper troposphere or lower stratosphere Explain what it shows and what it does not prove.

Detailed and higher-order guidance

1. A complete response should define the chapter's central system, explain at least two processes, use one accurate example and correct one misconception.
2. Answers should identify the missing evidence or condition, apply the relevant chapter principle and state any remaining uncertainty.
3. Prediction responses must describe the changed condition, the expected consequence and the scientific reason linking them.
4. Correct the statement explicitly, then replace it with an accurate explanation using chapter terminology.

Chapter 12: Rotation, Revolution, Seasons and Time

MCQ: 1-A 2-B 3-C 4-D 5-A 6-B 7-C 8-D 9-A 10-B

8. The axis does not repeatedly lean toward and away from the Sun; it maintains nearly the same direction in space
9. More direct sunlight concentrates energy over a smaller surface area and longer days increase heating time
10. When it is noon near one longitude, a point 15 degrees west is about one hour earlier by solar time

Short-answer guidance

1. Rotation: spinning about an axis. Axis: imaginary line about which a body rotates.
2. Expected points: Earth's orbital motion around the Sun once each year; gravity continually bends Earth's forward motion into an orbit
3. Expected points should use the definitions of Axial tilt and Season, then give one accurate link and one distinction.
4. Use this evidence: when it is noon near one longitude, a point 15 degrees west is about one hour earlier by solar time Explain what it shows and what it does not prove.

Detailed and higher-order guidance

1. A complete response should define the chapter's central system, explain at least two processes, use one accurate example and correct one misconception.
2. Answers should identify the missing evidence or condition, apply the relevant chapter principle and state any remaining uncertainty.
3. Prediction responses must describe the changed condition, the expected consequence and the scientific reason linking them.
4. Correct the statement explicitly, then replace it with an accurate explanation using chapter terminology.

Answer Key and Guidance (continued)

Chapter 13: The Moon: Phases, Eclipses and Tides

MCQ: 1-B 2-C 3-D 4-A 5-B 6-C 7-D 8-A 9-B 10-C

8. A solar eclipse is unsafe to view without proper protection except during the brief total phase at a location experiencing totality
9. Sunlight filtered and bent through Earth's atmosphere can make the Moon appear copper-red during totality
10. Many coasts experience about two high and two low tides each lunar day

Short-answer guidance

1. Lunar phase: appearance of the Moon's illuminated portion. Waxing: increasing visible illuminated fraction.
2. Expected points: waxing means the illuminated fraction seen from Earth is increasing; waning means it is decreasing; the synodic phase cycle takes about 29.5 days because Earth also moves around the Sun while the Moon orbits Earth
3. Expected points should use the definitions of Solar eclipse and Lunar eclipse, then give one accurate link and one distinction.
4. Use this evidence: many coasts experience about two high and two low tides each lunar day Explain what it shows and what it does not prove.

Detailed and higher-order guidance

1. A complete response should define the chapter's central system, explain at least two processes, use one accurate example and correct one misconception.
2. Answers should identify the missing evidence or condition, apply the relevant chapter principle and state any remaining uncertainty.
3. Prediction responses must describe the changed condition, the expected consequence and the scientific reason linking them.
4. Correct the statement explicitly, then replace it with an accurate explanation using chapter terminology.

Chapter 14: Natural and Artificial Satellites

MCQ: 1-C 2-D 3-A 4-B 5-C 6-D 7-A 8-B 9-C 10-D

8. Low orbit does not remain fixed over one point unless special motion and latitude conditions are involved
9. The altitude and direction must be precise; this position provides continuous view of a broad region
10. Earth's Moon stabilises aspects of Earth's axial motion and drives strong tides

Short-answer guidance

1. Satellite: object orbiting another body. Orbit: gravity-curved path around a body.
2. Expected points: motion controlled mainly by gravity, with support forces absent or very small; satellites and astronauts share nearly the same acceleration, so astronauts feel weightless relative to their spacecraft
3. Expected points should use the definitions of Low Earth orbit and Geostationary orbit, then give one accurate link and one distinction.
4. Use this evidence: Earth's Moon stabilises aspects of Earth's axial motion and drives strong tides Explain what it shows and what it does not prove.

Detailed and higher-order guidance

1. A complete response should define the chapter's central system, explain at least two processes, use one accurate example and correct one misconception.
2. Answers should identify the missing evidence or condition, apply the relevant chapter principle and state any remaining uncertainty.
3. Prediction responses must describe the changed condition, the expected consequence and the scientific reason linking them.
4. Correct the statement explicitly, then replace it with an accurate explanation using chapter terminology.

Answer Key and Guidance (continued)

Chapter 15: Communication, Navigation, Weather and Remote Sensing

MCQ: 1-D 2-A 3-B 4-C 5-D 6-A 7-B 8-C 9-D 10-A

8. A receiver usually does not send its position to the satellites in order to calculate it
9. Each measured range defines a sphere of possible positions, and several spheres intersect near one solution
10. Rapid image sequences reveal cyclone development and movement

Short-answer guidance

1. Uplink: signal sent from ground to satellite. Downlink: signal sent from satellite to ground.
2. Expected points: the area served by a satellite beam and the travel time of its signal; higher satellites cover larger areas but signals travel farther, increasing delay and reducing received strength
3. Expected points should use the definitions of Satellite navigation and Triangulation and trilateration, then give one accurate link and one distinction.
4. Use this evidence: rapid image sequences reveal cyclone development and movement Explain what it shows and what it does not prove.

Detailed and higher-order guidance

1. A complete response should define the chapter's central system, explain at least two processes, use one accurate example and correct one misconception.
2. Answers should identify the missing evidence or condition, apply the relevant chapter principle and state any remaining uncertainty.
3. Prediction responses must describe the changed condition, the expected consequence and the scientific reason linking them.
4. Correct the statement explicitly, then replace it with an accurate explanation using chapter terminology.

Chapter 16: Rocket Principles and Flight

MCQ: 1-A 2-B 3-C 4-D 5-A 6-B 7-C 8-D 9-A 10-B

8. Kilograms measure mass in SI usage; newtons measure force
9. Launch trajectories and throttling balance atmospheric stress, efficiency and safety
10. A model rocket with a forward centre of mass is usually easier to stabilise

Short-answer guidance

1. Thrust: force produced by accelerating exhaust. Nozzle: engine passage that directs expanding gas.
2. Expected points: when one object exerts a force on another, the second exerts an equal-magnitude opposite-direction force on the first; the engine pushes exhaust backward and exhaust pushes the engine forward
3. Expected points should use the definitions of Mass and weight and Drag and gravity loss, then give one accurate link and one distinction.
4. Use this evidence: a model rocket with a forward centre of mass is usually easier to stabilise Explain what it shows and what it does not prove.

Detailed and higher-order guidance

1. A complete response should define the chapter's central system, explain at least two processes, use one accurate example and correct one misconception.
2. Answers should identify the missing evidence or condition, apply the relevant chapter principle and state any remaining uncertainty.
3. Prediction responses must describe the changed condition, the expected consequence and the scientific reason linking them.
4. Correct the statement explicitly, then replace it with an accurate explanation using chapter terminology.

Answer Key and Guidance (continued)

Chapter 17: Stages, Payloads, Launch Sites and Reusable Rockets

MCQ: 1-B 2-C 3-D 4-A 5-B 6-C 7-D 8-A 9-B 10-C

8. Every mission cannot use the same direction because required orbital inclination differs
9. Launching eastward to suitable orbits can reduce the velocity the rocket must supply
10. Recovery can reduce manufacturing of repeated hardware when flight rate and operations support it

Short-answer guidance

1. Stage: rocket section used during part of flight. Stage separation: controlled release of a completed stage.
2. Expected points: the payload is the mission cargo; the fairing is a protective aerodynamic cover used during atmospheric flight; the fairing shields spacecraft from airflow, sound and heating, then is discarded when no longer needed
3. Expected points should use the definitions of Launch azimuth and downrange safety and Earth-rotation advantage, then give one accurate link and one distinction.
4. Use this evidence: recovery can reduce manufacturing of repeated hardware when flight rate and operations support it Explain what it shows and what it does not prove.

Detailed and higher-order guidance

1. A complete response should define the chapter's central system, explain at least two processes, use one accurate example and correct one misconception.
2. Answers should identify the missing evidence or condition, apply the relevant chapter principle and state any remaining uncertainty.
3. Prediction responses must describe the changed condition, the expected consequence and the scientific reason linking them.
4. Correct the statement explicitly, then replace it with an accurate explanation using chapter terminology.

Chapter 18: ISRO, Major Centres, PSLV and GSLV

MCQ: 1-C 2-D 3-A 4-B 5-C 6-D 7-A 8-B 9-C 10-D

8. Centres collaborate; their functions are not completely isolated boxes
9. Launch preparation, engine qualification and data services occur at different specialised facilities
10. PSLV launched Chandrayaan-1, the Mars Orbiter Mission and Aditya-L1

Short-answer guidance

1. Isro: Indian Space Research Organisation. Vssc: centre leading launch-vehicle development.
2. Expected points: use of space systems and data to meet scientific, economic or public needs; satellite services support communication, cyclone warning, mapping, resource monitoring, navigation and disaster response
3. Expected points should use the definitions of VSSC, URSC and SAC and SDSC SHAR, IPRC and NRSC, then give one accurate link and one distinction.
4. Use this evidence: PSLV launched Chandrayaan-1, the Mars Orbiter Mission and Aditya-L1 Explain what it shows and what it does not prove.

Detailed and higher-order guidance

1. A complete response should define the chapter's central system, explain at least two processes, use one accurate example and correct one misconception.
2. Answers should identify the missing evidence or condition, apply the relevant chapter principle and state any remaining uncertainty.
3. Prediction responses must describe the changed condition, the expected consequence and the scientific reason linking them.
4. Correct the statement explicitly, then replace it with an accurate explanation using chapter terminology.

Answer Key and Guidance (continued)

Chapter 19: Chandrayaan, Mangalyaan and Aditya-L1

MCQ: 1-D 2-A 3-B 4-C 5-D 6-A 7-B 8-C 9-D 10-A

8. An orbiter does not land, and a technology demonstration can also produce scientific observations
9. Launch timing uses the moving positions of Earth and the target so the spacecraft and planet arrive at the same region
10. Its instruments combine imaging and in-situ measurements of particles and magnetic fields

Short-answer guidance

1. Orbiter: spacecraft operating around a world. Lander: spacecraft designed to reach and work on a surface.
2. Expected points: India's mission that achieved a soft landing near the lunar south polar region on 23 August 2023 with the Vikram lander and Pragyan rover; the propulsion module delivered the lander system to lunar orbit, the lander controlled descent and the rover made local measurements
3. Expected points should use the definitions of Mars Orbiter Mission and Interplanetary trajectory, then give one accurate link and one distinction.
4. Use this evidence: its instruments combine imaging and in-situ measurements of particles and magnetic fields Explain what it shows and what it does not prove.

Detailed and higher-order guidance

1. A complete response should define the chapter's central system, explain at least two processes, use one accurate example and correct one misconception.
2. Answers should identify the missing evidence or condition, apply the relevant chapter principle and state any remaining uncertainty.
3. Prediction responses must describe the changed condition, the expected consequence and the scientific reason linking them.
4. Correct the statement explicitly, then replace it with an accurate explanation using chapter terminology.

Chapter 20: Gaganyaan and India's Recent Space Achievements

MCQ: 1-A 2-B 3-C 4-D 5-A 6-B 7-C 8-D 9-A 10-B

8. Docking is not a high-speed collision and matching position alone is insufficient without matching velocity
9. These capabilities support future servicing, station assembly and multi-spacecraft observations
10. Radar's long wavelengths and active illumination provide capabilities unavailable to ordinary visible-light images

Short-answer guidance

1. Gaganyaan: Indian human-spaceflight programme. Human-rated: designed and verified to protect crew with high reliability.
2. Expected points: a rapid system intended to pull or propel the crew module away from a failing launch vehicle during critical early phases; sensors and commands must act quickly, and the escape path must lead to safe descent and recovery
3. Expected points should use the definitions of Rendezvous and docking and Power transfer and formation flight, then give one accurate link and one distinction.
4. Use this evidence: radar's long wavelengths and active illumination provide capabilities unavailable to ordinary visible-light images Explain what it shows and what it does not prove.

Detailed and higher-order guidance

1. A complete response should define the chapter's central system, explain at least two processes, use one accurate example and correct one misconception.
2. Answers should identify the missing evidence or condition, apply the relevant chapter principle and state any remaining uncertainty.
3. Prediction responses must describe the changed condition, the expected consequence and the scientific reason linking them.
4. Correct the statement explicitly, then replace it with an accurate explanation using chapter terminology.

Answer Key and Guidance (continued)

Chapter 21: Astronauts and Life in Space

MCQ: 1-B 2-C 3-D 4-A 5-B 6-C 7-D 8-A 9-B 10-C

8. Life support is not only an oxygen tank
9. Food packages, airflow, wipes, toilets and sealed storage are designed for the cabin environment
10. Resistance and aerobic exercise help limit these changes

Short-answer guidance

1. Astronaut: trained person who travels or works in space. Microgravity: very small apparent weight during shared free fall.
2. Expected points: the resistance of mass to a change in motion remains even when weight is small; a massive object may float but still requires force to start, stop or turn, and can injure a crew member if moving
3. Expected points should use the definitions of Life-support system and Food, hygiene and waste, then give one accurate link and one distinction.
4. Use this evidence: resistance and aerobic exercise help limit these changes Explain what it shows and what it does not prove.

Detailed and higher-order guidance

1. A complete response should define the chapter's central system, explain at least two processes, use one accurate example and correct one misconception.
2. Answers should identify the missing evidence or condition, apply the relevant chapter principle and state any remaining uncertainty.
3. Prediction responses must describe the changed condition, the expected consequence and the scientific reason linking them.
4. Correct the statement explicitly, then replace it with an accurate explanation using chapter terminology.

Chapter 22: Space Stations, Moon Missions and Mars Missions

MCQ: 1-C 2-D 3-A 4-B 5-C 6-D 7-A 8-B 9-C 10-D

8. A flyby can still return major discoveries despite a short encounter
9. Mobility compares multiple sites, while returned samples can be studied with large laboratories for decades
10. Mars rovers use onboard autonomy to avoid immediate hazards

Short-answer guidance

1. Space station: long-duration crewed orbital facility. Microgravity research: study under greatly reduced apparent weight.
2. Expected points: experiments that study processes when buoyancy, settling and weight-driven convection are greatly reduced; researchers can separate gravitational effects from other influences in flames, fluids, crystals and organisms
3. Expected points should use the definitions of Flyby, orbiter and lander and Rover and sample return, then give one accurate link and one distinction.
4. Use this evidence: Mars rovers use onboard autonomy to avoid immediate hazards Explain what it shows and what it does not prove.

Detailed and higher-order guidance

1. A complete response should define the chapter's central system, explain at least two processes, use one accurate example and correct one misconception.
2. Answers should identify the missing evidence or condition, apply the relevant chapter principle and state any remaining uncertainty.
3. Prediction responses must describe the changed condition, the expected consequence and the scientific reason linking them.
4. Correct the statement explicitly, then replace it with an accurate explanation using chapter terminology.

Answer Key and Guidance (continued)

Chapter 23: Major Space Agencies and International Cooperation

MCQ: 1-D 2-A 3-B 4-C 5-D 6-A 7-B 8-C 9-D 10-A

8. Joint does not mean every partner performs the same task

9. Sharing reduces duplication and increases global coverage of tracking networks

10. Spacecraft components exchange commands only when voltage levels and message formats match

Short-answer guidance

1. Space agency: organisation managing public space activities. Nasa: United States civil space agency.

2. Expected points: the European Space Agency, an intergovernmental organisation through which participating European states cooperate on space programmes; member contributions support shared launch, science, Earth-observation and exploration missions

3. Expected points should use the definitions of Joint mission and Hosted instrument and tracking support, then give one accurate link and one distinction.

4. Use this evidence: spacecraft components exchange commands only when voltage levels and message formats match Explain what it shows and what it does not prove.

Detailed and higher-order guidance

1. A complete response should define the chapter's central system, explain at least two processes, use one accurate example and correct one misconception.

2. Answers should identify the missing evidence or condition, apply the relevant chapter principle and state any remaining uncertainty.

3. Prediction responses must describe the changed condition, the expected consequence and the scientific reason linking them.

4. Correct the statement explicitly, then replace it with an accurate explanation using chapter terminology.

Chapter 24: Force, Motion, Gravity and Newton's Laws

MCQ: 1-A 2-B 3-C 4-D 5-A 6-B 7-C 8-D 9-A 10-B

8. Motion does not require a continuing force in empty space; changing motion does

9. Each force acts on a different object, so the pair does not cancel on either object's individual force diagram

10. The Moon falls around Earth instead of moving away in a straight line

Short-answer guidance

1. Speed: distance travelled per unit time. Velocity: speed with direction.

2. Expected points: rate at which velocity changes; speeding up, slowing down or changing direction all count as acceleration

3. Expected points should use the definitions of First and second laws and Third law, then give one accurate link and one distinction.

4. Use this evidence: the Moon falls around Earth instead of moving away in a straight line Explain what it shows and what it does not prove.

Detailed and higher-order guidance

1. A complete response should define the chapter's central system, explain at least two processes, use one accurate example and correct one misconception.

2. Answers should identify the missing evidence or condition, apply the relevant chapter principle and state any remaining uncertainty.

3. Prediction responses must describe the changed condition, the expected consequence and the scientific reason linking them.

4. Correct the statement explicitly, then replace it with an accurate explanation using chapter terminology.

Answer Key and Guidance (continued)

Chapter 25: Work, Energy, Heat, Light and Sound

MCQ: 1-B 2-C 3-D 4-A 5-B 6-C 7-D 8-A 9-B 10-C

8. Temperature does not measure the total amount of energy in an object
9. Spacecraft use conduction through structures and radiation to space; ordinary convection inside liquids or gases depends on buoyancy and gravity conditions
10. A mirror reflects a beam and a lens refracts it

Short-answer guidance

1. Work: energy transfer by force through displacement. Joule: SI unit of energy and work.
2. Expected points: kinetic energy is energy of motion; potential energy is stored through position or configuration; a falling object converts gravitational potential energy mainly into kinetic energy, while engines transform chemical energy into vehicle motion and heat
3. Expected points should use the definitions of Temperature and Conduction, convection and radiation, then give one accurate link and one distinction.
4. Use this evidence: a mirror reflects a beam and a lens refracts it Explain what it shows and what it does not prove.

Detailed and higher-order guidance

1. A complete response should define the chapter's central system, explain at least two processes, use one accurate example and correct one misconception.
2. Answers should identify the missing evidence or condition, apply the relevant chapter principle and state any remaining uncertainty.
3. Prediction responses must describe the changed condition, the expected consequence and the scientific reason linking them.
4. Correct the statement explicitly, then replace it with an accurate explanation using chapter terminology.

Chapter 26: Electricity and Magnetism in Space Technology

MCQ: 1-C 2-D 3-A 4-B 5-C 6-D 7-A 8-B 9-C 10-D

8. A solar panel does not store energy by itself unless storage equipment is connected
9. Engineers prioritise essential loads and schedule high-power instruments when enough energy is available
10. A magnetorquer can slowly rotate a small satellite without expelling propellant

Short-answer guidance

1. Electric current: rate of charge flow. Voltage: energy transferred per unit charge.
2. Expected points: components in series share one path; parallel branches provide more than one path; spacecraft distribute power through carefully protected buses so one fault does not always disable every subsystem
3. Expected points should use the definitions of Solar array and battery and Power budget, then give one accurate link and one distinction.
4. Use this evidence: a magnetorquer can slowly rotate a small satellite without expelling propellant Explain what it shows and what it does not prove.

Detailed and higher-order guidance

1. A complete response should define the chapter's central system, explain at least two processes, use one accurate example and correct one misconception.
2. Answers should identify the missing evidence or condition, apply the relevant chapter principle and state any remaining uncertainty.
3. Prediction responses must describe the changed condition, the expected consequence and the scientific reason linking them.
4. Correct the statement explicitly, then replace it with an accurate explanation using chapter terminology.

Answer Key and Guidance (continued)

Chapter 27: Scientific Observation, Measurement and Evidence

MCQ: 1-D 2-A 3-B 4-C 5-D 6-A 7-B 8-C 9-D 10-A

8. More decimal places do not automatically make a result accurate
9. Reading scale divisions, reaction time and changing conditions contribute to uncertainty; repetition helps estimate spread
10. When testing how ramp height affects cart travel distance, ramp height is changed and distance is measured

Short-answer guidance

1. Observation: recorded information detected directly or by instrument. Inference: explanation drawn from evidence.
2. Expected points: a hypothesis is a testable proposed explanation; a prediction states an expected result under specified conditions; predictions link a hypothesis to evidence that could support or challenge it
3. Expected points should use the definitions of Accuracy and precision and Uncertainty, then give one accurate link and one distinction.
4. Use this evidence: when testing how ramp height affects cart travel distance, ramp height is changed and distance is measured Explain what it shows and what it does not prove.

Detailed and higher-order guidance

1. A complete response should define the chapter's central system, explain at least two processes, use one accurate example and correct one misconception.
2. Answers should identify the missing evidence or condition, apply the relevant chapter principle and state any remaining uncertainty.
3. Prediction responses must describe the changed condition, the expected consequence and the scientific reason linking them.
4. Correct the statement explicitly, then replace it with an accurate explanation using chapter terminology.

Chapter 28: Patterns, Data Interpretation and Numerical Aptitude

MCQ: 1-A 2-B 3-C 4-D 5-A 6-B 7-C 8-D 9-A 10-B

8. Many rules can fit a very short sequence, so context and answer choices matter
9. Use more than appearance when possible and state both inclusion and exclusion conditions
10. If a model uses 1 cm for 1,000 km, 4.5 cm represents 4,500 km

Short-answer guidance

1. Axis: reference line defining a graph variable. Scale: relationship between display intervals and values.
2. Expected points: a trend is an overall relationship; an anomaly is a point that differs markedly from the pattern; anomalies may reveal error, unusual conditions or new phenomena and should be investigated rather than deleted automatically
3. Expected points should use the definitions of Sequence rule and Classification pattern, then give one accurate link and one distinction.
4. Use this evidence: if a model uses 1 cm for 1,000 km, 4.5 cm represents 4,500 km Explain what it shows and what it does not prove.

Detailed and higher-order guidance

1. A complete response should define the chapter's central system, explain at least two processes, use one accurate example and correct one misconception.
2. Answers should identify the missing evidence or condition, apply the relevant chapter principle and state any remaining uncertainty.
3. Prediction responses must describe the changed condition, the expected consequence and the scientific reason linking them.
4. Correct the statement explicitly, then replace it with an accurate explanation using chapter terminology.

Answer Key and Guidance (continued)

Chapter 29: Analytical Reasoning and Problem Solving

MCQ: 1-B 2-C 3-D 4-A 5-B 6-C 7-D 8-A 9-B 10-C

8. Confusing necessary with sufficient produces overconfident conclusions
9. Test words such as all, never and must by searching for an exception
10. For a satellite speed question, list distance, time, unit conversion and the speed formula

Short-answer guidance

1. Constraint: rule limiting possible solutions. Case analysis: systematic testing of possible situations.
2. Expected points: division of a problem into a small number of possible situations; choose a condition with few possibilities, explore each case and reject cases that violate another rule
3. Expected points should use the definitions of Necessary and sufficient conditions and Counterexample, then give one accurate link and one distinction.
4. Use this evidence: for a satellite speed question, list distance, time, unit conversion and the speed formula Explain what it shows and what it does not prove.

Detailed and higher-order guidance

1. A complete response should define the chapter's central system, explain at least two processes, use one accurate example and correct one misconception.
2. Answers should identify the missing evidence or condition, apply the relevant chapter principle and state any remaining uncertainty.
3. Prediction responses must describe the changed condition, the expected consequence and the scientific reason linking them.
4. Correct the statement explicitly, then replace it with an accurate explanation using chapter terminology.

Chapter 30: Space Awareness: Missions, Discoveries and People

MCQ: 1-C 2-D 3-A 4-B 5-C 6-D 7-A 8-B 9-C 10-D

8. Data volume alone does not measure scientific importance; calibration, analysis and questions matter
9. A second integrated air-drop test in April 2026 used a simulated crew module released over sea to examine parachute and recovery performance
10. Sarabhai argued for advanced space technology serving national development

Short-answer guidance

1. Mission status: dated stage of a mission or programme. Information cut-off: latest date covered by a publication.
2. Expected points: an official mission page, technical report, dataset or statement produced by the responsible organisation or research team; primary sources give direct status and methods, though they still require careful reading and may later be updated
3. Expected points should use the definitions of Active and recent Indian missions and Gaganyaan preparation, then give one accurate link and one distinction.
4. Use this evidence: Sarabhai argued for advanced space technology serving national development Explain what it shows and what it does not prove.

Detailed and higher-order guidance

1. A complete response should define the chapter's central system, explain at least two processes, use one accurate example and correct one misconception.
2. Answers should identify the missing evidence or condition, apply the relevant chapter principle and state any remaining uncertainty.
3. Prediction responses must describe the changed condition, the expected consequence and the scientific reason linking them.
4. Correct the statement explicitly, then replace it with an accurate explanation using chapter terminology.

Unit Review Answers

Unit 1: Fundamentals of Astronomy

Mixed MCQ: 1-B 2-C 3-D 4-A 5-B 6-C 7-D 8-A 9-D 10-A 11-B 12-C

Olympiad Challenge: 1-B 2-C 3-D 4-A 5-B 6-C 7-D 8-A

Written 1: A valid response names both chapter ideas and gives an accurate causal, structural or measurement link.

Written 2: Definitions must be accurate and the comparison should identify both a shared feature and a difference.

Written 3: Apply the relevant concept, identify needed evidence and avoid guessing beyond the given information.

Written 4: State the changed condition, expected result and scientific reason.

Written 5: The model must represent a real relationship and explicitly state what is not to scale or omitted.

Written 6: The claim should be specific, evidence relevant, and uncertainty honestly stated.

Unit 2: The Solar System

Mixed MCQ: 1-C 2-D 3-A 4-B 5-C 6-D 7-A 8-B 9-A 10-B 11-C 12-D

Olympiad Challenge: 1-C 2-D 3-A 4-B 5-C 6-D 7-A 8-B

Written 1: A valid response names both chapter ideas and gives an accurate causal, structural or measurement link.

Written 2: Definitions must be accurate and the comparison should identify both a shared feature and a difference.

Written 3: Apply the relevant concept, identify needed evidence and avoid guessing beyond the given information.

Written 4: State the changed condition, expected result and scientific reason.

Written 5: The model must represent a real relationship and explicitly state what is not to scale or omitted.

Written 6: The claim should be specific, evidence relevant, and uncertainty honestly stated.

Unit 3: Stars, Galaxies and the Universe

Mixed MCQ: 1-D 2-A 3-B 4-C 5-D 6-A 7-B 8-C 9-B 10-C 11-D 12-A

Olympiad Challenge: 1-D 2-A 3-B 4-C 5-D 6-A 7-B 8-C

Written 1: A valid response names both chapter ideas and gives an accurate causal, structural or measurement link.

Written 2: Definitions must be accurate and the comparison should identify both a shared feature and a difference.

Written 3: Apply the relevant concept, identify needed evidence and avoid guessing beyond the given information.

Written 4: State the changed condition, expected result and scientific reason.

Written 5: The model must represent a real relationship and explicitly state what is not to scale or omitted.

Written 6: The claim should be specific, evidence relevant, and uncertainty honestly stated.

Unit 4: Earth and Moon

Mixed MCQ: 1-A 2-B 3-C 4-D 5-A 6-B 7-C 8-D 9-C 10-D 11-A 12-B

Olympiad Challenge: 1-A 2-B 3-C 4-D 5-A 6-B 7-C 8-D

Written 1: A valid response names both chapter ideas and gives an accurate causal, structural or measurement link.

Written 2: Definitions must be accurate and the comparison should identify both a shared feature and a difference.

Written 3: Apply the relevant concept, identify needed evidence and avoid guessing beyond the given information.

Written 4: State the changed condition, expected result and scientific reason.

Written 5: The model must represent a real relationship and explicitly state what is not to scale or omitted.

Written 6: The claim should be specific, evidence relevant, and uncertainty honestly stated.

Unit Review Answers

Unit 5: Satellites and Space Technology

Mixed MCQ: 1-B 2-C 3-D 4-A 5-B 6-C 7-D 8-A 9-D 10-A 11-B 12-C

Olympiad Challenge: 1-B 2-C 3-D 4-A 5-B 6-C 7-D 8-A

Written 1: A valid response names both chapter ideas and gives an accurate causal, structural or measurement link.

Written 2: Definitions must be accurate and the comparison should identify both a shared feature and a difference.

Written 3: Apply the relevant concept, identify needed evidence and avoid guessing beyond the given information.

Written 4: State the changed condition, expected result and scientific reason.

Written 5: The model must represent a real relationship and explicitly state what is not to scale or omitted.

Written 6: The claim should be specific, evidence relevant, and uncertainty honestly stated.

Unit 6: Rockets and Launch Vehicles

Mixed MCQ: 1-C 2-D 3-A 4-B 5-C 6-D 7-A 8-B 9-A 10-B 11-C 12-D

Olympiad Challenge: 1-C 2-D 3-A 4-B 5-C 6-D 7-A 8-B

Written 1: A valid response names both chapter ideas and gives an accurate causal, structural or measurement link.

Written 2: Definitions must be accurate and the comparison should identify both a shared feature and a difference.

Written 3: Apply the relevant concept, identify needed evidence and avoid guessing beyond the given information.

Written 4: State the changed condition, expected result and scientific reason.

Written 5: The model must represent a real relationship and explicitly state what is not to scale or omitted.

Written 6: The claim should be specific, evidence relevant, and uncertainty honestly stated.

Unit 7: ISRO and India's Space Missions

Mixed MCQ: 1-D 2-A 3-B 4-C 5-D 6-A 7-B 8-C 9-B 10-C 11-D 12-A

Olympiad Challenge: 1-D 2-A 3-B 4-C 5-D 6-A 7-B 8-C

Written 1: A valid response names both chapter ideas and gives an accurate causal, structural or measurement link.

Written 2: Definitions must be accurate and the comparison should identify both a shared feature and a difference.

Written 3: Apply the relevant concept, identify needed evidence and avoid guessing beyond the given information.

Written 4: State the changed condition, expected result and scientific reason.

Written 5: The model must represent a real relationship and explicitly state what is not to scale or omitted.

Written 6: The claim should be specific, evidence relevant, and uncertainty honestly stated.

Unit 8: Space Exploration

Mixed MCQ: 1-A 2-B 3-C 4-D 5-A 6-B 7-C 8-D 9-C 10-D 11-A 12-B

Olympiad Challenge: 1-A 2-B 3-C 4-D 5-A 6-B 7-C 8-D

Written 1: A valid response names both chapter ideas and gives an accurate causal, structural or measurement link.

Written 2: Definitions must be accurate and the comparison should identify both a shared feature and a difference.

Written 3: Apply the relevant concept, identify needed evidence and avoid guessing beyond the given information.

Written 4: State the changed condition, expected result and scientific reason.

Written 5: The model must represent a real relationship and explicitly state what is not to scale or omitted.

Written 6: The claim should be specific, evidence relevant, and uncertainty honestly stated.

Unit Review Answers

Unit 9: Basic Physics for Space Science

Mixed MCQ: 1-B 2-C 3-D 4-A 5-B 6-C 7-D 8-A 9-D 10-A 11-B 12-C

Olympiad Challenge: 1-B 2-C 3-D 4-A 5-B 6-C 7-D 8-A

Written 1: A valid response names both chapter ideas and gives an accurate causal, structural or measurement link.

Written 2: Definitions must be accurate and the comparison should identify both a shared feature and a difference.

Written 3: Apply the relevant concept, identify needed evidence and avoid guessing beyond the given information.

Written 4: State the changed condition, expected result and scientific reason.

Written 5: The model must represent a real relationship and explicitly state what is not to scale or omitted.

Written 6: The claim should be specific, evidence relevant, and uncertainty honestly stated.

Unit 10: Scientific Reasoning and Logical Aptitude

Mixed MCQ: 1-C 2-D 3-A 4-B 5-C 6-D 7-A 8-B 9-A 10-B 11-C 12-D

Olympiad Challenge: 1-C 2-D 3-A 4-B 5-C 6-D 7-A 8-B

Written 1: A valid response names both chapter ideas and gives an accurate causal, structural or measurement link.

Written 2: Definitions must be accurate and the comparison should identify both a shared feature and a difference.

Written 3: Apply the relevant concept, identify needed evidence and avoid guessing beyond the given information.

Written 4: State the changed condition, expected result and scientific reason.

Written 5: The model must represent a real relationship and explicitly state what is not to scale or omitted.

Written 6: The claim should be specific, evidence relevant, and uncertainty honestly stated.

Unit 11: General Space Awareness

Mixed MCQ: 1-D 2-A 3-B 4-C 5-D 6-A 7-B 8-C 9-B 10-C 11-D 12-A

Olympiad Challenge: 1-D 2-A 3-B 4-C 5-D 6-A 7-B 8-C

Written 1: A valid response names both chapter ideas and gives an accurate causal, structural or measurement link.

Written 2: Definitions must be accurate and the comparison should identify both a shared feature and a difference.

Written 3: Apply the relevant concept, identify needed evidence and avoid guessing beyond the given information.

Written 4: State the changed condition, expected result and scientific reason.

Written 5: The model must represent a real relationship and explicitly state what is not to scale or omitted.

Written 6: The claim should be specific, evidence relevant, and uncertainty honestly stated.

Cumulative Revision Answers

Objective questions: 1-B 2-C 3-C 4-D 5-D 6-A 7-A 8-B 9-B 10-C 11-C 12-D 13-D 14-A 15-A 16-B 17-B 18-C 19-C 20-D

Data and calculation tasks

1. 96 minutes
2. 15 cm
3. 2 °C per hour
4. 600 kN upward
5. about 2.6 seconds
6. observation, inference, observation, inference
7. answer depends on supplied table; cite values
8. multiply by 2; 48

Extended tasks should be assessed for correct principles, evidence, units and status language. Accept alternative scientifically valid designs when assumptions are stated.

Model Paper Answer Keys

Practice Model Paper 1

1-C 2-B 3-A 4-B 5-C 6-B 7-C 8-B 9-B 10-C 11-D 12-B 13-D 14-B 15-C 16-A 17-B 18-D 19-A 20-D
21-D 22-A 23-C 24-D 25-B 26-D 27-A 28-C 29-A 30-D 31-A 32-D 33-D 34-B 35-C 36-B 37-A 38-A 39-B 40-C
41-D 42-D 43-B 44-A 45-B 46-B 47-A 48-A 49-D 50-C 51-C 52-D 53-C 54-C 55-D 56-A 57-A 58-A 59-C 60-C

Explanations for selected challenge items

3. choose a condition with few possibilities, explore each case and reject cases that violate another rule
4. these achievements concern different capabilities: solar science, Earth radar observation and orbital operations
5. proper front-mounted solar filters block almost all visible, infrared and ultraviolet radiation before it enters an instrument
7. launch timing uses the moving positions of Earth and the target so the spacecraft and planet arrive at the same region
9. collisions, resonances and moon gravity shape gaps and narrow structures
12. stars orbit a common galactic mass distribution while gas can form new stars and feedback can reshape the galaxy
13. greater force gives greater acceleration for the same mass, while greater mass gives smaller acceleration for the same force
14. modern boundaries ensure every sky position belongs to one constellation region

Practice Model Paper 2

1-C 2-D 3-A 4-A 5-C 6-B 7-B 8-B 9-B 10-A 11-D 12-A 13-C 14-A 15-A 16-C 17-D 18-B 19-C 20-B
21-A 22-D 23-B 24-C 25-B 26-C 27-C 28-D 29-A 30-C 31-D 32-B 33-A 34-A 35-B 36-D 37-B 38-D 39-C 40-C
41-B 42-C 43-A 44-C 45-B 46-A 47-B 48-A 49-D 50-C 51-A 52-B 53-A 54-D 55-D 56-D 57-D 58-D 59-D 60-C

Explanations for selected challenge items

1. clearing does not mean the orbit is literally empty
6. Venus is not Earth's twin in present surface conditions; size alone does not determine habitability
8. "spring" here means springing up, not the season
9. rotation advantage does not replace rocket thrust and is not useful for every orbit
11. a reflector is not a telescope that merely uses any flat mirror
12. differentiation is not the same as the planet splitting into separate bodies
13. the horizon is an apparent circle, not a physical edge of Earth
15. colour is a temperature clue, not a simple sign that a star is approaching or moving away

Glossary

acceleration	rate of change of velocity (Chapter 24)
accretion	growth by gathering smaller material (Chapter 4)
accuracy	closeness to a correct value (Chapter 27)
Aditya-L1	Indian solar-observatory mission (Chapter 19)
altitude	angular height above the horizon (Chapter 2)
anomaly	value differing strongly from a pattern (Chapter 28)
aperture	diameter of the light-collecting opening (Chapter 3)
apparent brightness	brightness received by an observer (Chapter 8)
apparent motion	motion as it seems from an observer's viewpoint (Chapter 1)
assumption	unstated idea taken as true (Chapter 29)
asterism	recognisable informal star pattern (Chapter 2)
asteroid	small rocky or metallic Sun-orbiting body (Chapter 7)
asteroid belt	small-body region between Mars and Jupiter (Chapter 4)
astronaut	trained person who travels or works in space (Chapter 21)
astronomy	scientific study of space and celestial objects (Chapter 1)
axial tilt	inclination of a rotation axis (Chapter 12)
axis	imaginary line about which a body rotates (Chapter 12)
azimuth	horizontal direction of a launch path (Chapter 17)
battery	device storing chemical energy for electrical use (Chapter 26)
carbon-dioxide scrubbing	removal of exhaled carbon dioxide (Chapter 21)
cardinal directions	north, south, east and west (Chapter 2)
case analysis	systematic testing of possible situations (Chapter 29)
celestial equator	projection of Earth's equator on the sky (Chapter 2)
celestial object	natural object beyond most of Earth's atmosphere (Chapter 1)
celestial pole	projected direction of Earth's rotation axis (Chapter 2)
celestial sphere	imaginary sphere used to map the sky (Chapter 2)
centripetal acceleration	inward acceleration in curved motion (Chapter 24)
circuit	complete path for electric current (Chapter 26)
CNSA	Chinese national space administration (Chapter 23)
coma	cloud around an active comet nucleus (Chapter 7)
comet	volatile-rich body that can become active near the Sun (Chapter 7)
commissioning	testing and calibrating a spacecraft after launch (Chapter 30)
communication delay	signal travel time between locations (Chapter 22)
constellation	officially defined region of the sky (Chapter 2)
constraint	rule limiting possible solutions (Chapter 29)
control	actuation that changes motion to follow a command (Chapter 16)
counterexample	example disproving a universal claim (Chapter 29)
coverage footprint	ground area reached by a satellite beam (Chapter 15)
crew escape system	emergency system separating crew from a failing launcher (Chapter 20)

Glossary (continued)

crew module	re-entry spacecraft containing crew (Chapter 20)
crust	thin rocky outer layer of Earth (Chapter 11)
deconditioning	loss of physical capacity during reduced loading (Chapter 21)
differentiation	internal separation by density and composition (Chapter 4)
docking	mechanical connection of spacecraft (Chapter 20)
downlink	signal sent from satellite to ground (Chapter 15)
drag	force opposing motion through a fluid (Chapter 16)
dwarf planet	round Sun-orbiting body that has not cleared its neighbourhood (Chapter 6)
electric current	rate of charge flow (Chapter 26)
electromagnet	magnet produced by electric current (Chapter 26)
elimination	rejection of options using evidence (Chapter 29)
elliptical galaxy	smooth rounded or oval galaxy (Chapter 10)
emission nebula	gas cloud that emits light after being energised (Chapter 9)
ESA	European intergovernmental space agency (Chapter 23)
estimation	approximate calculation used for checking (Chapter 28)
exosphere	extremely thin outer atmosphere (Chapter 5)
fairing	protective cover around payload (Chapter 17)
false colour	display colours assigned to measured data bands (Chapter 15)
flyby	mission passing a target without orbiting (Chapter 22)
free fall	motion mainly under gravity (Chapter 14)
frost line	early-disc region beyond which ices could condense (Chapter 4)
fusion	combining light nuclei to release energy (Chapter 5)
Gaganyaan	Indian human-spaceflight programme (Chapter 20)
galaxy	vast gravitational system of stars, gas and other matter (Chapter 10)
gas giant	large planet dominated by hydrogen and helium (Chapter 6)
geostationary orbit	equatorial daily orbit appearing fixed (Chapter 14)
globular cluster	dense spherical generally old cluster (Chapter 9)
gravity	attraction between masses (Chapter 24)
greenhouse effect	warming caused when an atmosphere slows heat loss (Chapter 5)
GSLV	Geosynchronous Satellite Launch Vehicle family (Chapter 18)
guidance	selection of desired path (Chapter 16)
habitability	ability of an environment to support life as defined (Chapter 5)
heat	energy transferred due to temperature difference (Chapter 25)
horizon	apparent boundary between Earth and sky (Chapter 1)
human-rated	designed and verified to protect crew with high reliability (Chapter 20)
hydrostatic balance	balance of inward gravity and outward pressure (Chapter 8)
hypothesis	testable proposed explanation (Chapter 27)
ice giant	giant rich in heavier volatile materials (Chapter 6)
inclination	tilt of an orbit to a reference plane (Chapter 14)

Glossary (continued)

inertia	resistance to change in motion (Chapter 21)
inference	an explanation based on observations (Chapter 1)
information cut-off	latest date covered by a publication (Chapter 30)
inner core	solid metallic centre (Chapter 11)
interface standard	agreed rule for connecting systems (Chapter 23)
interplanetary trajectory	path between planetary regions (Chapter 19)
irregular galaxy	galaxy without a simple regular form (Chapter 10)
ISRO	Indian Space Research Organisation (Chapter 18)
IST	Indian Standard Time, UTC+5:30 (Chapter 12)
JAXA	Japanese space agency (Chapter 23)
joint mission	mission with shared partner responsibilities (Chapter 23)
joule	SI unit of energy and work (Chapter 25)
kinetic energy	energy associated with motion (Chapter 25)
Kuiper Belt	icy-body region beyond Neptune (Chapter 4)
L1 region	Sun–Earth gravitational region useful for solar viewing (Chapter 19)
lander	spacecraft designed to reach and work on a surface (Chapter 19)
launch site	facility and region supporting launch operations (Chapter 17)
life support	systems maintaining a safe crew environment (Chapter 21)
light pollution	artificial brightening of the night sky (Chapter 1)
light-year	distance light travels in one year (Chapter 10)
line of sight	direction from observer to an object (Chapter 9)
longitude	east-west angular coordinate on Earth (Chapter 12)
low Earth orbit	relatively low-altitude Earth orbit (Chapter 14)
luminosity	total energy a star emits per second (Chapter 8)
lunar eclipse	Moon passes through Earth's shadow (Chapter 13)
lunar phase	appearance of the Moon's illuminated portion (Chapter 13)
LVM3	heavy-lift Indian launch vehicle family (Chapter 18)
magnetic field	region in which magnetic forces act (Chapter 5)
magnification	apparent enlargement of an image (Chapter 3)
main sequence	core hydrogen-fusing life stage (Chapter 8)
mantle	thick rocky layer that flows slowly (Chapter 11)
mass	measure of matter and inertia (Chapter 16)
meteor	visible atmospheric streak caused by an incoming body (Chapter 7)
meteor shower	period when many meteors seem to radiate from one sky region (Chapter 3)
meteorite	surviving material found on the ground (Chapter 7)
meteoroid	small natural solid travelling in space (Chapter 7)
microgravity	very small apparent weight during shared free fall (Chapter 21)
microgravity research	study under greatly reduced apparent weight (Chapter 22)
Milky Way	galaxy containing the Solar System (Chapter 10)

Glossary (continued)

mission status	dated stage of a mission or programme (Chapter 30)
NASA	United States civil space agency (Chapter 23)
natural satellite	natural body orbiting another non-stellar body (Chapter 6)
navigation satellite	satellite broadcasting precise timing and orbit data (Chapter 15)
near-Earth object	small body with an orbit approaching Earth's orbital region (Chapter 7)
nebula	interstellar cloud of gas or dust (Chapter 9)
necessary condition	condition required for a result (Chapter 29)
net force	combined effect of all forces (Chapter 24)
Newton	SI unit of force (Chapter 24)
NISAR	NASA-ISRO Synthetic Aperture Radar mission (Chapter 20)
nozzle	engine passage that directs expanding gas (Chapter 16)
nucleus	solid central body of a comet (Chapter 7)
observable universe	region whose signals can reach us (Chapter 10)
observation	careful collection of information (Chapter 1)
open cluster	loose generally young cluster (Chapter 9)
operational	performing intended routine work (Chapter 30)
orbit	gravity-curved path around a body (Chapter 14)
orbital neighbourhood	region around an orbit influenced by a body's gravity (Chapter 6)
orbiter	spacecraft operating around a world (Chapter 19)
Orion Spur	small spiral feature containing the Sun (Chapter 10)
outer core	liquid metallic layer (Chapter 11)
payload	mission equipment carried by a spacecraft (Chapter 14)
peer review	expert evaluation before scholarly publication (Chapter 30)
photosphere	visible layer of the Sun (Chapter 5)
planet	round Sun-orbiting body that has cleared its neighbourhood (Chapter 6)
planetesimal	small early body that could grow into a planet (Chapter 4)
plasma	ionised state of matter with free charged particles (Chapter 8)
potential energy	energy associated with position or configuration (Chapter 25)
precision	closeness among repeated values (Chapter 27)
prediction	expected result under stated conditions (Chapter 27)
primary source	original official or research record (Chapter 30)
proportion	equality of two ratios (Chapter 28)
protoplanet	large developing planetary body (Chapter 4)
PSLV	Polar Satellite Launch Vehicle (Chapter 18)
radiation	energy transfer by electromagnetic waves (Chapter 25)
radio wave	electromagnetic wave used for communication (Chapter 26)
ratio	comparison by division (Chapter 28)
red giant	expanded late stage of a lower-mass star (Chapter 8)
reflection nebula	dust cloud seen by scattered starlight (Chapter 9)

Glossary (continued)

reflector	telescope using a main mirror (Chapter 3)
refractor	telescope using a main lens (Chapter 3)
remote sensing	measurement without direct contact (Chapter 15)
rendezvous	controlled approach with matched relative motion (Chapter 20)
resolution	ability to separate fine detail (Chapter 3)
reusable launcher	vehicle with recoverable flight hardware (Chapter 17)
revolution	orbital motion around another body (Chapter 12)
ring	orbiting band of many particles (Chapter 6)
Roscosmos	Russian state space corporation (Chapter 23)
rotation	spinning about an axis (Chapter 12)
rover	mobile surface exploration vehicle (Chapter 19)
SAC	centre developing payloads and applications (Chapter 18)
sample return	mission returning extraterrestrial material (Chapter 22)
satellite	object orbiting another body (Chapter 14)
scale	relationship between display intervals and values (Chapter 28)
SDSC SHAR	launch centre at Sriharikota (Chapter 18)
season	yearly period with characteristic sunlight conditions (Chapter 12)
seismic wave	wave carrying earthquake energy through Earth (Chapter 11)
sequence	ordered set following a rule (Chapter 28)
simulator	training system reproducing selected mission conditions (Chapter 21)
sky journal	dated record of sky observations (Chapter 1)
soft landing	controlled landing at survivable speed (Chapter 19)
solar cell	device converting light to electricity (Chapter 26)
solar eclipse	Moon blocks the Sun from part of Earth (Chapter 13)
solar filter	certified filter designed for safe Sun viewing (Chapter 3)
solar nebula	cloud from which the Solar System formed (Chapter 4)
solar wind	stream of charged particles from the Sun (Chapter 5)
sound	mechanical wave requiring a medium (Chapter 25)
space agency	organisation managing public space activities (Chapter 23)
space awareness	current and historical knowledge checked against reliable evidence (Chapter 30)
space debris	non-functional human-made material in orbit (Chapter 14)
space station	long-duration crewed orbital facility (Chapter 22)
space weather	space-environment conditions influenced by solar activity (Chapter 19)
spacewalk	crew activity outside a spacecraft (Chapter 21)
SpaDeX	Indian Space Docking Experiment (Chapter 20)
speed	distance travelled per unit time (Chapter 24)
spiral galaxy	disc galaxy with spiral structure (Chapter 10)
spring tide	large tidal range near alignment (Chapter 13)
stability	resistance or correction of unwanted motion (Chapter 16)

Glossary (continued)

stage	rocket section used during part of flight (Chapter 17)
stage separation	controlled release of a completed stage (Chapter 17)
star	self-gravitating luminous sphere of hot plasma (Chapter 8)
star cluster	physical group of stars (Chapter 9)
stratosphere	layer above the troposphere containing much ozone (Chapter 11)
sufficient condition	condition that guarantees a result (Chapter 29)
supernova	powerful explosion marking the death of some massive stars (Chapter 8)
supernova remnant	expanding material from a stellar explosion (Chapter 9)
telescope	instrument that collects and focuses radiation (Chapter 3)
temperature	measure related to average particle motion (Chapter 25)
terrestrial planet	planet with a rocky solid surface (Chapter 5)
thrust	force produced by accelerating exhaust (Chapter 16)
tidal heating	internal heating caused by changing gravitational stretching (Chapter 6)
tide	periodic sea-level change driven mainly by gravity (Chapter 13)
time zone	region using a common standard time (Chapter 12)
trade-off	balance in which improving one factor may cost another (Chapter 17)
trend	overall pattern in data (Chapter 28)
trilateration	position finding from measured distances (Chapter 15)
troposphere	lowest atmospheric layer and main weather region (Chapter 11)
umbra	dark central part of a shadow (Chapter 13)
uncertainty	estimated limit of a measurement (Chapter 27)
uplink	signal sent from ground to satellite (Chapter 15)
URSC	centre leading spacecraft development (Chapter 18)
variable	factor that can change (Chapter 27)
velocity	speed with direction (Chapter 24)
verification	checking a result against evidence and rules (Chapter 29)
voltage	energy transferred per unit charge (Chapter 26)
VSSC	centre leading launch-vehicle development (Chapter 18)
waning	decreasing visible illuminated fraction (Chapter 13)
waxing	increasing visible illuminated fraction (Chapter 13)
weather satellite	satellite observing atmospheric conditions (Chapter 15)
weight	gravitational force on an object (Chapter 16)
work	energy transfer by force through displacement (Chapter 25)
zenith	point directly overhead (Chapter 2)

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Scientific Source Notes

These sources support factual review and changing mission status. Stable school-level explanations were cross-checked against standard astronomy and physics references. URLs are printed for editorial traceability; access dates should be recorded during final pre-publication review.

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Information cut-off 29 June 2026. Future mission dates and outcomes must be rechecked before reprinting.

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